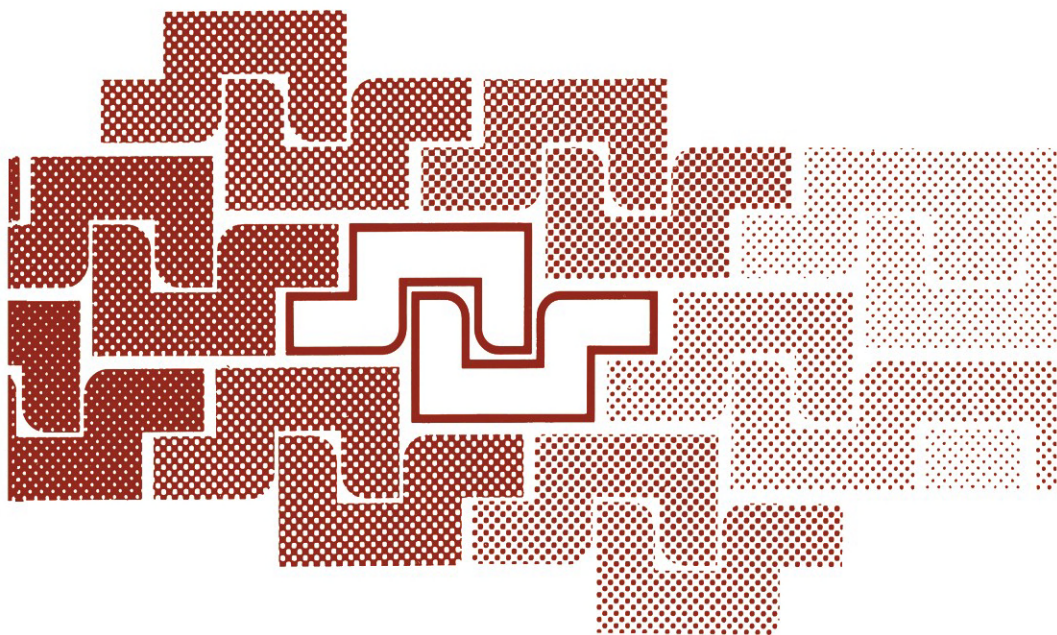




Social Support: Theory, Research and Applications

Edited by

Irwin G. Sarason and Barbara R. Sarason

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Theory, Research and Applications

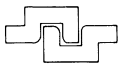
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Social Support: Theory, Research and Applications

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Preface

"No one is rich enough to do without a neighbor."
Traditional Danish Proverb

This bit of Danish folk wisdom expresses an idea underlying much of the current thinking about social support. While the clinical literature has for a long time recognized the deleterious effects of unwholesome social relationships, only more recently has the focus broadened to include the positive side of social interaction, those interpersonal ties that are desired, rewarding, and protective. This book contains theoretical and research contributions by a group of scholars who are charting this side of the social spectrum.

Evidence is increasing that maladaptive ways of thinking and behaving occur disproportionately among people with few social supports. Rather than sapping self-reliance, strong ties with others - particularly family members - seem to encourage it. Reliance on others and self-reliance are not only compatible but complementary to one another. While the mechanism by which an intimate relationship is protective has yet to be worked out, the following factors seem to be involved: intimacy, social integration through shared concerns, reassurance of worth, the opportunity to be nurtured by others, a sense of reliable alliance, and guidance.

The major advance that is taking place in the literature on social support is that reliance is being placed less on anecdotal and clinical evidence and more on empirical inquiry. The chapters of this book reflect this important development and identify the frontiers that are currently being explored.

The book has five parts. Part One is concerned with several theoretical and methodological issues. What is the relationship between actual and perceived support? How can social support be assessed? Can it be investigated from an experimental standpoint?

The papers in Part Two look at social support in a developmental context and consider some of the individual difference variables, such as personality and sex, that may be related to social support. In addition, they consider the role of network variables that bear upon the quality and quantity of social support and may also be correlated with personality and developmental experiences.

Part Three deals with what happens when there are deficiencies in either the quality or quantity of support or when there are deficiencies in both. The topic of loneliness is an example of many human experiences (social support is another example) which are neglected or ignored by researchers for a long time and then suddenly become exciting fields of inquiry. It is encouraging to see the increasing theoretical and methodological sophistication being brought to bear in the study of loneliness. The chapters of Part Three are especially valuable because of

their attempts to link the social support and loneliness concepts and their analysis of social relationships from the perspective of the individual's social perceptions.

Part Four considers the interrelationship between stress and social support in personal maladaptation. Why is it that some people are able to resist being overwhelmed by untoward events, while others succumb to them? In what ways are social support deficiencies related to coping failures that result in unhappiness, and disorder? The chapters of Part Four delineate a number of interpersonal factors (for example, loss and other blows received in the course of living) that play important roles in human disorder.

The chapters of Part V reflect the complexity that almost always becomes apparent when a phenomenon receives careful study. While the idea that interventions might be designed to make up for deficiencies in social support makes perfectly good sense, the chapters of Part Five make clear that the ways to implement such interventions are by no means obvious. One impediment to easy implementation is the fact that there are limits to how much support a person can give. One reason for these limits is that there are costs attached to caring about and for other people. Social support can provide a powerful buffer against stress but the communication of support in a way that does not unduly tax the communicator and nurturer needs to be better understood.

All of the chapters of this book are based on papers presented at an Advanced Research Workshop held at Chateau de Bonas, France, September 19-23, 1983. This international meeting, attended by leading researchers actively studying social support, was sponsored and supported by NATO's Scientific Affairs Division. While this volume is the most tangible product of the meeting, the interchanges among the participants were highly stimulating and have led to many contacts and collaborations among those who attended the meeting.

We are indebted to Liz Klein and Gregory Pierce for their help in the production of this book.

Irwin G. Sarason
Barbara R. Sarason

TABLE OF CONTENTS

Part I THEORETICAL AND METHODOLOGICAL ISSUES

Chapter 1	CONCEPTUAL AND THEORETICAL DILEMMAS FACING SOCIAL SUPPORT Brian L. Wilcox and Eric M. Vernberg	3
Chapter 2	SOCIAL SUPPORT: THEORETICAL ADVANCES, RECENT FINDINGS AND PRESSING ISSUES Toni C. Antonucci	21
Chapter 3	SOCIAL SUPPORT - INSIGHTS FROM ASSES- MENT AND EXPERIMENTATION Irwin G. Sarason and Barbara R. Sarason	39
Chapter 4	SOCIAL SUPPORT AND PSYCHOLOGICAL WELL- BEING: THEORETICAL POSSIBILITIES Peggy A. Thoits	51
Chapter 5	MEASURING THE FUNCTIONAL COMPONENTS OF SOCIAL SUPPORT Sheldon Cohen, Robin Mermelstein, Tom Kamarck and Harry M. Hoberman	73
Chapter 6	SOCIAL SUPPORT AND SOCIAL HEALTH Robert M. Kaplan	95

Part II HUMAN DEVELOPMENT, PERSONALITY AND SOCIAL NETWORKS

Chapter 7	SOCIAL NETWORKS AND THE ECOLOGY OF HUMAN DEVELOPMENT: THEORY, RESEARCH AND APPLICATION Barton J. Hirsch	117
Chapter 8	LONGITUDINAL COURSE OF SOCIAL SUPPORT AMONG MEN IN THE BALTIMORE LONGITUDINAL STUDY OF AGING Paul T. Costa, Jr., Alan B. Zonderman and Robert R. McCrae	137
Chapter 9	INTIMACY, SOCIAL SUPPORT, AND LOCUS OF CONTROL AS MODERATORS OF STRESS Herbert M. Lefcourt	155

VIII

Social Support

- Chapter 10 COPING STYLES, SOCIAL SUPPORT AND SEX-DIFFERENCES 173
P.B. Defares, M. Brandes, C.H.Th. Nass and J.D. van der Ploeg
- Chapter 11 A CONCEPTUALIZATION OF PROFESSIONAL WOMEN'S INTERPERSONAL FIELDS: SOCIAL SUPPORT, REFERENCE GROUPS, AND PERSONS-TO-BE-RECKONED-WITH 187
Jeanne M. Plas, Kathleen V. Hoover-Dempsey and Barbara Strudler Wallston
- Chapter 12 FROM SOCIAL SUPPORT TO SOCIAL NETWORK 205
Barry Wellman, with the assistance of Robert Hiscott

Part III LONELINESS AND PERCEIVED SUPPORT

- Chapter 13 THE PSYCHOLOGY OF LONELINESS: SOME PERSONALITY ISSUES IN THE STUDY OF SOCIAL SUPPORT 225
Warren H. Jones
- Chapter 14 THE FUNCTIONS OF SOCIAL BONDS: PERSPECTIVES FROM RESEARCH ON SOCIAL SUPPORT, LONELINESS AND SOCIAL ISOLATION 243
Karen S. Rook
- Chapter 15 LONELINESS RESEARCH: BASIC CONCEPTS AND FINDINGS 269
Letitia Anne Peplau
- Chapter 16 PERCEIVED SUPPORT AND SOCIAL INTERACTION AMONG FRIENDS AND CONFIDANTS 287
Kenneth Heller and Brian Lakey

Part IV STRESS, COPING AND MALADAPTION

- Chapter 17 LIFE STRESS AND HUMAN DISORDER: CONCEPTUALIZATION AND MEASUREMENT OF THE DISORDERED GROUP 303
Richard A. Depue and Scott M. Monroe
- Chapter 18 LIFE EVENTS, SOCIAL SUPPORT AND CLINICAL PSYCHIATRIC DISORDER 321
E.S. Paykel

Table of Contents

Chapter 19	SOCIAL SUPPORT, LIFE EVENTS AND DEPRESSION	349
	George W. Brown and Antonia Bifulco	
Chapter 20	SOCIAL SUPPORT AND CHILDREN OF DIVORCE	371
	Irwin Sandler, Sharlene Wolchik and Sandy Braver	
Chapter 21	LIMITATIONS OF SOCIAL SUPPORT IN THE STRESS PROCESS	391
	Stevan E. Hobfoll	

Part V HELPING AND THE COSTS OF CARING

CHAPTER 22	THEORY INTO PRACTICE: ISSUES THAT SURFACE IN PLANNING INTERVENTIONS WHICH MOBILIZE SUPPORT	417
	Benjamin Gottlieb	
Chapter 23	SOCIAL SUPPORT AND THE ALLEVIATION OF LOSS	439
	Margaret S. Stroebe and Wolfgang Stroebe	
Chapter 24	REACTIONS TO VICTIMS OF LIFE CRISIS: SUPPORT ATTEMPTS THAT FAIL	463
	Camille B. Wortman and Darrin R. Lehman	
Chapter 25	THE COSTS OF CARING: A PERSPECTIVE ON THE RELATIONSHIP BETWEEN SEX AND PSYCHOLOGICAL DISTRESS	491
	Ronald C. Kessler, Jane D. McLeod, and Elaine Wethington	
	Author Index	507

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PART I

***THEORETICAL AND METHODOLOGICAL
ISSUES***

CONCEPTUAL AND THEORETICAL DILEMMAS FACING SOCIAL SUPPORT RESEARCH¹

Brian L. Wilcox and Eric M. Vernberg

University of Virginia

The past two decades have witnessed an explosion in the number of studies investigating the relationships between the occurrence of a wide variety of social stressors and both physical and psychological symptomatology. These studies come from a number of different fields and a variety of perspectives. Many of these studies have examined the relationships between a variety of specific social stressors and health outcomes. Investigators have focused on reactions to a variety of disasters, both natural (Erikson, 1976) and man-made (Baum, Gatchel, & Schaeffer, 1983), as well as more common stressors such as marriage (Rausch, Goodrich, & Campbell, 1966), divorce (Hetherington, Cox, & Cox, 1977), job loss (Cobb, 1974), retirement (Bell, 1975), and bereavement (Lindemann, 1944). A second line of research has focused on the joint or accumulated contributions of a variety of life events to both physical and psychological distress (Dohrenwend & Dohrenwend, 1974; 1981; Rabkin & Struening, 1976). The relatively modest relationship between measures of social stressors and measures of symptomatology and illness behavior has led many researchers to explore the ways in which a variety of biological, psychological, behavioral, and situational factors might moderate the relationship between life stressors and health outcomes (Cohen, Horowitz, Lazarus, Moos, Robins, Rose, & Rutter, 1982; Jenkins, 1979; Johnson & Sarason, 1979). One particular factor which has received more attention than all others combined is social support. Several reviews which evaluate the empirical literature and consider conceptual, methodological, and theoretical issues have been published in the past few years (Cohen & McKay, in press; Cohen & Wills, 1983; Heller & Swindle, 1983; Silver & Wortman, 1980; Thoits, 1982).

The intent of the present paper is to contribute further to

this set of evaluative reviews. More specifically, we will attempt to summarize some of the conceptual and methodological dilemmas confronting social support research and offer suggestions concerning what we believe to be promising directions for research and theory. We will argue that social support research has become mired in the same trap that psychotherapy outcome research found itself in during the early to mid-1950's. A necessary starting point for such a discussion is the definition of social support. In developing this definition, we will consider the mechanisms by which social support is presumed to affect health and well-being. We will review definitions of both the conceptual and operational variety. Following this rather lengthy presentation, we will examine a series of questions which are concerned with the parameters constraining the consequences of varying levels of social support.

What is social support?

The term "social support" has been around for quite some time, although it has only recently entered the lexicon of the social scientist (cf. Cobb, 1976; House 1981). Dozens of conceptual definitions of social support have been offered. For example, Caplan (1974) suggests that social support systems consist of "continuing social aggregates that provide individuals with opportunities for feedback about themselves and for validations of their expectations of others" (p. 4). These supportive others are said to provide information and cognitive guidance, tangible resources and aid, and emotional sustenance in times of need. Cobb (1976) provides a definition of social support which differs markedly from Caplan's. Excluding tangible aid and resources, Cobb (1976) defines social support as "information leading the subject to believe that he is cared for and loved. . .esteemed and valued. . .[and] belongs to a network of communication and mutual obligation" (p. 300). In a later paper, Cobb (1979) offers descriptions of three additional forms of (non-social) support: instrumental support (counseling), active support ('mothering'), and material support (goods and services). House (1981) suggests that "social support is an interpersonal transaction involving one or more of the following: (1) emotional concern (liking, love, empathy), (2) instrumental aid (goods or services), (3) information (about the environment), and (4) appraisal (information relevant to self-evaluation)" (p. 39). Many definitions offered are rather circular ('support is behavior that is supportive'), and far less explicit than the above definitions, and thereby fail to lend themselves to adequate operationalization. Indeed, Pearlin and Schooler (1978) have noted that "the very term social supports prejudices an effect of social ties that empirically is still only putative" (p. 204).

In spite of the diversity of approaches taken to the definition of social support, some clear commonalities and

differences in orientation have emerged in the past few years. These differences and commonalities are particularly obvious when one also considers the operational definitions most commonly used. One commonality apparent in the above definitions is that social support is frequently considered to be a multidimensional construct. While some investigators have failed to reflect this multidimensionality in their measures of support, most researchers involved in the development of social support measures identify and attempt to assess several different categories of support. Some of the category labels which appear in a number of scales include emotional support, esteem support, belonging support, network support, appraisal support, tangible support, instrumental support, and informational support. The need for multidimensional measurement of social support has been extensively developed elsewhere (Cohen & McKay, in press; Wilcox, 1979). Briefly, it has been hypothesized that support functions in a stressor-specific fashion. Stressors vary in the types of adaptational demands they make, and the various categories of social support differ with respect to the type of adaptational demands they can moderate. Social support is effective in minimizing the negative effects of stressors only when there is congruence between adaptational demands and support resources.

Before further developing our definition of social support, we would like to briefly consider the mechanisms through which support is believed to influence health. Our thinking on this issue has been heavily influenced by the work of Richard Lazarus and his colleagues at the Berkeley Stress and Coping Project (Lazarus, 1966; 1980; Lazarus, Cohen, Folkman, Kanner, & Schaefer, 1980; Lazarus & Launier, 1978). This orientation holds that psychological stress is experienced by people when they confront a situation that requires some form of adaptation on the part of the person which seriously taxes or exceeds his or her adaptational capabilities. The perception (or cognitive appraisal) of the objective stressor plays a central role in the stress and coping process. The cognitive appraisal process is frequently divided into two processes: primary appraisal and secondary appraisal. We shall use the terms "event appraisal" and "resource appraisal", respectively, to describe these processes. Event appraisal involves the evaluation of the stressfulness of the environmental event according to whether it involves threat, harm-loss, and/or challenge. Resource appraisal, on the other hand, involves the evaluation of one's coping repertoire. The appropriateness of the event appraisal is evaluated with reference to the coping options the individual believes are available. "The resources a person believes are available are arrayed psychologically against the dangers and harms being faced" (Lazarus & Launier, 1978, p.308). This is the first stage in the stress and coping process at which social support operates, and many researchers speculate that this is the point at which support plays its major role (cf. Cohen & McKay, in press; Wilcox, 1979).

Lazarus and Launier (1978) provide an example in which a person preparing for a job interview appraises the situation in light of tangible social support believed to be available: "As things stand now I will probably be rejected. This is a very damaging outcome because I have no other job opportunities. If I had the ability to deal effectively with the interview I could be hired, but I don't. However, I have a good friend who knows the personnel manager, and I think he will help me" (p. 306-307). Thus, the initially threatening appraisal is reappraised as moderately benign.

Even when the processes of event and resource appraisal lead to stress reactions, these initial appraisals may be reevaluated as new information about the stressor or the individual's coping options comes to light. A person may recall, after experiencing considerable emotional turmoil, that an old friend has had extensive experience in dealing with similar situations and is likely, if called upon, to offer valuable assistance. Thus, the perceived availability of social support may result in the reappraisal of an initially threatening situation as less problematic. While such a reappraisal does not alter the nature of the stressor, it may reduce the person's initial emotional reaction to the event. Cobb (1974) has proposed that social support plays a major role in the coping process by keeping the person's affect under control, thereby allowing the person to focus attention on the tasks necessary to deal with the objective situation.

The experience of stress in response to an environmental event will not invariably lead to maladjustment. When a person-environment transaction is defined as stressful, a person will typically engage in coping activities (cf. Lazarus & Launier, 1978, for a discussion of the dilemmas involved in defining coping). Seeking and receiving (or simply receiving without seeking) social support is clearly a major form of coping activity. A handful of studies have sought to investigate the types of interactions which are viewed as supportive and contributing to coping (Gottlieb, 1978; Wilcox, in press). Not surprisingly, the empirically-derived categories are quite similar in nature to the conceptually-based categories presented earlier. Respondents in these studies suggest that coping is often facilitated by communication of trust, reassurance, love, empathy, and caring. There are a number of mechanisms by which communication of these types of information might have salubrious effects. Pearlin and his colleagues (Pearlin, Lieberman, Menaghan, & Mullan, 1981) have suggested that emotional support seems to protect individuals indirectly from the negative consequences of stress by reinforcing their sense of mastery and self-esteem. Presumably, the individual with a strong sense of self (facilitated in part by the presence of emotional support) will be better able to mobilize other coping resources than the person who must also deal with a diminished sense of self. Others

have proposed that emotional support may decrease the reactivity of the neuroendocrine apparatus, making the person less physiologically reactive to psychological stress (Henry & Stephens, 1977; Moss, 1973).

Coping can be facilitated in a variety of ways by other forms of social support. In many situations, the provision of instrumental support (money, task assistance, direct intervention on behalf of the recipient) can lessen the load of coping with the stressor or alter the nature of the stressor itself (such as when the stressor involves a loss of material resources). Informational support (advice, directives, or information communicated directly or via social comparison) can facilitate coping by encouraging forms of cognitive or behavioral coping which might increase stress resistance, redirect inappropriate coping activities, and result in the ability to tolerate increased levels of stress (House, 1981; House & Wells, 1978). For example, studies have consistently found that the appropriate utilization of formal helping services is frequently preceded by a referral from a family member or friend (Gottlieb & Hall, 1980). Appraisal support (feedback relevant to self-evaluation through processes such as social comparison) may, like emotional support, result in enhanced self esteem which can facilitate coping in a variety of ways. Additionally, researchers have pointed to several other interpersonal mechanisms not fitting nicely into the above categories which appear to facilitate coping. Interactions which distract one's attention away from the stressor (Gottlieb, 1978; Wilcox, in press) or focus one's attention on the positive (or less negative) aspects of the stressor (Pearlin & Schooler, 1978) may reduce the magnitude of the stress reaction simply by distracting the individual's attention from internal sensations of symptomatology (Pennebaker, 1982; Pennebaker & Skeleton, 1981).

The foregoing analysis of the potential mechanisms of social support points to another important feature of many definitions of support. Many researchers have focused their attention on what has come to be termed perceived or subjective social support, defining the construct in largely cognitive terms. Cobb's definition of social support in terms of different categories of information is one example. Turner (1981; Turner, Frankel, & Levin, in press) also defines support in cognitive terms. "In our view, social support, like crises (Miller & Iscoe, 1963) can usefully be regarded as a personal experience rather than as a set of objective circumstances or even a set of interactional processes" (Turner et al., in press, p. 14). Researchers maintaining this view tend to view the cognitive appraisal process as the major means by which support influences stress and coping.

Others place more weight on the actual nature of interpersonal transactions which reduce stress and enhance coping. Gottlieb (1981) has referred to social support as "the help that helpers extend (p. 209)," although he clearly views support in both objective and subjective terms (Gottlieb, 1983). Henderson

and his colleagues (Henderson, Byrne, & Duncan-Jones, 1981; Henderson, Duncan-Jones, Byrne, & Scott, 1980) have argued that researchers should attend to actual deficits in social relationships rather than deficits as perceived by the individual.

The empirical literature clearly indicates that health and well-being are influenced by both objective and perceived characteristics of supportive relationships. Theoretical models of the stress and coping process also suggest that perceived support and the actual provision of support will influence responsiveness to environmental events. Nevertheless, some writers have argued that social support researchers should focus on either objective or subjective support (cf. Henderson et al., 1981; and Turner et al., in press). The arguments offered for focusing solely on perceived support are couched in both theoretical and pragmatic terms. Champions of the objective viewpoint base their stance primarily on theoretical issues. Psychologists, speaking from a neo-Lewinian perspective, and sociologists, speaking from a symbolic-interactionist perspective, view social support as being primarily determined by the individual's interpretation of the environment.

Even when from the standpoint of the physicist, the environment is identical or near identical for a child and an adult, the psychological situation can be fundamentally different. . . the situation must be represented in a way in which it is real for the individual in question, that is, as it affects him (Lewin, 1936, pp. 24-25).

Those arguing for an objective definition of support feel that the assessment of perceptions gets one necessarily tangled up in the issue of the effect of individual differences (personality traits, cognitive styles, etc.) on perception of the interpersonal environment. Pragmatically, it is far simpler to develop measures of individual's perceptions of support than it is to the actual exchange of support. A number of researchers have developed instruments which collect information on the individual's perceptions of support (Barrera, Sandler, & Ramsay, 1981; Cohen & Hoberman, 1983; Procidano & Heller, 1983; Sarason, Levine, Basham, & Sarason, 1983; Turner et al., in press; Wilcox, 1981a); fewer attempts have been made to develop methods of assessing actual support provided (Henderson et al., 1981; Sandler & Barrera, 1980; Schaefer, Coyne, & Lazarus, 1981).

Our perspective on this issue (perceived vs. objective support) is that social support research should be viewed as a subset of social influence research and stress and coping research. That is, support research should represent the study of the influence the interpersonal environment has on health. As such, since social influence processes can be direct as well as cognitively mediated, we feel that both avenues of research should be pursued and that researchers should attempt to examine the relationships between these differing conceptualizations of

support. This advice, however, is more easily offered than implemented. Particularly problematic is the development of an adequate operational definition of actual (or behavioral) social support. One is tempted to define actual support as "behavior that is supportive," or "help that is helpful." That is, social support is behavior that helps the person cope with difficulties and develop new competencies. The problem with this definition, aside from its circularity, is evident when we move from the abstract discussion of "supportive behavior" to the concrete task of classifying behaviors, or classes of behavior, as supportive or nonsupportive. Is it possible to determine whether a particular interaction is supportive without observing the consequences of that interaction? Probably not. Behaviors which might appear to be supportive, which might seem to reflect empathy and understanding, may in fact be counter-therapeutic under certain conditions. A good example of such an occurrence is found in Mechanic's (1962) study of students' efforts to cope with the stress of Ph.D. examinations, in which he found that some attempts by spouses to provide support actually increased the pressure on the students. Meyerowitz (1980), in her review of psychosocial correlates of breast cancer, notes that the process of coping with a mastectomy is sometimes impeded by well-intentioned offers of support and advice from family members, friends, other patients, and the medical staff. These "supportive others" sometimes disrupt the process of selective denial, which is thought by many to aid post-operative adjustment, by encouraging the patient to deal with the problem head on. Wortman and Dunkel-Schetter (1980) review evidence suggesting that persons attempting to communicate support to cancer victims often communicate the negative affect they feel towards the victim as well. The stress and coping literature is unfortunately replete with examples of cases in which behaviors of others which are helpful under one set of circumstances are counterproductive when those circumstances change (Averill, 1979). Actions which are intended to be supportive and which, on the surface, may appear to be helpful to an observer of the interaction as well as to the recipient and the provider, may actually promote dependence, increase pressure and stress, and dilute self-responsibility and perceived self-efficacy (Coates, Renzaglia, & Embree, 1983; Fisher, DePaulo, & Nadler, 1981). The dilemma here is that the very same actions have positive effects under other circumstances. Thus the a priori determination of a particular set of behaviors as indicative of social support may be inappropriate.

A second dilemma facing the person who wishes to devise a measure of objective support concerns the inaccuracy often involved in the retrospective recall of past events. This is particularly problematic when the interval over which the past supportive behaviors are to be recalled is greater than a few (3-4) months. Indirect evidence for this contention comes from several studies which demonstrate the inaccuracies involved in the

retrospective recall of stressful life events (Jenkins, Hurst, & Rose, 1979; Monroe, 1982; Nelson, Mensch, Hecht, & Schwartz, 1972; Uhlenhuth, Haberman, Balter, & Lipman, 1977; Wilcox, 1983a; Yager, Grant, Sweetwood, & Gerst, 1981). A recent study indicates that similar distortions are found when supportive behaviors are recalled over long time periods, and that the types of recall biases displayed are systematically related to the diagnostic status (depressed vs. nondepressed) of the respondent (Wilcox, 1983b).

Clearly, then, developing an adequate means of assessing actual social support presents researchers with a substantial challenge. Perhaps we will be able to do no better than to assess variables such as satisfaction with past support. This dilemma is similar in nature to the one faced by psychotherapy outcome researchers who have struggled with the problem of defining their independent variable, psychotherapy. Indeed, the course of social support research has paralleled the early development of psychotherapy research. Until very recently, most studies seemed to be directed toward answering the question "Does social support buffer stress?" much in the same way that early psychotherapy research addressed the question "Does psychotherapy work?" Psychotherapy researchers have found that the answer to their question is "It depends!" It depends on the type of psychotherapy, who is providing it, who is receiving it, what problem is being addressed, etc. The answer to a complex question is likely to be, not surprisingly, equally complex. Social support research has, likewise, been attacking an overly broad question. While the studies directed toward the testing of the buffering hypothesis have been illuminating in certain respects, they have, to a degree, obscured the complexity of the issue. The available evidence should lead us to conclude that social support directly influences health, moderates the relationship between stressors and health, and has no influence on health. Numerous studies can be cited in support of each of these positions. The answer to the question "Does social support work?" and its corollary, "How does it work?" is "It depends!" The 'new wave' of studies investigating the relationship between social support and health have begun to exhibit an appreciation for the necessity of asking more fine-grained questions. We will turn now to a brief consideration of some of the parameters which shape the relationships between environmental stressors, health, and social support.

What kind of support? This is an issue we have already elaborated on to some degree. Social support is not a unitary construct. Cohen and McKay (in press) develop this issue extensively, and argue persuasively for what might be called a stressor specificity model of support processes. This model would suggest that experienced stress which is caused by a loss of self-esteem brought about by a personal failure, such as failing

an important examination, will probably not be moderated by the provision or perception of tangible support, whereas emotional and appraisal support may facilitate coping. The focus of this model is on the fit between adaptational demands and support resources.

The major research agenda with respect to this problem centers around the development of measures of social support which reflect the multidimensionality manifested in most conceptual formulations of support. Recent attempts to develop such multidimensional scales appear promising (Cohen & Hoberman, 1983; Turner et al., in press; Wilcox, 1979).

What kind of problem? This question raises the flip side of the issue presented above. Further development of this body of knowledge necessitates a move away from studies in which social support and health are examined in relationship to measures of accumulated stressful life events. Life event measures lump together events which make very different types of adaptational demands on individuals. Studies using such measures make it difficult, if not impossible, to detect the types of stressor-support relationships hypothesized by the specificity model. Instead, investigators should design their studies as to maximize the possibility of detecting specific stressor-support relationships. Such studies might capitalize on naturally occurring stressors (Cobb, 1974) or create a stressful situation under more controlled circumstances (Sarason et al., 1983). This line of work would be greatly facilitated by the development of taxonomies of social stressors (Magnusson, 1982).

The nature of the stressor may also dictate whether social support will be a viable coping option. Some stressors, such as marital disruption, result in a major breakdown of the social support system (Wilcox, 1981b). Some very intense stressors may disrupt support processes in a somewhat more complex fashion. For example, a number of studies report that the birth of a handicapped child in a family brings about a substantial decrease in the perceived and actual availability of support (Kazak & Marvin, in press; Kazak & Wilcox, in press; Korn, Chess, & Fernandez, 1980; McAndrew, 1976). This constriction of support resources seems to be due to a feeling on the part of the support system members that they are incapable of offering effective aid and a feeling of general discomfort in interaction with the family. Wortman and Dunkel-Schetter (1979) report similar findings in their studies of the coping behavior of cancer patients. Finally, other stressors may be so intense as to render virtually all coping resources, including social support, ineffectual.

Who is the source of support? A number of recent studies have indicated that all sources of support are not equally effective for a given problem. For example, Gottlieb (1978) found that single, low-income mothers found different sources of support

more helpful for different types of problems. In a recent study, Wilcox and Birkel (1983) found a very strong relationship between the type of problem the divorced women in their sample were coping with and the sources from which they preferred to receive support. In a laboratory study, Procidano and Heller (1983) found differential effects for perceived support from friends and family members. Additionally, a substantial corpus of studies have documented the powerful influence that the social structure within which one's support resources are embedded can have on the distribution of support (Hirsch, 1981; Moos & Mitchell, 1982).

What are the characteristics of the recipient? Another set of parameters which influence the nature of the support process can be discussed under the general rubric of recipient characteristics or individual differences. Both theory (Antonovsky, 1979) and research (Nadler, 1983) suggests that individuals vary in their need for support. Additionally, a variety of individual differences appear to influence the propensity to seek support from informal sources. These factors include sex (McMullen & Gross, 1983), age (Kahn & Antonucci, 1980; Nelson-LeGall, Gumerman, & Scott-Jones, 1983), race (Dovidio & Gaertner, 1983), and psychological constructs such as locus of control and self-esteem (Nadler, 1983). Individual differences also affect individual's reactions to aid which, in turn, influence future help-seeking and offers of aid (see Fisher, Nadler, & DePaulo, 1983, for reviews).

One last parallel between psychotherapy research and social support research bears comment. Researchers attempting to assess the efficacy of various forms of psychotherapy often pondered the effect of factors outside the therapeutic relationship on therapy outcome. Interactions with family members and friends, for example, were believed by many to have some effect, positive or negative, on the outcome of the therapy process. Analogously, social support researchers might do well to attend to the interactions of social support with other coping processes. Social support is merely one amongst a variety of potential coping resources available to an individual. Although it may prove to be a centrally important coping mechanisms, a fuller appreciation of the role of social support will best be gained by viewing support within the context of the stress and coping process in toto.

These are just a few of the issues social support researchers must confront if the field is to maintain its newly-achieved status as a legitimate domain of social science research. Psychotherapy outcome research made little headway until investigators began pursuing the question of who does what to whom for what types of problems. Similarly, we must investigate the interactions between provider, support, recipient, and problem characteristics. Doing so will necessitate our studying the relationships between social support and other coping resources. This should not be taken to imply that nothing has been or can be

learned from large scale epidemiological investigations. The epidemiological perspective is oriented towards asking broader questions whose answers are particularly important to social program planners and policy makers. Our primary concern, however, rests with the explication of the basic psychological and social processes underlying the social support phenomenon.

We have argued that support research should be placed within the context of social influence research and the stress and coping paradigm. Doing so will not necessarily solve many of the problems facing researchers, but it may make us more aware of the directions our research needs to take.

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SOCIAL SUPPORT: THEORETICAL ADVANCES, RECENT FINDINGS AND
PRESSING ISSUES

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The topic of social support has received increased attention over the last several years. At this point in time, with the benefit of much preliminary research, it is possible to move forward considering, and frequently correcting, limitations of the past and delineating the issues which are in most need of attention in the future. With this in mind, the purpose of this chapter is threefold: To outline the convoy model of social support which has benefited from previous theoretical approaches to social support but represents a life span extension of the concept. Second, to present a unique national study of the supports of older adults which has recently been conducted and which represents the source of much of the data reported in the present chapter. And, finally, to delineate several issues and provide relevant information under a general rubric which can best be summarized as present knowledge/pressing issues. By this is implied some of the intriguing, interesting, puzzling, and/or contradictory findings in the literature. Where possible, relevant data from our national study will be provided which address these issues.

THE CONVOY MODEL

In some ways the topic of social support has been pursued atheoretically. Possibly as a result of the face validity of the concept and its colloquial popularity, many people have pursued empirical research in this area without a specific theoretical perspective. At the same time, those researchers who have considered the theoretical basis of social support have done so in an age specific, cross-sectional manner (cf. Gottlieb, 1981).

Although this work has advanced the theoretical basis of social support and for that reason has made important contributions, it does not address the serious issue of continuity and change in social support over the life course. The convoy model is designed to address this issue directly.

The convoy model of social support builds upon the work of Bowlby (1969) and other attachment theorists and notes the importance of interpersonal relationships over the entire life course. First articulated several years ago (cf. Kahn, 1979; Kahn & Antonucci, 1980), this view of social support incorporates the individual's life time of experiences in the interpersonal domain and emphasizes the developmental importance of these accumulated experiences. Beginning with the infant's attachment relationship with a primary caregiver, the individual begins to learn about interpersonal relationships. Evidence already exists which indicates that these early attachment relationships have a significant impact on later childhood development. For example, using Ainsworth's (Ainsworth, Blehar, Waters & Wall, 1978) terminology of secure, anxious-avoidant, and ambivalent attachment, Arend, Gove and Sroufe (1979) have documented a relationship between type of infant attachment and later childhood competence. Other research with older individuals has traced consistency in patterns of interpersonal relationships, i.e. attachments or social support (Mussen, 1962; Antonucci & Wethington, 1981). For example, Mussen (1962) found that adolescent boys who were very masculine stereotyped in their sex role interactions were often similarly stereotyped in their behaviors as adults. Unfortunately the behaviors were considerably more adaptive in high school than they were in middle age. Therefore, although there was continuity in interpersonal style it was not similarly adaptive over the life course. Antonucci and Wethington (1981) found that adolescents who exhibited what they considered insecure attachment behaviors in adolescence, e.g. early and frequent dating, reported unsatisfactory interpersonal relationships as adults twenty years later. They were significantly more likely to report lower levels of marital and life satisfaction. At another point in the life cycle, Neugarten et al (1964), in the now famous Kansas City study, reported considerable consistencies in pre- and post-retirement behaviors for many of the individuals in their unique longitudinal sample. People who were active and involved in a great number of social interactions prior to retirement were likely to remain, proportionately speaking, more actively involved post-retirement.

Of course, these data are general in nature. However, they provide enough of a basis to assume that there is some consistency in interpersonal relationships and therefore social support over the life course. It is this assumption upon which the convoy model of support is based.

The convoy model emphasizes that the individual grows up and

matures, for the most part, surrounded by people who are close and important to him or her. Beginning with the primary caregiver, usually the mother, the tone of these interpersonal relationships is set. Infant theorists have ranged from the psychoanalytic, emphasizing instincts, to the social learning theorist, emphasizing the reinforcement contingencies provided by interpersonal interactions. Regardless of the theoretical basis, the joint influence of both inherited and environmental factors is considered important for setting the tone for future relationships.

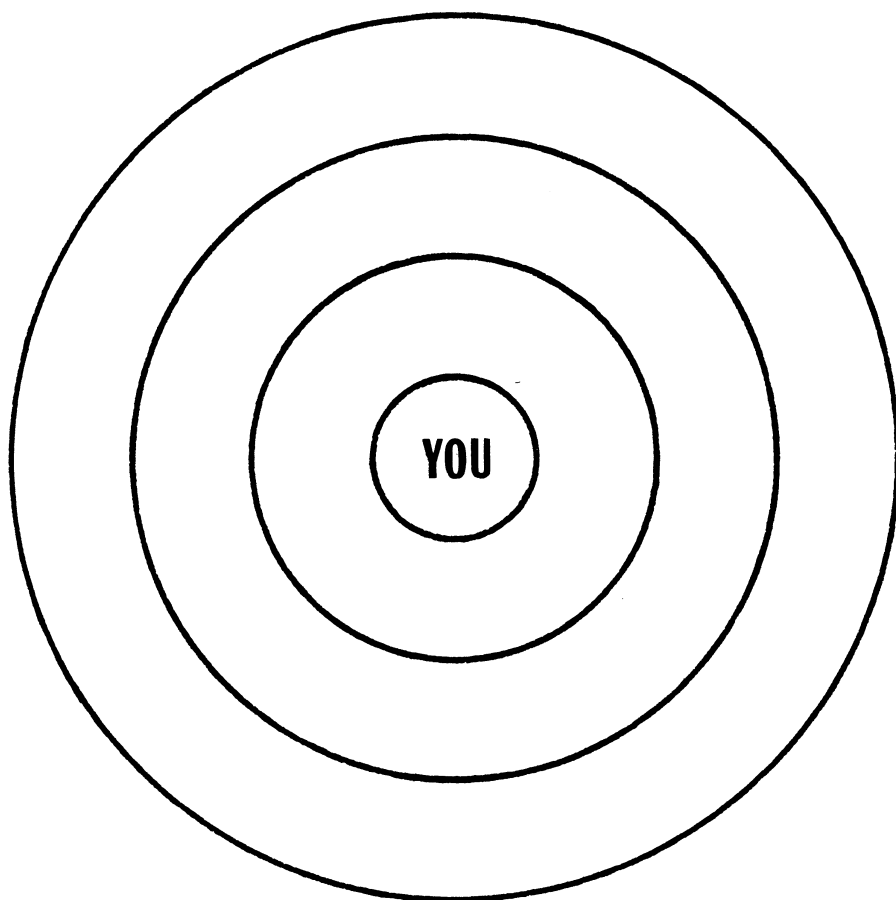
The maturing infant is exposed first to a select few primary caregivers and then to an ever broadening range of family and friends. These people represent a convoy or protecting shield through which the individual interacts with the world at large. The properties of the convoy are both stable and dynamic. For example membership in the convoy may be stable, i.e. certain relatives will remain in the convoy for an individual's entire lifetime. But even this stability is dynamic: the young child whose parent is in his or her convoy has a very different relationship with that parent at four, fourteen, and forty. On the other hand, some members of the convoy will be lost and others gained. The school mates or friends of childhood may be very important in these early years but may become less important or slip away from the convoy altogether with adulthood. At the same time, perhaps the most important interpersonal relationship in adulthood, that of marital partner, is of course not usually represented in the convoy of the young child.

What is not clear, however, is how much the earlier interactions influence the later selection and interaction with adult friends and new family members. In the convoy model we assume that there is an important etiological relationship which can be empirically examined. According to the model a close accounting of the early interpersonal interactions and the role transactions of an individual will provide insights into the individual's present and future course of interpersonal relationships and social support.

We (Kahn & Antonucci, 1980, 1981) have operationalized the convoy both developmentally and conceptually. To emphasize the various levels of importance convoy members might attain, we have pictorially represented the convoy as three concentric circles with the target person in the middle (Figure 1). Individuals to be interviewed about their networks are shown this diagram, and told that people who are in the innermost circle are those who are close and important to them and without whom life would be difficult to imagine. The remaining two circles are described as involving people who are successively less close. As will be detailed later, our work and the work of others using the same instrument indicates that people have no difficulty conceptualizing their network in this hierarchical fashion.

The convoy model of social support is designed to provide a

FIGURE 1. Network Diagram - Social networks in adult life.



broad perspective on the determinants and outcomes of social support as well as an understanding of the structure and function of support over the life course. As Figure 1 illustrates, personal and situational characteristics are considered determinants of the support convoy; structure and function of the support convoy provide basic information about the network itself and the kinds of support that are exchanged; and finally adequacy and outcomes describe the results of the support convoy as it exists for each individual and affects him or her over the life course. The term convoy is used to separate this model from the more typical network model which tends to be more static and does not consider the change and continuity that takes place over the lifetime of an individual. Only a brief summary of the model can be presented here. The interested reader is referred to Kahn and Antonucci (1980) and Antonucci (in press) for a more detailed presentation of the convoy model. Below some illustrative examples of each aspect of the model are presented to help the reader operationalize the model and become familiar with the variables explored in the program of research to be described in this chapter.

Personal characteristics are those properties of the individual which are thought to have a determining influence on both the structure and function of an individual's support network. Although not limited to these, some of the personal characteristics which we have explored include age, sex, income, marital status, other demographic characteristics, personality, and abilities. Similarly, situational characteristics which are more external to the individual but also likely to affect the structure and function of the network were explored as possible determinants of an individual's support convoy. Examples of situational characteristics include role expectations, opportunities, demands, resources, residence, organizational membership and life events. The term convoy structure refers to network composition and structural characteristics of the network over the life course. Common examples of structural characteristics include size, connectedness, stability, symmetry, complexity and homogeneity. The reader not familiar with these terms is referred to other chapters in this book and to Gottlieb (1981). Some have referred to structure as the vehicle through which social support is exchanged. The actual giving, receiving and exchange of support is commonly referred to as the function of social support. This refers to what most people probably would consider the actual social support, i.e. the commodity or thing which one person may give to another. Examples of support functions which we have explored include: confiding, respect, reassurance, sick care, talk when upset and talk about health.

The terms convoy adequacy and outcomes are used to describe the effects of the support convoy on the individual. First, it is hypothesized that as a result of the structure and function of the individual's support convoy, he or she accesses the adequacy of

the convoy's supportiveness. People may come to think of their network as too demanding; not understanding; as consisting of too few people or of people who are not dependable. In general, people develop a sense of satisfaction or dissatisfaction with their convoy of support. This sense of convoy adequacy is then translated into outcomes. In this model the term outcomes is used to provide a general rubric under which both mental and physical health measures might be included. Common examples of outcome measures include life satisfaction, well being, health, negative affect, and happiness.

SUPPORTS OF THE ELDERLY: A NATIONAL STUDY

With the convoy model in mind but necessarily (at least initially) limited to a cross-sectional sample, the national study Supports of the Elderly: Family/Friends/Professional (SSE) was designed. The study title is something of a misnomer since the sample included middle aged as well as elderly people. Seven hundred eighteen non-institutionalized adults aged 50 and over from the coterminous United States were interviewed by the Survey Research Center staff of The University of Michigan. The sample was roughly divided into three age groups: 50-64; 65-74; and 75+. Since the original research program did focus on the elderly, two interesting aspects of the study should be noted. In order to ensure an adequate number of respondents in the 70 and over age bracket, this age group was oversampled. In addition, two to three network members of respondents 70 years of age and over were also interviewed. These are labeled network interviews. The age and sex distribution of the main principal respondent sample is presented in Table 1.

The interview, which averaged approximately one hour in length, assessed all six aspects of the model--although some in greater detail than others. Half of the interview consisted of questions concerning social support. Each respondent was provided with the diagram of three concentric circles described earlier. The specific instructions regarding circle placement were as follows. "This is you in the middle. The first circle should include only the one person or persons that you feel so close to that it would be hard to imagine life without them. People you don't feel quite that close to, but who are still important to you would go in the second circle. People to whom you feel less close but who are still important to you, would go in the third circle. Circles can be empty, full, or anywhere in between."

After the circle information was ascertained, the interviewer then assessed the structural and functional characteristics of the network as well as the personal and situational characteristics outlined above. In addition, the interview included sections on the perceived adequacy of the network and numerous outcome measures such as life satisfaction, happiness, health and

TABLE 1

Age and Sex Distribution of SSE Sample

	50-64	65-74	75-95	Total
	% (N)	% (N)	% (N)	% (N)
Men	50 (150)	31 (92)	19 (56)	42 (298)
Women	44 (183)	32 (135)	24 (102)	59 (420)
Total	46 (333)	32 (227)	22(158)	

Kahn and Antonucci, 1984

negative affect. The extensive data concerning social support acquired from this study will be used to address some of the important issues that have developed in the field. These are discussed below.

PRESENT KNOWLEDGE/FUTURE GOALS

In this section, some of the interesting and sometimes puzzling results that have been reported in the literature are considered with some speculation about what the underlying etiological factors may be.

Qualitative vs. Quantitative Support

One of the most intriguing questions that is asked in the social support literature concerns the relative importance of qualitative versus quantitative support. The general question asks, "Is there some quantity of support that must be provided either totally or in specific areas to produce a positive outcome or is the amount of support provided irrelevant as long as the quality of that support is high?" Although this question is not frequently addressed, the majority of studies which explore the relationship between quality versus quantity of social support and

positive outcomes indicate that quality of support is the more critical variable (e.g. Duff & Hong, 1982; Porritt, 1979). Of course, to obtain "quality support" some minimum quantity of support is necessary.

In a recent investigation, Antonucci and House (1983) used the Supports of the Elderly data to explore the relationship between social support and health. In this study support from three sources; spouse, child and friend, was examined in relation to subjective health. In addition, the respondent's global feelings of satisfaction with the network and the relationship between this variable and subjective health was examined. Comparing the counts of support provided by different sources (spouse, child and friend) as a quantitative assessment of social support and the overall evaluation of network satisfaction as a qualitative assessment of social support, the differences are clear. The quantitative analysis indicates that 3 to 18% of the variance in subjective health can be accounted for by the amount of support received from spouse, child and friends. If however, these quantitative assessments are replaced by the more qualitative assessment of network satisfaction, the percent of variance accounted for in subjective health is greatly increased and sometimes doubled. Thus, it seems clear that "qualitative" support is a better predictor of positive outcomes than "quantitative" support.

Of course, the question of the relative impact of quantity versus quality of support points to a more fundamental question. If one accepts the assumption that quantitative data concerning social support is important as a predictor of positive outcomes only because it frequently represents an indirect assessment of the quality of support an individual is receiving, the next question must be what constitutes qualitative support and why is it important. One possibility is that what we have come to consider "qualitative support" is actually a subjective assessment by the individual. Since most of the measures of social support are self-report, there is some element of subjective interpretation and assessment involved with these measures. When asking questions concerning a quantitative assessment, e.g. does your spouse do this for you, your child, your friend, etc., the responses, although self-report, may be somewhat objective. On the other hand, if asked how well does your network do this for you or how satisfied are you with the way your network does this for you, a larger element of subjective assessment is introduced. Thus one might argue that a person who evaluates their support network positively is also more likely to report positive outcomes such as life satisfaction, well-being or even physical health.

The relationship between the subjective nature of the support variables and the outcome variables is not understood etiologically. Researchers in this field have for the most part assumed that people who receive social support are better off as a result of it and therefore are more likely to report positive

outcomes. However, the underlying etiology of this relationship is not clear. It may be, for example, that a certain degree of social competence is required for a person to seek or receive social support from others (Heller and Swindle, 1983). This social competence in turn is related to a positive assessment of the support network. People who are socially competent may therefore score better on assessment outcomes either because they are generally more competent or because that competence permits them to receive social support which then influences outcomes in a positive way. A similar argument could be developed using Rotter's (1966) notion of trust in people. Those who trust in people are more likely to receive support, to positively evaluate that support and to score positively on general outcome measures.

In sum, it seems clear that the distinction between quality and quantity of support is an important one. But even more clear is the lack of knowledge concerning the basic etiology of the relationship between social support and other variables.

Negative Support and Support Reciprocity

Another important question concerning social support has to do with negative support. Many people feel this term is counterintuitive. Social support is "supposed" to be positive. However, as many clinicians can attest, this is not always the case. Support which is intended by the provider to be positive, may be negative either because the objective outcome of the support provided is negative or the recipient of the support perceives the support negatively. Examples might include overprotection, reinforcement of health damaging behaviors or assistance provided on demeaning or debilitating terms. Although the Supports of the Elderly study does not lend itself to the exploration of this issue it is clearly an important area which requires further research. In the same way that social support is heralded as a way of helping people with a life crisis, it is probably true that negative support can have the opposite effect. Thus the smoking or drinking friends of the person who is trying to quit are probably "supportive" but of these maladaptive behaviors. Similarly, the overprotective parent who refuses to permit an adolescent child to engage in any social activities with opposite-sexed peers, provides him or her with no opportunity to prepare for adult heterosexual relationships and thus interferes with normal development. It has been shown that non-support or negative support relationships especially among family members can increase the probability of negative outcomes including ill health (Nuckolls, Cassel, & Kaplan, 1972; Berkman & Syme, 1979; House, Robbins & Metzner, 1982; Medalie & Goldbourt 1976). These types of relationships must be better understood so that we can intervene and prevent their development and maintenance.

One type of negative support that has been explored is non-reciprocity. Is it important to both give and receive

support? Does non-reciprocity have a negative effect on well-being? It has been suggested and some preliminary data indicate, that in order for social support to have a positive effect reciprocity is necessary. However, several issues need to be addressed. For example, is in-kind reciprocity necessary from the same people and/or over time? Wentowski (1981) has noted that there is a norm of reciprocity and that this varies depending upon the level of intimacy between the two parties. For the people who are not close, reciprocity is expected both directly and immediately. On the other hand, for closer more intimate relationships there appears to be a less formal accounting of the reciprocity. It is not as necessary to reciprocate immediately or in kind. The intimacy of the relationship allows some latitude and it appears that reciprocity is assumed over time. It is as if a distant network member feels they must be reimbursed immediately, but with more intimate network members it is acceptable to assume that reciprocity will occur over the years and as needed. The importance of a life span perspective in understanding the support convoy becomes particularly clear with this issue. If an intimate relationship exists it is possible that the individual will feel more comfortable receiving support from that person. They may feel they have built up a "support reserve" because they have provided the other person with support in the past or because they feel that they will be an important source of support in the future.

Two sets of analyses have been conducted on the Supports of the Elderly data which are relevant to this issue (Ingersoll & Antonucci, 1983; Antonucci & Israel, 1984). In the original interview of principal respondents each person was asked if they provided six types of support to any of the specific members of their network they had previously mentioned. In addition, principal respondents were also asked if their network members provided any of the same six types of support to them. Thus, data are available which document reciprocity according to our principal respondents. It should be noted that this is reciprocity as perceived by our principal respondents. Thus far we (Ingersoll & Antonucci, 1983) have examined this reciprocity by three relationships, spouse, child and friend. The results are quite encouraging indicating that for the most part people perceive a great deal of reciprocity with their network members, especially spouse. Spouses perceive between seventy-one and eighty-six percent reciprocity depending upon the specific type of support. Our respondents perceive less reciprocity with children, between forty-seven percent and seventy percent and even less reciprocity among friends, between fifty-three and sixty-three percent. Overall, the percentage of perceived reciprocity is quite high.

People who report that their relationships are non-reciprocal are most likely to indicate that they provide more support than they receive. This type of non-reciprocity is especially existent

among children and to a lesser extent among friends. Age, need for functional care, and retirement status tend to predict non-reciprocity. As one might predict older respondents and people requiring functional care, are more likely to report non-reciprocity. They report receiving more support from spouse, children and friends than they provide. On the other hand, retirees report that they provide more support for their children than non-retirees suggesting a different type of non-reciprocity.

Respondents who indicate that their network is non-reciprocal are likely to perceive their network as not adequate and to report less positive outcomes. Our specific analyses indicate that they are more likely to report that their network is too demanding and to indicate less satisfaction with their network. They are also more likely to report higher levels of negative affect. Additional analyses indicate that reciprocity with spouse is associated with higher levels of happiness and that reciprocity with friends is associated with higher levels of life satisfaction (Kahn & Antonucci, 1984). These data provide solid evidence that the perception of reciprocity is an important element in the role and effect of social support on an individual.

I have emphasized thus far that the above analyses represent the respondent's perception of reciprocity. This emphasis has been made because additional analyses comparing the report of our principle respondents concerning support received from them by their network members with network respondents' reports concerning the support they provide to the principal respondents indicate substantially different results. Approximately 500 network respondents were interviewed. Matching their responses with the principal respondent who nominated them, it is interesting to note that the agreement rate was only around 50%. This includes 80 people nominated by the principal respondents but not even mentioned by the network respondent as part of their network. As the 50% agreement figure suggests, when the amount of veridicality, agreement between principle and network respondents, is used to predict principal outcomes, these relationships are predominantly non-significant.

Thus, these data suggest that perceived reciprocity is an important component in understanding the relationship between social support and outcome. However, actual or at least less subjective reports of reciprocity: a) do not seem to be significantly related to perceived reciprocity and b) do not have a significant impact on outcomes.

Sex Differences

There is much available data which indicates that there are fundamental sex differences in both the nature and function of social support. Women clearly have more extensive and more varied networks than men. Men generally maintain a close and intimate tie with only one person, their spouse. Similarly women report

both providing and receiving more support than men. In the study mentioned earlier by Antonucci and House (1983) examining the relationship between social support and health, women were consistently more positively influenced by social support than men and consistently reported receiving more support from others than men. This finding is not unique in the literature, Troll and others (Troll & Turner 1979; Troll, Miller & Atchley, 1979) have noted that women tend to be kinship carriers. They maintain the familial and other social obligations. However, the ramification of these sex differences and the effects of possible changes in women's role as kinship keepers is yet to be explored. Several people (Hess & Markson, 1980; Block, Davidson & Grams, 1981; Powers & Bultena, 1976) have hypothesized that women's greater diversity of support networks provides them with a necessary cushion to cope with the long years of widowhood. Men, on the other hand, seem to do quite well with one primary support person as long as that person is available. This makes meaningful the findings that widowers, unless remarried, and men experiencing late life separation or divorce have a much more difficult time adjusting (c.f. Chiriboga, 1982).

This sex difference needs to be further understood for several reasons. First it has practical implications for the kind of support networks one might expect to activate in times of need. Men would rely primarily on their spouse while women would rely on more varied sources of support. In addition, some understanding of the etiology of this difference would be helpful to anticipate and deal with future changes. Thus, if women have maintained these diverse networks and the role of kinship keeper because of their traditional role as homemaker, the increased tendency for women to work outside the home most of their adult lives will have a very negative effect on the maintenance of this role. It may be that special interventions are warranted and should be targeted.

Friend vs. Family Support

And finally the role of friends versus family in the support network must be more fully understood. The research alluded to previously on both non-reciprocity and the relationship between health and social support suggests that family and friendship support are different in some fundamental ways. Reciprocity analyses (Ingersoll & Antonucci, 1983) indicate that there is considerably less perceived reciprocity among friends than between marital partners in the two types of support explored, confiding and care when ill. Similarly in the paper exploring the relationship between health problems and social support, support from family was considerably more important than support from friends, especially for men. On the other hand, Arling (1976), Wood and Robertson (1978) and others have reported that support from friends is a better predictor of well-being than support from family. Such results in light of the previous findings may seem

counterintuitive. Further research is warranted so that the mechanisms by which social support has these differential effects can be better understood.

I hypothesize that friends and family are judged by different standards and their effectiveness as support providers is based on these different standards. For example, family are "supposed" to provide support especially in times of crisis. An extreme interpretation of this would therefore suggest that the family providing support in times of crisis is perceived as doing what they are obligated to do by virtue of their life time relationship to the family member. Therefore, although of course grateful for the support received, the support is expected and does not warrant too much attention. However, if support were not provided under these circumstances it would have a very negative effect since it would be judged as an obligation not met, a debt not paid. Thus for family there is a relatively strict code of expectations. Meeting this code is expected, not meeting it is assessed quite negatively. Friends, on the other hand, are judged by totally different standards. They have no obligations to the individual, therefore not performing support functions would not be assessed negatively. It is neither obligated nor required. However, when support is provided since it is non-obligated it is evaluated very positively. This interpretation would explain the frequent finding that support provided by friends is more positively related to well-being than support provided by family. I expect that a corollary finding which to my knowledge has not been documented, can be anticipated. Support not provided by family has a much more negative effect on well-being than support not provided by friends.

As families get smaller a better understanding of the family versus friendship support will be helpful in meeting the challenges of changing structures of support networks.

SUMMARY AND CONCLUSIONS

Recent advances and empirical research on the topic of social support have provided us with a basis upon which to build. A main focus for research in the future must be a theoretical understanding of the concept of social support and the mechanisms through which it operates. The convoy model of social support outlined in this paper proposed such a theory within a life span framework. In addition, several major issues that must be addressed theoretically have been considered empirically. First, the distinction between quality and quantity of social support is well recognized as is the general finding that quality of support is more predictive of positive outcomes. Future research needs to focus on the underlying basis for this and the nature of the etiological relationship between qualitative support and positive outcomes. A second focus of support research suggested in this

paper focuses on the notion of reciprocity. Is it necessary for social support to be both given and received or is unidirectional support, i.e. only giving or only receiving, support, sufficient? Our data suggest that reciprocity, that is perceived rather than objective reciprocity, is more effective. However, since the relationship between perceived reciprocity and more objective measures of reciprocity is not high, future research should explore the factors that contribute to perception of reciprocity. Third, it is important, especially in light of the changing roles of women in this society to understand the well documented differences in social support between men and women. Is the wider support network of women likely to shrink with the acquisition of the work role or is the difference based on something other than traditional role differences? Is the broader, more multifaceted network of women necessarily more adaptive than the narrower, focus of male networks? The quality versus quantity argument considered above suggests not. And finally in a similar vein the role of family versus friendship networks needs to be understood more fully. Data thus far available suggest, as proposed in this chapter, that different expectations and evaluations are associated with two types of support. This hypothesis can be empirically tested. If supported, the effect of support from the two sources can be better understood and better predicted.

Attention to the four issues discussed in this chapter building upon the life span model of social support and the empirical evidence outlined above will constitute an important step forward in the field of social support. These emphases will provide a basis upon which to extend and implement the concept of social support in practical and applied settings.

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SOCIAL SUPPORT - INSIGHTS FROM ASSESSMENT AND EXPERIMENTATION

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Although the social support literature is growing rapidly, this field of inquiry like so many others is marked by fuzziness, a multitude of definitions of its major variables, and a lack of information about which characteristics are likely to be typical of those high or low in social support (Heller, 1983; Sarason, Sarason, Hacker & Basham, in press). Another unclear area concerns the effects of socially supportive interventions. What constitutes an effective supportive intervention? What are its important elements? Whom does it help? Is it helpful to everyone or only to those individuals who are lacking in social support?

This paper is aimed at pointing out the progress being made in a number of these areas especially with regard to the characteristics of persons high and low in social support. One of the important things to remember as work on all these questions progresses is that social support, important as it seems to be, is likely not to be a unitary concept. Investigators will profit from keeping this point always in view.

One important distinction that must be made in speaking of social support is the difference between perceived and administered social support. The idea of perceived support comes from the long time emphasis in social psychology and in anxiety research on the appraisal process. Perceived support refers to the individual's belief that he or she can obtain help or empathy when it is needed. Another aspect of perceived support is the degree to which the person is satisfied with the type and/or amount of support available. For example those who have an ambivalent or negatively valenced relationship with certain family members or with a partner may view themselves as able to obtain support but only at considerable psychological cost to themselves. Perceived support and especially satisfaction with

the support perceived to be available appear to have strong links to personality characteristics.

Administered social support, on the other hand, involves the actual provision of support. Much of the original interest in social support arose from the field of community psychology and primary prevention. (Iscoe, Bloom & Spielberger 1977; Heller, 1979; Cowen, 1980). Support available from community resources and from families backed or strengthened by community services was envisioned as a key to prevention of mental illness and maladaptive behavior. The social support network made up of family members and friends has been the subject matter of sociological study for some time. In particular the use of network analysis has revealed different patterns of relationships that characterize healthy and troubled adjustment and also patterns that are effective in some situations but not in others (Hammer, 1981; Mitchell & Trickett, 1980; Hirsch, 1980; Cohen and Sokolovsky, 1978). Many questions remain about the characteristics of effective administered support and how individuals' personality characteristics interact with particular types of support received. Another important issue is whether administered support is helpful in general or whether it should be directed only to those who see themselves as lacking in support.

So far, most of the work on social support has been directed toward one of two topics. The first is the relationship of social support and health with support viewed as a moderator of stress (particularly stress experienced in the form of negative life events). The second has to do with the assessment of social support through the creation of scales for its measurement. Investigators have attempted to objectify some of the theoretical ideas by construction of a number of measurement devices, mainly in the form of questionnaires that might then be used as criteria in further investigation (Barrera, Sandler & Ramsay, 1981; Cohen & Hoberman, 1983; Henderson, Duncan-Jones, Byrne & Scott, 1980; Procidano & Heller, 1983; Sarason, Levine, Basham & Sarason, 1983).

The concentration on clinical studies and measurement devices leaves untouched a number of questions. Do individuals high and low in social support differ in personality characteristics or behavior? What are the effects of different ways of administering social support and are they the same for all individuals? What techniques are useful in helping an individual raise his or her social support level? Some preliminary answers to these questions are probably more effectively obtained in a laboratory setting. Although researchers are beginning to deal with social support in this way, the number of studies is as yet small.

This paper describes the beginning of one program of assessment and experimental studies designed to provide preliminary answers to questions such as these. The research emphasizes both perceived social support and the effects of

administering support. Perceived social support in the studies to be described has been defined by the individuals' scores on the Social Support Questionnaire (Sarason, Levine, Basham & Sarason, 1983). This questionnaire yields two scores, a number or perceived availability score (SSQN) that indicates the total number of individuals named by the subject as available for support in a variety of specified situations and a satisfaction score (SSQS), the degree of satisfaction (on a six point scale) with the support perceived as available for each item.

Defined in this way, differences in perceived social support can be related to a variety of personality measures and retrospective views of experiences, such as parent-child relationships. Perceptions of lack of support and related feelings of isolation and loneliness may be associated with childhood losses of support and related feelings of inadequate or insecure relationships to important figures in childhood.

Table 1 illustrates some of the relationships between both number and satisfaction measures of the SSQ and several personality measures. Two impressions arise from an inspection of these relationships. The first, borne out by data to be discussed later in this paper is that social support may have somewhat different meanings and/or mechanisms for men and women. The second impression is that the relationships among these measures make some intuitive sense.

TABLE 1

Correlations of Social Support Questionnaire (SSQ) With Three Multiple Adjective Affect List (MAACL) Scales, and the Eysenck Personality Inventory (EPI) Scales.

	MAACL			EPI	
SSQ					
<u>Males</u>	Anxiety	Depres- sion	Hos- tility	Extra- version	Neuro- ticism
Number	-.14	-.24*	-.23*	.13	-.25
Satisfaction	-.17	-.22*	-.17	-.03	-.29
N	100	100	100	28	28
<u>Females</u>					
Number	-.30**	-.31**	-.26**	.35*	-.15
Satisfac- tion	-.39**	-.43**	-.36**	.09	-.37*
N	127	127	127	38	38

*p < .05

**p < .001.

These include the positive relationship of number of supportive persons listed and the Extraversion scale of the Eysenck Personality Inventory and the inverse relationship between the Eysenck Neuroticism scale and both SSP number and satisfaction scores. Individuals high in neuroticism might be described as neither initially attractive to others nor the source of particularly rewarding experiences in an ongoing relationship. In addition the way individuals describe their characteristic affect patterns bears a logical relationship to their self described support. Table 1 shows that depression and hostility are inversely related to both number and satisfaction scores. Neither of these affective states seem attractive to others. Depressed individuals have consistently been described as unpleasant to relate to (Coyne, 1976; Lewinsohn & Talkington, 1979). The negative feelings toward others that are characteristic of hostility are logically inconsistent with the attraction of others and the development of supportive relationships. Both depression and hostile feelings also are logically consistent with a lack of satisfaction in most kinds of interpersonal relationships. The negative relationship of anxiety to both number and satisfaction scores also helps us flesh out a picture of the kinds of personality characteristics associated with high levels of perceived support. This is especially true if we think of the cognitive side of anxiety --- the tendencies toward ruminating and worry. Why these tendencies reach significance only for females is unclear.

Another personality measure related to perceived social support is the Loneliness Questionnaire (Russell, Peplau and Cutrona, 1980). College undergraduates differing in social support level differ significantly in their scores on the Loneliness Questionnaire. (Sarason, Sarason, Hacker and Basham, in press). For both number and satisfaction measures, those low in social support also described themselves as more lonely. This relationship between measures clearly indicates that most people low in social support perceive their condition as one of isolation and discomfort rather than simply as one of autonomy and independence.

Some of the roots of the social support concept lie in John Bowlby's work on attachment (1969, 1980). In this view early childhood experiences, particularly the experience of a secure non-threatening relationship with a parental figure are required to make possible satisfying relationships in later life. Some definitions of social support depend heavily on this idea. One promising measure of social support, the Interview Schedule of Social Interaction (ISSI) (Henderson, Duncan-Jones, Byrne & Scott, 1980) is based on attachment theory. Assessment measures used in our research also suggest that memories of early experiences with parents are related to Social Support Questionnaire scores in ways consistent with attachment theory.

The Parental Bonding Instrument (PBI) (Parker, Tupling &

Brown, 1979) is a self report measure designed to investigate perceived parent-child relationships in the first 16 years of the child's life. It yields a "care" score and an "overprotection" score for the relationship with each parent. In our research this instrument has yielded only mild to moderate correlations with a variety of measures of current adjustment including a 12 item scale designed to measure general happiness and freedom from worry. Subjects who rated their parental relationships in a positive light also scored higher on their own life satisfaction and happiness, but the difference was not a large one. In contrast the SSQ yielded robust relationships with the PBI care scale for both parents. This strong relationship remained even after the level of present life satisfaction was controlled statistically to reduce any distortion of the data because of possible response sets.

The pattern of SSQ-PBI relationship was consistent with attachment theory. The importance of the relationship lay in the feelings of empathy and support and a tolerance of the child's mistakes, not in the emphasis on protection of him or her from harm. Another assessment device, the Lack of Protection Scale (Sarason, 1958) produced somewhat similar findings. Subjects high in social support (both N and S scores) reported less separation anxiety in childhood than those low in social support. However, this relationship was significant only for women (Sarason, et al., in press).

These results suggest that the hypothesized relationship between the concepts of social support and attachment may have an empirical basis. While the assessment measures used are retrospective and no statements about causality are possible, nevertheless the results are supportive of the theoretical view that both the ability to perceive a supportive network and to feel satisfied with what is perceived may be related to a specific type of early experience.

INDIVIDUALS' PERFORMANCE AND PERCEPTION BY OTHERS

One question barely touched on by researchers is whether certain behavioral characteristics distinguish individuals who rate themselves high or low in social support. In a recent study, Sarason, et al., (in press) compared the social skills and physical appearance of males and females who were high or low in SSQN score. The subjects were videotaped in same sex dyads in two different interaction situations. One situation simply involved the dyad spending five minutes getting acquainted with each other. The second situation required that the pair agree on tactics to solve a hypothetical problem concerning a difficult roommate. The subjects rated their own performance and that of their partners, and the videotape of their interaction was also evaluated by trained raters both on a global basis and in terms of

specific behavioral characteristics. In addition, the experimenter made a global rating of each subject's social skills on the basis of his brief initial contact with them before the dyadic interaction began. Two other measures of social skill were obtained. The subjects completed a brief specially constructed social competence questionnaire (COMQ) and also performed a story completion task designed to elicit knowledge of appropriate and skilled social behavior. In addition the physical attractiveness of each subject was rated by his or her partner and by raters using a color snapshot. The attractiveness ratings were included because earlier work on interpersonal attraction has indicated that physical appearance is a strong factor at least in early stages of interpersonal relationships.

The findings of the study demonstrated clearly that individuals high and low in social support differ in their social skills no matter how these are measured. Self evaluation, either by formal questionnaire or rating scale; the partner's rating; the experimenter's global assessment; the rating of the videotape raters and the story completion task all were in agreement and all characterized those high in self-described social support as being more skilled than those who saw themselves as low in this characteristic. In addition to indicating this consistent relationship between social skills and social support, this study makes it clear that certain types of response tendencies in self description are not themselves sufficient to account for the relationship. At the same time, the study does not make possible a causal statement about the relationship of the two characteristics. It might be that those who have many supportive relationships have more chance to practice and develop skills or to develop self esteem which may facilitate performance. On the other hand, the existence of the skills may have made the building of the perceived network possible.

Prior work on social skill training often regarded assertiveness as an important asset and perhaps the main defining characteristic of social skill. In this study, those who were high in social support satisfaction spent more time talking in the dyadic situations than those who were low in SSQS. The same tendency was present when SSQN scores were used to divide the subjects but the difference did not reach statistical significance. Total time speaking in a dyadic situation seems to be at least one aspect of assertive behavior. Another aspect of social skill, knowledge of appropriate behavior, was also a characteristic differentiating those high and low in perceived support. This finding, discussed below, demonstrates that assertiveness is not the only aspect of social skill that is related to social support.

Contrary to the suggestions of earlier research on interpersonal attraction, physical attractiveness did not appear to be an important factor in skill and attraction ratings at least in these same sex dyads. Although the high social support

subjects tended to be more attractive, the difference was not significant except when the groups were later divided into those high and low in satisfaction score.

Another provocative finding from this study was the difference in social skills level and rated attractiveness between male and female groups. On all these measures, except partner evaluation and story completion, the female subjects were rated as more socially skilled and more physically attractive than the male subjects. Females also seemed to have a somewhat different style of interaction than males. When measures were made of the duration of various behaviors traditionally associated with social skills in the psychological literature, the two sexes behaved quite differently. The principal difference was that females spent much more time than males looking at their partner both while speaking and while listening. Looking at one's partner during an interaction is a behavior often emphasized in social skills training. Apparently the social norms for men and women differ in this respect.

There is also evidence that the behavior of those high and low in social support engender different reactions in the observer. Subjects high in social support were rated as more likeable and more competent. (Sarason, et al., in press). In another study male and female raters watched videotapes of same-sex pairs engaged in the two tasks described earlier (Sarason & Sarason, 1983). The raters, half male and half female, rated each subject on ten items. These dealt with the raters' impressions of the personal qualities of the subjects including such things as their degree of considerateness, their interest in others, their success in social relationships and the likelihood that they would be a good friend. In addition ratings of probable effectiveness in difficult situations were made. Raters estimated the subjects' good judgement, how desirable they would be as leaders and, the confidence the rater would feel in the abilities of each to function effectively in a dangerous situation. The results showed that all of these items correlated highly with number of social supports for men but had no relationship to this variable for women. At least as far as this group of questions was concerned the results suggest that men are evaluated on different criteria than women. These ratings stressed dependability, judgement, and effectiveness. It may be that culturally these are considered more relevant characteristics for men than for women and thus are unrelated to female social skills in the raters' generalized assessments.

In addition to descriptions of behavioral differences, the thoughts those high in social support (SSQN) have about their social interactions seem to differ from those of individuals low in perceived social support. Subjects who have low SSQN scores report spending time thinking about their inability to make friends and difficulty in being noticed by others. They also report discomfort in maintaining eye contact in interpersonal

situations. This discomfort is reflected in their behavior because their timed duration of eye contact is significantly less than that for high SSQN subjects. Although these differences in cognitions relating to social comfort of those high and low in social support existed for both men and women, men who were low in social support reported less discomfort than women in the low group. This was in spite of the fact that observers rated women at all levels of social support as being higher in social skills than men at comparable social support levels. These results may be linked with the greater relationship of test scores for anxiety, depression, and neuroticism to social support for females than for males. Women low in social support seem to worry more and be more self critical than men. The self-described cognitive activity of women low in social support in many ways resembles the type of self-statements used by depressed individuals as they are described in the work of researchers such as Beck and Lewinsohn and their coworkers (Beck, 1967, Beck, Rush, Shaw & Emery, 1978, Lewinsohn & Arconad, 1981). It may be that this difference in reporting reflects a sex difference in willingness to admit negative things about oneself that may be a result of differing socialization experiences for men and women. This finding is parallel to the frequently reported tendency of women to admit to more physical and psychological symptomatology than men (Weissman & Klerman, 1977).

ADMINISTERED SOCIAL SUPPORT

In addition to the perception of social support in individuals' lives and the effect of this perception on their behavior, another important area for investigation is the effect of receiving support in specified situations on the behavior of the recipient. This question is important if one aspect of social support, its role in primary or secondary prevention, is to be addressed. A problem in this line of research is how social support can be unambiguously defined.

In one experimental study, social support was defined by the experimenter's offer of help if needed to students who were about to take a story completion test (Lindner, 1983). The task used was similar in format to the Means-Ends Problem-Solving procedure developed by Platt and Spivack (1975). The experimenter told the subjects that she would be available to them throughout their work to answer any questions that might come up. This statement followed her reassurance that many people felt uneasy about writing stories, so the subjects should not worry if they felt this way. Although no subject requested help, those subjects who had low SSQS scores performed significantly better after receiving the instructions than a comparable group of low SSQS subjects that did not receive these instructions. Their performance was within the same range as that of high SSQS subjects. The administered

support did not raise the performance of the high SSQS subjects, compared to that of an untreated high SSQS group. Thus, in this study, the interactive effect of the administered support was clear; administered support was helpful only to the group whose self-evaluated support was low.

Another way of defining social support is analogous to the kind of response people often have in a group therapy situation. The group interaction appears to engender feelings of mutuality of experience which often decrease anxiety, guilt and worry. In addition, the element of helping or sharing or giving useful input to others is often thought to be therapeutically important. In a recent experiment (Sarason & Turk, 1983) subjects differing in test anxiety were exposed to several different experimental conditions before performing in an evaluative situation. The conditions represented various combinations of approaches that had been found helpful in previous work in improving the performance of high test anxious subjects. One group received a written summary of five coping strategies. Subjects in a second group were distracted by asking them to write essays about problems related to school but not including tests or test taking. A third group discussed the same five coping strategies presented to the first group. A fourth group held a general discussion on their feelings about tests and a final group served as a control. After the treatment condition, all subjects worked on moderately difficult anagrams. The subjects who discussed the specific coping strategies in a group setting showed the most improved performance compared to the controls. Presenting the coping strategies in written form had no effect on performance. The freewheeling group discussion improved performance to a limited degree. For high anxious subjects, the distraction condition (writing an essay) also improved performance but not as much as the group discussion of coping strategies. Thus, the combination of support from the group and knowledge of coping techniques produced the best performance especially for highly test anxious subjects.

Social support has frequently been characterized as a buffer to protect an individual from high stress levels. In a testing situation highly test anxious individuals should be more stressed than low test anxious individuals. The group interaction was more effective in improving performance (and reducing stress) if information to improve skills was also included. Mere presentation of this information in written form was not sufficient to facilitate performance. The written presentation may not have been emphatic enough to enable subjects to utilize the skills described. More likely, however, the discussion provided validation of the coping strategies and combined emphasis on the particular points with a supportive interaction that also provided suggestions for positive behaviors. This was in contrast to the freewheeling group discussions which were likely to stress gripes, fears, and other negative thoughts and feelings without

any impetus toward greater efficacy. Therapists, as well as parents, have observed that a combination of support and coping strategies is much more effective than empathy alone in producing a change in behavior. This study illustrates how the nature of the support provided is important in influencing performance. One of the benefits of studying supportive behaviors in a laboratory setting is the role such studies can play in understanding how and why supportive intervention is effective or not effective.

CONCLUDING COMMENTS

Like many other concepts, that of social support is in need of better definition. Not only is consensus needed about the meaning of the term but also a better understanding is required of characteristics that distinguish those who describe themselves as high and low in support. Knowledge of these differences is important in working toward the goal of helping people attain more support. More information about how support can be provided and the effects of its provision are also important. Laboratory studies, although they cannot supplant studies in the natural environment, have a unique role in increasing our understanding of the concept of social support.

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SOCIAL SUPPORT AND PSYCHOLOGICAL WELL-BEING: THEORETICAL POSSIBILITIES

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Considerable controversy has centered on the role of social support in the stress process. Some theorists (Cassel, 1976; Cobb, 1976; Kaplan, Cassel, & Gore, 1977) have argued that support acts only as a resistance factor; that is, support reduces, or buffers, the adverse psychological impacts of exposure to negative life events and/or chronic difficulties, but support has no direct effects upon psychological symptoms when stressful circumstances are absent. Several studies confirm this buffering-only view of social support influences (see Turner, 1983, for a review). Others (Thoits, 1982a, 1983c) have argued that lack of social support and changes in support over time are stressors in themselves, and as such ought to have direct influences upon psychological symptomatology, whether or not other stressful circumstances occur. A number of studies now confirm this main-effect view of social support influences (e.g., Andrews, Tennant, Hewson, & Vaillant, 1978; Aneshensel & Frerichs, 1982; Lin, Ensel, Simeone, & Kuo, 1979; Thoits, 1983b; Turner, 1981; Williams, Ware, & Donald, 1981). These studies report an inverse association between measures of support and indicators of psychological disturbance, and no stress-buffering effects at all. A handful of other studies report both types of effects; support reduces symptoms directly and reduces the disturbing impacts of stressful circumstances (Dean and Ensel, 1982; Henderson, Byrne, Duncan-Jones, Scott & Adcock, 1980; Husaini, Newbrough, Neff & Moore, 1982).

Given the incomparability of research designs, of measures of stressors, support, and disturbance, and of analytical strategies across these studies, the conditions under which main effects only, buffering effects only, or both types of effects will be observed are presently unknown.¹ But these cumulative

empirical findings clearly indicate that neither the buffering-only view nor the main-effect only view is fully valid. Social support can have a direct impact upon psychological symptomatology and can mediate the psychological consequences of stressful circumstances. The combination of these approaches yields a new causal model of social support influences for exploration.

But, as Brown (1979) has pointed out, causal models of social support influences, regardless of their validity, are not enough. We currently lack an understanding of supportive processes. That is, the mechanisms through which supportive relationships can result in psychological benefits are presently unknown. Consequently, the conditions under which social support will produce impacts upon symptoms differentially cannot easily be identified.

To illustrate this problem, consider the practical difficulties involved in developing a supportive intervention program for clients of a mental health agency. The stress-buffering literature implies that a "volunteer friends" program might be a simple and expedient way of reducing the symptomatology of clients. A trained community volunteer might be assigned to each outpatient as a "supportive friend." But the hitch lies in the selection and training of volunteers. Just who might the most effective volunteers be? Should they be similar to the outpatients in age, race, marital status, and life style? Should they be persons who have or have not had past experience with the patients' problems themselves? And exactly what does one train these volunteers to do to be supportive? Give advice? Listen sympathetically? Bolster self-esteem? Offer help with practical tasks? What? And more importantly, why? A "volunteer friends" program is unlikely to be successful without an explicit theoretical understanding of how support works to reduce symptoms of anxiety and despair both directly and in the face of difficult circumstances.

To date, we have no theories of social support processes. That is, we know little about what aspects of support are really supportive, and from whom, through what mechanisms, and under what conditions support can be beneficial (or harmful). The purpose of this paper, therefore, is to focus on these theoretical issues. But presenting a theory of social support processes that addresses all of these issues would require a small monograph. Consequently, this paper selectively focuses upon one major problem, namely, the possible mechanisms through which support may have impacts upon psychological well-being. In particular, main effects of support upon well-being will be explicated, drawing upon existing social psychological theory. Extensions of these theoretical possibilities to stress-buffering processes will be discussed briefly, as well. Although the focus here is admittedly quite limited in scope, it is my position that the identification of supportive mechanisms is crucial to answering more complex

questions, such as from whom and under what conditions support will have effects. In my view, continued ad hoc, trial-and-error modelling efforts will not easily or rapidly yield this information. Rather, efforts to test and refine theoretically-derived processes should more quickly produce cumulative findings that may have direct applications in clinical and field settings.

The processes described here are based upon two related observations. First, effective social support appears to consist primarily of emotional support from significant others. Second, psychological distress and disorder are most often identified by the persistence of undesirable or unpleasant emotional states in the individual, and well-being at minimum by the absence of such affective symptoms. Consequently, this paper attempts to explicate the support-disturbance relationship by examining the social determinants of emotional outcomes.

THE NATURE OF SOCIAL SUPPORT: EMOTIONAL SUPPORT

Social support most commonly refers to helpful functions performed for an individual by significant others such as family members, friends, co-workers, relatives, and neighbors. These functions typically include socioemotional aid, instrumental aid and informational aid. Socioemotional aid usually refers to assertions or demonstrations of love, caring, esteem, value, empathy, sympathy, and/or group-belonging (e.g., Cobb, 1976; House, 1981; Kaplan et al., 1977). Instrumental aid refers to actions or materials provided by others that enable the fulfillment of ordinary responsibilities, such as household, childrearing, financial, and job-related obligations (House, 1981). Informational aid refers to communications of opinion or fact relevant to a person's current difficulties--advice, personal feedback, notifications of job openings, of available medical assistance, or of other opportunities that might make an individual's life circumstances easier (House, 1981). Measures of these supportive functions typically assess the objective utilization or availability of such aids, or the subjective perception that such aids are, or could be, available. Social support has also loosely referred to the structure of a person's supportive network (Mueller, 1980; Wellman, 1981). Measures of structure assess the nature, number, frequency, density, multiplexity, and social characteristics of a person's social contacts, from whom supportive resources are presumed to flow.

Clearly, social support is a multidimensional concept (Thoits, 1982a). Unfortunately, most studies utilize indicators that combine a variety of these dimensions (e.g., Gore, 1978; Lin, Ensel, Simeone, & Kuo, 1979), so that the relative efficacy of each aspect for reducing the impacts of stress or for directly reducing disturbance cannot be easily determined. However,

despite this difficulty, several reviewers of the support literature have come to similar conclusions: Socioemotional support from significant, or primary, others appears to be the most powerful predictor of reduced psychological distress or disorder, whether stressful circumstances are present or absent (Cohen & McKay, 1983; Heller, 1979; House, 1981; Turner, 1983). Something about being married (Eaton, 1978), having an intimate, confiding relationship, particularly with a spouse or lover (Brown & Harris, 1978; Kessler & Essex, 1982; Lowenthal & Haven, 1968; Pearlin, Lieberman, Menaghan, & Mullan, 1981; Thoits, 1982b), the receipt of instrumental aid, or the provision of practical information.

So a key aspect of social support lies in its positive emotional functions. This conclusion, of course, can only be viewed as tentative, as studies rarely compare the efficacy of various dimensions of support to one another directly. But because empirically, socioemotional aid appears to be the most beneficial aspect of support, and connotatively, emotional aid is closest to the intended meaning of the support concept (see Caplan, 1974), the assumption is made here that social support consists essentially of emotional assistance from significant others.

THE NATURE OF PSYCHOLOGICAL WELL-BEING AND DISTURBANCE: EMOTIONAL STATES

In research practice, psychological well-being and disturbance are usually measured in one of two ways. Symptoms are reported by individuals themselves, responding to standard indices such as the Langner 22-Item Index (Langner, 1962) and the CES-D Scale of depression (Radloff, 1977). These scales are designed to distinguish between psychiatric patients and "known well" individuals in the community, and are generally believed to assess mild to moderate levels of anxiety, depression, distress, or "demoralization" (Link & Dohrenwend, 1980). Well-being is indicated by the relative absence of such symptoms.

Alternatively, psychological well-being and disturbance are assessed by admitting staff, clinicians, or trained interviewers using more formal diagnostic criteria (e.g., Brown & Harris, 1978). Although criteria can vary from study to study, generally accepted criteria are summarized in the Diagnostic and Statistical Manual of Mental Disorders (American Psychiatric Association, 1980) or in the ninth revision of the International Statistical Classification of Diseases, Injuries and Causes of Death (World Health Organization, 1977: Section V). Psychological well-being is again usually indicated by the relative absence of symptoms, as assessed by clinician-researchers.

For the purposes of this paper, the most striking similarity between assessments of distress and disorder lies in the

predominance of affective criteria. Distress scales typically ask directly about feelings of nervousness, anxiety, depression, and upset, or assess the presence of physiological and bodily symptoms associated with these affective states (e.g., shaking hands, trouble getting breath, poor appetite, loss of sexual interest). In essence, these scales measure the degree to which individuals' feelings deviate from an implicit standard of affective neutrality, positive contentment, or emotional comfort and stability.

Similarly, an examination of the criteria for mental disorders in the most recent Diagnostic and Statistical Manual of Mental Disorders, DSM III (American Psychiatric Association, 1980) indicates that excessive or inappropriate affect and affect displays are important indicators for several types of disorder. My informal analyses of the diagnostic criteria in DSM III reveals that inappropriate, usually negative, emotional states or emotional displays are an essential defining feature of 81 out of a total of 228 disorders (35.5%) and an "associated feature" of 64.9% of these disorders. These percentages would be even higher if disorders due to genetic or organic causes (e.g., mental retardation, substance abuse, organic disorder) were excluded. These observations suggest that socially inappropriate or undesirable feelings and feeling displays play an important part in the recognition and labelling of disturbance. Psychological disturbance might usefully be conceptualized, then, as persistent or recurrent emotional deviance (Pugliesi, 1981; Thoits, 1983e).

It cannot be denied that unusual cognitions, sensory perceptions, and behaviors also play a part in the formal diagnosis of psychological disorder. My belief is that these unconventional thoughts, sensations, and behaviors are not regarded as problematic unless they are accompanied by affect; for example, we do not take a stated persecutory delusion seriously unless the person him/herself invests that belief with emotional significance. Although suggestive evidence is available in support of this hypothesis (Pugliesi, 1981), it has not yet been adequately substantiated. Consequently, for the purposes of this paper, attention will be confined to self-reported psychological distress and only to certain classes of disorder as defined in DSM III (American Psychiatric Association, 1980). These include anxiety disorders, affective disorders, and adjustment disorders (situational stress reactions). The classes of disorder targeted here are identified primarily by the predominance of persistent and/or recurrent emotional deviance, usually excessive or unwarranted anxiety, fear, irritability, hostility, depression, or "stress."

It is no accident, I think, that negative feeling states--e.g., anxiety, anger, depression--are most often indicators of distress and disorder. Although the concept of emotional deviance might also include "flattened affect," or the persistent lack of emotion (as in schizophrenia), and excesses of

positive feeling (as in mania), prolonged or recurrent negative feelings tend to predominate as symptoms on distress scales and in diagnostic criteria. This is probably due to the problematic nature of negative feelings; the personal and social consequences of such feelings are often undesirable, or dysfunctional.

Displays of negative affect can disrupt the flow of ordinary interaction and can elicit sanctions. The arousal of similar unpleasant emotions in overly empathetic others can cause those others to withdraw from further contact. And strong negative arousal can disrupt the concentration, skill, and/or motivation necessary for the performance of role-related activities, resulting in what psychiatrists term "impaired functioning" or "maladjustment." In short, negative feelings may be socially appropriate or "normal" responses to persistent difficult circumstances initially but, nevertheless, have serious problematic consequences. It is quite likely that prolonged or recurrent stress reactions are viewed as evidence of disturbance on the basis of these possible (perhaps actualized) consequences. Given these potential consequences, reactions to prolonged or recurrent stressful circumstances can be justifiably viewed as evidence of distress or disorder due to their undesirable nature, despite their initial normative appropriateness.³

To rephrase the original problem at this point, emotional support from significant others somehow reduces the likelihood that individuals will experience and display inappropriate or undesirable affect. How is this possible? I will discuss the processes producing apparent main effects of emotional support upon psychological well-being first, then briefly turn attention to stress-buffering processes.

SUPPORTIVE PROCESSES: MAIN EFFECTS

One undisputed aspect of socioemotional support is its social foundation; that is, emotional support originates in social relationships. Since most interpersonal relationships (other than fleeting contacts) can be characterized as role relationships, one can fairly safely say that emotional support is obtained from ongoing role relations. So to explain how emotional support helps sustain well-being, one must understand how role relationships can affect psychological states. That is, one must start with more abstract conceptions of the link between society and the person, more specifically, with aspects of the self that are based in role relations (the "social self"). From "social self" processes can be derived dimensions of support usually identified as important by researchers (e.g., security, belonging, love, esteem). In essence, I will argue that aspects of regularized social interaction, and not emotional support dimensions per se, are responsible for maintaining well-being. What we recognize as dimensions of emotional support and main effects of support are

simply byproducts of these more abstract social-psychological processes.

I will propose here that role relationships can be psychologically beneficial in three major ways: by providing a set of identities, as sources of positive self-evaluation, and as the bases for a sense of control or mastery. These views are grounded in the symbolic interactionist approach (Cooley, 1902; James, 1890/1950; Mead, 1934; Stryker, 1980). This approach assumes that social interaction is essential to normal personality development and to appropriate social conduct. In brief, the ability to see oneself from the eyes of others--or to "take the role" of others (Mead, 1934)--produces the social and socialized self.

Identity and Belonging

By taking the role of specific and then of "generalized" others (i.e., the community), the individual perceives that he/she has been placed into recognized and meaningful social categories, or social positions, such as male, female, parent, child, Catholic, Jew, husband, wife, employer, employee (Mead, 1934; Stryker, 1980; Thoits, 1983d). The individual obtains a view of him/herself as a meaningful object, or set of objects, in the eyes of others. Attached to these social positions are sets of behavioral expectations, ways of enacting those positions, called roles. When the individual accepts the positional designations assigned to him/her by others and behaves as expected in these role relationships, he/she can be said to have taken on a set of identities (Stryker, 1980; Thoits, 1983d). Identities, then, emerge from and are sustained in role relationships.

How is psychological well-being related to identity enactment? At minimum, role-identities provide answers to the existential questions, "Who am I?" and "Why am I here?" (Thoits, 1983d). Role involvements give meaning and purpose to one's self and to one's life, and thereby reduce the likelihood that profound anxiety or despair will be experienced (for similar views, see Bart, 1974; Frankl, 1959; Erikson, 1963, 1968; Sarbin, 1968; Sieber, 1974). In essence, social identities provide "existential security," preventing the persistent or recurrent undesirable feelings which constitute distress or disorder as conceptualized here. Additionally, because role relationships are governed by behavioral expectations, identities guide one's behavior, reducing the likelihood of "disordered" (aimless, disorganized) conduct, often viewed as further evidence of disturbance.

Role relationships can be characterized as emotionally supportive, then, because they provide meaning and purpose in life and thus prevent anxiety and despair. Conversely, the lack or loss of such relationships should promote disturbance through a sense of meaninglessness and purposelessness, or existential insecurity (see Bart, 1974; Thoits, 1983d, for supporting

evidence). A sense of security has been identified by some researchers as an important aspect of emotional support (e.g., Kaplan et al., 1977). In this formulation, security is based in ongoing identity enactment; one knows who one is in relation to others and, therefore, knows how to conduct oneself appropriately as well.

It is important to note that this argument is similar to Durkheim's (1951) explanation of suicide rates in the general population. Durkheim proposed that the traditional and stable rules of conduct characteristic of socially integrated groups give members a sense of certainty and purpose in living. According to Durkheim, normative regulation "controls the passions" and thus "preserves" persons from suicide. Conversely, social disintegration, or anomie, facilitates suicide, since the "passions" are no longer channeled and regulated.

In my view, involvement in role relationships is the essence of social integration at the individual level. The person is tied to the norms of society because he/she is embedded in a system of regularized relationships with others. And because these social ties are defined by reciprocal rights and obligations, they provide the individual with a sense that "he or she belongs to a network of communication and mutual obligation" (Cobb, 1976, p. 300). A sense of belonging is thought by many to be an essential feature of emotional support (Caplan, 1974; Cassel, 1976; Cobb, 1976; Dean & Lin, 1977; Kaplan et al., 1977). In this formulation, the individual "belongs" because he/she fulfills others' needs and in turn has the right to ask that his/her own needs be met; these reciprocal duties and privileges (roles) simultaneously define who the individual is and provide a sense of belonging. Role relationships are how individuals belong, and also provide a sense of security.

Reflected Self-Esteem

Through taking the role of others individuals not only acquire a set of meaningful identities, but evaluations of themselves as well (Cooley, 1902). That is, evaluations of one's overall worth, lovability, importance, and competence depend, at least in part, upon the perceived appraisals of others with whom one regularly interacts (see Webster & Sobieszek, 1974, for supporting evidence). Role relationships, then, are a source of reflected self-esteem.

Self-esteem is intimately tied to psychological well-being. This point requires little belaboring, as it is well-accepted by psychologists and sociologists (e.g., Abramson, Seligman, & Teasdale, 1978; Brown & Harris, 1978; Kaplan, 1980) and is well-established empirically (e.g., Kaplan, 1980). Reflected positive self-evaluations generate satisfaction, contentment, or other positively-toned affective states in general. Reflected negative self-evaluations are a primary source of anxiety and

depression (and possibly, guilt or shame [Shott, 1979]), which, when persistent or generalized across situations, become recognized symptoms of psychological disturbance. In short, to the extent that positive appraisals are perceived or received from role partners, role relationships can be characterized as emotionally supportive. Positive evaluations from others (love, caring, esteem, value, prestige, attributed competence) have been a crucial element of most conceptions of sociemotional support (Cobb, 1976; House, 1981; Kaplan et al., 1977).⁴

It is important to recognize that role partners do not invariably communicate approval to the individual. Some partners (e.g., parents) may disapprove of a person's other role involvements, if those roles are culturally or subculturally devalued (e.g., ex-con, divorcee, homosexual). And more importantly, positive appraisals from reciprocal partners depend upon the adequacy of the individual's role performance--the degree to which he/she meets behavioral expectations. That not all role relationships can be characterized as emotionally supportive is a possibility which is often overlooked by social support researchers beguiled by the positive connotations of the term (Heller, 1979; Wellman, 1981). Disapproval from primary others may be instead a source of ongoing strain or conflict, generating shame or guilt, anxiety, frustration, and/or despair. Reflected positive self-esteem probably cannot come from others who are directly involved in or affected by a person's role failures.

Comparative Mastery

Although individuals are motivated to meet role expectations for social approval, obviously other sources of motivation are involved as well. For example, promises of money, power, or punishment are extrinsic motivators. But intrinsic factors can also operate, and these, I think, are more closely tied to psychological well-being. Individuals obtain intrinsic gratification and self-approval from successful attempts at environmental control (Abramson et al., 1978; Bandura, 1977; White, 1959). The adequate or above-adequate performance of role-related tasks is a central source of what some have termed a sense of mastery (Pearlin et al., 1981) or efficacy-based self-esteem (Bandura, 1977; Franks & Marolla, 1976; Gecas, 1979). Successful efforts to control the outcomes of role-related tasks generate feelings of pleasure, satisfaction, pride, perhaps elation; the individual also awards approval to him/herself for these competencies. Failures to control performance outcomes can generate frustration, anxiety, hopelessness, shame, as well as self-awarded disapproval.

But as Abramson and her colleagues have pointed out (1978; see also House, 1981), these feelings in response to controllable and uncontrollable task outcomes are mediated by social comparisons. Objective standards for most role activities are not

available, so individuals must look to others involved in similar activities for standards against which to evaluate their performances (Festinger, 1954). To the extent that others are easily able to perform the same tasks, the pleasures of success are dampened and the stings of failure exacerbated. If others often prove unable, however, successes are more gratifying and failures less dismaying. Comparison others--role models, if you will--provide standards against which persons evaluate their own competencies. Moreover, by careful observation of comparison others the individual can extract information or methods that enhance his/her own ability to perform; vicarious learning is possible (Bandura, 1977). Although one would be hard pressed to characterize such exemplary others as emotionally supportive, the presence of these others in fact influences the emotional outcomes of role-related efforts. However, to the extent that role partners or role models offer direct encouragement for mastery attempts, these others can be said to provide emotional support.

In sum, an individual's sense of mastery, control, competence, or efficacy-based self-esteem depends heavily upon the ability to meet role expectations, particularly expectations attached to important identities. Comparatively successful performances produce positive feelings contributing to well-being; comparative failures produce emotional disturbance when failures are prolonged, recurrent, or generalized to other situations (Abramson et al., 1978). These psychological outcomes are primarily a function of intrinsic gratification (or frustration) and self-awarded evaluations, and not of emotionally supportive aspects of these role relationships. But encouragement from role partners or role models, thought to be an aspect of emotional support by some (e.g., House, 1981), can enhance a sense of mastery, and thus psychological well-being.

STRESS-BUFFERING PROCESSES: MEDIATING EFFECTS OF EMOTIONAL SUPPORT

The processes described in the preceeding section can be applied rather straightforwardly to stress-buffering phenomena, if certain conceptual equivalences are pointed out.

"Stressors" or "stressful circumstances" usually refer to the experience of major life events (Holmes & Rahe, 1967) and/or chronic strains (Brown & Harris, 1978; Pearlin, 1983). These are conditions that disrupt usual activities either on a short-term or daily basis. It is important to recognize that only culturally or personally undesirable events and strains are associated with indicators of distress and disorder; socially or subjectively desirable experiences have little influence upon psychological disturbance (see Pearlin, 1983; Thoits, 1983a, for reviews). These well-established findings suggest that only objective circumstances that are perceived as personally threatening may

have negative psychological effects. Several researchers have argued this explicitly (e.g., Brown & Harris, 1978; Lazarus & Launier, 1978; Mechanic, 1962; Pearlin et al., 1981). This argument is accepted here for reasons discussed earlier. Theoretically, only negative events or ongoing difficulties that are perceived to have undesirable implications for the self should have negative emotional impacts through potential or actual loss of identity and belonging, reflected self-esteem, and/or efficacy-based self-esteem. Specifically, undesirable changes in role relationships and role-related difficulties may be most likely to threaten these aspects of the self, and thus, to influence well-being.⁶

These stressors are presumed to cause "stress reactions," usually indicated by symptoms of psychological arousal and typically identified by the person or by observing others as a state of anxiety, nervousness, or distress. Certain stressors (in particular, the loss of loved ones) produce depressive reactions instead--sadness, apathy, depression, or despair. Note that these are negatively-toned, undesirable feeling states. In short, role changes and role difficulties (stressors) that are perceived as self-threatening produce undesirable feelings (stress reactions) that, if prolonged or recurrent, are primary indicators of psychological disturbance as conceptualized here.

If the processes outlined in the previous section are valid, then "stress-buffering influences of support" may be attributable to reductions in perceived threats to the self produced by the deliberate interventions of others. That is, threat reduction may be how support operates to lessen symptoms of disturbance in response to stressors. In particular, the words or deeds of others reassure the individual that he/she is meaningful and needed, loved and valued, and/or comparatively efficacious in the face of distressing or depressing evidence to the contrary. Significant others actively manipulate perceptions of identity and belonging, reflected self-esteem, and/or comparative mastery to counteract the damaging psychological (emotional) impacts of stressful circumstances (for similar suggestions see Brown & Harris, 1978; Pearlin et al., 1981). Emotional support, in short, may consist of words and deeds intended to alter the self-perceptions of distressed individuals, and altered self-perceptions are the mechanisms through which support operates to buffer, or reduce, symptoms.

Three implications follow from this explanation of stress-buffering processes. First, supportive others may offer reassurances that do not address the self-referent concerns of distressed individuals. That is, mismatches between offered support and perceived threats to self-image are possible. For example, a spouse may reassure a seriously ill person that things could be worse, that his/her survival is not really at stake, while the patient is concerned with other threatening issues (Will I be able to return to my job? Will I be able to perform as

adequately as before, given these new health limitations?). Or a spouse may reassuringly address one major concern, while ignoring, failing to perceive, or denying others. In such cases, proffered support is unlikely to have much beneficial effect and may even "boomerang," leaving the distressed person more upset than before (Coates & Wortman, 1980; see also Wortman, this volume). In short, words and deeds intended to be supportive in actuality may be nonsupportive.

Second, implicit in these examples is a related point. Negative events and chronic difficulties may have multiple effects upon self-perceptions. Divorce, for example, may damage identity, lower esteem in the eyes of others, and eliminate one area of efficacious action entirely, resulting in complex feelings of despair, humiliation, and frustrated anger. Interventions by others may need to address all of these self-referent issues to be efficacious. This implies that to observe stress-buffering effects or to apply buffering findings we must first identify the various threats engendered by particular stressors.

Finally, the possibility that others may misperceive or fail to identify self-referent issues suggests a precondition for the seeking and/or acceptance of emotional support, and the likelihood of efficacious support being offered. That precondition may be empathetic understanding of an individual's plight. Empathy refers to the imaginative sharing of another's situation and affective reactions (Shott, 1979; Stotland, Mathews, Sherman, Hansson, & Richardson, 1978). Empathetic understanding is the product of "taking the role of the other," in this case, vicariously sharing the experience of the stressed individual. Empathetic understanding is most likely to come from socially similar others who have faced in the past or who are currently facing the same stressors (Cohen & McKay, 1983). Similar others not only have more detailed knowledge of the situation but are more likely to identify and address the full range of threats perceived by the distressed individual. The importance of social and experiential similarity is underscored by the existence of a wide variety of self-help groups in Western society that are focused on specific shared problems (e.g., Parents Without Partners, Recovery Inc., Alcoholics Anonymous, divorce groups, widows groups, and so on).

In sum, emotional support may buffer, or reduce, distress by bolstering one or more aspects of self that have been threatened by objective difficulties (most often, negative role changes and role-related strains). Efficacious support may depend upon the degree of "fit" between reassurances needed and those offered. A possible precondition for "fit" may be empathetic understanding based upon social and experiential similarities to the distressed person. These hypotheses are derived from theoretical processes that connect aspects of social relationships to emotional disturbance through self mechanisms.

SUMMARY AND DISCUSSION

To summarize, social relationships--specifically, role relationships--affect psychological well-being directly through several mechanisms recognizable as dimensions of emotional support. Roles as social identities simultaneously provide existential security and a sense of belonging, thus reducing the likelihood of anxiety and despair (as well as disordered conduct). Role partners' positive evaluations determine, at least in part, one's self-evaluations, and reflected self-esteem in turn has positive emotional consequences. And the comparative adequacy of one's role performances influences a sense of mastery over environmental circumstances, with direct and indirect consequences (through self-awarded evaluations) upon emotional well-being. In brief, role relationships provide security, belonging, self-esteem, and a sense of comparative mastery, and each of these self-referent factors has positive affective consequences, constituting evidence, if these conditions and emotional responses persist, of psychological well-being. Conversely, to the extent that a person possessed few roles or loses roles, perceives or receives negative evaluations from role partners, or observes his/her relative inadequacy in role performance (each types of stressful circumstances), senses of security, belonging, and so on will decrease, and negative feeling states will result, constituting evidence, if these conditions and responses are persistent, of psychological disturbance. It is straightforward to extend and elaborate these processes as benign or vicious cycles, since emotional responses have subsequent influences on role behaviors and thus on-going relationships. Such vicious cycles can be disrupted or reversed through the intentional words or deeds of others aimed at bolstering damaged aspects of self-conception. Such intentional bolstering is what we mean, I think, by emotional support, in its "active assistance" sense.

These theoretical processes are not at all mysterious nor unfamiliar. They have been repeatedly explicated in one form or another in the social psychological literature for several decades. But they have been detailed here for three reasons.

First, social support researchers generally have failed to see the relevance of these processes for explaining apparent main effects of socioemotional support upon psychological well-being; it seems necessary and useful to point them out. Other well-known processes linking social relationships to the psyche might also be relevant; I have simply selected those which apply to aspects of emotional support thought particularly important by researchers in the area.

Second, explications of these main-effect processes help lay bare several difficulties with the term "emotional support." As mentioned earlier, the positive connotations of the term have led researchers to overlook possible negative consequences of presumably "supportive" relationships. The theoretical mechanisms

described here help clarify that relationships with significant others do not necessarily result in desirable psychological outcomes. Beyond this, a more important terminological difficulty emerges.

"Support" connotes intentional action--words said or deeds done with a helpful purpose in mind. Yet many of the supportive aspects of role relationships described as main effects in this paper are essentially unintentional byproducts of regularized interaction. For example, the individual obtains existential security and a sense of belonging from the simple acquisition and enactment of role relationships. Meaning, purpose, and inclusion are not offered explicitly or deliberately to him/her by others. Similarly, evaluations from others are most often perceived implicitly--in body language, tone of voice, or the sheer fact of continued or noncontinued interaction--rather than in explicit feedback (although periodic assertions and demonstrations of love may be an exception to this observation). "Support" seems to lie in the benefits extracted by individuals from role involvements (not necessarily consciously or intentionally), rather than in the benefits purposefully offered by role partners as aid. And the qualifier "emotional" simply seems to specify the nature of the benefits extracted.

This commentary suggests that emotional support is any aspect of role relations that results in beneficial emotional outcomes, i.e., psychological well-being as conceptualized here. Such a statement is clearly tautological and unacceptable; it is not intended here, and a clarification will be offered momentarily. The point is that emotional support has ambiguous empirical referents when its main effects are under considerations. To state that emotional support has a direct effect upon psychological well-being is to suggest that support in any of its forms (belonging, esteem, etc.) operates as an independent variable. Yet when one examines how security, belonging, esteem, and a sense of comparative mastery are obtained when troubles are not present to prompt their direct offer, support ceases to be an independent variable and becomes the (often unintentional) byproduct of a social interactional process, or set of processes. That is, support appears to be an intervening variable (or set of variables) between contacts with significant others and psychological outcomes. Those contacts are usually termed "emotionally supportive." And when it becomes clear that not all such contacts have beneficial outcomes, one feels compelled to specify that emotionally supportive relationships are only those that produce positive emotional outcomes, creating the tautological difficulty just described. Emotional support in this case becomes synonymous with the dependent variable. Where in this confusion of possible terminological uses does emotional support lie?

With respect to main effect processes, I would reply, "Nowhere, but everywhere." That is, emotional support should be

considered a descriptive phrase that guides our search for the social origins of psychological well-being, much like "stress" is a descriptive term guiding the search for the social, psychological, and/or biological origins of ill health or psychological disturbance. To me, "socioemotional support" connotes a process or set of processes linking social life to emotional experience. The term can be used as a shorthand reference to or characterization of certain antecedent aspects or subsequent products of role relations that have emotional consequences. But I would hesitate to state that emotional support is this aspect or that product, when describing main effect processes. (In fact, throughout the "main effects" section I have attempted to avoid such statements.) In my view, the phrase "main effects of emotional support upon psychological well-being" signifies "main effects of on-going social involvements upon emotional well-being." A conceptual equation is made here between emotional well-being and psychological well-being, but no tautological claim is made that emotional support is that which produces emotional well-being. In essence, I have tried to argue that what empirically appear to be main effects of security, belonging, etc., on psychological states are actually superficial indications of underlying processes that connect social relationships to emotional experiences and behaviors. Role-identities, positive evaluations from role partners, and comparatively successful role performances (or the lack or loss of these) affect perceptions of belonging and security, esteem, and efficacy, respectively, which in turn have emotional consequences that are central indicators of psychological well-being or disturbance. Nowhere in this summary proposition does the term "emotional support" or the phrase "emotionally supportive" appear, but they are implied throughout, as they guided the search of the origins and mechanisms by which individuals can benefit affectively from social relations.

However, when significant others intentionally offer reassurance or encouragement to a distressed individual, I think the term "emotional support" takes on concrete meaning and becomes a measurable independent variable. In this case, I would define emotional support as words or deeds that are intended to alter the damaged self-perceptions of an individual facing stressors. Although one again runs the risk of implying a tautology by using the term (i.e., emotional support is that which produces positive emotional outcomes), if one's focus is on the mechanisms through which such assistance operates, I believe it will be possible to examine the intended and unintended emotional consequences of self-concept manipulation by others.

NOTES

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¹House (1981; see also Turner, 1983) suggests that main effects of support may be found when stressors have occurred several months prior to the measurement of psychological symptoms. Adjustments to those stressors may have already been completed, so support may simply appear to have a main effect and no buffering effect. When stressful circumstances are more recent and adjustment processes are still operating, buffering influences may predominate over main effects.

²Instrumental support often implies emotional support (Thoits, 1982a). That is, offers of material or practical aid suggest that the person is cared about by the provider. Consequently, measures of instrumental support can be strongly correlated with indicators of emotional support (e.g., House, 1981). The most parsimonious inference when this occurs is that emotional support is the primary factor underlying such correlations.

³One could simply assume that prolonged or recurrent negative feelings are always socially inappropriate (deviant) and thereby simplify this conception of disturbance. In fact, there may be a pervasive norm against the experience and display of negative feelings, at least in Western culture; it is a short conceptual step from undesirable feelings to inappropriate feelings. However, the assumption that undesirable feelings are always viewed as inappropriate (deviant) glosses over the implied consequences of such feelings (e.g., impaired functioning) which often play an important part in clinical judgments.

⁴In this formulation, the appraisals of others include their love or hate; these can be viewed as relatively intense evaluative attitudes upon which an individual's self-perceptions partly depend. Although some view love from significant others as a crucial and distinct aspect of social attachment affecting psychological well-being (e.g., Bowlby, 1969, 1973) I assume that self-love is acquired through the same process as other aspects of self-evaluation (prestige, competence, worth): by seeing oneself reflected in the eyes of others. The term self-esteem in this paper therefore includes reflected love for oneself.

⁵Recall that the discussion in this section is grounded in the symbolic interactionist tradition. Most symbolic interactionists emphasize the "looking-glass" nature of the self, implying a passive, conformist view of human nature and self-development. But early symbolic interactionists (e.g., James, 1890/1950; Mead, 1934) placed considerable importance upon the creative,

spontaneous aspects of human nature as well. Mead distinguished between the "I" and the "me." The "I" refers to the spontaneous, impulsive, active self; the "me" to aspects of self acquired and known through interaction with others. The discussion of efficacy-based self-esteem in this section is consistent with a view of the self as both reflector of and actor upon the social environment.

⁶ Notice that role changes and role difficulties often entail the loss or lack of social relationships. For this reason, I have previously argued (Thoits, 1982a, 1983c) that the loss or lack of social support are stressors, and as stressors should have direct impacts upon psychological well-being.

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MEASURING THE FUNCTIONAL COMPONENTS OF SOCIAL SUPPORT¹

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MEASURING THE FUNCTIONAL COMPONENTS OF SOCIAL SUPPORT

In the last several years, we have been interested in the role social supports play in protecting people from the pathogenic effects of stress. By social supports, we mean the resources that are provided by other persons (cf. Cohen & Syme, 1985). Although others have investigated and in some cases found evidence for a "buffering" hypothesis--that social support protects persons from the pathogenic effects of stress but is relatively unimportant for unexposed individuals, there are difficulties in interpreting this literature. First, there are almost as many measures of social support as there are studies. Hence it is difficult to compare studies and to determine why support operates as a stress buffer in some cases, but not in others. Second, in the vast majority of work, support measures are used without regard to their psychometric properties or their appropriateness for the question under study. For example, studies using measures assessing the structure of social networks (e.g., how many friends do you have?) are seldom distinguished from those addressing the functions that networks might serve (e.g., do you have someone you can talk to about personal problems?). In fact, in many cases, structural and functional items are thrown together into single support indices

resulting in scores that have little conceptual meaning. In the context of the limitations of earlier work, we developed our own social support instrument to study the support buffering process. This chapter describes the questions we wanted to address, the assumptions we needed to make in order to develop an instrument that addressed these questions, the instrument itself, its psychometric properties, and data on the relationship between support and well-being that has been collected by ourselves and others with this instrument.

Scale selection. The issue of how to choose a social support measure for any particular study is a complex and controversial one that cannot be fully addressed in this chapter (cf. House & Kahn, 1985). What is central, however, is that the instrument provides the information necessary to answer the question that is being posed. In general, psychologists interested in the relationship between social support and health pose questions about the cognitive and/or behavioral mechanisms that link the demographic fact that one has friends, relatives and acquaintances, with improvement in health and well-being. These mechanisms are presumed to be elicited by resources provided by one's social network. Hence social support instruments used in studies posing psychological questions need to assess the functions that others may serve.

Although some of the early studies assessed individual support functions (confidant measures are the principle example) and others used indices combining multiple functions into one index, there is little work comparing the relative impact of different kinds of support functions on well-being. In the case of the buffering issue, it seemed plausible that only certain kinds of resources provided by others would operate as buffers. In fact, our own theoretical work argues that one's interpersonal relationships function as stress buffers only when the type of support resources that are provided by one's relationships match the coping requirements elicited by the stressor(s) (Cohen & McKay, 1984). For example, while a person who is temporarily out of a job may benefit from a monetary loan, this same resource would be totally ineffective as a response to the death of his or her child. This stressor-support specificity model suggests the importance of assessing a range of available support resources.

Our first task was to develop a typology of functions served by interpersonal relationships. In particular, we were concerned with ways in which others could affect persons' responses to stressful events. Four categories of support functions were proposed: tangible support, appraisal support, self-esteem support and belonging support. (See theoretical justification for these categories in Cohen & McKay, 1984). Tangible support refers to instrumental aid; appraisal support to the availability of someone to talk to about one's problems; self-esteem support to the availability of a positive comparison when comparing oneself

with others; and belonging support to the availability of people one can do things with.

Perceived or objective support? The next question was whether we were interested in the objective existence of these interpersonal resources or subjects' perceptions that they would be available if needed. Our assumption was that the buffering effect of social support is primarily cognitively mediated, that is, support operates by influencing one's appraisal of the stressfulness of a situation (cf. Cohen & McKay, 1984; House, 1981). Potentially stressful events could be assessed as less stressful or even benign if support affected interpretation of the threat the stressor posed, influenced perceived ability to cope, or inflated self-concept (cf. Lazarus, 1977). As a consequence, a measure of perception of the availability of support would be a more sensitive indicator of its buffering effects than objective existence of that resource (e.g., Blazer, 1982). This is so because the appraisal of stress is based on a person's beliefs about available support as opposed to its actual availability.

Below we describe the Interpersonal Support Evaluation List (ISEL), the instrument we developed in response to the questions and assumptions discussed above. We also discuss its psychometric properties, and present data on the relationship between these functions and health behavior and symptomatology. Our discussion emphasizes the role played by each of the support functions in the buffering process.

Interpersonal Support Evaluation List [ISEL]

The ISEL consists of a list of 40 (48 in college student form) statements concerning the perceived availability of potential social resources. The items are counterbalanced for desirability; that is half of the items are positive statements about social relationships (e.g., "There are several different people with whom I enjoy spending time."), while half are negative statements (e.g., "I feel that there is no one with whom I can share my most private worries and fears."). Items were developed on theoretical grounds to cover the domain of supportive social resources that could potentially facilitate coping with stressful events. Respondents were asked to indicate whether each statement is "probably true" or "probably false" about themselves. The ISEL is scored simply by counting the number of responses indicating support. (See appendix for key.)

The ISEL was designed to assess the perceived availability of the four separate functions of social support discussed above as well as providing an overall functional support measure. Thus, the items which comprise the ISEL fall into four 10-item (12-item in student form) subscales. The "tangible" subscale is intended to measure perceived availability of material aid; the "appraisal" subscale, the perceived availability of someone to

talk to about one's problems; the "self-esteem" subscale, the perceived availability of a positive comparison when comparing one's self with others; and the "belonging" subscale, the perceived availability of people one can do things with. Subscale independence was maximized by selecting items (from a larger pool) which were highly correlated with items in their own subscale and at the same time minimally correlated with other subscales. A copy of the general population form of the ISEL appears in the appendix. The college student version of the scale is published in Cohen & Hoberman (1983).

Description of Samples

Data presented in this chapter derive from seven studies employing the student version of the ISEL and five studies employing the general population version. Three of the studies using the student scale were conducted by the authors at the University of Oregon. Subjects in Oregon studies I (27 males and 43 females) and III (60 males and 52 females) were undergraduate students enrolled in introductory Psychology classes. Subjects in Oregon study II (120 males and 210 females) were freshman undergraduates living in university dormitories.

The remaining studies using the student form of the ISEL with undergraduates are the work of Michael Scheier (other data from this sample reported in Scheier & Carver, 1983) at Carnegie-Mellon University (CMU study I; 79 males and 62 females), Larry Cohen and his colleagues (Cohen, McGowan, Fooskas & Rose, in press) at the University of Delaware (Delaware study; 47 males and 45 females), Donald Graham & Ben Gottlieb (1983) at the University of Guelph (Guelph study; 59 males and 59 females), and Irwin Sandler (1982) at the Arizona State University (Arizona State study; 118 subjects). Four other studies used the general population scale with student samples. This work was done by Maryann Jacobi at the University of California at Irvine (Irvine study; 31 females), and the authors at Carnegie-Mellon University (CMU II; 154 males and 62 females) and the University of Oregon (Oregon IV; 14 males and 18 females). Finally, a study of a nonstudent (community) sample going through the Oregon Smoking Cessation program was done by Robin Mermelstein and her colleagues (Mermelstein, Cohen & Lichtenstein, 1983) and will be referred to as the Oregon Smoking Study (27 males and 37 females).

PSYCHOMETRIC PROPERTIES OF ISEL SCALES

This section provides detailed information on the psychometric properties of the ISEL and its subscales. Readers who are not interested in the technical characteristics of the scale are advised to skip ahead to the section entitled "ISEL as a Predictor of Health and Health Behavior".

Means and Standard Deviations

The general population form of the ISEL was administered three times to the Oregon Smoking sample with intervals of six weeks and six months. Across the three measurement periods the mean scores for all respondents ranged from 32.9 to 34.4 with standard deviations ranging from 4.96 to 5.98. There is a tendency for females to have slightly higher scores than males but this difference only occasionally reached significance. Mean scores for the College student form of the ISEL for all respondents (Oregon studies II and III and Delaware study) range from 34.33 to 38.80. The only available standard deviations are 7.3 (Oregon II study) and 7.5 (Oregon III study). Again there is a tendency for females to have higher scores than males.

Validity of the ISEL Scales

Correlations between the ISEL and other social support measures. As noted above, the ISEL was designed to provide a measure that was somewhat different than any of the existing social support scales. Even so, we assumed that the ISEL would moderately correlate with the existing structural, past support, and perceived availability measures. That in fact was the case for both student and general population forms.

The student scale was found to correlate .46 with a measure of the perceived receipt of social support during the last month, the Inventory of Socially Supportive Behaviors (ISSB; Barrera, Sandler & Ramsay, 1981), in both the Oregon I and the Arizona State studies. It also correlated .62 with the involvement and emotional support subscales of the Moos University residence environment scale, and .39 with network size in the Guelph study.

The general population scale correlated .30 with the total score of the Moos Family Environment Scale (FES; Moos & Moos, 1981) in the CMU II study. Correlations with subscales of the FES were .21 with expressiveness, .46 with cohesiveness, and .19 with conflict. In the same study, the general population ISEL also correlated .46 with number of close friends and .42 with number of close relatives. The general population ISEL also correlated .31 with the Partner Adjustment Scale (Mermelstein, Lichtenstein & McIntyre, 1983)-- a measure of the quality of marital or living partner relationships, in the Oregon Smoking study.

Correlations with Self-esteem Measures. Since trait self-esteem is strongly influenced by the feedback one receives from others, we expected that the self-esteem support subscale would be highly correlated with trait self-esteem. Although it is likely that just having a social network contributes to self-esteem, relatively small correlations were expected between self-esteem and the other ISEL subscales. In the Irvine study, the self-esteem support subscale from the general population form

of the ISEL was correlated .74 ($p < .001$) with the Rosenberg Self-esteem Scale. When one overlapping item was dropped from the ISEL, the correlation was .58 ($p < .001$). Although the self-esteem support subscale was not used in the Guelph study, they report that tangible, belonging and appraisal support (student scale) were correlated with a trait self-esteem measure .14, .32 ($p < .001$), and .26 ($p < .001$) respectively.

Correlations with self-disclosure. Since the appraisal support subscale assesses the availability of interpersonal transactions which allow self-disclosure, we expected that the appraisal scale would correlate with a self-disclosure measure. Again, the remaining scales were expected to show some association as well but of a lesser magnitude. In the Guelph study, the desire for verbal intimacy subscale of the Colwill and Spinner Privacy Measure correlated .40 ($p < .001$) with the appraisal scale and .08 and .24 ($p < .01$) with the tangible and belonging scales respectively.

Test-retest and Internal Reliability of the ISEL Scales

Adequate internal and test-retest reliabilities have been found for both student and general population scales and subscales in several samples. Internal reliability (Alpha Coefficient) of the total student ISEL has been reported as low as .77 (Oregon I study) and as high as .86 (CMU-I study). Ranges for student ISEL subscales are .77-.92 for appraisal, .60-.68 for self-esteem, .75-.78 for belonging and .71-.74 for tangible support.

Internal reliability (Alpha Coefficient) of the total general population ISEL ranges from .88 (CMU II study) to .90 (Oregon Smoking Study and Oregon IV study). Ranges for general population ISEL subscales are .70-.82 for appraisal, .62-.73 for self-esteem, .73-.78 for belonging, and .73-.81 for tangible support.

The student scale was taken twice with a four week interval by the CMU I sample. Correlations for the two periods were .87 for the entire scale, .87 for appraisal, .82 for belonging, .71 for self-esteem and .80 for tangible support. The general population version was taken twice by students in the Oregon IV study. The interval between testings was two days. For this sample, the test-retest correlation for the ISEL was .87. Correlations for subscales were .78 for tangible support, .74 for self-esteem support, .67 for belonging support and .84 for appraisal support. A six week interval between retests of the general population scale is available in the smoking sample. These data also suggest a reasonable stability across time. The test-retest correlations were .70 for the ISEL, .63 for appraisal, .65 for belonging, .68 for self-esteem and .69 for tangible support.

Long-interval (6 month) test-retests are also available for both the student and general population scales. These data are

not presented in regard to the psychometric validity of the scale but rather to illustrate the stability of support across time and some of the patterns of shifts in support that can occur across time for different samples. The student scale was taken twice by about 1/3 (122) of the Oregon II sample approximately six months apart. This sample is particularly interesting since it consists of college freshman who, for the most part did not have social networks at school when the first scale was completed. For this sample, the correlation for the two testings for the total scale was .72. Correlations for the subscales were .73 for tangible support, .66 for self-esteem support, .47 for belonging support and .45 for appraisal support. One interpretation of these data is that tangible support is largely provided by students' parents and hence is stable over time. Similarly, self-esteem support is largely affected by stable personality factors that lead one to see oneself as better off than others. However, appraisal and belonging support are more dependent on the existence of a proximate social network which changes substantially during the first six months of college. These changes in support level for those placed in a new social environment suggest that at least the appraisal support and belonging support scales are not just proxies for some underlying personality factor but in fact reflect changes in available support.

The general population form was also taken six months apart by persons in the Oregon Smoking Study. Six month test-retest correlations were .74 for the entire ISEL, .49 for the tangible subscale, .54 for self-esteem, .68 for belonging, and .60 for appraisal. There is a striking difference in comparing these six month correlations to the six month correlations reported above for the student sample. While students showed the most stability for tangible and self-esteem support, the smoking sample showed the least stability in these two areas. Less stability in tangible support than found in the student sample is easy to understand. As noted above students probably obtain most of their tangible support from their parents. This older sample may well lack such a stable source of material aid. Reasons for the shifts in self-esteem support are less clear. One might argue that success or failure in the smoking treatment program and followup may alter self-esteem level. In any case, these data suggest that self-esteem support may not be influenced as much by stable personality characteristics as suggested above.

In short, long-term test-retest data suggest that support changes over time and that the stability of particular types of social support may differ across populations. Further data on long-term retests is especially important in light of the possibility that most support findings are subject to interpretations that suggest they are artifactual; i.e., actually attributable to stable personality factors that cause both support and health outcomes (cf. Cohen & Syme, 1985).

Correlations between Subscales

One of our major goals in developing the ISEL scales was to create four subscales that were reasonably independent of one another, i.e., subscales that did not measure the same thing. Table 1 presents the correlations between the subscales of both the student and general population ISEL as reported in a number of studies. In general, these, correlations are in the .30-.50 range indicating a moderate degree of independence. Subscales on the general population scale tend to be more highly intercorrelated than those on the student scale. It is noteworthy that complete independence of these scales is neither desirable nor possible since people receive different kinds of resources from the same persons in their network.

TABLE 1

Correlations Between Subscales of the ISEL

	Student Scale				
	OR I	OR II	OR III	CMU I	DE
Appraisal-Belonging	.48	.51	.36	.46	.56
Appraisal-Tangible	.26	.41	.30	.42	.54
Appraisal-Self-Esteem	.19	.33	.38	.36	.46
Belonging-Tangible	.56	.49	.51	.42	.45
Belonging-Self-Esteem	.26	.46	.50	.48	.40
Tangible-Self-Esteem	.19	.30	.48	.24	.43

	General Population Scale			
	Smoking	CMU II	Irvine	OR IV
Appraisal-Belonging	.56	.53	.57	.73
Appraisal-Tangible	.40	.50	.44	.31
Appraisal-Self-Esteem	.50	.48	.51	.67
Belonging-Tangible	.61	.59	.81	.70
Belonging-Self-Esteem	.61	.64	.78	.61
Tangible-Self-Esteem	.46	.46	.76	.33

Discriminant validity

The ISEL and social desirability. In the Oregon IV study, the Crowne-Marlowe Social Desirability Scale was administered along with the general population ISEL. Social desirability was not correlated with the ISEL or any of its subscales.

Social support or social anxiety. Even prospective studies of the relationship between social support and various outcomes do not eliminate the possibility that the support concept as measured in the study is not merely a proxy for a personality factor such as social anxiety or social skills. For example, it is possible that persons who are socially anxious have difficulty forming and maintaining interpersonal relations and are generally anxious, depressed and otherwise psychologically impaired. Hence what appears to be an effect of social support may merely be an effect of a highly correlated personality trait. In the Oregon II sample, we attempted to determine whether the ISEL accounts for significant variance in the prediction of depression after controlling for social anxiety as measured by the Social Anxiety and Distress Scale (SADS) (Watson & Friend, 1969). Hence we partialled social anxiety out of the correlation between the ISEL and depressive symptomatology. The remaining correlation was $-.19$, $p < .001$. Hence the ISEL accounted for significant variance in the prediction of depression above and beyond social anxiety.

ISEL AS A PREDICTOR OF HEALTH AND HEALTH BEHAVIOR

The ISEL has been used in studies simply examining the relationship between social support and well-being as well as in studies specifically investigating the buffering hypothesis, the idea that social support protects persons from the pathogenic effects of stress but is unimportant for unexposed individuals. These data are reported below to both bolster the theoretical considerations on which the scale is based as well as providing further evidence for the validity of the scale itself.

Correlations between the ISEL Scales and Psychological Symptomatology

Table 2 reports correlations between the ISEL scales and psychological symptomatology from seven studies. The symptom measures used in these studies include the Center for Epidemiological Study of Depression Scale (CES-D), the Beck Depression Inventory (BDI), the Langner Symptom Checklist (L-22), and the Kobassa symptom scale. As is apparent from the table, increases in the ISEL total score are consistently associated with decreases in symptomatology. The self-esteem subscale is most strongly associated with symptom reporting, with the tangible subscale the least predictive. It is possible that the

TABLE 2

Correlations between ISEL and Psychological Symptomatology

	Student Scale					
	CED-D OR I	CES-D OR II	CES-D OR III	BDI CMU I	BDI DE	L-22
ISEL	-.47	-.37	-.43	-.38	-.51	-.51
Appraisal	-.33	-.23	-.28	-.23	-.18	-.20
Belonging	-.38	-.35	-.30	-.34	-.36	-.34
Self-Esteem	-.37	-.43	-.45	-.43	-.30	-.35
Tangible	-.22	-.14	-.29	-.14	-.10	-.15

	General Population Scale		
	CES-D CMU II	HOPKINS IRVINE	KOBASSA IRVINE
ISEL	-.52	-.57	-.60
Appraisal	-.29	-	-
Belonging	-.48	-	-
Self-Esteem	-.53	-	-
Tangible	-.39	-	-

self-esteem-symptomatology correlation is inflated by an overlap between what is measured by the self-esteem support scale and the symptom measures. These correlations are all cross-sectional and hence direction of causality is not implied.

In addition to the cross-sectional correlations reported in the table, there have been two longitudinal studies of the relationship between the ISEL and psychiatric symptomatology. Analysis of the Delaware study data indicated a significant negative relationship between the ISEL and both depressive (BDI) and psychiatric (L-22) symptomatology at the second testing with Time 1 criteria partialled out. Moreover, a prospective analysis of the CMU I data, partialling out time 1 depression, indicated a $-.21$ ($p < .05$) correlation between Time 1 social support and Time 2

depressive symptomatology. In sum, there is substantial evidence for a relationship between the ISEL and psychological symptomatology. Moreover, a significant part of this relationship can not be attributed to the effect of symptomatology on support level. The ability of the ISEL to predict subsequent changes (i.e., controlling for initial level) in depressive symptomatology also indicates that the ISEL is not itself measuring symptomatology.

Correlations between the ISEL Scales and Physical Symptomatology

Six studies (five with the student scales and one with the adult scale) report correlations between the ISEL and self-reported physical symptoms. All of these studies used the 39-item Cohen-Hoberman Inventory of Physical Symptoms (CHIPS; Cohen & Hoberman, 1983). As apparent from table 3, while there are only occasional small correlations between the ISEL and physical symptomatology in the student samples, moderate cross-sectional correlations are found in the community sample. However, prospective data analysis from the CMU I study and the Oregon smoking study both indicate a small relationship between the general population ISEL and physical symptomatology. In the Oregon Smoking study, correlations between the pretreatment ISEL and physical symptomatology three weeks later and six weeks later with pretreatment physical symptomatology partialled out were $-.21$ ($p < .06$) and $-.19$ ($p < .07$) respectively. In the CMU study, the same partial correlation with a four week interval was $-.18$ ($p < .05$). In sum, the ISEL does predict changes in physical symptomatology,

TABLE 3

Correlations between ISEL and Physical Symptomatology

	OR I	Student Scale		CMU I	DE	SCALE SMOKING
	OR II	OR III				
ISEL	-.13	-.08	-.22*	-.09	.02	-.34*
Appraisal	-.08	-.01	-.09	-.06	-.07	-.39*
Belonging	-.10	-.10	-.16	-.17	-.15	-.23*
Self-Esteem	-.07	-.19*	-.22*	-.07	-.09	-.26*
Tangible	-.12	-.01	-.22*	-.06	-.05	-.11

* $p < .05$

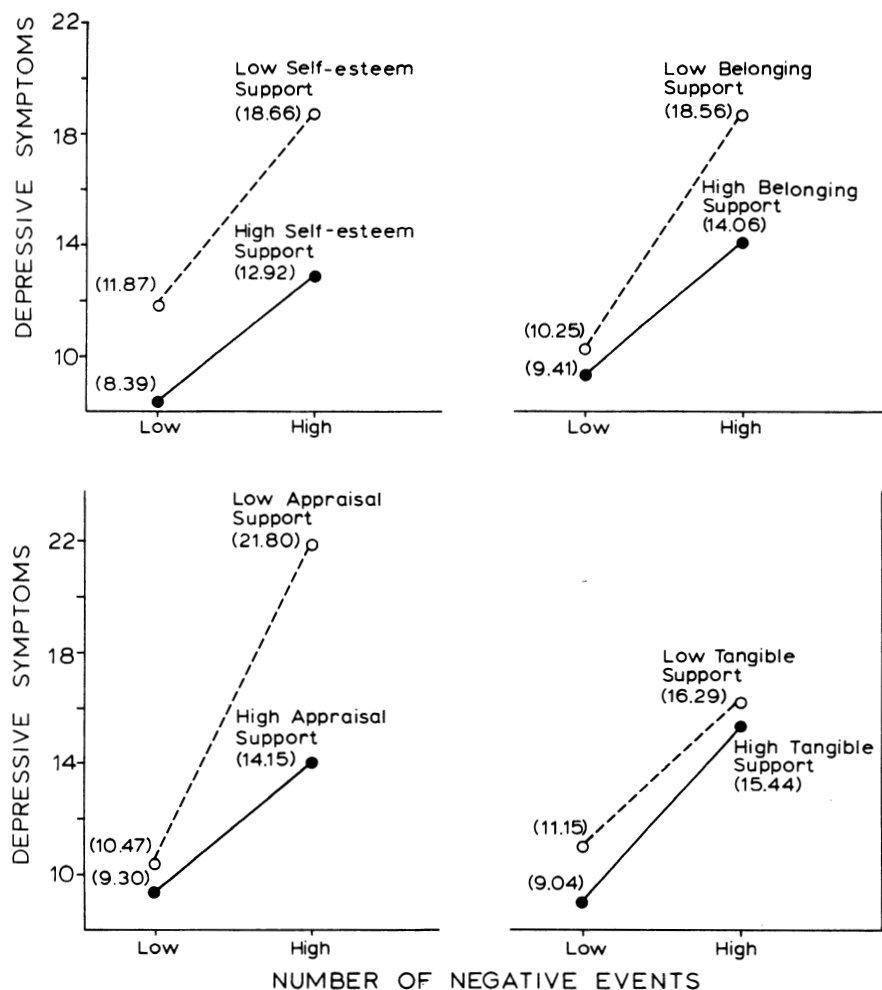
even though cross-sectional correlations between support and physical symptomatology are small.

Buffering Effects and the ISEL

The four subscales of the ISEL reflect four types of resources that may protect people from the pathogenic effects of stress (Cohen & McKay, 1985). The "buffering" hypothesis predicts a statistical interaction between social support and stress. The form of the interaction is that support is beneficial to persons under high levels of stress and is either less helpful or ineffective for persons under low levels of stress. Four studies (Oregon I, Arizona State, Delaware, Oregon Smoking) have used the ISEL in testing the buffering hypotheses. In the Oregon I study, Cohen and Hoberman (1983) found a buffering interaction between life events and the total ISEL in the prediction of depressive symptomatology. When we examined the contributions of the separate subscales to these regression analyses, we found that the interaction effect occurred in the case of self-esteem, belonging and appraisal support but not tangible support. Moreover, when all the life event X support interactions were entered into the regression equation simultaneously, only appraisal and self-esteem support made significant independent contributions to accounted variance. Figure 1 depicts the data from this study. Although lacking any detail, a summary of the Arizona State study (Sandler, 1982) reports a life events X ISEL buffering interaction for depressive symptomatology with life events X ISEL subscale interactions that "were similar to those reported by Cohen and Hoberman (1983)". In the case of physical symptoms, the Oregon I study data only partially supported the buffering hypothesis. Particularly, the data suggest that social support protects one from the pathogenic effects of high levels of stress but harms those (i.e., is associated with increased symptomatology) with low levels of stress. This crossover interaction may suggest that the increased responsibilities that are part and parcel of the interpersonal relationships that provide support, themselves contribute a small increment in one's stress level and consequently in symptomatology. However, since this form of the interaction occurred only in the prediction of physical symptoms, and is not commonly found in the literature, we are not very confident that it is reliable.

The Delaware study provides both a cross-sectional and longitudinal replication of the buffering effect of the ISEL reported in the Oregon and Arizona State studies but indicates slightly different results in regard to subscale analyses. At the beginning of the semester, undergraduate students completed a life events scale, the Langer Symptom Checklist, and the Beck Depression Inventory. About two months later, subjects completed these same three scales as well as the student form of the ISEL. During this second testing, they reported the occurrence of life

FIGURE 1



events from the time of the first measurement period. Cross-sectional analysis of time 2 data indicate significant buffering interactions in the case of both the Beck and Langner checklist. Analyses of life event X ISEL subscale interactions in predicting the Beck indicated independent contributions to variance for the interaction of life events with belonging and (nearly significant) for the interaction between life events and

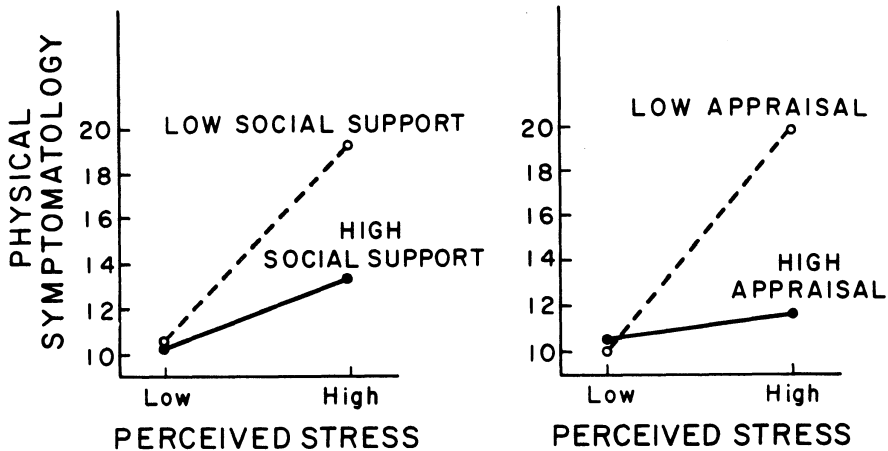
the appraisal scale. The L-22 analysis revealed significant independent interactions between life events and belonging, appraisal, and self-esteem subscales. Each subscale effect was consistent with the stress-buffering effect found for the total ISEL. A longitudinal analysis of the same data with time 1 criteria forced into the equation first indicated the same buffering interaction for the ISEL. Although none of the interactions between life events and the ISEL subscale made significant independent contributions in predicting the Beck for this longitudinal analysis, the interaction between life events and the self-esteem scale was significant in the prediction of the L-22.

Although it is difficult to integrate the data from these studies into some clear picture of which support resources are responsible for buffering effects in college students, there are some strong hints. First, none of the studies found a buffering interaction with the tangible support subscale. The lack of any evidence for a tangible buffer suggests that material aid is not an important resource for buffering the stress-psychological symptom association in college students. Second, there is a tendency for the interaction between life events and self-esteem support and between life events and appraisal support to appear across studies. This might occur either because all stressors inherently elicit needs for appraisal (information on defining and coping with stressful situations) and self-esteem (information that would help one cope with threats to self-esteem) or that college students are especially prone to stressors that elicit these specific needs (cf. Cohen & McKay, 1984; Cohen & Wills, 1985).

Analyses of the buffering effectiveness of the general population ISEL in the prediction of physical symptoms were conducted on Oregon Smoking study data. A prospective analysis predicting end of treatment (six weeks after pretreatment testing) physical symptomatology from pretreatment life event and social support did not find a significant life events X ISEL interaction. However, a similar analysis replacing life events scale scores with scores from the Perceived Stress Scale, a measure of appraised level of stress (Cohen, Mermelstein & Kamarck 1983), did predict buffering effects on physical symptomatology at the end of treatment. In both cases, these effects were primarily attributable to the role of appraisal support. Figure 2 depicts the form of this interaction for the entire ISEL and for the appraisal scale.

These data again suggest the importance of appraisal support in protecting people from the pathogenic effects of stressful events. As noted above, it is possible that appraisal support (having persons to talk to about your problems, suggest coping strategies, etc.), by definition, plays an important role in facing stressful events. That is, almost all such events require one to assess the demands, threats and/or challenges they pose and

FIGURE 2



determine one's ability to cope (cf. Lazarus, 1977).

ISEL as a Predictor of Smoking Reduction & Cessation Maintenance

In an ongoing project, we have been examining the role of social support in the maintenance of smoking abstinence and reduction. In a recently completed study (Mermelstin, Cohen & Lichtenstein, 1983), smoking clinic clients participated in a 6-week treatment program including self-monitoring of smoking, self-management training, nicotine fading, and cognitive behavioral relapse prevention. An assessment battery which included the general population form of the ISEL was completed during the week prior to the first treatment session, at the end of treatment, and at a six-month follow-up meeting. Smoking rate was assessed by telephone interview at 1, 2, and 3 months after treatment, and by face to face interview at 6 months. Two smoking outcomes, percent of baseline smoking rate and smoking status (abstinent or smoking) were used. Smoking status at the end of treatment and during follow-up was confirmed by a combination of self-report, expired air carbon monoxide and saliva thiocyanate measurements, and reports by significant others.

It was hypothesized that social support would aid in the maintenance of a behavioral change. Indeed, the ISEL proved to be a good predictor of smoking during the follow-up period. However, the relationship between social support and maintenance differed for subjects who were abstinent at the end of treatment and for those who were still smoking when the treatment ended. For

subjects abstinent at the end of treatment, the pretreatment ISEL significantly predicted six-month follow-up percent rate and smoking status. High levels of support predicted lower percent rates (-.35) and the maintenance of abstinence (-.28). For the end-of-treatment smokers, on the other hand, the pretreatment ISEL predicted both rate (.41) and status (.52), but in the unpredicted direction. For the end-of-treatment smokers, high perceived support was associated with increases in smoking over follow-up. When one examines the correlations between the subscales of the ISEL and six-month follow-up smoking for the end-of-treatment abstainers and smokers, it becomes apparent that the results described above are due largely to the effects of appraisal support. Correlations between the pretreatment appraisal subscale and six-month smoking rate and status are presented in Table 4. Thus, it is possible for smokers to have highly supportive environments in general, but for these environments not to be supportive of quitting or reducing smoking. It is particularly noteworthy that the appraisal subscale was clearly responsible for the predictive effects of the pretreatment ISEL. These data suggest that whether one's confidants (as opposed to casual friends, work peers, and more formal family acquaintances) are supportive of one's attempts to quit smoking is an important determinant of maintenance. We are presently following up this work by assessing the appraisal networks of persons entering the cessation program and determining which and how many people in these networks smoke. We have developed an appraisal network measure, the Social Network Inventory for Tobacco Smokers (SNITS), to address this issue.

COMMENTS

The data presented in this chapter provide strong support for

TABLE 4

Correlations Between Pretreatment Appraisal Support and
Six Month Smoking Rate and Smoking Status
for Quitters and Nonquitters

		6 month Smoking Rate	6 month Smoking Status
Smoking Status At Termination	Abstinent	-.45	-.43
Of Treatment	Smoking	.63	.59

the validity and reliability of the ISEL scales. As noted above, the ISEL has proved to be a good predictor of both symptomatology and smoking cessation behavior. It has also consistently been found to interact with stress measures in a manner that is consistent with the hypothesis that social support protects people from the pathogenic effects of stressful events.

We feel, however, that the most important contribution of the scale is its ability to indicate the type of resources that operate to improve health and well-being in any particular situation. For example, in the studies of college students, data from the subscales clearly exclude the possibility that tangible support operates as a stress-buffer. This work also suggests that appraisal and self-esteem support are generally effective buffers for college students under stress. Although these patterns of data are suggestive, they are not to be interpreted as definitive evidence for the operation of some versus other support resource in this population. The effectiveness of any particular support resource may well depend on the context of the situation. For example, our own work (Cohen & McKay, 1984) argues that the match between the needs elicited by the stressful events one encounters and available support is central to understanding when a particular kind of support will be a successful buffer. It is possible that the stressors that college students confront typically elicit needs for appraisal and self-esteem support but not for tangible support. As noted above, it is also possible that appraisal support may be inherently linked to dealing with stress since it provides generic kinds of resources for coping with stressful experiences. Further work should assess these needs as well as various available support resources.

The usefulness of the scale in determining the operative support resource is also exemplified by the smoking cessation study. This work suggests that only appraisal support is an important determinant of the maintenance of abstinence and of smoking reduction and that what one's confidants say may determine the success or failure of a behavioral change program of this sort.

Further research with the ISEL and other new scales that assess different support functions will provide us with a better understanding of the process by which support is linked with improved health and well-being.

NOTES

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Scheier, Chuck Carver, Larry Cohen, Donald Graham, Ben Gottlieb, Irwin Sandler and Maryann Jacobi for allowing us to report data from their studies, and to Dru Sherrod for his comments on an earlier draft.

²In the College Student studies, item 2 of the appraisal scale reads "there is someone who will give me suggestions about activities for recreation or entertainment". Item 5 of the appraisal scale was reversed and read "there is someone who...". We prefer the revised version of the scale (as listed), but the changes have little effect on the scales' psychometric qualities.

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Appendix: The General Population Form of the ISEL

For clarity, each subscale is listed separately. The scale presented to subjects consists of all 40 items listed in random order. T or F indicates response coded as social support.

Instructions

This scale is made up of a list of statements each of which may or may not be true about you. For each statement we would like you to circle probably TRUE (T) if the statement is true about you or probably FALSE if the statement is not true about you.

You may find that many of the statements are neither clearly true nor clearly false. In these cases, try to decide quickly whether probably TRUE (T) or probably FALSE (F) is most descriptive of you. Although some questions will be difficult to answer, it is important that you pick one alternative or the other. Remember to circle only one of the alternatives for each statement.

Please read each item quickly but carefully before responding. Remember that this is not a test and there are no right or wrong answers.

Appraisal

- | | |
|---|---|
| T | 1. There is at least one person I know whose advice I really trust. |
| F | 2. There is really no one I can trust to give me good financial advice. |
| F | 3. There is really no one who can give me objective feedback about how I'm handling my problems. |
| T | 4. When I need suggestions for how to deal with a personal problem I know there is someone I can turn to. |
| T | 5. There is someone who I feel comfortable going to for advice about sexual problems. |
| T | 6. There is someone I can turn to for advice about handling hassles over household responsibilities. |
| F | 7. I feel that there is no one with whom I can share my most private worries and fears. |
| F | 8. If a family crisis arose few of my friends would be able to give me good advice about handling it. |
| F | 9. There are very few people I trust to help solve my problems. |
| T | 10. There is someone I could turn to for advice about changing my job or finding a new one. |

Belonging

- T 1. If I decide on a Friday afternoon that I would like to go to a movie that evening, I could find someone to go with me.
- F 2. No one I know would throw a birthday party for me.
- T 3. There are several different people with whom I enjoy spending time.
- F 4. I don't often get invited to do things with others.
- T 5. If I wanted to have lunch with someone, I could easily find someone to join me.
- F 6. Most people I know don't enjoy the same things that I do.
- T 7. When I feel lonely, there are several people I could call and talk to.
- T 8. I regularly meet or talk with members of my family or friends.
- F 9. I feel that I'm on the fringe in my circle of friends.
- F 10. If I wanted to go out of town (e.g., to the coast) for the day I would have a hard time finding someone to go with me.

Tangible

- T 1. If for some reason I were put in jail, there is someone I could call who would bail me out.
- T 2. If I had to go out of town for a few weeks, someone I know would look after my home (the plants, pets, yard, etc.)
- F 3. If I were sick and needed someone to drive me to the doctor, I would have trouble finding someone.
- F 4. There is no one I could call on if I needed to borrow a car for a few hours.
- T 5. If I needed a quick emergency loan of \$100, there is someone I could get it from.
- F 6. If I needed some help in moving to a new home, I would have a hard time finding someone to help me.
- F 7. If I were sick, there would be almost no one I could find to help me with my daily chores.
- T 8. If I got stranded 10 miles out of town, there is someone I could call to come get me.
- T 9. If I had to mail an important letter at the post office by 5:00 and couldn't make it, there is someone who could do it for me.
- F 10. If I needed a ride to the airport very early in the morning, I would have a hard time finding anyone to take me.

Self-Esteem

- F 1. In general, people don't have much confidence in me.
- T 2. I have someone who takes pride in my accomplishments.
- F 3. Most of my friends are more successful at making changes in their lives than I am.
- T 4. Most people I know think highly of me.
- F 5. Most of my friends are more interesting than I am.
- T 6. I am more satisfied with my life than most people are with theirs.
- F 7. I have a hard time keeping pace with my friends.
- F 8. I think that my friends feel that I'm not very good at helping them solve problems.
- T 9. I am closer to my friends than most other people.
- T 10. I am able to do things as well as most other people.

SOCIAL SUPPORT AND SOCIAL HEALTH

Is it Time to Rethink the WHO Definition of Health

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For nearly thirty-five years, physicians, psychologists, sociologists, and epidemiologists have been attempting to include social support and social function in the definition of health status. Despite relentless efforts, it has been difficult to meaningfully define social support as a component of health. In this paper, I argue that the effort to define "social health" should be abandoned. Instead, efforts should be directed toward identifying the role of social support as a mediator of health status.

WHO DEFINITION

Although the term "social health" has been used for centuries, it emerged in a most important way when the World Health Organization outlined its mission in 1948. They defined health as. . . "a state of complete physical, mental, and social well-being and not merely an absence of disease and infirmity" (WHO, 1948). In addition, they suggested that a fourth dimension of health status was physiologic health which describes the functioning of specific organ systems. For example, physiologic health includes the function of the heart, the lungs, and the kidneys. In identifying the dimensions of health, WHO neglected to provide any operational definitions. Thus, different investigators have taken different approaches in their attempt to capture physical, mental, and social dimensions.

ADHERENCE TO THE WHO DEFINITION

Since the publication of the WHO statement, many investigators have tried to develop measures to operationalize concepts of health status. Nearly all of the methods attempt to quantify separately physical, mental and social health. With surprising consistency, authors quote the WHO definition and then present their methods measuring each of the three components of health (Bergner, Bobbitt, Carter & Gilson, 1981; Chambers, Macdonald, Tugwell, Buchanan & Kraig, 1982; Liang, Cullen & Larson, 1982; Neuman, 1982; and many others). For example, in describing the rationale for their McMaster Health Index Questionnaire (MHIQ) Chambers et al stated, "The MHIQ team began reviewing the health status measurement literature along the lines of the WHO definition of health. . ." This led to the development of "separate indices of physical function, emotional function, and social function (Chambers et al, 1982, p. 783). So prevalent is the notion that health measures must include these three components that many reviews now negatively evaluate any measure that does not conform to the WHO definition. Meenan (1982) disapproved of several health measures because, "These approaches fall far short of conceptualizing or measuring health in the WHO sense of a physical, psychological and social state" (p. 785). Given the wide acceptance for the WHO definition, it is not surprising that so many authors have tried to measure social health.

Many of these authors noted great difficulty in finding a measure of social health. For example, Kane & Kane (1981) devoted a substantial section of their monograph to describe problems in the quantification of social health. These problems included vague concepts, lack of norms, the interactive nature of variables, difficulty in constructing a continuum, the subjective nature of social health, socioeconomic and cultural determinants, lack of societal role expectations and the multiplicity of patterns of satisfactory social functioning. Despite these problems, they went on to describe the available methods for measuring social health. In summary, countless investigators have blindly followed the WHO definition of health without questioning whether "social health" is a meaningful and distinct entity.

Only Ware and colleagues have begun to question the meaning of social health (Donald, Ware, Brook & Avery, 1978; Ware & Donald, 1980; Ware, Brook, Davies & Lohr, 1981). Among the many components of their analysis they reviewed various definitions of health. They found that the dictionary definition of health status includes only physical and mental health. The Oxford English Dictionary does not include social health in its definition. In their analysis of data from the Rand Health Insurance Experiment, Ware and colleagues found little evidence that measures of social health contributed to general definitions of health status.

In 1978, Donald, Ware and colleagues reviewed seventy studies relevant to social health. They selected from these eleven studies for more detailed analysis. The great majority of these studies focused on what we now call social support. Yet, there were at least two separate components being assessed by the many investigators contributing to this literature. One component, dubbed by Donald et al as "social contacts", referred to performance of social role. They named the other component "social resources" because it was relevant to individual satisfaction with and comfort in using members of the social network.

In an empirical analysis, Ware et al (1980) identified some of the difficulties with including social support as a component of health status. For example, it became clear that two individuals at the same level of physical and mental health would have different overall health status scores if one of them lived in a strife torn community or was separated from family. They proposed an alternative model that considers social support as an external, predictor factor for health status rather than as a component in the definition of health.

In light of these problems, it is necessary to consider the definition of health in relation to social support.

What Is Health ?

Over the course of the last decade, a group of colleagues and I have attempted to develop operational definitions of health status. In a variety of different publications, we have argued that a single index of health status is both feasible and highly desirable (Kaplan, 1982; Kaplan & Bush, 1982; Kaplan, Bush & Berry, 1976, 1978, 1979; Kaplan & Ernst, 1983). This single index of health is very general and we believe that it includes physical, mental and social aspects. The argument against a separate measure for "social health" is rooted in the need for a single comprehensive numerical expression for health. In order to develop this argument, it is necessary to spend some time reviewing the rationale for a general health status measure.

The traditional approach to health status assessment focuses on measures of mortality. There are a variety of different mortality indexes, such as the crude mortality rate and the age-specific mortality rate. There are also a variety of disease-specific mortality rates. Sometimes, global comparisons of health care in different nations are made using extremely crude indexes, such as the infant-mortality rate or the number of infants that die before their first birthday for every 1,000 live births. There are many difficulties with focusing only on mortality as a measure of outcome. The most obvious problem is that the mortality rate ignores all those who are alive. Most of health care is oriented toward improving the quality of life in addition to extending the duration of life. Measures of quality of life typically consider morbidity. The other extreme from

mortality alone is the breakdown of morbidity into numerous specific disease categories. Considering different specific disease indicators makes it impossible to make rational comparisons between programs or treatments that have different specific objectives. For many treatments in medical and health care, it is possible to define an outcome that is clearly related to the objectives of treatment. For example, a treatment designed to control hypertension can be assessed through its effect upon blood pressure.

However, very specific measures may overlook the consequences or side effects of treatment (Jette, 1980; Mosteller, 1981). A treatment for hypertension, for example, may cause gastric irritation, nausea, and even bed disability. Surgery may have beneficial effects that are very specific. However, it may also be associated with increased risks and it may produce a variety of other symptoms. Even investigators that have used mortality as an outcome measure have often failed to consider the effects of treatments in a broader perspective. For instance, several investigators have attempted to show that specific interventions that lower serum cholesterol reduce the rate of death due to coronary heart disease. In most cases, decrease in deaths due to heart disease are associated with increases in deaths from other causes. It has been relatively rare to show that the interventions reduced total mortality (Kaplan, 1984).

Health care should not be considered effective if we exchange one symptom for another. We do not want to decrease heart disease if as a consequence we increase cancer. We have proposed that the objectives of health care be carefully defined and quantified. Then, various interventions can be evaluated to determine whether they are meeting the objectives of enhanced health status.

Our approach is to express the benefits of medical care, behavioral interventions, or preventive programs in terms of well-years. Others have chosen to describe the same outcome as quality adjusted life years (Weinstein & Stason, 1977). Well-years integrate mortality and morbidity to express health status in terms of equivalence of well-years of life. If a man dies of heart disease at age 50 and we expect him to live to age 75, it might be concluded that the disease caused him to lose 25 life years. If 100 men died at age 50 (and also had life expectancies of 75 years), we might conclude that 2,500 (100 men X 25 years) life-years had been lost.

Yet, death is not the only outcome concern in heart disease. Many adults suffer myocardial infarctions leaving them somewhat disabled over a long period of time. Although they are still alive, the quality of their lives has diminished. Our model permits various degrees of disability to be compared to one another and uses a continuum where death is given a value of 0 and optimum function is 1.0. A disease that reduces the quality of life by one-half will take away .5 years over the course of one year. If it affects two people, it will take away 1.0 well-years

(equal to $2 \times .5$) over a one year period. A medical treatment that improves quality of life by .2 (on a 0 to 1.0 scale) for each of five individuals will result in the production of one well-year if the benefit is maintained over one year. In summary, it is possible to express the benefits of very different programs and treatments by showing how many well-years of life they produce.

Creating a system that quantifies health status in this way requires several steps. During the early phases of our work, a set of mutually exclusive and collectively exhaustive levels of functioning were defined. This was accomplished by reviewing medical reference works and a variety of instruments used in epidemiological studies. This review, which was published by Patrick, Bush & Chen (1973), suggested that disease and disability impact upon three different dimensions of daily functioning; mobility, physical activity, and social activity. Scales were created that described the impact of disease or disability on each of these aspects of functioning. Table 1 shows steps from the

TABLE 1

Dimensions and Steps for Function Levels in the Quality of Well-Being Scale

Mobility	Physical Activity	Social Activity
Drove car and used bus or train without help (5)	Walked without physical problem (4)	Did work, school or housework and other activities (5)
Did not drive, or had help to use bus or train (4)	Walked with physical limitations (3)	Did work, school, or housework but other activities limited (4)
In house (3)	Moved own wheelchair without help (2)	Limited in amount or kind of work, school or housework (3)
In hospital (2)		
In special care unit (1)	In bed or chair (1)	Performed self-care but not work, school or housework (2)
		Had help with self-care (1)

Source: Kaplan and Bush, 1982

three scales. This same classification system has now been used by a variety of investigators to classify functioning (Reynolds, Rushing & Miles, 1974; Stewart, Ware, Brook & Davies-Avery, 1978). However, most other investigators choose to call this an index of physical functioning. We believe social, mental and physical problems can cause disruptions in functioning. For example, consider the person who is able to travel freely, walk without any limitations, and perform self care, but did not work on a particular day.

There are many explanations for this problem. It may be physical, such as an acute illness, mental, such as being depressed, or "social". An example of a social explanation might be death of spouse. We would simply say there has been a disruption in function. The next step would be to determine an explanation for the dysfunction. This is done by combining a symptom or problem with each level of function. For most of our work, we have used a list of 35 complexes of symptoms and problems. Examples of symptom/problem complexes that are used to explain dysfunction are shown in Table 2.

The next step is to determine the desirability of each of the function states. As was noted earlier, the health decision model requires that the impact of health conditions upon the quality of life be evaluated. The desirability of health situations can be evaluated on a continuum from death to completely well. An evaluation such as this is a matter of utility, value or preference. Combinations of function and symptom/problems can be scaled to represent precise degrees of relative importance. These values are obtained in community preference surveys. In scaling experiments, community members rate desirability of various health states on the continuum from 0 to 1.0. Mathematical models of this preference have been developed and cross validated in a variety of studies (Kaplan, Bush & Berry, 1978). These weights, then, describe the relative desirability of all function states on a scale from 0 (for death) to 1.0 (for optimum function). Thus, a state with a weight of .5 is viewed by members of the community as being about half as desirable as optimum function or about halfway between optimum function and death. It is important to note that community groups do not often vary greatly in their preference for health states. Various studies have compared graduate students and nurses, community groups and graduate students, health providers and health administrators, etc. They all show that different groups do not differ in their preference for health states (Patrick, Bush & Chen, 1973b). Similarly, various social groups do not differ in the preference for these states. Blacks, Whites and Hispanics differ only very slightly. The rich and the poor seem to agree on the desirability of different states. So do those with little education and those with many years of education. The percentage of variance accounted for by different social group characteristics after variance associated with the stimuli (or case descriptors) has been removed is very small

TABLE 2

Ten Sample Symptom or Problem Complexes and Adjustments (W) for level of Well-Being Scores

Complex Number	Symptom or Problem Complex	Adjustment
C 1	Any trouble seeing--includes wearing glasses or contact lenses	0.0190
C 9	Pain in chest, stomach, side, back, or hips	-0.0382
C 11	Cough, wheezing, or shortness of breath	-0.0075
C 13	Fever or chills with aching all over and vomiting or diarrhea	-0.0722
C 15	Painful, burning or frequent urination	-0.0327
C 19	Pain, stiffness, numbness, or discomfort of neck, hands, feet, arms, legs, ankles or several joints together	-0.0344
C 23	Two legs deformed (crooked), paralyzed (unable to move), or broken--includes wearing artificial limbs or braces	-0.0881
C 32	Loss of consciousness such as seizures (fits) fainting, or coma (out cold or knocked out)	-0.1507
C 33	Taking medication or staying on a prescribed diet for health reasons	0.1124
C 35	No symptom or problem	0.2567

Note. Adapted from Kaplan et al., 1976

(Kaplan, Bush & Berry, 1978).

A consensus is important because these preferences can be aggregated for use in policy analysis. Data suggest that different groups agree on the health objectives they are attempting to achieve. Many authors have argued that values differ so greatly that they cannot be used in aggregate policy models. However, it may not be all values that differ. For

instance, Rokeach (1973) distinguished between instrumental and terminal values. Instrumental values describe instrumental behaviors used to obtain some desirable end state of existence. Terminal values are the end states themselves. In other words, the instrumental-terminal distinction is analogous to means and ends.

Rokeach (1973) suggested that social groups differ in instrumental values but not in terminal values. There is agreement that it is desirable to be free of disease, to have enough food, to be happy, etc. Groups disagree on how to achieve these objectives (instrumental values). In summary, there seems to be little disagreement between social groups that certain levels of function are more desirable than others. For instance, different social groups agree that it is better to be able to walk without limitations than to walk with limitations. Traveling under one's own power without a wheelchair is more desirable than being limited to a wheelchair. Thus, people with very different social backgrounds agree on the rank order of these preferences. There has been some debate about the precise method used to scale these levels of function (Veit & Ware, 1982; Kaplan, 1982). Yet, this debate concerns the interval and ratio characteristics of the data. All scaling methods provide, essentially, equivalent data up to a monotonic transformation.

Later in this chapter, it will be suggested that comparable agreements cannot be obtained for social support. This contributes to the difficulty in finding an objective definition of social health. Before reviewing this problem, it will be important to consider transition among health states over the course of time.

Quality of life at a particular point of time is only one of two major components in our definition of health status. The other component is prognosis or the probability of transition among defined levels of function over the course of time. In order to illustrate this concept, consider two individuals who are in the same defined level of functioning at a particular point in time. Because of their particular disease or disability, both were confined to the house, could walk without physical limitations, but needed some assistance to perform their social role. However, the reason for these limitations was different. One person had cancer that had metastasized to the lymph glands while the other had the 24 hour flu. Although their level of functioning on a specific day was the same, they might have very different expected transitions to other levels of functioning over the course of time. The cancer victim would be expected to continue to convalesce or to get worse. The person with the flu would be expected to recover within a few days. When transition is considered and documented in empirical studies, a particular diagnosis is no longer needed. We fear diseases because they affect our current functioning or the probability that there will be a limitation in our functioning at some time in the future.

Several models of social support (Cobb, 1976) suggest that support may not affect current functioning but may affect these transition probabilities. For example, according to the buffering model, someone exposed to extreme stress who does not have social support might be expected to have a higher probability of illness at some point in the future.

When properly determined, health status can be expressed precisely at the expected value (product) of the preference associated with states of function at a point in time and the probabilities of transitions to other states over the remainder of the life expectancy. Quality of well-being (W) is a static or a time specific measure of function while the well-life expectancy (E) also includes the dynamic or prognostic dimension.

In this section, I have outlined our philosophy for the definition and measurement of health status. Health status is largely defined by determination of function at a particular point in time, attaching a preference or weight to those observable levels of function, and estimating the transition among the observable states over the course of time. Social health is absent from this system. Yet, the system does include a classification of the impact of disease and disability upon the performance of social role.

Ware & Donald (1980) suggested that social health includes at least two dimensions; social activities and social resources. The social activities portion is clearly included within this general health measure. If a disease, disability, or health problem interferes with social role performance, its impact is quantified. The system is not limited to "physical health", it includes the impact of health conditions upon "functioning" and functioning includes social activities. There is considerable agreement among different groups that it is undesirable to have social activities disrupted. One of the reasons health conditions are a concern is that they disrupt social function. If health conditions did not disrupt function or cause premature death they would not be of interest. A definition of social health or social function apart from "function" is unnecessary and, in fact, confuses predictor and criterion variables.

The portion of social health missing from our definition is the social resources or the social support component. The remainder of this paper will be devoted to outlining why we have not included this in our definition of health and why we do not feel it should be included in other definitions of health.

HOW DOES SOCIAL SUPPORT RELATE TO HEALTH?

Although social support does not seem to be a meaningful component of a health measure, it is still possible that social support mediates health status. Sometimes, investigators will accept as an outcome measure a variable that is consistently found

to correlate with the outcome. For example, some epidemiologic evidence suggests that certain lipid carrying lipoproteins mediate heart disease. For instance, a profile of high levels of low density lipoprotein (LDL) and low levels of high density lipoprotein (HDL) is considered by some to be predictive of future cardiovascular disease. Although this is an area of active debate (Mann, 1977) it is fair to say that a substantial number of physicians and epidemiologists believe this relationship is true (Kaplan & Criqui, 1984). Given this purported relationship, many investigators have accepted levels of cholesterol as outcome measures in research studies. Although they are not measures of health status, they are studied because they are believed to be related to health status.

By analogy, although social support is not a measure of health status, many believe that there is a connection between support and health outcomes (Wallston, Alagna, DeVellis and DeVellis, 1983). However, this relationship is very inconsistent across studies and the direction of the relationship is not always the same. Some of our own data from studies involving adults and juveniles afflicted with different forms of diabetes mellitus help illustrate this point.

The first example comes from a study examining the relationship between health status, control of blood sugar, and social support among adults suffering from Type II or Non-Insulin Dependent Diabetes Mellitus. There are approximately six million diabetics in the United States and more than 80% of them have this form of the condition. Diabetes is a problem in energy metabolism that may lead to serious complications including blindness, kidney failure, amputation of extremities, and heart disease. As many as 90% of adults affected with Type II Diabetes are overweight and evidence suggests that the most appropriate treatment is focused on weight loss. The core of this treatment involves behavior modification for weight loss and daily exercise. Stress is also believed to be an important factor in diabetes control (Surwit, Feinglos, & Scovern, 1983).

The primary responsibility for treatment is with the patient--self-care including diet, exercise and in some cases medication adherence are the treatment. Given the behavioral demands placed on a Type II Diabetic patient, we predicted that social support would be an important mediator of blood sugar control. Those with adequate support systems were expected to be more compliant with this difficult regimen and to have better control over their condition. Conversely, those without adequate social support were predicted to be in poor control of their metabolic state. Social support was measured using the Sarason SSQ (Sarason, Levine, Basham, & Sarason, 1983). Diabetes control was measured using the glycosylated hemoglobins blood test, which is an index of the level of blood sugar control over the preceding 80 to 120 days.

To our surprise, the results were different than anticipated.

Overall, there was no relationship between control of metabolic state and either the number of people in the social support network (SSQN) or the satisfaction with social support (SSQS). However, there was a strong interaction between sex and social support satisfaction scores for the diabetes control outcome. That interaction is pictured in Figure 1. Women were in better control if they had high satisfaction with social support. Conversely, men were in poorer control if they had high social support satisfaction scores (Haggerty, Heitzmann & Kaplan, 1984). These results are consistent with a variety of other papers showing sex differences in the relationship between social support and outcome variables (See Lefcourt, this volume). However, the direction of the relationship between social support, sex, and outcome was not consistent across studies. For some problems, women derive more benefit from support than men, while in other circumstances men benefit more from support than women.

Another important finding was that the correlations between support satisfaction and number in the network were in opposite directions for males and females. Women were more satisfied if their networks were large, while men were more satisfied if their networks were small. This result, which confirms the observations

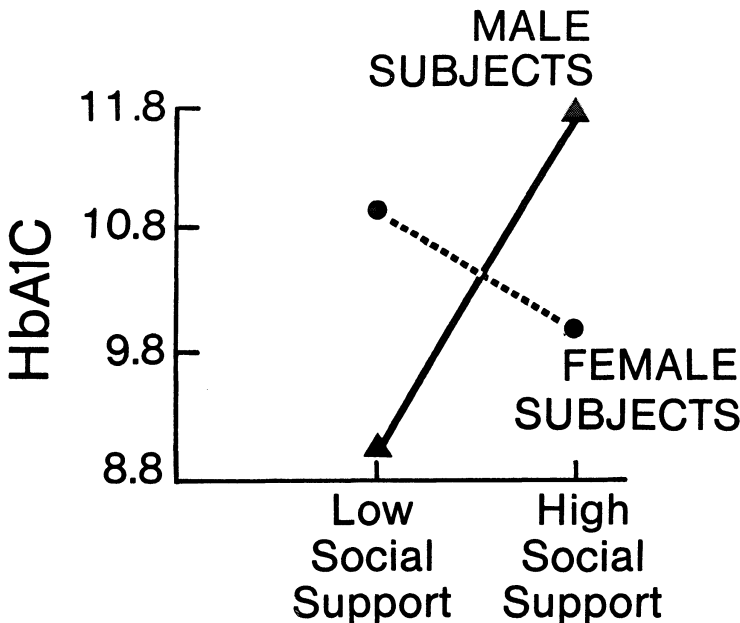


FIGURE 1. Graphic display of sex X support satisfaction for HbA1C.

of others (See Kessler et al., this volume) suggests that social preferences for social support may differ for men and women. As a result, it may be difficult to develop a social policy for network size that applies to both men and women. I will return to this point later in the discussion.

In another recent study, we examined the relationship between social support and self-care in teenage diabetics. Adaptation to peer norms is difficult for any group of teenagers. Yet, this problem is greatly compounded for teenagers with diabetes. Proper control of the condition requires strict adherence to a complex regimen of diet, exercise and self-medication. Several authors have argued that successful adaptation to Type I or Insulin Dependent Diabetes requires the teenagers be well integrated into their social group. Further, it has been suggested that stress and social isolation may be associated with poor diabetic control (Surwit, Feinglos and Scover, 1983). There are very few empirical studies that have tested these ideas.

During the summer of 1983, we conducted a three week summer school program for teenage diabetics. Twenty teenagers between the ages of 13 and 17 participated. While participating in the summer school program, each youth completed a variety of measures including the Sarason Social Support Questionnaire, self-reports of behaviors relevant to diabetes control and a wide range of biochemical measures including glycosolated hemoglobin or HbA_{1c}. High HbA_{1c} scores are indicative of poor diabetic control, while low scores suggest good control.

Contrary to our predictions, there was a significant relationship between social support satisfaction and poor control of diabetes ($r = .33$, $p < .05$). The relationship between diabetes control and other variables was also examined. It was shown that diligent self-care was associated with good control ($r = -.45$, $p < .05$; high hemoglobin A scores reflect poor control resulting in the negative correlation). Appropriate self-care was significantly correlated with Self-Control (Rosenbaum Scale; $r = .45$, $p < .05$) while Self-Control was negatively correlated with social support. Why would satisfaction with social support be negatively associated with good diabetes control? One explanation is that the behavior of teenagers is strongly influenced by their peer group. Those teenagers who are most socially adept may also be the ones most satisfied with their social relationships. We found that poor diabetes control was also highly correlated with social skills as measured by the Means Ends Problem Solving test or MEPS (Platt & Spivak, 1974). For these socially adept teenagers, the peer environment has a significant influence on self-care behaviors. For a diabetic teenager, this influence may not be associated with good metabolic control.

It is interesting to consider what happens to these socially adept teenagers when the social environment changes. In the intense diabetes summer school, the youths were removed from their regular peer group and immersed with other diabetic teenagers.

During this period, there was continual discussion and emphasis of the importance of adherence to the prescribed regimen of diet, exercise, and medication. All of these youths were assessed on the day they entered the program and reassessed on the day they left. Then, analyses were performed to determine predictors of change while in the program. Social support as measured by both the number and the satisfaction components of the Sarason SSQ were each very strong predictors of positive change in the program. Social support was a predictor of poor adherence and control of diabetes when the youths entered the program. However, it was also a predictor of change as a function of participation. One explanation for these apparently discrepant results is that the social environment creates a strong influence for socially oriented individuals. Prior to entering the program, socially oriented teenagers with diabetes may be most influenced by a peer group of non-diabetic friends. Upon entering a program with diabetic peers, the mores of the social group change and behavior change results. We will collect another Hemoglobin A sample from these teenagers in three months.

In summary, our studies with both adult and teenage diabetics suggest that social support may play an important role in diabetes control. However, the results are complex and not clearly predicted from existent theories. At present, we do not know what to recommend with regard to social support for diabetic individuals.

Contrast this with studies in health policy in which there is high consensus for what we are trying to achieve. There is agreement that high quality of life is desirable because quality of life is an endpoint in itself. Social support appears to be an interesting predictor of functioning, yet the direction of the relationship between social support and functioning is not clear and consistent from all studies. This suggests that social support should be considered an external mediator of health but not a measure of health status itself.

OTHER PROBLEMS

There are a wide variety of other reasons that social support should not be considered health. These include social control problems, policy problems, and the relationship between social services and the medical care system.

Social Control Problems

Since there is strong consensus that high quality of life is desirable, it seems reasonable to invest public resources in order to maximize these end states of existence. Some authors have argued for a "threshold" or "unmet needs" social support model that characterizes lack of social relationships as a stressor

(Kaplan, Cassel & Gore, 1977). This model suggests that there is a continual need for social interaction and when these interactions fall below some critical value, people become vulnerable to disease. For example, there does appear to be a relationship between social isolation and psychiatric difficulties (Leighton, 1959). Yet, the causal relationship has not been clearly established. At present, it would seem premature to construct a social policy that attempts to maximize social support.

An example may help clarify this point. First, consider health status according to our definition. There is widespread agreement that it is more desirable to be able to travel around the community than to be limited to the house because of a health condition. Thus, it seems a reasonable objective to develop programs that will prevent limitations or help those who have them. The same does not hold for social support. Social groups may differ greatly in the degree to which they find a large social network desirable. Studies of loners show that they are content to be alone. Yet it is rare for people to express a desire to remain in a dysfunctional state. A large familial network characterizes Mexican-American families, but does not characterize Black-American families. If we decide that a threshold level of social support is a desirable health state, should we then implement programs to make families larger? Any effort that attempts to maximize social support in the absence of choice would have serious social control implications. Including social support in the definition of health implies that there is some desirable level of support that a health care system should set as its goal.

A recent essay by Kiesler makes a similar point. He cited evidence from a study by House, Robbins and Metzner (1982) showing that people with no others in the support network had a higher prospective mortality rate than those with one or more social contacts. Yet, these isolated individuals may not be capable of maintaining relationships and may have been socially incompetent throughout their lives. Further, it is not clear any program will be helpful. A schizophrenic with extremely negative feelings about his relatives may not benefit from an effort to reunite the family (Kiesler, 1983).

Policy Implications

The issues of social control are related to policy problems. Considering the example of function, there is strong consensus that function is desirable. Thus, it seems reasonable to devote public resources to maximize the level of function and quality of life within a community. Optimized health status might be considered a common goal as would national defense, a strong educational system, etc. Including social support in the definition of health status would imply that community resources

should be used to obtain some defined level of social support. Yet, we might expect considerable public disagreement about what that objective should be.

Excluding social support from the definition of health makes policy analysis relatively straightforward. There is little disagreement about what states of health are desirable and achieving these states becomes the objective of health care. A major issue is in defining a mix of programs that most efficiently and effectively achieve these objectives. Programs that enhance social support might be considered in this mix. Yet including social support in the definition of health only confuses the definition of the objectives.

Problems In The Medical Care System

It is not clear why non-physicians have been attempting to force social support into the definition of health. Physicians dominate health care and claim everything that all health services should be under their control. Other workers within the health care system provide technical assistance and scientific support for the activity of physicians.

Most medical schools provide relatively little training in social science. As social support has been defined into health, many physicians have been nudged into the social science arena. They sometimes employ social scientists to advise them on issues relevant to social support and they may bill for the services of social workers under their medical licenses. Yet, the physicians themselves may not be well equipped to study social support or to effect modifications in lifestyle believed to enhance other aspects of health. Including social support in the definition of health legitimizes the medicalization of social relationships. To affect "social health" physicians may go beyond their medical and surgical training into an arena for which they are poorly equipped. There has been a similar medicalization of some legitimate topics in cognitive science. Medicalizations of social relationships may lead to the relegation of those with specific interest and training in social sciences to be assistants in the quest for social health.

CONCLUSION

Nearly thirty-five years ago, the World Health Organization included "social health" in its definition of health status. Since the introduction of this definition, physicians and scientists have been scrambling to find an operational definition of "social health". Most measures of social health can be classified as reflecting social functioning/role performance or social resources/social support. Functioning and social role are appropriate components of the definition of health status because

diseases and disability impact upon functioning. A restoration of functioning should be a major objective of the health care system.

Conversely, social resources and social support may be important mediators of functioning and health status. However, the relationships between social support and health are not clearly understood. They may be complex, and preference for different levels of social support may vary from individual to individual. It seems a reasonable objective to devote public resources to the enhancement of health status. Yet there are serious social and policy implications for programs designed to set objectives for social relationships.

The thirty-five year quest to force social support into the definition of health status has been a failure. It is time to give it up.

NOTES

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PART II

***HUMAN DEVELOPMENT, PERSONALITY
AND SOCIAL NETWORKS***

SOCIAL NETWORKS AND THE ECOLOGY OF HUMAN DEVELOPMENT: THEORY, RESEARCH AND APPLICATION¹

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Despite the upsurge of research on social support, we are only beginning to analyze the processes by which social support may affect mental health and well-being. Numerous studies do point to a positive association between support and mental health, sometimes as a "main effect," while at other times as a buffer against the effect of ongoing strain or discrete life events. These studies indicate that social support is worth studying. However, given the often superficial assessment of social support and the inattention to process issues, many of these studies contribute but marginally to our understanding of support phenomena.

The conceptual and theoretical weakness of much support research echoes previous failures in research on social stress. Endless studies demonstrated empirical associations between recent major life changes and psychological symptomatology. Few of these studies were able to help us to understand why this was so; indeed, few of the research designs even permitted us to address this question. It was hoped that research on social support would help to develop this understanding. Instead, a parallel path has been followed: there are numerous reports of empirical associations between support and mental health, but negligible concern with learning about the reorganization of one's life that is presumably required to successfully cope with major stressors--and to which social support presumably contributes. To understand how individuals attempt to maintain or repair the meaning and texture of their life does not call for the myopic research we have often witnessed.

Given the narrowness of much of the inquiry, it should hardly be surprising that specific social support findings have had such little impact on mental health practice. Psychotherapists and

community change agents need to be concerned with people's lives in ways that have typically been ignored by researchers, such as the goals and values that shape our coping efforts, and the fact that social networks can hurt as well as help. Practitioners need to know which network member can best provide the specific support needed to attain specific objectives. And they need to be aware of personal and environmental factors that aid us in or constrain us from seeking and getting that support.

In my own work, I am attempting to fashion a research strategy that addresses these theoretical and pragmatic shortcomings. The research is designed to contribute to our understanding of people and their lives more broadly, and to be useful in the development and evaluation of clinical and social-community interventions. The research focuses on the analysis of social networks and mental health within the broader context of the ecology of human development. It is concerned with how networks affect our ability to achieve critical developmental tasks over the life-span, particularly our ability to articulate viable social roles in the major spheres of life. It is ecological in that persons are located in a systematically differentiated social environment. After briefly outlining this framework, I present several ongoing studies with adults and adolescents as illustrations of how this kind of research can be designed.

FRAMEWORK FOR THE RESEARCH

This research is based on a family of theories relating social networks and social identities to mental health (e.g., McCall & Simmons, 1978; Mead, 1934; Miller, 1967; Sarbin, 1968; Sullivan, 1953; Thoits, 1983). We begin by considering our involvements in major spheres of life, such as work or school, family, friends, leisure interests, and so on. The quality of our participation in these spheres of life, and the satisfactoriness of the social identities which we thereby construct, are hypothesized to affect our mental health. Thus, I presume that individuals who have a happy marital and family life, enjoy their jobs, and have active and satisfying leisure pursuits will generally be mentally healthier than individuals who are dissatisfied in these areas. Interactions with members of our social network may affect the quality of our involvements; for example, spouse and coworkers can have an important impact on whether we are satisfied or dissatisfied with our job and the kind of work that we do. In some instances, the quality of our relationship with one or more network members (e.g., spouse) will define the quality of our involvement in that sphere of life (marriage); although interactions with other network members (e.g., parents, in-laws, children) can still affect the quality of that involvement (marital satisfaction). Thus, the overarching

hypothesis is that social networks affect the quality of our involvements and the satisfactoriness of our social identities in major spheres of life, which in turn affect our mental health.

Jules Henry's (1958) notion of a "personal community" is an apt metaphor for capturing this way of thinking about social networks. By expressing and embedding our social identities in a social network, we make our social network a personal community (Hirsch, 1981b). The ties we have formed, as well as those we have not established, reflect the nature of our lives, our interests and priorities, our obligations, loves, and disappointments. In creating and maintaining a particular personal community, we are at least implicitly choosing, from among feasible alternatives, how we seek to achieve meaningful participation in our culture and society (cf. Fischer, Jackson, Steueve, Gerson, Jones, & Baldessare, 1977). Whether we are successful in establishing a personal community that is viable and satisfactory should have an important bearing on our well-being.

To develop this framework for conceptualizing social networks into a scientifically and pragmatically useful theory requires much further work. Four complementary aspects of our roles and social identities, each at a particular "level of analysis" (Rappaport, 1977), will ultimately need to be addressed.

First, we must be sensitive to the sociocultural context of our roles and social identities. The meaning of particular role involvements varies among different cultures and subcultures, and over different historical periods. An excellent illustration is provided by the different expectations of women across cultures (e.g., U.S. vs. Saudi Arabia) and subcultures, and by the dramatic change in many norms over the course of the century. Such broad social factors contribute to defining the nature and meaning of our involvements, and provide both opportunities and constraints for obtaining satisfaction, support, and mental health.

Second, we need to know more about the ways in which specific roles are expressed in specific social environments. For example, while there may be generic similarities across families in being a parent, or across treatment units in working for a large hospital, there are also likely to be important differences in the ways in which these involvements are expressed and experienced between families and among hospitals and hospital units. What kinds of social environments have substantial positive or negative effects on support and role satisfaction? What creative solutions have been developed to provide for enriched or more supportive roles? What interventions work best to increase social support and decrease social rejection for specific people in specific roles in specific social environments?

Third, we must attend to the boundaries between different roles and spheres of life. To what extent do we integrate/segregate family, friends, and work or school? What effect does this boundary definition have on our sense of self and community? How do our multiple role involvements affect our

ability to cope with stress in one sphere or simultaneous stresses in several spheres? What are the limits of interventions targeted at only one sphere of life (e.g., family; work)? How would our ability to successfully manage multiple roles be affected by alternative social policies, for instance increasing the availability of day care centers?

Finally, we need to study relevant personality and individual difference variables. For example, we may have certain beliefs which constrain us from seeking out support or diminish the experience of support which is provided. What if our values are different from those of the dominant society or subculture? How may support be mobilized in those instances? More generally, how can we help given sets of people, with particular interests, values, and skills, to develop an ecological niche for themselves in which to enjoy and receive support for what they want to do?

Several of these concerns reflect long-term objectives. The present studies are considered as stepping-stones in the development of what can presently be no more than a preliminary, provisional analysis. These studies represent initial explorations in studying how social networks affect the quality of our involvements in and across major social spheres.

There are two principle populations being studied in this research: 1) women nurses and 2) high-risk and normal adolescents making the transition to high school. The study of women nurses allows us to consider developmental processes among adult women, a topic which until very recently (e.g., Gilligan, 1982) has been subject to much "benign neglect." As a field, nursing is currently in a state of upheaval and conflict regarding its professional identification and objectives, reflecting gender-role and technological changes in the larger society. A variety of broad social factors which may affect support and well-being may thus be addressed: the changing status of women in the workforce, particularly in what has traditionally been considered "women's occupations"; the viability of dual career families; the design of human service delivery systems; the quality of work life; and so on. Research designed with due regard for these contextual factors may also be used to evaluate social policies in those domains.

With respect to adolescents, the transition to high school may be considered a major developmental life event; indeed, some investigators argue that this age encompasses more life change than any other period (Newcomb, Huba & Bentler, 1981). Adolescence is the age par excellence for beginning to articulate a repertoire of social identities, a process considered of central theoretical importance in this model of social support and mental health. This period may thus provide a prime opportunity to study salient processes, as well as to implement preventive interventions based on the development of strong social networks. We hope to evaluate one or more such interventions as part of our research.

The nursing investigation involves extensive survey research with a sample of 350 women nurses. The research with adolescents making the transition to high school, though conducted with a much smaller n, involves the use of a multimethod assessment strategy and a short-term longitudinal design.

In what follows the research questions and hypotheses being addressed in both investigations are discussed. First, we consider the nature and quality of our involvement in specific spheres of life, with each sphere considered independently (by and large). This includes a consideration of aspects of our involvement in each sphere that may affect its quality, such as stressors, goals and values, and the subjective importance of the sphere. Next, there is a brief consideration of issues that emerge when we begin to consider these spheres in conjunction: this is referred to as multiple role phenomena. Given this formulation, we then consider a relevant conceptualization of social support and social rejection, and of person-network determinants of support/rejection. The analysis is concluded by discussing our assessment of mental health. These questions are considered first in the context of the nursing research, then with reference to the adolescent research, noting especially salient comparisons and contrasts. Occasional references will also be made to other ongoing investigations which serve to illuminate particular conceptual or methodological issues.

NURSES

Quality of Role Involvements

Work and family involvements are primary for most working adults. Our research is concerned with delineating and assessing qualitative dimensions of involvement in these roles and, then, the extent to which our involvements may be affected by social network interactions. Let us consider, first, work involvements, then, family involvements.

Work is certainly a major force in the lives of employed adults. We spend a considerable amount of time at our jobs, these jobs determine to a significant extent our status in society, much of our behavior and beliefs outside of work can be related to the job setting, our jobs often constitute a major source of our self-image, and various job characteristics have been related to our mental health (for reviews see, e.g., House, 1981; Kanter, 1977; Kasl, 1978). In the research with nurses, we are assessing several sources of job satisfaction: overall satisfaction with the job; satisfaction with the actual work that is done; and satisfaction with other specific dimensions of the job such as physician-nurse relationships, pay, staff cohesiveness, and so on. Several of these job satisfactions clearly relate to network interactions at work (e.g., physician-nurse relationships, staff

cohesiveness); for illustrative purposes, let us instead consider networks and satisfaction with the actual work that we do, especially as issues of job design are currently a major focus of analysis and change efforts in organizational settings (Hackman & Oldham, 1980).

Network interactions can positively or negatively affect our satisfaction with the job itself via several mechanisms. Coworkers can give advice, guidance, or training that will help us to learn the job, to do the job well, and thereby to increase our opportunity to gain intrinsic satisfaction from the performance of the task. "Rest and recreation" breaks with coworkers can help prevent job burnout and enhance our ability to derive satisfaction from our work over the long term. We may share our ideas and experiences with coworkers, which may enhance our satisfaction with and reinforce the value we place upon the intrinsic features of the task. By not acting aversively and by not engaging us in conflict on the job, coworkers can provide us the opportunity to participate in rewarding aspects of the job. Interactions with network members outside of the work place also probably contribute both positively and negatively to our satisfaction with the job itself. We intend to use path analytic procedures to determine the extent to which specific forms of network interactions are related to specific job satisfactions and, thereby, to mental health.

We are also investigating the extent to which network ties can affect levels of marital satisfaction. Our focus here is on how interactions with relatives and friends can affect marital satisfaction, insofar as there would be considerable overlap between measures of general (nonspecific) marital support and of marital adjustment (potential correlations here suggesting convergent validity rather than a causal linkage). As the discussion of networks in relation to job satisfaction emphasized positive effects of networks, let us here consider possible negative effects of networks. Parents, other relatives, and friends can make it quite clear that they do not like one's spouse. They may fail to support us during times of marital stress or suggest interpretations or give advice which proves highly detrimental. Friends may have different interests or interpersonal styles than one's spouse, which may lead them to avoid socializing with us as a couple, which in turn may increase our vulnerability to depression during stress. Most of the nurses in our study are married and we are assessing the extent to which network interactions with several different network members are related to a standard measure of marital satisfaction.

Network interactions may also effect the quality of our parenting. In her dissertation research, Cynthia Zarling (in progress) is investigating the social networks of mothers of a sample of high risk premature infants. This research will consider the effect that interactions with specific network members have on the mother's interaction with her child, as

measured by behavioral coding of laboratory interactions, as well as on objective measures of the child's development.

Stressors Within Specific Spheres of Life

A central issue in social support research has been whether support moderates or "buffers" the effect of stress on mental health, or as reformulated by Thoits (1982), whether the effects of support on health are only or especially significant under high conditions of stress. While it is necessary to move beyond this single issue focus in support and network research, it nonetheless remains of importance and is addressed in our research.

Within the framework being developed, life difficulties and events are conceptualized as ongoing or acute challenges to our attempts to achieve and maintain satisfactory involvements in major spheres of life. Coping with stress is thus seen as one particular facet of our more general effort to develop satisfactory role involvements and a rewarding quality of life. This approach permits us to continue to study stress, coping, social support, the etiology and prevention of disorders, and so on, but frames these efforts in terms of an overarching orientation to well-being and healthy development (cf. Hirsch, 1981b, in press; Hirsch & David, 1983; Hirsch & Jolly, 1983).

In the research with nurses, an assessment is made of ongoing stressors in their major social environments. We intend to analyze the relation of these various stress measures to indices of mental health, job and marital satisfaction. These analyses will permit a more detailed examination of the pathways by which stress, and perhaps thereby social support, may impact mental health.

One possible source of job stress of particular interest concerns participation in decision making with physicians about the patient's medical care. This is an area of much contention between physicians and nurses. Several experimental organizational innovations are currently being explored which attempt to involve nurses and physicians in collaborative decision making (National Joint Practice Commission, 1981). These experiments have considerable potential, at least at the organizational level, to enhance the quality of nurses' (and physicians') worklife. The social support provided to nurses (and physicians!) engaged in such an enterprise may well have a critical impact on the success of such interventions. Conversely, the absence of participatory decision making may be stressful for nurses and thus enhance the importance of social support. To address these and related issues, we have developed a measure of nurses' participation in decision making about the patient's medical care. We will analyze the support available to nurses at varied levels of participation and consider the association between support and the different outcome measures at each level as well.

Goals And Values Within Each Sphere of Life

In order to understand how individuals make use of, cope with, adapt to, or modify environmental opportunities and environmental constraints, we need to specify the specific objectives they are trying to achieve. Access to social support may be determined in part by what one wants social support for--e.g., whether or not support is provided may reflect the congruence between our goals and those of network members. Our appraisal of particular interactions as supportive may also vary as a function of the relevance of that interaction to our goals, the subjective importance of that specific goal, and the difficulty or ease with which the goal may be obtained. We intend to analyze the relationship between specific goal preferences and the provision of social support. Goals and values are assessed in relation to nursing as a profession and to feminism more generally.

Differences in the Importance of Each Life Sphere

There are certainly individual and group differences in the importance attached to any particular sphere of life, such as the importance of work. The salience of social support for a particular life domain may well vary as a function of the importance of value of that specific domain. It appears reasonable to hypothesize that the greater the value and importance we place on the quality of our life in a particular sphere, the greater will be the effect of the social support we are or are not able to obtain relevant to that particular domain. In regards to work involvement, for example, we would expect a higher correlation between work social support and mental health among those for whom work is a central rather than a peripheral life interest. In the research with nurses, we have assessed several complementary dimensions of the importance of work which will enable us to test this hypothesis. We also intend to examine whether the effects of work stress upon mental health are moderated by the importance of work.

Multiple Role Issues

To this point, our analysis has emphasized the quality of individual involvements in separate and distinct spheres of life. Occasional allusion has been made to cross-sphere phenomena, such as support or rejection provided by family members for work issues, and the impact of friends on marital adjustment. In this section we consider several such cross-sphere or multiple role phenomena (for a more extensive analysis, see Hirsch & Jolly, 1983).

On those occasions when goals or demands in two or more spheres are conflicting, the kind of support (rejection) we obtain

and the salience of this support (rejection) may well be related to the manner in which we have prioritized our involvement in one sphere or the other. Just as the goal we have within a sphere may affect the quantity or importance of our interactions, so may the extent to which we value one sphere over the other. This issue is being explored most thoroughly in an M.A. Thesis being conducted by Thomas M. Reischl (in progress). This research is concerned with a group that is experiencing considerable difficulty in making a successful developmental transition: college freshmen on academic probation. Two potentially competing goals that such students may have at this time are, first, to get better grades so that their probationary status may be lifted and, second, to have a satisfactory social life in a new environment. Reischl is comparing one group of students that is primarily oriented toward academic success rather than the quality of their social life with another group that is primarily concerned with the quality of their social life rather than their academic activities. The assumption is that both groups will want to achieve both objectives, but that the differential weighting of these goals may result in different coping strategies, different judgements about the supportiveness of particular interactions, and different associations between specific network interactions and measures of academic, social, and mental health outcomes.

Another critical multiple role issue is how to successfully combine simultaneous involvements in different spheres of life. In the nurses research, we have identified clusters of nurses who have good mental health, high job satisfaction, and high marital satisfaction, and will contrast this group with nurses who have not been able to combine work and family life as satisfactorily. We will determine whether there are particular patterns of social support and social rejection, as well as particular network structures (e.g., the extent of ties between work associates and family members), that are conducive toward achieving these dual objectives. We will then consider whether there are particular characteristics of their work environment that are conducive to eliciting support for multiple role demands. Dual career families are proliferating; we hope that findings from our research will contribute to the formulation of social policies that will enable more couples to satisfactorily combine work and family life.

Perceived Social Support and Perceived Social Rejection

The prominence given in this model to specific work objectives and the quality of work life suggests that our assessment of social support should be, at least in part, similarly work specific. We have therefore constructed a scale of 16 representative work-specific social interaction items. Nurses indicate the frequency with which each of those 16 interactions have occurred with each of 5 different network members: their

closest coworker at work, their supervisor, their closest friend outside of work, their spouse or person with whom they have an intimate relationship, and an adult family member. Thus, there are potentially 80 separate ratings which are made in this section. The items are divided equally between positive and negative interactions. The positive items are conceptualized as perceived social support, the negative items as perceived social rejection. Perceived work related social support involves those interactions which are believed to promote or to be congruent with a desired state of affairs concerning work, while perceived work related social rejection involves those interactions which are believed to promote or to be congruent with an undesired state of affairs concerning work. This assessment of social interactions reflects our hypothesis that network members may both help and hinder attaining work objectives, and that interactions from both work and nonwork network members can have this effect.

In addition to this work-specific network assessment, we have also included a more traditional assessment of nonspecific social supports and social rejection. This scale is heavily weighted toward assessing the communication of positive or negative affect, or what may be termed the provision of emotional support and rejection. Ten kinds of interactions are assessed, half expressing social support and half expressing social rejection, from the same five network members specified earlier. Potentially, therefore, 50 ratings may be made on this subscale (130 potential ratings across both the work-specific and nonspecific interaction items).

This assessment of social support and social rejection will enable us to examine the link between varied kinds of social interactions and multiple outcomes, the relative effect of social interactions with different social network members on these multiple outcomes, and the extent to which specific network members are associated with particular kinds of social support and social rejection. The assessment of work-specific social interactions will permit us to explore the utility of conceptualizing the mental health consequences of social ties in terms of their effects on the quality of our involvements in specific spheres of life.

Network Determinants of Social Support and Social Rejection

Many of the previous sections focused on how aspects of our work life, including personality variables relating to work, could affect the level or importance of social support. We now consider specific features of the social network that may have such an effect.

Results from several investigations suggest that the structure of a social network can have a considerable impact on the quality of social support provided and on mental health. For example, in my earlier research with recent younger widows and

with mature women (age 30 or over) returning to college full-time (Hirsch, 1980), the structure of ties among family and friends was a major factor related to their satisfaction with five possible social supports, as well as the level of symptoms, mood, and self-esteem. The specific variable of interest was the perceived nuclear family-friendship boundary density, which is defined as the proportion of actual to potential ties that are perceived to exist between a subject's nuclear family members and a subject's friends. In this particular instance, a lower nuclear family-friendship boundary density proved significantly more adaptive for most women, though some women were able to make adaptive use of networks with a higher boundary density (Hirsch, 1981a). In research with divorced women, Wilcox (1981) found that fewer ties between a woman's ex-husband and her friends facilitated a more successful adaptation. In the research with nurses, we are considering how perceived ties between work associates and family members, family members and nonwork friends, and work associates and nonwork friends are related to the provision of social support and social rejection, as well as the various outcome measures (e.g., job satisfaction, mental health). The extent to which nurses socialize with work associates outside of work is being considered in a similar manner.

Mental Health

Our assessment of mental health in the nurses research is limited to administration of the Brief Symptom Inventory (Derogatis & Spencer, 1982). We have include subscales from this self-report instrument measuring depression, obsessive-compulsive rumination, interpersonal sensitivity, anxiety, somatization, and anger-hostility. As will be seen, the adolescent research includes a much more thorough mental health assessment.

ADOLESCENTS

Let us now consider these issues in the context of the research with adolescents making the transition to high school. Some important developmental differences are reflected in the conceptualization of role involvements and of social networks. Some issues are also addressed in greater detail in this investigation given the availability of an interview format. This is a short-term longitudinal study, with time one assessment taking place the summer before entrance into high school, time two assessment occurring at the end of the first semester of high school, and time three assessment at the end of the first year (second semester) of high school.

Quality of Role Involvements

The need to develop new kinds of life involvements and social

identities is generally considered a major task of adolescence (e.g., Conger, 1977; Douvan & Adelson, 1966; Erikson, 1950). At school, adolescents must acquire knowledge of specific content areas, develop work skills, establish more mature relationships with nonparental authority figures, and lay the basis for formulating post-high school vocational plans. With friends, they must learn to share a variety of new activities, perspectives, and emotions, and explore alternative forms of closeness and conflict at both dyadic and group levels. With family members, they must develop or maintain valued relationships based on greater independence and changing interests and values.

The basic hypothesis in this research is that adolescents need to develop social networks that reflect and embed these new involvements and social identities. Our research emphasizes in particular how developmentally appropriate social involvements and identities are expressed in behavioral interactions with friends. We will thus be examining the ways in which specific kinds of friendships and friendship networks reflect and affect adolescents' ability to establish satisfactory, developmental appropriate involvements and identities across multiple spheres of life. Given the presumed intensity and significance of friendships in adolescence (e.g., Berndt, 1982; Douvan & Adelson, 1966), it will be interesting to compare the form and function of adolescent friendships to those maintained by adult women.

In this research we examine the relationship between friendships and already available measures of the quality of school and of family life. We are attempting to develop measures of social identities that are sensitive to the changes we expect to occur during the time of the transition of high school, although reviews of prior research in the area of self-concept suggest that this will not be an easy task (e.g., Wylie, 1974).

Stressors

While the research with nurses involved an assessment of ongoing stressors at work and at home, the adolescent research includes a more wide-ranging assessment of possible stressors. The study is designed, first and foremost, to take advantage of a naturally-occurring major change in their life: the transition to high school. We assess how this transition may change their involvements with respect to family, friends, and school. For each sphere, we assess the amount of change they expect will take place, the desirability and importance of those changes, the control they report having over both the occurrence and outcome of the changes, and how stressful they expect the changes to be. Each adolescent also completes a specially designed inventory which assesses specific positive and negative events that may have occurred in their life over the year preceding the transition to high school. Overall, this assessment permits us to consider the impact of this change in their life considered as an "event", as

well as to break down the component parts within and across critical life spheres. A much more differentiated framework is thereby provided for analyzing links between stress, support, and well-being.

Goals and Values Within Each Sphere of Life

While the research with nurses assesses goals and values relating to nursing as a profession, in the adolescent research we assess the importance of specific individual goals the students hope to achieve with respect to their own family, school, and friendship involvements. In contrast to the nurses research, the adolescent research tends to emphasize goals rather than values, and does so within the context of a specific, limited timeframe (9 months). We also assess adolescents' expectancies regarding how likely they are to achieve each possible goal. In the follow-up assessment, we will determine how successful they consider themselves to have been in achieving each possible objective.

Differences in the importance of each life sphere are not currently a major focus in the adolescent research.

Multiple Role Issues

We consider this to be an ecological transition (Bronfenbrenner, 1979) with respect to changes in roles, activities, and other aspects of friendships, school, and family life. Consideration will be given not only to changes within a particular sphere, but also to how changes in one sphere affect involvements in other spheres. We hope to identify different patterns expressed during this transition and to relate these to our other variables.

Perceived Social Support and Social Refection

Our assessment of adolescent social support has been modified to account for their different developmental position. An essential feature of adolescence is the need to develop new social involvements and identities. To have a social network that reflects and embeds these new identities, it is in the first place necessary to share and participate in such activities with one's friends. We assume that developing increased competence in varied life spheres and the construction of viable and satisfactory identities and involvements will be related to adolescents' abilities to have these involvements expressed and reflected in their relationships with their friends. We are therefore assessing the extent to which adolescents engage in diverse activities with their friends, the degree to which they confide in their friends, and the range of emotions they have expressed over the course of those relationships. Conflicts with friends are also being assessed. Preliminary findings from earlier research

with adolescent children of a depressed, arthritic, or normal parent indicate that several of these friendship dimensions are related to mental health (Hirsch & Reischl, in preparation). In the present research, an unusually detailed examination is being conducted in particular of each adolescent's best friend, given the hypothesized importance of this relationship in Sullivan's theory of mental health (Renders, in progress).

Person-Network Determinants of Social Support and Social Rejection

We are studying how friendship interactions are related to the extent of perceived ties (boundary density) between friends and family, as well as ties among friends themselves. Adolescent social networks are much denser than those of adults (Hirsch & Reischl, in preparation) and may well function in a different manner given the different context.

Another dimension of interest here is beliefs held by adolescents about the advisability of discussing personal issues or problems with network members. Tolsdorf (1976), in one of the earliest network investigations, emphasized the importance of one's "network orientation" as a factor affecting the utilization of a social support system. Tolsdorf found considerable differences between psychiatric and non psychiatric inpatients regarding their network orientation, and hypothesized that this difference accounted for part of the discrepancy in their use of network members for support. In my earlier research with adolescent children of a depressed, arthritic, or normal parent, an assessment was made of the extent to which each adolescent received helpful cognitive guidance, tangible assistance, or emotional support from family and friends for both the major school and the major family problem they had experienced during the preceding year. In analyzing these data, it was striking how many adolescents reported receiving no social support of any sort for either of these stressors.

In the present research, in addition to assessing the topics about which adolescents self-disclose with their friends, we are also systematically inquiring as to the reasons they do not share certain experiences. Similarly, in a separate study, Darlene DeFour (in progress) is investigating why Black graduate students do not talk with specific members of their academic and nonacademic social networks when they consider dropping out of school.(2) Information gained from these investigations may enable us to design interventions targeted at those factors that constrain individuals from making use of potentially available social support. Such a strategy would be consistent with what is a common axiom of much clinical practice, namely, that the initial target of therapeutic interpretation and intervention is the "resistance". However, not all resistance is in the person; environments can resist as well. This assessment will also

identify aspects of the environment that inhibit individuals from seeking out support or preclude the development of social support in the first place. This ecological assessment is consistent with our earlier suggestion that interventions should be conceptualized in terms of specific people in specific roles in specific social environments.

Mental Health Assessment

While this research program emphasizes elaborating the social context of mental health, an unusually detailed assessment of psychopathological processes is being undertaken in the adolescent research. In addition to the usual symptom inventory which populates social support research (the study with nurses included), the adolescent study will include the use of a structured diagnostic interview procedure and indices of subsyndromal symptomatology. Interest in subsyndromal dysfunctions has emerged recently in the mental health literature (e.g., Chapman, Chapman, & Miller, 1982; Depue, Slater, Wolfstetter-Kausch, Klein, Goplerud, & Farr, 1981). Subsyndromal symptoms are those that are not flagrant enough to qualify for full syndromal or disorder classification. These scales may be particularly useful for high risk research insofar as they may reveal early manifestations of psychopathological disorder. As our research includes a comparison of vulnerable and nonvulnerable adolescents, such an assessment may be especially important for considering the potential impact of support and rejection (Renders, in progress). Given that individuals may develop diverse symptoms upon exposure to the same life event (Bloom, 1979), a comprehensive assessment of mental health is required if we are to understand how social networks may effect the range of potential psychopathological outcomes.

IMPLICATIONS FOR PRACTICE

Implications of this framework for clinical-community practice have been alluded to in several prior sections. I wish here to state briefly the overarching relationship with respect to social-community interventions.

As much prior research has been directed solely at whether support buffers individuals from the effects of stress, much of our intervention thinking has been in terms of providing increased levels of emotional support to sustain coping efforts. There is a considerable danger in conceptualizing support interventions exclusively or even predominantly in terms of providing emotional support for stress (Hirsch & David, 1983). As House (1981) notes with respect to work social support, "some of the current enthusiasm for social support stems from a desire to avoid the often more difficult problems of restructuring work organizations and environments to reduce levels of occupational stress (p.

114). Furthermore, by restricting our concern to the alleviation of stress or negative experiences, we may fail to address the importance of developing support for achieving positive goals. A decrease in negative experiences need not imply an increase in positive experiences. Some individuals or groups may want to change both, while others may differentially prioritize their objectives. We need to be sensitive to each possibility and not blindly assume that minimizing stress is our *raison d'être*.

The framework being developed in this paper suggests that efforts to increase support and to enhance the quality of life should be inseparable. If the level of social support and rejection affects our ability to attain important objectives and achieve a satisfactory quality of life, then support interventions should be part and parcel of any intervention effort. Correspondingly, because support and rejection occur within the context of a complex person-environment ecology, support interventions need to be conceptualized and implemented as part of comprehensive programs and policies for enhancing the quality of life.

SUMMARY

This paper has provided an initial, provisional statement of a research strategy being developed to study the relationship between social networks and mental health. The strategy emphasizes linking network and support research to other domains, specifically: life-span development, with a particular focus on our ability to elaborate satisfactory roles in major spheres of life; social ecology, with its emphasis on systematically differentiating the social environment; personality variables that are defined with reference to salient aspects of the micro to macro environment; and social issues, such as the status of women in the work force, viability of dual career families, quality of work life, and so on. Social networks are seen as an integral element in understanding these other topics, though networks are obviously not the sole variable of interest. For example, many workers will be dissatisfied with their jobs because the actual work they do is boring and monotonous, or because they have little input into important policies affecting them, regardless of their level of social support or rejection. This research strategy is clearly still in an early stage of development and much conceptual, theoretical and empirical work remains to be done.

NOTES

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²It should be noted that while this research program is concerned with developing a theory of how networks may affect mental health, networks are conceived more fully as both cause and effect in an interactive manner; furthermore, while both longitudinal designs and "causal modeling" procedures will be employed, I remain skeptical about our ability to understand causal patterns until methods are developed to examine processes that extend over months and sometimes years.

³This investigation is concerned more broadly with how social networks relate to the academic and personal adjustment of Black graduate students. It is probably the clearest example among the present studies of how networks can be studied in relation to an important social policy question, in this instance, the affirmative action objective of increasing minority representation in the professions. DeFour's research will also be among the first to consider how the racial composition of networks is related to the provision of social support.

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LONGITUDINAL COURSE OF SOCIAL SUPPORT AMONG MEN IN THE BALTIMORE
LONGITUDINAL STUDY OF AGING

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National Institute on Aging
National Institutes of Health

Despite the attention to social supports as moderators between stress and stress outcomes, the picture we have to date is essentially static, showing only the relations among these three variables at a single point in time or over a limited follow-up period. Little is known about the longitudinal course of social supports; especially their vicissitudes or stability over the life span. Even less is known about the relations among social supports, enduring personality dispositions, and the course of life. This gap is particularly apparent to psychologists trained in a lifespan orientation.

For those concerned about the aging and elderly knowledge on the course of social supports is particularly important. If prevalent notions are to be trusted--and the reader should be forewarned that they aren't--the older adult is vulnerable to a host of age-related declines in health, life-satisfaction, and economic resources, as well as to a shrinking social world. Outside gerontological circles, the notion of an inevitable, if gradual, disengagement from productive social and personal roles and relations is still quite widespread, despite the wealth of evidence supporting an activity or continuity perspective (Costa & McCrae, 1980; Costa, McCrae & Norris, 1980; Havighurst, McDonald, Maeulen & Mazel, 1979; Maddox, 1968).

The stability of social supports was identified by House and Robbins (1983) as an important issue for aging research. The stereotypic view is that as people grow older, their children grow up and leave home; they retire and contact with former co-workers decreases; parents, friends, and spouses die. Among institutionalized people this characterization seems apt because those low in social support--the single, childless, divorced and widowed--are very much in evidence. An important question the

present study seeks to address is whether aging is synonymous with social losses and declines in social support in the far larger non-institutionalized segment of the elderly population.

One aspect of social support--the family--has been the subject of extensive cross-section research, and the findings generally contradict the view that family bonds are destroyed or greatly weakened when children leave the home and parents age (Costa & McCrae, 1983; Troll, Miller & Atchley, 1979). Hill (Hill, Foote, Aldous, Carlson & MacDonald, 1970) reported that 40-70% of adult children in a Minneapolis sample saw their parents weekly. Older adults frequently receive help from their children, and just as frequently give it, particularly in the form of money and child care (Riley & Foner, 1968). Finally, a number of studies have shown that positive emotional feelings tend to characterize the parent-child bond at all ages. In summary, most older people are actively involved in their family life and value it highly (Seelbach & Hansen, 1980); and they in turn are highly valued by their families (Bengtson & Cutler, 1976).

There has been less research on friendships in the elderly, and few longitudinal studies on any aspect of social support. The focus of the present paper is on an examination of the temporal stability of three indices relevant to social support: Quality of Family and Marriage, Social Participation, and Friendships. The data that we will be examining come from the Activities and Attitudes questionnaire (AAQ), an instrument developed in the 1940's (Caven, Burgess, Havighurst & Goldhammer, 1949). Although rather primitive compared to sophisticated measures recently developed, (e.g., Henderson, 1981; Sarason, Levine, Basham & Sarason, 1983) it has the merit of having been administered approximately every six years to a large sample of normal, community-living men over the past 25 years. Consequently, the data may provide some preliminary empirical answers to the question of change versus stability in the dimensions of social support.

After showing the derivation of the three indices of social support from AAQ items and considering some of their correlates, we will examine stability and change in mean levels of support over six and twelve-year intervals. We will also consider the degree to which levels of support characterize individuals over these time spans, as assessed by retest stability coefficients. Our findings may be due to characteristics of the individual, or of the environment. In a preliminary attempt to sort out these possibilities, we examine the influences of changes in residency on the stability of social supports.

DESCRIPTION OF THE SAMPLE

The AAQ was administered approximately every six years to subjects of the Baltimore Longitudinal Study of Aging.

Participants in the BLSA are a community-dwelling, generally healthy group of volunteers, 96% white, who have agreed to return for medical and psychological testing at regular intervals. Until 1978 all participants were male, and only data from men will be discussed here. The majority (80%) work in or are retired from scientific, professional or managerial positions. Almost all (93%) are high school graduates, and 71% are college graduates; nearly one-fourth have doctorate degrees. Because one of the measures of social support included adequacy of marriage as a major component, we limited all analyses to the 88% of the total sample who were married. Subjects have been recruited continuously since 1958, so first administration data ($N = 1093$) come from that entire time interval. As in all longitudinal studies, some attrition has occurred. In consequence, second and third administration data come from successively smaller groups of individuals ($N = 478$, 196) who have been in the study longer. Subjects who remained in the study tend to be psychologically better adjusted than those who dropped out.

Mean ages for the groups at the first test administration were 39.3 years ($N = 662$) and 67.0 years ($N = 431$) respectively. At the time of the third testing the mean ages were 56.3 ($N = 115$) and 75.2 ($N = 81$). The average interval between test administrations ranged from 6.6 to 7.1 years for the younger group and 4.9 to 5.8 years for the older group.

FACTOR ANALYSIS: ITEMS AND LOADINGS

After rationally examining the 158 AAQ items, 44 were selected as possible indicators of social support and used in an exploratory factor analysis of the first administration data. Based on a scree test (Cattell, 1966), three factors were extracted. Table 1 displays the final items selected with their loadings following varimax rotation.

The first factor, interpreted as Quality of Family and Marriage (QFM), contains items regarding marital satisfaction and satisfaction with familial social relations. The second factor, Social Participation (SP), contains items about social activities and participation in social organizations. The third, Friendships (F), contains items regarding quantity and intimacy of friendships. These three factors were transformed to social support scales by summing the items loading on each factor.

These three scales bear some resemblance to social support scales proposed by others. Both the QFM and the F factors appear to combine in some degree perceived adequacy and availability of support (Sarason, et al. 1983; Henderson, 1981), although adequacy is more pronounced in the former factor, and availability in the latter. The second, SP, factor resembles a Social Network Index used by Schaefer, Coyne, and Lazarus (1981).

TABLE 1

Factor Loadings and Alpha Reliabilities for Quality of Family and Marriage (QFM)

Loading	Item
.64	How do you rate the happiness of your (last) marriage?
.66	If you had your life to live over, would you marry the same person?
.48	If you have a family or close relatives, do they neglect you? (reversed)
.45	My family does not really care for me. (reversed)
.50	I wish my family would pay more attention to me. (reversed)
.45	My family likes to have me around.
.72	I am perfectly satisfied with the way my family treats me.
.43	I think my family is the finest in the world.
.37	I get more love and affection now than I ever did before.
.31	If you have a family or close relatives, do they try to interfere in you affairs? (reversed)

Alpha reliability = .65

Factor Loadings and Alpha Reliability for Social Participation (SP)

Loading	Item
.65	Participate in community or church work.
.54	Write letters.
.67	Attend clubs, lodges, other meetings.
.50	Visit or entertain friends.
.51	Plan to visit children, relatives or friends.

- .59 To how many organizations, such as clubs, lodges, unions, and the like, do you now belong?
- .64 How many club meetings do you usually attend each month?

Alpha reliability = .67

Factor Loadings and Alpha Reliability for Friendship (F)

Loading	Item
.40	I have no one to talk to about personal things. (reversed)
.42	I have so few friends that I am lonely much of the time. (reversed)
.54	I have all the good friends anyone could wish.
.36	I never dreamed that I could be as lonely as I am now. (reversed)
.58	How many friends do you have?
.34	How many close friends do you have such that you can talk to them about almost anything?
.63	I have more friends now than I ever had before.
.66	My many friends make my life happy and cheerful.

Alpha reliability = .61

CONSTRUCT VALIDITIES OF SOCIAL SUPPORT MEASURES

In an ideal design, these scales would be validated by correlating them with contemporary measures of social support. Since these data are unavailable, we looked for indirect evidence of construct validity by correlating them with other measures of physical and psychological well-being taken from the AAQ and with measures of personality. Table 2 shows the correlations between the social support scales and measures of psychological and physical well-being from AAQ first administration data. In general, happiness and satisfaction were positively correlated with both QFM and with F; perceived health problems were

TABLE 2

Correlations of Social Support Scales with Measures of Psychological and Physical Well-being

	Quality of Family & Marriage	Social Participation	Friendships
	(QFM)	(SP)	(F)
Happiness	.45***	.07*	.38***
Usefulness	.06	.00	.24***
Work Satisfaction	.16***	-.04	.27***
Life Satisfaction	.32***	.15***	.29***
Satisfaction with Accomplishments	.26***	.07*	.29***
Health			
Overall	.17***	-.06	.25***
Serious problems	-.03	.04	-.10
Symptoms	-.20***	-.01	-.23***

* $p < .05$

** $p < .01$

*** $p < .001$

negatively correlated with these two factors. Although life satisfaction, happiness, and satisfaction with accomplishments showed statistically significant correlations with SP, the size of the associations were trivial. SP does not appear to be related to well-being—a finding which parallels that of Schaefer, Coyne, and Lazarus with their Social Network Index.

The construct validities of the social support scales were also examined by correlating them with several measures of personality and well-being which we have administered during the past four years (Costa & McCrae, in press; Costa, McCrae & Norris, 1980). Correlations with the Eysenck Personality Inventory (EPI; Eysenck & Eysenck, 1964), Affect Balance Scales (Bradburn, 1969), Profile of Mood States (POMS; McNair, Lorr & Droppleman, 1971), Marlowe-Crowne S. D. (Crowne & Marlowe, 1964) and NEO Inventory and NEO Rating Form (Costa & McCrae, 1980; McCrae & Costa, 1983,) are shown in Table 3.

Perceived adequacy of family support is consistently related to neuroticism and to its attendant psychological distress; availability and adequacy of friendships is related to both extraversion and neuroticism. Well-adjusted extraverts have more friends and are more satisfied with them. SP does not appear to be

TABLE 3

Correlations of Social Support Scales with Personality Measures

	Quality of Family & Marriage	Social Participation	Friendships
	(QFM)	(SP)	(F)
EPI:			
Extraversion	.05	.00	.17**
Neuroticism	-.23***	-.07	-.24***
Bradburn			
Positive Affect	.08	.15**	.30***
Negative Affect	-.28***	-.08	-.20***
POMS Total Mood	-.22***	-.09	-.25***
NEO:			
Neuroticism	-.22***	-.04	-.26***
Extraversion	.04	.10	.27***
Openness	-.20**	.08	-.01
NEO Ratings by spouse:			
Neuroticism	-.32***	-.01	-.22**
Extraversion	.02	.05	.17*
Openness	-.05	-.05	.02
Marlowe-Crowne	.26***	.12	.19**

* p < .05

** p < .01

*** p < .001

related to these personality predictors; and the dimension of openness to experience is not consistently related to any aspect of social support measured here.

This pattern of correlations essentially replicates the results reported by Sarason et al., (1983) who found that satisfaction with supports was related to neuroticism, whereas number of supports was related to extraversion (at least in women). Since the F scale in the present study includes aspects of both satisfaction and availability, it is related to both neuroticism and extraversion.

STABILITIES OF INDIVIDUAL DIFFERENCES IN SOCIAL SUPPORT

In order to estimate the consistency of individual differences in social support, stability coefficients were calculated for the second and third administration samples for six year intervals between testings, and for the third administration sample for a twelve year interval. These coefficients are shown in Table 4. In the complete sample, all of the stability coefficients were greater than .45, and the majority were greater than .50. The QFM scale was as stable after a twelve year interval (.54) as it was after a six year interval (.54). The stability of the SP scale declined only slightly from .54 after a six year interval to .46 after a twelve year interval. The stability of F after a six year interval was slightly lower in the third administration sample (.48) than it was in the second administration sample (.54), though in the latter, the stability after twelve years was virtually identical to the six year coefficient (.47).

These results suggest that social support remains stable over at least a twelve year interval. Considering the moderate reliability of the social support scales, it is clear that levels of social support are quite stable within individuals. Like personality, the social world of the individual seems to change little in the latter half of life.

The present data may be used to test the hypothesis that major role changes in later life should lead to changes in social support. Stability coefficients were computed separately for subjects who were younger than 55 at time one and for subjects who were 55 or older at time one. These coefficients are also displayed in Table 4. Contrary to the hypothesis, the social supports of subjects who were 55 or older at time one were generally more stable than those of subjects who were younger than 55. For example, older subjects showed more stability in the perceived adequacy of marriage and family after twelve years (.80) than younger subjects (.37). Similar, though less disparate results were found for SP. The opposite pattern, however, was seen for F. Given the limitations of these scales and the relatively small sample sizes, it probably isn't wise to make too much of the differences between these age groups, but the general conclusion that social supports are consistent for adults of all ages over intervals of several years seems clear.

MAINTENANCE OR DECLINE WITH AGE

It is commonly asserted that social support declines with age, despite the evidence we cited earlier for stability in family ties. Our data forcefully contradict that stereotype, at least in

TABLE 4

Average Stability Coefficients
(sample sizes in parentheses)

Quality of Family and Marriage (QFM)

	Full Sample		Younger than 55		55 or Older	
	Interval		Interval		Interval	
	6 yrs	12 yrs	6 yrs	12 yrs	6 yrs	12 yrs
2 visits	.54 (308)		.46 (194)		.72 (114)	
3 visits	.57 (137)	.54 (127)	.43 (78)	.37 (78)	.79 (59)	.80 (52)

Social Participation (SP)

	Full Sample		Younger than 55		55 or Older	
	Interval		Interval		Interval	
	6 yrs	12 yrs	6 yrs	12 yrs	6 yrs	12 yrs
2 visits	.54 (308)		.54 (210)		.58 (137)	
3 visits	.54 (155)	.46 (147)	.50 (88)	.41 (79)	.58 (65)	.52 (66)

Friendships (F)

	Full Sample		Younger than 55		55 or Older	
	Interval		Interval		Interval	
	6 yrs	12 yrs	6 yrs	12 yrs	6 yrs	12 yrs
2 visits	.54 (350)		.51 (222)		.60 (127)	
3 visits	.48 (155)	.47 (154)	.52 (90)	.56 (90)	.45 (64)	.29 (62)

our sample of community-dwelling married men. Figures 1 and 2 show the mean levels of each of the three scales for our two groups of men. (The numbers of subjects are given in Table 4.)

FIGURE 1. Mean levels of social support for men younger than 55 and 55 or older after two administrations.

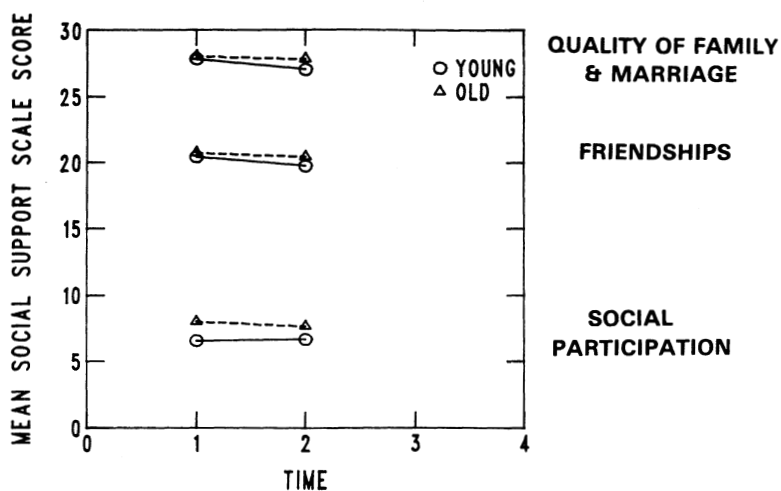
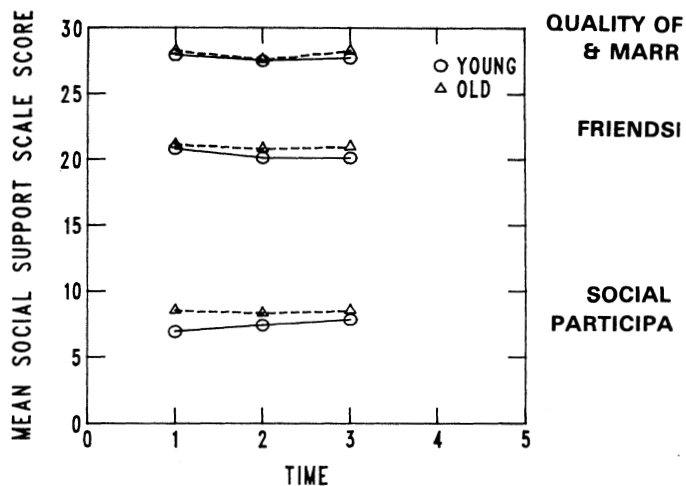


FIGURE 2. Mean levels of social support for men younger than 55 and 55 or older after three administrations.



Cross-sectional comparison of the two groups shows that there are significant differences, albeit trivial in magnitude, for SP. Most important however, is the direction of effect: older subjects are uniformly higher in social support than are the younger men.

Repeated measures analyses of variance for the two and three administration data showed only one significant, replicated effect: F declined in both analyses. As the figures show, however, the magnitude of the effect is very small--certainly not the dramatic decline often depicted for aging men and women. There were no consistent interactions between age and time.

SOCIAL SUPPORT STABILITY AFTER CHANGES IN RESIDENCY

Considerable evidence has been presented for the stability of social supports. To what do we attribute this stability? Do stable personality dispositions continually create a stable social network? Or is environmental consistency responsible for the stability, as social learning theorists sometime suggest? An examination of AAQ responses suggests that the environments of BLSA married men are extremely stable, and the present results may be due to this. As one test of this hypothesis, subjects were divided into those who had moved to a new city between test administrations and those who had lived in the same city for at least ten years. Moving is perhaps one of the more disruptive and potentially stressful events in life. Moving to a new city entails breaks in social networks and changes in neighbors, clubs, schools, and co-workers. Two hypotheses are suggested: first, since every move entails at least a temporary disruption of social relationships, we might predict that the mean level of social supports would decline for movers. Second, since the social opportunities provided by the new residence might be either greater or fewer than those afforded by the old residence, there should be little correlation between support before and after a move. Thus, we would predict a lower mean level and a lower retest stability for movers.

Note that these predictions apply primarily to the SP and F scales, since most individuals move with their nuclear family. Some disruption might be expected even in the QFM scale, however, since moves may take the individual away from parents, children, siblings, and other relatives.

Table 5 shows the stability coefficients within each group for the second and third administration groups over six and twelve year testing intervals. The pattern of correlations suggests that moving diminishes the stability of QFM somewhat, but not the stability of SP and F. Although these results may appear somewhat counterintuitive, it is likely that the stresses imposed by a change in residence may produce changes in marital and familial relations. However, a change in residence does not appear to

TABLE 5

Average Stability Coefficients in Movers and Nonmovers
(sample sizes in parentheses)

<u>Quality of Family and Marriage (QFM)</u>				
	Nonmovers		Movers	
	Interval		Interval	
	6 yrs	12 yrs	6 yrs	12 yrs
2 visits	.59 (263)		.21 (31)	
3 visits	.60 (110)	.55 (104)	.44 (19)	.40 (18)
<u>Social Participation (SP)</u>				
	Nonmovers		Movers	
	Interval		Interval	
	6 yrs	12 yrs	6 yrs	12 yrs
2 visits	.59 (263)		.52 (31)	
3 visits	.60 (120)	.44 (113)	.48 (24)	.54 (23)
<u>Friendships (F)</u>				
	Nonmovers		Movers	
	Interval		Interval	
	6 yrs	12 yrs	6 yrs	12 yrs
2 visits	.56 (291)		.41 (42)	
3 visits	.56 (120)	.46 (119)	.47 (25)	.60 (24)

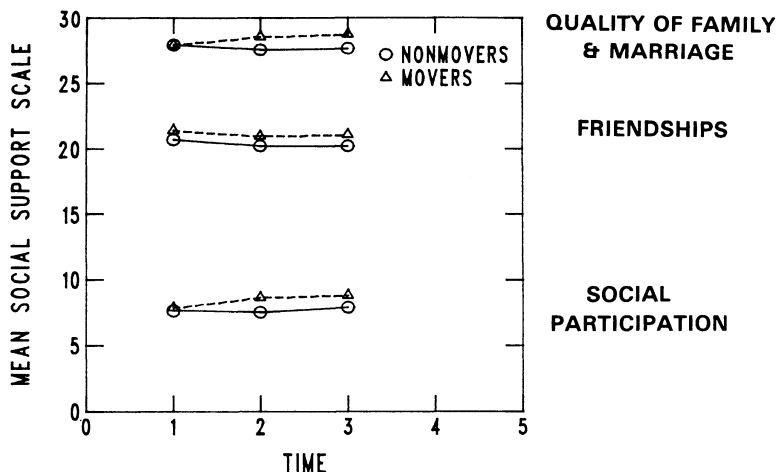
influence the extent of social participation or friendships, perhaps because these aspects of social support are quickly renewed after a move by individuals desiring them.

In addition to the evidence for stability of individual differences after a change in residence, repeated measures analyses made it apparent that mean levels of social support did not differ because of moving. Disruption due to moving should be seen as an interaction between group and time; none of these interactions were statistically significant. As shown in Figure 3 both movers and nonmovers were virtually unchanged after six year and twelve year intervals. There appears to be little difference in mean levels between movers and nonmovers before or after the move. All three forms of social support are maintained even after disruptions in social networks.

It is important to note that the present analyses examine changes over relatively long intervals: subjects may have moved shortly after filling out their first AAQ, and have had several years to readjust by the time they completed their next AAQ. It is reasonable to assume that moving to a new city does disrupt social networks in the short run, perhaps for as long as a few years. However, the present results suggest that previous levels of support are renewed within six years.

It is also worth recalling that all these subjects are men, and that a very different picture might emerge if women were

FIGURE 3. Mean levels of social support for movers and nonmovers after three administrations.



investigated. Most moves occur when the husband takes a new job elsewhere. Thus, men are provided with a ready-made social network in their new work setting (what Kahn and Antonucci (1980) call a convoy of support). In addition, men rely heavily on their wives for support, and the major burden of resettling may well fall on women.

GENERAL DISCUSSION

The evidence presented in this paper disconfirms the commonly accepted notion that social support is unstable and declines with age. Although the AAQ is not ideal for the present purposes, it yields three acceptable social support scales which show meaningful if indirect evidence of construct validity. Even though this study doesn't provide definitive answers, particularly since it did not include women, it does provide a glimpse of what longitudinal studies are likely to show.

Perhaps the most important theoretical issue in this area concerns the sources of stability in social support. A likely candidate for stability is the social environment. The stability of SP, in particular, is probably a product of the individual's social position. Professionals must maintain memberships in organizations; business men must maintain social contacts; and religious backgrounds lead to particular patterns of community affiliations. We can speculate that variables such as occupation, education, social class, race, religion, and ethnicity are the primary determinants of social participation, and that their stability accounts for the stability of SP. The likelihood of this explanation is strengthened by the finding that SP remains stable over the long term even after moving to a new city.

We might also suspect that social participation is related to personality dispositions such as extraversion. However, the present study, as well as previous research (Sarason et al., 1983) yields little support for this hypothesis. It is likely that personality dispositions play a relatively minor role in the extent of social participation given the other social demands placed on an individual by his occupational and cultural background.

A far better case can be made for the link between personality dispositions and QFM, and F. The finding in the present study, as well as by Sarason et al., that individuals high in neuroticism were lower on the QFM and F scales suggests a key role for this disposition. One possibility is that anxiety, hostility, and depression, and other symptoms of neuroticism may result from inadequate social supports. Certainly these states, as states, are likely outcomes of social isolation and rejection. But the documented stability of neuroticism as a trait across many decades of life (Costa & McCrae, 1980) makes this explanation unlikely. Surely something in the individual is responsible for

this lifelong pattern; and these personality traits may well account for the perceived adequacy of social support.

Since neither neuroticism nor social support changes markedly over the period covered by this study, it is impossible to determine the nature of the causal link. However, several hypotheses can be suggested that would account for the influence of neuroticism on social support. First, the distress which the individual high in neuroticism chronically feels despite support may make him devalue the support he gets. Here, support is "inadequate" in the sense that it is insufficient to reduce his anxiety, anger, and depression. Second, the neurotic may simply be a complainer, ready to perceive the worst side of every situation and to emphasize the failures and shortcomings of all aspects of life. Dissatisfaction with work, family, neighborhood, income, and government are characteristic of neurotics; perhaps dissatisfaction with social support is simply one more complaint.

These explanations presume that neuroticism produces differences merely in the perception of social support. An alternative explanation is that there are objectively verifiable differences in support between persons high and low in neuroticism. Individuals high in neuroticism may in fact receive less support, perhaps due to the social stimulus value of their personality. The emotionally unstable individual is likely to overburden friends and relatives who offer support. In addition, his hostility and impulsiveness may alienate others, and self-consciousness or low self-esteem may prevent him from seeking the support he needs. In general, individuals high in neuroticism may have characteristics which prevent them from forming and maintaining consistently adequate networks of social support.

The distinction between F and QFM is best seen in their divergent correlations with extraversion. Both of these scales correlate with neuroticism, but only F correlates with extraversion.

Regardless of their degree of extraversion, most individuals marry (all of them in a study of married men!) and have a family. Thus, availability of family support does not depend on extraversion. The number of friends one has, however, does depend on extraversion, and the more friends one has, the more likely one is to find support from them when needed. In addition, introverts are temperamentally reserved. Even among the friends they do have, they may be reluctant to ask for emotional support, and may depend more on the intimacy of their family. Again, extraversion may act as a social stimulus: we may be more ready to offer support to a warm, outgoing individual than to one who seems to value his or her privacy.

In summary, these longitudinal data add a temporal dimension to current knowledge of social supports. As with intelligence, personality and vocational interests, continuity rather than change seems to characterize adulthood. Even advanced age does not materially affect the kind or quality of perceived support, at

least for a community-dwelling, married sample. Further, individual differences in social support appear to be maintained or restored after potentially disruptive events such as moving to a new city. We argue from this that enduring characteristics of the individual are among the important determinants of support.

All of these findings are taken from an instrument designed years ago when notions like social support were informal at best. If the improved methodology and theoretical advances of recent research are employed in a longitudinal design which includes the measurement of key personality variables, the prospects for increased understanding are exciting indeed.

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INTIMACY, SOCIAL SUPPORT, AND LOCUS OF CONTROL AS MODERATORS OF STRESS

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Research into the effects of stressful life events has been prominent in psychological literature for many years. Though there has been sufficient documentation of the fact that an individual's state of well being is effected by the frequency of stressful life events that he or she has experienced, in most cases the magnitude of these relationships has not been overwhelming. Consequently, it is evident that there are some people who appear to be resilient in the face of stress and there are others who seem to readily succumb to the same pressures.

In more recent years, investigators have begun to explore differences between persons in the hope of explaining why it is that stressors vary as much as they do in their impacts upon individuals. Among these differences, access to social support has often been cited as a crucial moderator of stress. The fact that social support can help to soften the impacts of stressful experiences would seem to be self-evident, and research has substantiated this sensible thesis.

In the same way, however, that investigators felt compelled to seek out stress moderators by the varied responses that individuals made to stressful events, certain researchers have been raising questions concerning the often noted variability in effects of social support. These investigators have concluded that social support research would benefit from attention to the multidimensionality of social support and from greater specificity in hypothesis construction concerning the relationship between types of support and adaptive outcomes. To this end, Cohen (Cohen & Hoberman, 1982; Cohen & McKay, 1983) has constructed a measure of social support in which each of four functions of social support is assessed separately.

A second question concerning limitations on the effectiveness

of social support as a stress moderator has focussed upon differences among persons in their ability to make use of their available social supports. Sandler & Lakey (1982) for example, have found that social supports are more effective stress moderators among persons who are internal with respect to locus of control. That is, among internals, those who had greater access to social support showed lower magnitude relationships between life stress, anxiety, and depression than did those with lesser access to social support. In contrast, externals showed minimal variation in the relationship between stress and dysphoria as a function of social support. Similar results have been reported by Husaini, Neff, Newbrough, & Moore (1982) with a large sample of rural, married persons, and by Dean & Ensel (1982) among female subjects.

In this paper, I will be describing several studies that my colleagues and I have conducted at the University of Waterloo which focus upon both questions regarding social support: the particular type of social supports that moderate stress, and the differences between people in their ability to make use of social support. The aspect of social support that we have emphasized is social intimacy, and the individual difference variable that we have pursued is the locus of control.

RESEARCH WITH INTIMACY

Our contribution to the social support literature began with the creation of what we hoped would be a good measure of social support, the Miller Social Intimacy Scale (MSIS - Miller & Lefcourt, 1982).

Intimacy is a word with a long history of changing meanings. Etymologically, it derives from the Latin word *intimus*, meaning innermost. According to Webster's dictionary, intimacy refers to a very close association marked by familiarity and affection. Throughout the research literature the terms "close" and "intimate" occur frequently but without explicit operational definitions. In trying to construct an operational tool to measure intimacy, keeping the flavor of "innermost", "familiar", and "affectionate", we chose to assess the degree to which individuals experienced intimacy with their friends. In essence, our intimacy measure focusses upon the belongingness function that Cohen & Hoberman (1982) described as one aspect of social support. The scale included items that reflect mutuality or the give and take between friends. Given that subjects are instructed to describe their closest relationship, the sum of their ratings yields a measure of the maximum level of intimacy currently experienced. It was our contention that having an intimate relationship with someone functions as a stress buffer since the friends may provide each other with access to moral support in the face of challenge, with nurturance when it is needed, and with an

opportunity for the sharing of feelings and information that facilitate coping behavior. In other words, a close intimacy may be a cornucopia of social support, possibly offering all of the functions discussed by Cohen and Hoberman (1982).

In the two investigations that will be described we have tried to examine the buffering effects of social intimacy. In the first study use was made of conventional methodology. The intimacy measure was used along with a life stress scale in the prediction of mood disturbance. In the second investigation, a novel experimental test of stress buffering was developed which involved a quasi-stressful social situation.

Life Stress and Social Intimacy

In the more conventional investigation, a class of 47 university students completed three measures. First was a modified version of the Coddington Life Events Checklist (Coddington, 1972), a scale requiring subjects to note the occurrence of events throughout various eras of their lives. For our purposes, the high school era was the period of interest. Subjects checked the events that had occurred during that era and rated them as being positive, neutral or negative in impact. Both positive and negative life change scores were derived from this scale. The second measure was the Profile of Mood States (POMS - McNair, Lorr, & Droppleman, 1971) was administered during four consecutive weeks with subjects recording their affective experiences for each week. The average of Total Mood Disturbance scores derived from the POMS was used in this study. Third, all subjects completed the Miller Social Intimacy Scale (MSIS).

RESULTS

The Means of the Total Mood Disturbance scores comprised the dependent measure in two regression analyses. In the first analysis, negative life event scores were entered as the first independent variable, intimacy as the second, and the product of the two, representing the interaction term, as the third independent variable. The second analysis was identical in form with the first analysis except that instead of negative life events, the number of positive life events scores was entered as the first variable in the regression equation.

As commonsense might suggest negative life events were positively associated with mood disturbance ($r = .41$), and positive life events were negatively related to mood disturbance ($r = -.21$). Intimacy, was also related to mood disturbance in a negative direction ($r = -.49$). However, only in the analysis with positive life events as the first independent variable did the interaction term, or product, produce an increment in the variance accounted for. With negative life events as the

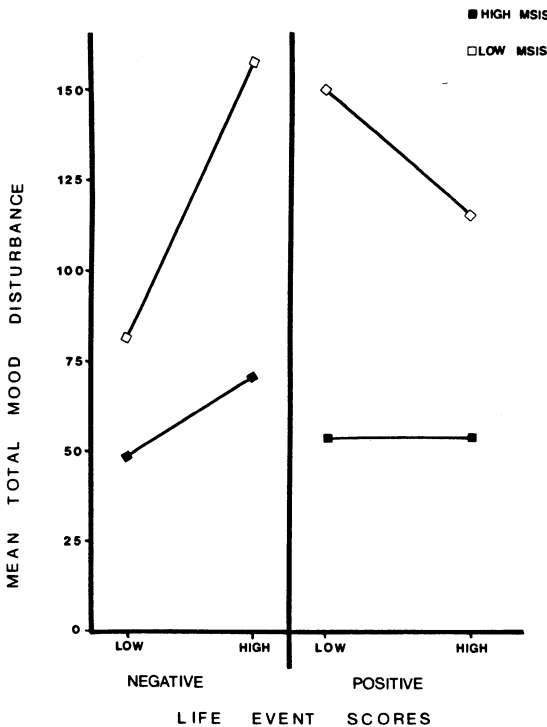
independent variable the product term failed to add information beyond that already contributed by each independent variable separately.

It is evident from Figure 1 that subjects who had experienced a great number of negative life events during their high school years, and who were not enmeshed in a highly intimate relationship at the present time, were the most likely to report mood disturbances. However, this tendency was overshadowed by the more robust main effects present in these data such that the relationships with mood disturbance were additive rather than interactive.

In contrast, when positive life events scores were entered as the first variable, the interaction term proved to be statistically significant, providing an independent contribution beyond that already evident in the two significant main effects.

In common parlance, stress refers to the experience of negative life events. If intimacy were a true buffer of these

FIGURE 1. Total Mood Disturbance as a Function of Negative and Positive Life Events and Intimacy.



experiences, the relationships between stress and mood disturbance should differ, depending upon the status of one's intimate relationships. This was not the case. Negative life events had deleterious effects upon everyone, regardless of their intimacy scores. On the other hand, intimacy was not irrelevant. Persons who had closer relationships with their friends than did others, generally evinced less mood disturbance; and, for those persons whose memories of their high school years are comprised of negative events, and a dearth of positive events, the absence of a close intimate relationship seemed to have strong negative portents.

We can conclude, therefore, that closeness or intimacy has an obvious impact upon mood disturbance but that our data do not encourage us with regard to the stress buffering function that we had hypothesized. However, it is also possible to fault our own data because we were not actually assessing current stress while we were measuring current intimacy status. In other words, to more fairly test the buffering hypothesis we should be looking at contemporaneous experiences. The second, experimental study, allowed for a better test of the buffering hypothesis focussing as it did on the manner in which subjects coped with an immediate albeit weak social stress.

Social Stress and Social Intimacy

For the experimental investigation, we borrowed a procedure that had been used some twenty years ago in a study of the social effects of reinforcement withdrawal (Adams & Hoffman, 1960). In that study, an interviewer ceased providing "generalized reinforcers" midway through an interview which resulted in a decline of verbalization and some displays of anger from among the subjects.

In our study (Miller & Lefcourt, 1983) university student subjects were taken into a small lab room where they were interviewed by an experimenter who was ostensibly interested in students' views about marriage. As a preliminary task, subjects had completed the MSIS and other measures pertinent to marriage. During the interview which lasted 7 minutes, subjects were questioned about their own expectations of marriage in the future, the merits of traditional marriage as opposed to "living together", and their views of extramarital sex.

During the initial 3 minutes, the experimenter maintained eye contact with the subjects, uttered mm-hm's and nodded his head in response to the subject's verbalizations, or appeared to be taking notes about what the subjects were saying. In the control condition (N=73) the experimenter exhibited this same display of interest throughout the entire 7 minute period. In the experimental condition (N=45), the experimenter ceased showing interest in the subject's commentary during the latter 4 minutes of the interview. In this condition, the experimenter began to

look at his notes instead of at the subject, ceased vocally responding to the subject, and engaged in some slow note taking while the subject continued to talk.

The interview was videotaped from an adjoining room. The taped interactions were then rated for the degree of verbal disclosure exhibited during each interval (pre and post extinction) of the interview. Verbalizations were rated from 0 to 5, ranging from 0 for "non-communicative" to 5 for "personal, revealing of feelings, involved, open".

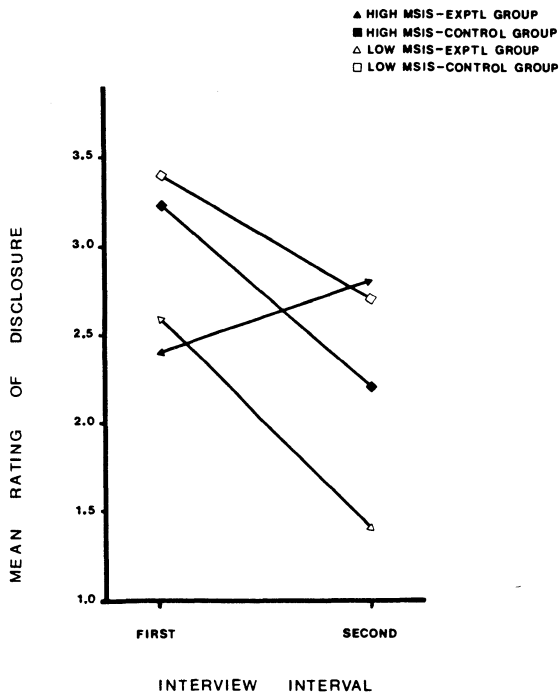
Our hypothesis was simply this: that subjects who currently enjoyed a close intimate relationship with another person would be less vulnerable to the slights and rejections that are commonly experienced in everyday interactions. We, therefore, anticipated that persons who had low MSIS scores would be more easily hurt by our slight. That is, they would take umbrage with us more quickly than would persons with high MSIS scores. As a consequence, we hypothesized that disclosure would decline most noticeably among low MSIS subjects when they experienced our quasi-social rejection. Though disclosure should decline for all subjects as the session progressed simply because many would have exhausted the topic of conversation in the latter half of the interview, the decline in self disclosure was expected to be most precipitous among the low MSIS subjects given the withdrawal of the experimenter's attention.

RESULTS

In the four way analysis of variance where MSIS, condition (control vs. experimental), sex of S, and interval comprised the predictor variables, significant main effects were found for condition ($F = 6.63$), $p < .025$) and interval ($F = 19.04$, $p < .001$). It was evident that disclosure was less in the experimental than in the control conditions, and less in the second than in the first interval. Most importantly, a significant interaction was found between MSIS, condition, and interval ($F = 13.28$, $p < .001$). Subject's sex proved to be irrelevant in this analysis.

The form of the interaction (Figure 2) was in exact conformity with our hypothesis. In the control condition all subjects, regardless of MSIS scores, decreased significantly in self-disclosure from the first to the second interval. In the experimental condition, the decrease in self-disclosure was even greater among low MSIS subjects than it was among subjects in the control condition. However, in contrast, high MSIS subjects increased, albeit not significantly, in disclosure from the first to the second intervals in the experimental condition. In essence, high MSIS subjects were not dissuaded from discussing the topics by the examiner's apparent disinterest. If anything, our display of disinterest seemed to have acted as a goad to keep them

FIGURE 2. Self Disclosure as a Function of Intimacy, Social Rejection and Conditions.



talking. Since high MSIS subjects had decreased as much in their disclosure scores in the control condition as had low MSIS subjects, it would seem that the experimental condition had facilitated the continuance of disclosure rather than discouraged it for high MSIS subjects. Since the mean disclosure scores for the high MSIS subjects in the second interval of the experimental condition were not different from subjects' disclosure scores in the control condition, we may conclude that the buffering effect is evident in these data.

In this experiment, then, we found evidence to the effect that persons who did not currently enjoy a highly intimate relationship with another person seemed to be more sensitive to a social rebuff. On the other hand, persons who had a more intimate relationship with someone seemed to be less deterred by social slights. While the data from both studies indicate that intimacy does play a role in predicting how people will respond to life stress, there is no further information in these data to suggest how intimacy comes to have a protective effect.

Our own data do not allow us to more finely examine the buffering effect. It is evident, however, that a closer examination of the functions of intimacy may reveal something about man's basic needs which are echoed throughout classic literatures concerned with aloneness, loneliness, and the desire to be "home" among close intimates.

Locus of Control and Social Support as Moderators of Stress

A second direction we have pursued concerns the question of whether some persons are better able to make use of social support as a stress buffer than others. It is reasonable to assume that the pseudo-friendliness of the bar habitue' or compulsive conversationalist would not provide the same relief from stress as would a heart-to-heart talk between two close friends. Likewise, extremely gregarious individuals might have too much need of others to feel capable of sharing unpleasant experiences as that might repel their would-be friends. In other words, some persons might be in a better position than others, due to their personal proclivities, to be able to draw benefits from their social interactions for dampening the effects of stress.

One such proclivity that we have examined along with social support pertains to subjects' beliefs regarding causality, or in social learning terms, the locus of control. In brief, locus of control refers to a person's beliefs about how his or her experiences have come about. At one extreme, persons may believe that their experiences or outcomes are more a matter of chance or other persons' machinations than of their own actions or abilities. At the other extreme, persons may believe that whatever occurs to them is in response to some acts of their own, that in effect, responses are contingent upon their characteristics and/or actions. Such persons are said to have internal control expectancies. Needless to say, most people are not that extreme in either direction. Extreme externality would seem to be the hallmark of depression or paranoia whereas extreme internality would suggest grandiosity or megalomania. Somewhere between these extremes, people may be thought of as more or less external, or more or less internal.

Most pertinent to this paper are a series of studies in which locus of control has been used as a predictor of the ways in which people find and use information to help interpret tasks in which they are engaged. Internals have been found to be better able to seek out and make use of information than externals (Lefcourt, Gronnerud, & McDonald, 1973; Phares, 1968; Wolk & DuGette, 1974); and more ready to ascertain what is involved or to discover what is the meaning inherent in different tasks and situations than are externals (Baron & Ganz, 1972; Dollinger & Taub, 1977; Gregory, Chartier, & Wright, 1979; Lefcourt, 1967; and Stewart, & Moore, 1978).

If the sharing of information and perspectives with

consequent reinterpretations of experiences constitutes one of the major benefits of friendship, it would seem as if internals would stand a better chance than would externals of being able to use these benefits of friendship in helping to defuse the effects of stress. That is, the research literature indicates that internals are better discerners and users of information. Consequently, they should be more ready to perceive in their interactions with friends the information that would enable them to cope with or reinterpret their experiences, thus helping to defuse potential stressors.

Sandler and Lakey (1982) were the first to test this hypothesis. In their investigation, interactions between negative life events and social supports were found in the prediction of both anxiety and depression among internals but not among externals. Subsequent analyses revealed that both anxiety and depression were moderated for internals by the presence of social support, which was not the case for externals.

The studies that I will describe here represent a replication of, and an expansion upon, the Sandler and Lakey findings. Where Sandler and Lakey made use of the personal control factor of Rotter's locus of control scale (Mirels, 1970), we have used the affiliation and achievement locus of control scales from our own Multidimensional-Multiattributational Causality Scale (MMCS - Lefcourt, von Baeyer, Ware, & Cox, 1979; Lefcourt, 1981). These scales assess the degree to which persons believe that they can exercise control over two areas which, it may be assumed, are of high value to university students.

The subjects for our investigations were all university students, either in their first or third year of undergraduate training. With some differences in the means of administration and source of subjects, the procedure consisted of gathering scores on Sandler and Lakey's (1982) College Student Recent Life Events Schedule, Barrera, Sandler, & Ramsey's (1981) Inventory of Socially Supportive Behaviors, our own MMCS, and as our dependent measure, the POMS. Similar to the previous study, the Total Mood Disturbance scores derived from the POMS provided the dependent variable for this investigation.

Our hypothesis was that the relationship between negative life events and mood disturbance would be most attenuated among internals who also had a high degree of social support. Among externals social support was expected to have much lesser stress buffering effects.

RESULTS

It was evident from the intercorrelation matrix that each variable was associated with mood disturbance in predictable ways. Sex of subject was the only variable to produce a surprising result. Where we might have expected parity, men had higher mood

disturbance scores than females. Otherwise, sex was unrelated to any of the other predictor variables. The strongest effect upon mood disturbance was produced by negative life events, followed closely by affiliation externality. Externality for achievement was also related to mood disturbance, followed lastly by social support. The latter variable proved to be the weakest of all the correlates, of a lesser magnitude even than subject's sex.

In the regression analysis with affiliation locus of control as a predictor variable social support exerted an equal influence upon all subjects regardless of their classification as internals or externals.

On the other hand, in the regression analysis with achievement locus of control as the predictor, the results appeared to be as we had predicted. Among achievement externals there was a borderline main effect for social support but no interaction with negative life events. Among achievement internals, however, there was a highly significant interaction between negative life events and social support. For those high in social support, an increase in negative events was accompanied by a lessened degree of mood disturbance. For subjects low in social support, an increased number of negative events was linked with increased mood disturbance. It was evident that social support played a moderating role for achievement internals which it did not for achievement externals. Among achievement internals, the more available was social support, the less impact did negative events have upon mood disturbance.

Why affiliation locus of control should not have produced as powerful results as had achievement locus of control was not answerable with these data. My suspicion, however, is that achievement was valued more highly than was affiliation by our subjects, such that achievement control expectancies were more salient differentiators than were affiliation control expectancies. That is, to feel helpless about achievement when one is already in one's third year of university education (about 65% of our subjects) may have more prognostic significance than helplessness about affiliation, since such externality would seem to be aberrant or at least non-modal within this age group. It is interesting to note that when only first year students comprised the subject sample, locus of control for affiliation did produce the anticipated interaction effects, with affiliation internals showing the greater benefits of social support. During the first year of university life, the chores of achievement and friendship-making may be of equivalent importance with affiliation, perhaps being even more important to morale than achievement. Thus again, it maybe the salience of the area of concern that accounts for which locus of control measure will produce the hypothesized results. Regardless of these intricate differences, the results did provide some replication of the Sandler and Lakey findings that we had anticipated.

While we were examining our data we had become suspicious of

the possibility that our male and female subjects were producing somewhat different results. As other investigators have noted (Husaini, Neff, Newbrough, & Moore, 1982; Sarason, Levine, Basham, & Sarason, 1983) females have more often been found to exhibit the beneficial effects of social support than have males. Since our sample sizes were limited, we had not included sex within the previous analyses as a predictor variable. What we proceeded to do instead was to divide our samples into male and female groups and perform regression analyses with negative life events, social support, and their products, as predictors of mood disturbance as we had with the locus of control measures.

The regression analyses for males and females revealed that, as we had suspected, social support interacted with negative life events only within the female sample; the interaction reflects the fact that among females the higher was the social support, the less impact did negative life events have upon mood disturbance. For males, social support seemed to have very little import, there being no trends evident among either main effects or interactions.

That females were the prime beneficiaries of social support confirms the findings of other researchers and is congruent with the literature concerned with sex differences pertaining to sociability (Maccoby, 1966). Though it seems inconceivable that social support should be totally irrelevant to the well being of males, it would seem that in comparison to females at any rate, social support plays a minimal role in offering protection from stress. In one study with a largely male sample (Kobasa, 1982) social support has even been found to contribute to distress rather than being a stress buffer.

Although our sample was too limited in size to allow us to examine the hypothesis directly, it is likely that among those females who were internal for achievement, the moderating effects of social support were the most pronounced.

Several years ago when we were conducting our laboratory study concerned with intimacy as a moderator of social rejection (Miller & Lefcourt, 1983), we were simultaneously evaluating the role of affiliation locus of control as a moderator. We had hypothesized that affiliation internals would be more resilient than externals in the encounter with social rejection. The results we obtained at that time were perplexing and have since been languishing in a remote file. Given the sex specific findings with regard to social support, however, these earlier results seem more interpretable now than they were then.

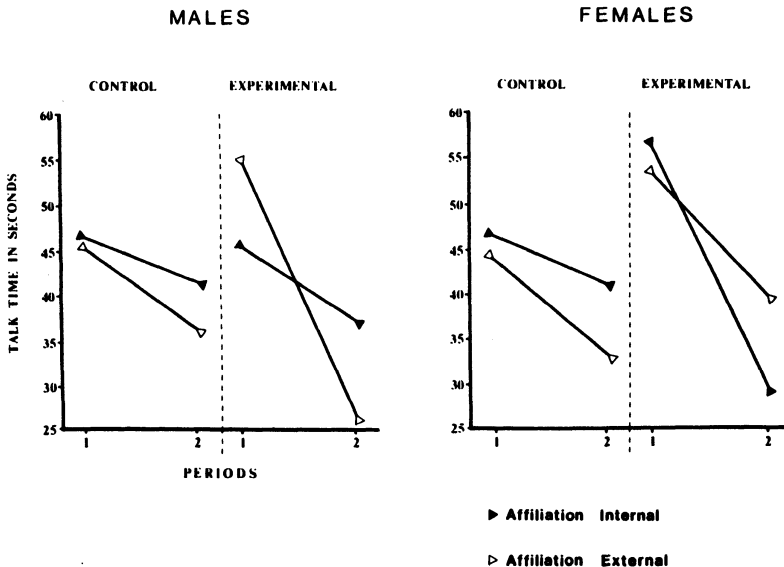
As described previously, we had withdrawn our attention from the subjects during an interview while they were expressing their viewpoints concerning marriage as opposed to living together. One of our major dependent variables was the length of time that subjects had continued to talk after we had withdrawn our attention from them. The analysis of variance results produced confirmatory evidence for the impact of conditions and intervals (experimental versus control, and early versus later intervals) as

well as for the interaction between affiliation locus of control and interval. The perplexing term, however, was a highly significant four-way interaction ($p < .001$) between conditions, intervals, affiliation locus of control, and sex.

On graphing the interaction, we found that among males the results were exactly as predicted. In the control condition both internal and external males declined in the time they had spent talking from the first to the second interval. However, as can be seen in Figure 3, during the second interval (the social rejection) in the experimental condition, external males spoke much less than they had during the first interval, showing the effect of this quasi-rejection. Among females, however, the behavior in the experimental condition was the opposite of that for males. Females who were internals for affiliation resembled males who were external for affiliation. Female internals, provided the withdrawal of social attention condition, seemed least able and/or willing to continue discussing the queried topic.

At the time, this paradoxical result seemed to be a cul-de-sac that was puzzling and definitely inconclusive. Given the above findings, however, concerning a possibly elevated need for social support among internal females, these findings may be viewed as a reflection of the greater social sensitivity to be

FIGURE 3. Talk Time as a Function of Affiliation Locus of Control, Sex, Social Rejection and Condition.



found among internal females. In other words, females who perceive themselves as responsible for their outcomes and experiences may be in particular need of social support if they are to weather potentially aversive experiences. Social rejection may be less tolerable among such persons than it would be among those who are more fatalistic about their experiences.

CONCLUSION

Social scientists have not been the sole propagandizers for the value of social support. Newspaper columnists, advocates of the various "anonymous" groups, as well as grandmothers, have long spoken of the benefits to be derived from social support. The contributions of the social science literature to an understanding of the effects of social support, however, have been bountiful and manifold. First, sociologists and psychologists, as well as epidemiologists and physicians, have demonstrated the fact that persons with social assets, confidants, good marriages, etc. seem better equipped to survive the ravages of stressful experiences. Secondly, a more recent set of contributions have focussed upon delimiting characteristics -- those elements that help to define when, how, and for whom social support becomes effective as a stress moderator.

Most particularly, we have been led to consider the specific functions of social support, and to be aware that social support is an omnibus term referring to many different kinds of relationships, and varying ways in which people may be able to help each other. Our own studies in this area have concerned the value of a particular kind of relationship, that of an intimate friendship, in helping to lessen an individual's duress. We have not addressed the question of what it is exactly that intimate friends do for each other. What we have been able to demonstrate, however, is that close intimacy works to reduce our vulnerability in social interactions with others; and the absence of such intimacy coupled with an absence of positive events leaves individuals in a markedly dysphoric mood state. In essence, intimacy is a desired experience that contributes to feelings of euphoria. Its absence is experienced as a deficit state that people would choose to rectify if it were possible to do so. At the same time, intimacy seems to protect people from social slights, perhaps because the person with very close friends is not as active in his or her search for closeness among strangers, and is therefore, less concerned about their judgments.

The other delimiting characteristic we have examined is that of the individual differences that bear upon the person's ability to make use of social support. Here we have found evidence in support of the hypothesized interaction between locus of control and social support. Persons who perceive themselves as the active determiners of their fates seem to need social support more than

their fatalistic counterparts, benefitting more from its presence, and suffering more from its absence. Since internals more often exert effort in the pursuit of their goals, given their belief that outcomes primarily reflect their efforts, it would seem reasonable that they would be in greater need of solace if their efforts came to naught than would persons who more readily blame others, chance, or the nature of tasks for their misfortunes.

In addition to locus of control, we have also found evidence to the effect that it is largely females who derive benefits from social support. This isn't terribly surprising in that reviews of the sex difference literature reveal that females are more apt to be nurturant, affiliative, and interested in others than are males (Maccoby, 1966). That they are also better able to derive benefit from social support than are males has likewise been reported in the literature as noted earlier. Our findings suggest one further possibility, that it is the more internal females who are the principal beneficiaries of support. Whether these sex differences are inherent or derive from socialization, it would seem as if females who regard themselves as responsible for their fates are particularly vulnerable to the effects of stress if they lack in social support. On the other hand, if these internal females should be graced with sufficient social support they appear to be well buffered from the exigencies of stress.

Overall, our research lends support to those who have advocated a closer examination of the operations of social support. The specific functions of relationships, the kinds of relationships, and characteristics of persons in those relationships, should interact and reveal to us more in the way of explaining how humans provide sustenance to each other as they encounter the inevitable stressors of daily living.

NOTE

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COPING STYLES, SOCIAL SUPPORT AND SEX-DIFFERENCES

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The main focus of our study is on coping strategies men and women demonstrate in coping with environmental stressors. Substantial differences between the sexes were evident in several variables which are thought to influence the coping capability of the individual. In particular, it was found that women resort to social support as a salient coping strategy to a much higher degree than men.

It is generally recognized that socialization processes play a major role in explaining differences between the sexes. Historically three basic theories seem to have gained prominence in the realm of gender specific socialization:

1. Psycho-analytic theories with strong emphasis on the concept of identification.
2. Social learning theories
3. Cognitive theories with reference to developmental stages.

Freud's classic psycho-analytic view on the topological structure of the personality has been severely criticized, for its lack of empirical evidence. Nevertheless his basic contention, that Ego Identity formation is partially mediated by identification with the parent of the same gender, is still generally accepted in modern psycho-dynamic theories. With respect to psycho-social determinants in Freudian theory, it is interesting to consider differences which appear to arise at a very early age in basic feelings of anxiety in response to the potential loss of a beloved parent. The strategy the child adopts in order to cope with threats of this kind may very well be sex related.

Freud, in his famous essay on womanhood and sexuality contends that for girls, in contrast to boys, the emotional attachment to the mother has strong repercussions for separation anxiety (Freud, 1951). The boy who feels strongly attracted to the mother, while desperately opposing the father, is not likely to experience separation anxiety. Instead he anticipates

punishment inflicted upon him by the target of his hatred. The boy later reconciles this conflict by identifying with the father, and the primal source of his basic security (the mother) has been preserved. By contrast, the girl is confronted with a loyalty conflict. Tempted to experience strong affection towards the father at the cost of being disloyal to the mother, she runs the risk of losing her sense of basic security. The uncertainty and ambivalence that goes with this loyalty conflict may, according to Freud, greatly enhance separation anxiety.

According to social learning theory children develop gender specific characteristics because of differential reinforcements of behaviors which a particular society favors for either sex. Bandura's modeling theory with its strong emphasis on the role of imitation and vicarious reinforcement is used to explain why gender appropriate behaviors are usually acquired. Avoidance of behaviors specific to the opposite gender is enhanced by punishment of these behaviors (Bandura, 1963, 1977). Rotter's social learning theory is highly relevant in this respect. It states that the conjoint impact of expectancy and preference for alternative reinforcements (reinforcement value) determines the behavior of an individual in a particular situation (Rotter and Hochreich, 1975).

Cognitive theories, such as the Piagetian approach launched by Kohlberg, relate sex differences to cognitive developmental stages (Kohlberg, 1969). According to Kohlberg sex identity is mediated by cognitive information processing, based on accumulative experience, which signals to the child that he is to acknowledge the specific nature of his gender characteristics. This cognitive "coding" determines the child's awareness of his own sex identity and the structure of his social interactions.

Whatever the implications of gender specific determinants, as hypothesized in the afore mentioned theories, remarkably the ultimate outcome seems to be the same: the boy assimilates and demonstrates typical masculine behavior and the girl feminine behavior. In our view the psycho-analytic emphasis on identification does not necessarily contradict the effects of social learning principles and cognitive information processing. The impact of these may very well be superimposed upon the resolution of emotional conflicts concerning problems of identification and may be related to differential significance of social support for the sexes. The Freudian stand that for girls uncertainty and ambivalence may enhance separation anxiety has been supported by data from a large longitudinal study that has an interactional rather than a Freudian perspective. Magnusson and his coworkers from Stockholm University have found that girls show far more separation anxiety than boys (Magnusson and Olah, 1981). If this is true, in view of the impact of early experiences on future development, one might suspect that the use of social support as a coping style to master environmental stress would be of much greater significance for girls than for boys.

The tendency for girls to resort to social support as the major palliative to separation anxiety, might be reinforced in later stages of development by various socializing agents. Girls are expected to act in a gentle manner in their dealings with others (Moulton, 1980), and the expression of personal feelings in their social interactions is acceptable. This attitude is further facilitated by modeling processes in the imitation and internalization of maternal caring behavior (Bandura, 1963). Boys, on the other hand, learn through socialization to repress emotion and strongly inhibit the expression of personal feelings. The inhibition of emotional expression, and an urge to behave autonomously ultimately cripples their capability to ask for social support. In the present research, it was indeed shown that in stressful situations women tend to seek social support while men engage in active coping. These findings are in line with a recent study reported by Sarason, Sarason, Hacker & Basham (in press), indicating that females report greater perceived social support and greater satisfaction with the support available than males.

METHOD

The aim of the present study was to describe differences between men and women who are engaged in the same organizational work settings. Comparisons of this kind require that job demands, educational level, payment and average age should be within the same range for both groups. The research to be reported pertains to group workers in residential settings who are responsible for children ranging from 12 to 18 years of age. A representative sample of workers in Child Guidance Clinics in the Netherlands was obtained which permitted a systematic comparison between the sexes. The research design was based on a multi-variate prediction model in which work satisfaction, physiological strain and psychosomatic complaints were the main dependent variables.

French and Caplan's model of organization stress was used as a theoretical frame of reference (French & Caplan, 1972; French, 1974, 1976; Dijkhuizen, 1980). The approach of French and Caplan describes a sequential model in which (so-called) Person-Environment Fit variables are utilized as predictors of strains, the latter being moderators of defective health states. In this view "adjustment" is conceived as the "Goodness of Fit" between the characteristics of the person and the properties of the environment. Two aspects of the environment are distinguished: the subjective environment and the objective environment. The objective environment exists independently of the person's perceptions, while the subjective environment reflects the person's assessment of environmental circumstances.

The model is explicitly labelled "Person-Environment Fit", and analogous sub-distinctions pertain to the Person component of the model. The subjective Person-Environment encompasses two

different matching processes. The first refers to the degree of correspondence between the subjectively perceived sources of fulfillment (supplies) and the subjectively perceived motives for working. A low discrepancy between subjectively perceived supplies and subjectively perceived motives implies a good fit or positive emotional adjustment. The reverse holds in case of negative adjustment. The second aspect refers to the discrepancy between the perceived job demands and the perceived abilities to meet these demands. Coping strategies, labeled as "environmental mastery", represent deliberate efforts to change the environmental conditions. By contrast coping strategies aimed at engendering changes within the person are called "personal adaptation". Cognitive defense" may play a role in warding off information detrimental to one's self-esteem, for example when a discrepancy between demands and capabilities is sensed. The model strongly emphasizes the value of subjectively perceived social support from supervisors and others at work or at home. This model is used as a frame of reference for presenting the results.

RESULTS

A purely descriptive comparison is presented first for each of the separate rubrics of the model. This allows a diagnostic overview of the extent to which the scores of men and women converge or differ. Although specific differences between men and women were expected, directionality was not predicted in advance. In all cases two-tailed probability testing was applied. Few sex differences show up with regard to variables pertaining to the subjective perception of the environment regarding perceived social support from a variety of sources or from tensions in the organization workgroup. The only, albeit extremely significant, difference relates to the positive impact of life experiences. These are shown in Table 1. Inspection of the means indicated that men seem to be far more optimistic than women in this respect. Scores for trait anxiety tended to be higher for women than for men, but self-esteem and locus of control did not differ. For the two interpersonal measures, beliefs as to the relative significance of support, and leadership in interpersonal relations, there are extremely significant differences (see Table 1). Women show a preference for social support and men for leadership. Sex differences were also obvious in the data of coping strategies. Taking into account that negative scores indicate frequent use of this coping strategy, men resort to a far greater extent than women to cognitive-active coping, suggesting an assertive outgoing attitude towards problem solving. By contrast, women resort to a far greater extent to social support in seeking solutions for their problems. Although as mentioned above men and women did not differ in their perceptions of social support received, these findings suggest that they do differ considerably with regard to the tendency to resort to social

TABLE 1

Means, standard deviation and 2-tailed probability for T-test between sexes

SUBJECTIVE ENVIRONMENT					
	Men (N=100-102)		Women (N=62-65)		
	Mean	SD	Mean	Sd	p
<u>Life experiences</u>					
Sarason et al (1978)					
Brandjes et al (1982)					
positive impact	15.42	9.98	10.45	7.44	.00
negative impact	7.97	9.45	8.63	8.46	.65
PERSON					
<u>Interpersonal values</u>					
(Gordon, 1960;					
Drenth et al, 1973)					
support	18.81	4.44	21.57	3.76	.00
leadership	11.42	5.92	8.87	4.18	.00
recognition	7.63	3.73	8.17	2.35	.29
<u>Coping</u>					
(Westbrook, 1979;					
Schreurs, 1980					
cognitive-active	-0.14	0.85	0.21	0.83	.01
optimistic attitude	-0.09	0.84	0.14	0.85	.10
social support	0.20	0.77	-0.30	0.86	.00
escape	0.08	0.91	-0.12	0.73	.11

High scores imply retrospective positive appraisal of the impact of life events.

support for problem solving.

In passing it should be noted, that for women, relatively high anxiety levels are associated with a preference for social support as a personal value and with resorting to social support in real-life situations. This outcome could be interpreted as support for the suggestion that there might be a strong relationship between separation anxiety and seeking for social support. With regard to psychological strains, the two variables which relate to satisfaction with the working conditions show different patterns of effect (see Table 2). Dissatisfaction regarding autonomy and participation, is considerably higher among men than women. Satisfaction pertaining to the function of the organization in general, implying an assessment on a more abstract and symbolic level, is affected differently: women indicate far more dissatisfaction than men in this respect. Overall this may indicate that women might question the efficiency or effectiveness of the organization as a whole, but at the same time may not feel deprived of genuine participation. Their focus is at a concrete practical level: interpersonal communication is perceived more positively than abstract impersonal organizational structures.

TABLE 2

Means, standard deviations, 2-tailed probability for T-tests between sexes for strain and health-illness measures.

STRAINS	Men (n=100-102)		Women (N=65)		
	Mean	SD	Mean	SD	p
<u>Psychological</u>					
work dissatisfaction: (Groenier et al, 1978)					
autonomy and participation	0.12	0.97	-0.19	0.74	.02
general organiza- tion	-0.12	0.90	0.19	0.84	.02
relationship with co-workers	0.01	0.85	-0.01	1.00	.88
general: life dissatisfaction	2.15	0.58	2.15	0.46	.98
state anxiety (Spielberger et al, 1970; van der Ploeg, et al, 1980)	35.15	8.16	35.83	8.31	.60
<u>Behavioral</u>					
smoking	1.64	1.59	1.46	1.35	.44
drinking	1.69	1.31	0.82	0.99	.00
<u>Physiological</u>					
systolic blood pressure	134.16	10.96	124.97	12.20	.00
HEALTH ILLNESS					
psychomatic complaints (Dirken, 1967)	0.17	0.15	0.23	0.16	.03
absenteeism	13.98	25.61	13.49	23.00	.90

Except systolic blood pressure n=90, 58.

This seems consistent with a strong preference for social support in the personal sphere.

Table 2 also shows exceedingly significant sex differences in the drinking and systolic blood-pressure data; in both cases women showed lower levels than men. Men and women, however, appear no different with regard to smoking. Though women seem to be less susceptible to health risks, such as elevated blood-pressure and excessive drinking, they do show more psychosomatic complaints than men.

Overall the difference between men and women were most obvious in relation to the Person, Strains and Health-Illness variables; we computed the correlations between these variables, for men and women separately (see Table 3). The most interesting findings pertain to psychosomatic complaints. This variable may be the most obvious indication of poor health, and is related in the case of men to practically all strain variables, except for state-anxiety and blood pressure. This outcome strongly contrasts

TABLE 3

	Strains - Health and Illness									
	1	2	3	4	5	6	7	8	9	10
autonomy and participation	-									
general organization	-									
relationship with co-workers	-									
life dissatisfaction	08	-16	22*	-						
	-26	-02	00	-						
state anxiety	23*	-06	25**	24**	-					
	09	-18	23*	25*	-					
smoking	00	-17	04	00	19*	-				
	06	-21*	-11	23*	15	-				
drinking	-04	-16	13	07	12	42**	-			
	27*	-10	-16	-06	24*	29*	-			
systolic blood pressure	21*	11	-07	-03	20	06	16	-		
	02	08	12	-01	-11	08	-06	-		
psychosomatic complaints	21*	-30**	28**	28**	14	21*	21*	-01	-	
	01	-12	-03	10	23*	18	25*	-02	-	
absenteeism	09	-06	19*	15	39**	14	21*	05	28**	-
	-18	-10	-03	21*	27*	20	14	-02	22*	-

Intercorrelations strains and health x 100

first row men (N=100-102, except systolic blood pressure N=90)

second row women (N=65, except systolic blood pressure N=58)

*: p<.05

** : p<.01

with the findings for women; besides state anxiety only two other variables: drinking and absenteeism were significantly correlated with psychosomatic complaints. Interestingly the correlation with state-anxiety exclusively holds for women.

In order to determine which relationships might be relevant to sex differences, we first screened our data and selected those variables for which there were substantial differences in correlation for men and women. These variables were then examined using ANOVA's. In computing the ANOVA's, the men-women dichotomy and the specific variable representing a Person, Social-Environment (P-E-Fit) category were treated as input-variables and either Strain or Health-Illness measures as dependent variables. Though a great number of the ANOVA's showed significant interactions, we will restrict our presentation to the results which may enlighten the predictive power of independent variables (see Figure 1, a-c). For the trait anxiety and relationship with co-workers variables, a curvilinear relationship holds for women, suggesting that certain levels of anxiety may

Figure 1 Sex Differences

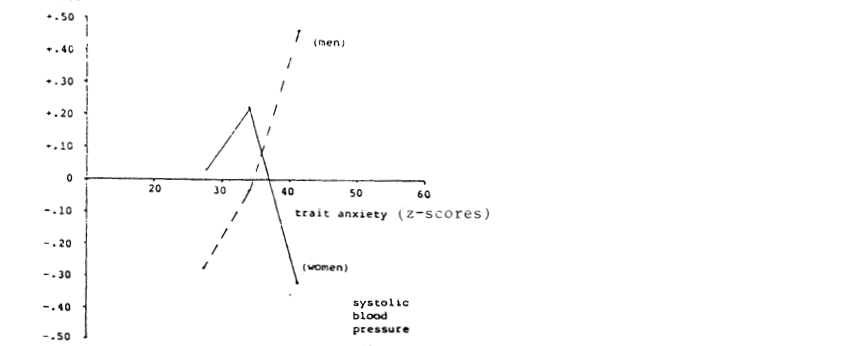


Figure 1A

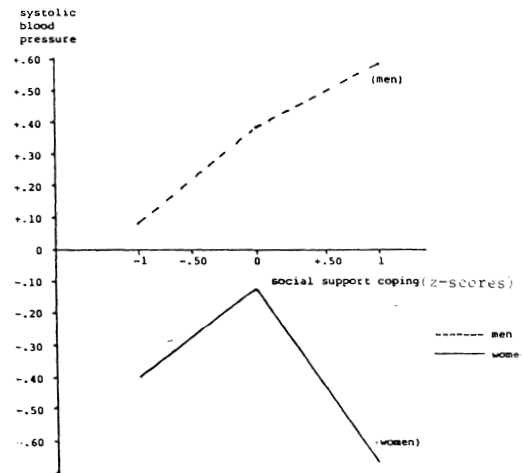


Figure 1B

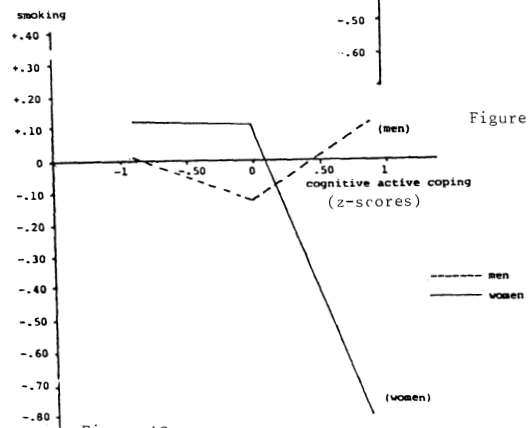


Figure 1C

The vertical axes represent the dependent variable, transformed into Z-scores; evidently strains or Health-Illness-scores.
The horizontal axes represent the independent variables.

enhance their relationship with co-workers. Although this may seem unexpected, a plausible interpretation might be that at certain levels train anxiety wards off potential threats from negative relationships with co-workers; in order to prevent more painful experiences, one over-estimates the potential support of co-workers. For men in contrast, a linear relationship is shown.

For the blood-pressure and social support coping variables, the ANOVA also suggests that a curvilinear relationship exists for women. Figure 1b shows, as has been reported previously, that overall blood-pressure levels are lower for women. For men a linear relationship indicates that the more social support is sought for, the steeper the increase of blood-pressure. For women, low blood-pressure levels are associated with both very low levels and with relatively high levels of the tendency to seek social support. The association with low levels of social support coping is in correspondence with the linear trend for men. The association with high levels may once again suggest need for social support. This may imply that the anticipation of veridical support from other people is perceived as soothing and strongly reassuring and ultimately preserves low blood-pressure levels. This again suggests a sensitivity for social support, and indicates that women recognize the potential benefits of social support to a much greater extent than men.

For the smoking and cognitive active coping variables, the ANOVA reveals that for men cognitive active coping is related to moderate levels of smoking behavior. It also shows that women possessing passive (non-active) coping strategies resort to smoking to a much higher degree than women using active coping strategies. The stress-model outlined in the Introduction has served the purpose of presenting a frame of reference for the assessment of sex differences. One should bear in mind, however, that the model is essentially predictive. The independent variables (Person, P-E-Fit and Social Environment rubrics) should be conceived as potential predictors of the dependent variables (Strains and Health-Illness). In order to assess the predictive power of the independent variables, pair-wise Multiple Regression Analyses were carried out for each of the dependent variables and for men and women separately.

As Table 4 shows, in addition to P-E-Fit, lack of social support of staff contributes substantially to the prediction formula. For men on the other hand, the perception of tension in the organization offers an additional contribution to feelings of dissatisfaction with regard to autonomy and participation. Interestingly, this finding seems to indicate that women are most sensitive to social support. Men, however, are more prone to over-emphasize the impact of organizational stressors. Table 4 also suggests that social support is the sole substantial predictor of relationships with co-workers for women. Support of co-workers is, to a much lesser degree, correlated with co-worker relationship for men and an additional variable trait anxiety also

TABLE 4

Multiple Regression Analysis Results for Men and Women

<u>Men</u>					<u>Women</u>				
<u>predictors</u>	<u>Beta</u>	<u>standard error B</u>	<u>r</u>	<u>r(2)</u>	<u>predictors:</u>	<u>Beta</u>	<u>standard error B</u>	<u>r</u>	<u>r(2)</u>
AUTONOMY AND PARTICIPATION									
Participation					Participation				
P-E fit	.41	.57	.66	.44	P-E fit	.29	.32	.44	.19
tensions in									
organization	.29	.55	-.47	.50	support	-.21	-.24	-.38	.30
other variables					other variables				
Total				.65	Total				.56
RELATIONSHIP WITH CO-WORKERS									
support					support				
co-workers	-.31	-.59	-.40	.16	co-workers	-.84	-.16	-.62	.39
trait					other				
anxiety	.23	.27	.37	.24	variables				
Total				.45	Total				.74
SMOKING									
recognition	.21	.90	.26	.07	leadership	-.38	-.12	-.43	.18
other variables					negative impact				
					life exper-				
					iences	.27	.43	.32	.25
					other variables				
Total				.29	Total				.50
DRINKING									
escape	.30	.44	.32	.10	escape	.36	.49	.45	.20
other variables					support				
					directors	-.54	-.58	-.41	.33
					esteem P-E				
					fit	.24	.55	.30	.39
Total				.30	Total				.66
SYSTOLIC BLOOD PRESSURE									
					tensions in				
					work group	.42	.15	.24	.06
					soc. sup.	.22	.30	.18	.11
					other variables				
					Total				.46
PSYCHOSOMATIC COMPLAINTS									
trait					trait				
anxiety	.67	.13	.59	.34	anxiety	.57	.14	.55	.31
					negative impact				
					life events	.69	.41	.10	.41
Total				.55	Total				.55

plays a role. This may signify, in view, of the linear relationship found in the ANOVA analysis, that deterioration of perceived relationships with co-workers is to some degree determined by increasing levels of trait anxiety. As we may recall from the ANOVA data, a curvilinear relationship holds for women between anxiety and relationships with co-workers. This may have balanced out the potential influence of trait anxiety on the perceived relationship with co-workers in the regression analysis.

In examining the prediction of the other dependent variables, we found differences between men and women, but in some cases the outcomes were either hard to interpret, or did not deliver very substantial multiple correlations. This may be partially due to the complexity and broadness of definition of the independent variables involved. In order to exclude contaminations of this kind, we will concentrate on the more concrete and less complex variables, such as smoking, drinking, blood-pressure and psychosomatic complaints. It should be noted from Table 4 that the explained variance with regard to smoking is much greater for women than for men: (50 per cent vs. 29 per cent). Regarding the data for men, only one variable contributes to the prediction, namely recognition. Women with low scores on leadership, who do not rate the importance of leadership, commit themselves to smoking. Once again referring to the correlational data, it was suggested that for women, low leadership scores are associated with excessive P-E Fit discrepancies for work-load, passive coping and escape. Taking into account that life events also contributed substantially to the prediction of smoking behavior for women, the conclusion seems plausible that women resort to smoking because of two simultaneously operating sources of frustration and distress: actual stressors in the work setting, and negatively impacting events in their personal life.

Table 4 indicates that drinking alcoholic beverages seem to be a simple function of escape for men, but is more complicated for women. Though women share the same coping strategy of escape, lack of support by superiors and the discrepancy between perceived needs for esteem and actually received esteem also substantially influence drinking behavior. This, once again, suggests the importance of social support for women.

The prediction of blood-pressure for women is most interesting. As Table 4 shows both variables of the predictor for systolic blood-pressure have strong interactional and communicative connotations. The analysis for men is not presented because the first predictor did not reach the five per cent criterion.

"Psychosomatic complaints" is a potent and central dependent variable. Table 4 shows that for men the main predictor was trait anxiety. Many other predictors pertaining to work conditions also contributed, though slightly in each case. Trait anxiety is also an important predictor for women but to a much lesser extent. For women only, negative impact of life events also presents as a

second major determinant. This may suggest that women might be more susceptible to the detrimental effects of aggravating circumstances in their personal life, and are thus more vulnerable in coping with these events.

FINAL COMMENTS

Utilizing a stress model for organizational settings, originally proposed by French and Caplan and adapted to our data, the following major findings were ascertained. Women show much higher anxiety levels and suffer less from health threats such as hypertension and excessive drinking, but are more vulnerable to the effects of detrimental life events. Inasmuch as coping strategies are concerned, men resort to a far greater extent to cognitive active coping, suggesting an assertive outgoing attitude. In contrast, women resort to a far greater extent to social support in seeking solutions for their problems. These findings were further corroborated in a multiple regression analysis utilizing Health Risks and Strains as the dependent variables.

In an introductory section it was contended that the strong urge to resort to social support with women is partially due to separation anxiety. This reasoning is in line with Freud's assertion that contrary to boys, girls are much more highly susceptible to worry over the potential loss of a beloved parent. Remarkably Magnusson, using a very different perspective from Freud, has obtained concurring empirical results that girls indeed show far more separation anxiety than boys. This conclusion does not exclude the possibility, however, that other socializing factors such as modeling of maternal behavior and gender specific information processing may have played a role in inducing the preference for social support as a major coping strategy.

In commenting on the findings of our study we feel tempted to dwell briefly on their implications for interpersonal relationships. The divergencies among the sexes, as reported in this paper, call for fundamental changes for the betterment of intersex relationships.

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A CONCEPTUALIZATION OF PROFESSIONAL WOMEN'S INTERPERSONAL FIELDS:
SOCIAL SUPPORT, REFERENCE GROUPS, AND PERSONS-TO-BE-RECKONED-WITH

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In recent years, as social scientists have sought clearer understanding of environmental factors that influence a variety of personal outcomes, the role of social support in the lives of adults has come under increasing scrutiny.

In our initial efforts to understand more about the occupational choices and career paths of adult women, we turned to this literature to see what support-related variables one might reasonably expect to influence career decisions. We found that levels of aspiration, career orientation, and involvement in role-innovative fields have been related to support from college faculty members (Dickerson, 1974; Tangri, 1972) and friends (Tangri, 1972), particularly male friends (Parsons, Frieze & Ruble, 1978). Role models--women who have been successful in the career of choice and who are accessible or known to the individual--have also been linked to career aspirations, career choice, and success (Almquist & Angrist, 1971; Douvan, 1976), although the absence of female role models has been noted repeatedly in many career fields (e.g., Kashket, Robbins, Lieve, & Huang, 1974; Widom & Burke, 1978). The salience of support in the achievement of career success has been observed throughout the life span. For example, it has been observed that most successful women experience highly supportive family environments early in their lives (Anderson, 1973). It also has been shown that successful women experience strong positive identification with their fathers and a variety of female models, who stimulate professional careers and often alleviate the guilt frequently attending efforts to combine work and family responsibilities (Douvan, 1976; citing work by Walum, 1974). Support factors on the job are particularly salient: participation in informal

networks, involvement with a "sponsor," and being taken under the wing of a mentor are all mentioned as critical to success in many careers (e.g., Laws, 1980).

In our own work, too, we have found limits to the explanatory power of support. For example, we examined selected social network variables in relation to self-valuing in a group of women oriented toward male-dominant careers (Plas & Wallston, 1983) and found that perceived support captured less of the variance than did valuing of women, encouragement of women, and number of women named as important. The pattern of findings was similar for the male-oriented predictor set. In a subsequent comparison of women in male- and female-dominant careers, we found the same pattern of results (Wallston, Hoover-Dempsey, & Plas, 1983).

Thus, we have turned from a consideration of support as the major influence on women's career development to a consideration of other forces which might enhance its explanatory power. This effort has led to the development of a transactional model of the interpersonal field, of which support is one component. We developed the model for specific use in understanding more about women's career choices and patterns, but we offer it here in its more generic sense, a tool we believe to be helpful in understanding personal functioning within varied environments.

THE INTERPERSONAL FIELD: A TRANSACTIONAL MODEL

It is our contention that the failure of existing approaches to the issues identified above resides in the tendency to search for variance in either the person or the environment. We believe that a model which incorporates variables representing person-environment transactions is likely to be more powerful in capturing the dynamics that operate within the daily lives of human beings.

Transactional theory is guided by a philosophical perspective that seeks to safeguard the integrity of natural phenomena (Dewey & Bentley, 1949; Plas & Dokecki, 1982). A person-in-environment constitutes such a natural phenomenon. In contrast, within classical psychological science, person variables are separated from environmental variables in order to identify causal sequences of events. Thus, models of behavior that are based on the classical paradigm must establish the temporal priority of either person or environmental variables. One set of variables belonging either to the person or to the environment is seen logically to be activated prior to the activation of the other. For example, in a nursery situation, such a model might seek to establish a causal connection between a mother's responsive behavior and a baby's cry (or, between the cry and a perceived antecedent environmental event such as an accidentally misused diaper pin). Thus, the classical paradigm is bound by an a priori appreciation of temporal sequence; theoretically, it is an action-reaction model.

Within a transactional framework, the concept of the temporal priority of either subject or object is disregarded at several stages of model construction (cf. Gibbs, 1979). Temporal priority may be introduced at final stages of logical analysis such as path modeling, but the core of a transactional approach requires theoretical identification of variables that represent a transaction between person and environment; and such a transaction must be identifying causal relationships. For example, in the diaper-pin-cry-responsive mother illustration mentioned above, the variable of experienced distress might be the critical concern from a transactional perspective. The distress of a mother who may have been harried and hurrying prior to the unfortunate misapplication of the pin need not be temporarily isolated from the distress of the infant. Rather, the distress experience is of concern, as are those person-environment variables that enhance or mitigate it.

The theoretical model presented here is a product of a transactional orientation. It attempts to capture the dynamic nature of relationships among those experiences of stress and experiences of nourishment which arise from the transactions of career-oriented women with their interpersonal fields (i.e., the variety of persons identified as important throughout the individual's life space).

Interpersonal Field

The interpersonal field is comprised of three major components representing categories of persons defined by salient relational functions rather than traditional role relationships (see Figure 1). Together, the three components contain all the essential functions performed by those persons who are perceived by an individual to be generally important for the conduct of interpersonal living. Thus, as the definitions and discussion below reveal, there are no relevant relational functions operative within the life space that are not captured within this model. There may be events within one's life space that are potential stressors or nourishers and that do not originate in specific human relationships (e.g., a tornado, economic depression, a sweepstakes win). While the experience of these events as stress or nourishment is a function of the interpersonal field, it is important to keep in mind that the source of some stressors and nourishers does not lie within the web of human relationships.

Lewin and Sherif Influences

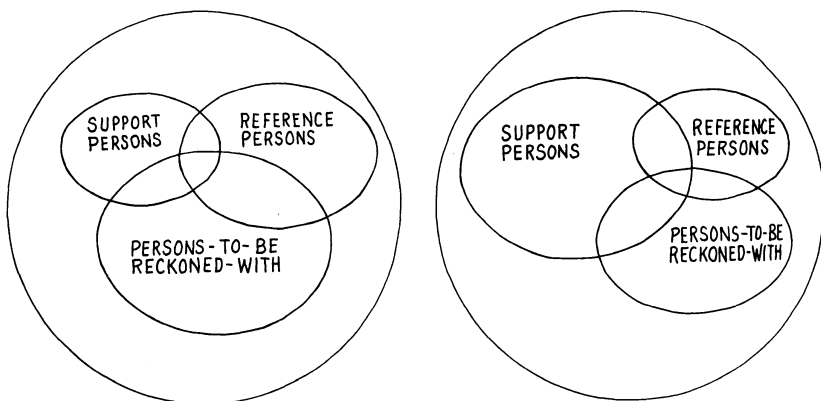
The transactionally-oriented perspectives that provide the theoretical base for the model presented here have been formed through a synthesis of portions of the action research work of Lewin (Marrow, 1969) and Sherif (1982). Both of these "practical theorists" sought to identify behavioral outcomes that are

dependent upon variables which represent a transaction between person and context factors.

Lewin has inspired our identification of the interpersonal field, an array that contains all the significant persons with whom the professional person interacts. In Lewinian thinking, the concept of life space is used as a tool for the theoretical construal of all those physical and psychological dynamics from which motivated behavior arises. The interpersonal field described in our model can be thought of as a life space. In order to protect the integrity of the Lewinian concept, those categories of persons that compose the interpersonal field have been identified with the model according to the perceived functions that the relationships offer; i.e., roles (family members, friends, co-workers, etc.) have been superseded by functional relationships (support, reference, and gate keeping) in delineating the components of the field. The categories are thus identified as these persons providing support, those providing comparison possibilities, and those persons providing gatekeeping functions with respect to such things as time, influence, and so forth.

The Sherif concept of self system (or, attitudinal schemata) has been particularly important to the development of the model of person-environment transaction described below. The self system is defined as: ...a constellation of attitudinal schemata, formed

FIGURE 1. Examples of interpersonal fields within psychological life spaces.



during development through interaction with physical and social realities. Centered on experiences of "me," "I," and "mine," these schemata relate the individual in fairly consistent ways with the body, its parts and capabilities; with others in the same and different social categories, with social objects, places, activities, and time schedules; with groups and institutions, each with status criteria, values or norms and belief systems--hence to varied situations and activities composing that person's environment (Sherif, p. 381). This concept fits productively with Lewinian constructs. For example, Sherif views gender as an attitudinal schemata that emerges from the social environments in which individuals develop. What individuals experience as gender-related is a psycho-social question. Thus, this concept can be thought of as a transactional variable that represents a person-environment dynamic. Through use of the attitudinal schemata construct, we are able to represent experience within the interpersonal field model, thus providing for measurable variables that constitute relevant person-environment factors within the life space. The use of this construct also permits us to move away from Lewin's reliance on the functions of conflict and tension within the lifespace; it offers a viable theoretical alternative.

In addition to this basic construal of the self system, Sherif has influenced our thinking by guiding our identification of two of the function-related categories of the interpersonal field: the need for inclusion of reference persons has been argued cogently by Sherif; our conception of persons-to-be-reckoned-with flows from her consideration of social power.

Model Components

Support Persons. Exchange of support may or may not be reciprocal; it may be a one-way relationship or may entail mutual offerings of supportive resources that differ either quantitatively or qualitatively (e.g., one offers affective support, the other, instrumental). According to our model, a supportive relationship can be identified when one individual perceives herself to receive support from another. As illustration of one possible variation of a supportive relation, consider the support transaction involving a person who infrequently offers substantial instrumental support of a monetary nature to another who quite frequently relates to her by offering rather small amounts of encouragement for a variety of endeavors.

The subcategories of instrumental and affective support have been further delineated conceptually through construction of a list of types of supportive resources that typically are exchanged within interpersonally supportive relationships (see Table 1). These delineations have resulted from a literature review (e.g., Bell, 1981; Sarason, Levine, Basham & Sarason 1983), and pilot

TABLE 1

Support Functions

Affective

Positive Regard. Affirmation of all aspects of self. May include unconditional positive regard, or positive regard linked to evaluation of one's self, one's attributes or functioning.

Validation. Affirmation and confirmation of one's ideas, efforts, and/or self.

Love. Broad and deep feelings of positive attachment.

Affirmative Physical Contact. Physical demonstrations of emotional support and positive regard, such as hugs, caresses, arm around the shoulder, etc.; involves physical actions of support, and comradeship.

"Being There". The state of being fully present and available for help and support. It implies positive and willing accessibility and a constancy of availability.

Listening. Willing and available to hear what another says and means; giving time and full attention to an interaction.

Constructive Confrontation. Positive challenge to one's ideas and perceptions. It may involve disagreement and argument, but is based consistently on a valuing of the person and is focused on constructive feedback.

Encouragement. Active stimulation of hope and belief in one's self and abilities.

Consolation. Provision of emotional and physical comfort in response to another's loss, dismay, setback, unhappiness.

Instrumental

Information. Response to a request (explicit or implicit) from another; oriented toward enabling the other to perform more effectively, gain more effective access to resources, etc.

Physical Task Assistance, Related to Life Maintenance, Personal and Family Issues. The giving of specific, material help in response to specific needs arising in the course of everyday life (e.g., a ride to work when the car breaks down, help in fixing a broken faucet, help with child care, the loan of a heating pad, assistance in preparing meals, etc).

TABLE 1 (continued)

Physical Task Assistance, Related to Professional Maintenance and Development. The giving of specific, concrete help in relation to professional matters and issues (e.g., reading and commenting on a report or paper, "filling in" at a meeting, brainstorming on a particularly puzzling concept or problem, advising or helping with a task temporarily beyond the capacities of the individual, etc.).

Intercession. Actions that involve interceding on behalf of another, facilitating another's well-being or progress in a system, defending the actions or progress in a system, defending the actions and character of another in response to doubts or attacks expressed within a system. It may involve active defense of another, and participation as a mediator or "friend of the process."

Financial Help. The giving of financial assistance when needed, through outright gifts, loans, etc.

work of our own occurring primarily at the Association for Women in Psychology conference held at Seattle, Washington, in March, 1983. The subdivisions of support functions can be found in Table 1.

Reference persons. These are persons whose professional or personal characteristics provide a standard for evaluative comparisons; there may be little or no reciprocity of reference. Hyman (1942, 1960) developed the concept of reference groups which has been utilized by sociologists (e.g., Merton & Kitt, 1950) and social psychologists (e.g., Kelley, 1952; Shaw, 1981; Sherif, 1976). Festinger's (1954) theory of social comparison provides a similar framework for understanding our self-evaluation in relation to others (e.g., Latane, 1966; Suls & Miller, 1977).

This component of the interpersonal field has been divided conceptually into three areas of reference possibility; personal, professional, and general resources. A person may compare her personal characteristics to those of another on the following dimensions: interpersonal skills, personality characteristics, values and attitudes, physical appearance, life style, and physical skills. Four professional reference dimensions have been identified: competence, status and recognition, productivity, and professional style. The third category incorporates those assets perceived in themselves to be at least somewhat self-generative: financial resources, personal background, social support system, intellectual resources, and physical resources (e.g., health, energy).

Four qualities of a reference relationship have been identified and are characteristic of all three subcategories of this interpersonal field component. The first of these concerns

the degree to which the reference person is viewed as similar to the self (e.g., Bleda & Castore, 1973; Gruder, 1971; Mettee & Smith, 1977). The second quality concerns the direction of the evaluative judgement (i.e., is the person viewed as "worse," "same," or "better" on the given dimension?). The third reference quality represents the degree to which the reference dimension is considered to be of importance to the perceiving person. The fourth quality concerns reciprocity of comparison.

The theoretical construction of the interpersonal field component allows us to capture in a logical fashion the wide variety of reference relationships that are important for daily living. Use of the model permits a loosely alluded-to professional reference relationship to be specified, for example, as one that operates on the dimensions of competence and productivity with approximately equal reciprocity of reference within a relationship where the comparison person is viewed as highly similar to the self and "better" on both dimensions which are seen to be of moderately high importance to the perceiving person.

Persons-to-be-reckoned-with. A person-to-be-reckoned-with is part of an obligatory relationship that is characterized by gatekeeping functions. Such a person controls the flow of needed resources such as time, energy, money, working conditions, and so forth. When a given person is functioning as a person-to-be-reckoned-with, support and reference functions are minimal for the perceiving person. Of the three model components, the persons-to-be-reckoned-with group has been the most neglected within the literature. Yet we believe that relationships of this type represent a sizable proportion of daily transactions, especially for career-committed persons. These relationships are (a) required in that we do not choose to participate in functional relations with these people but are bound to honor their roles, and (b) characterized by some kind of power, either direct or indirect. Our relationship with such persons is characterized by mutuality of power over the other or by the unidirectional exercise of power on the part of only one member of the relation. By power we mean the ability to act upon, control, or influence persons or resources. Often, persons-to-be-reckoned-with influence us because they have influence over others who are close to us. The following list provides examples of required influential roles that may fall in the Persons-to-be-Reckoned-With category: Boss, secretary, housekeeper, best friend's best friend, child's teacher, auto repairperson, the ex-wife of a husband, and so forth. When a person-to-be-reckoned-with relation is functioning, it is not characterized by the giving or receiving of support and the person is not an important member of a reference group. However, such a relationship need not be a negative one. Even though we are forced to "deal" with such persons, the interactions may be pleasant.

The persons-to-be reckoned-with component is divided into

three construct domains: awareness of the presence of these functions in one's personal and professional worlds; perceptions of the members of this component, their functioning and interrelationship; attitudes toward self in relationship to them. Clearly, explication of this interpersonal field component is still at a preliminary stage of development. Questions that immediately arise relative to this component concern objective and experiential information in all three domains. For example, in the first domain, we want to know who a respondent's persons-to-be-reckoned-with are and how easily she can generate their names. Vis-a-vis the second domain, it is important to assess whether the influence of these persons on the respondent tends to be direct or indirect and what she perceives their effects to be. Relative to the third domain, it seems important to know whether the respondent tends to avoid persons in this category, and how she perceives her ability to function in relation to each one.

At this point in the development of the model, four qualities of persons-to-be-reckoned-with relations have emerged as important: reciprocity, relevance (defined by perceived importance and frequency), and relationship location (i.e., personal life vs. professional life) and types of resources controlled. Given that people have a satisfactory level of awareness of this kind of relation (still an empirical question), we are able to discern: Whether an individual's unique set of such relations tend to be nonreciprocal (and if so, in which direction), the degree of relevance of these relations (level of perceived importance and frequency of contact), whether the relevant relations are clustered within the personal or professional life and the primary resources affected. Each of these qualities can be described relative to functioning within the construct domain of direct versus indirect power and/or impact as well as the domain of individual perceptions of the self in relation to this interpersonal field component. For example, our current instrumentation can reveal that an individual is aware of this interpersonal field component; recognizes that most of her relations within it tend to be directly influential, involving direct rather than indirect power; that she tends to perceive herself to be moderately capable of effectively participating in such relations within the professional rather than personal sphere; that most of the people-to-be-reckoned-with relations that she considers relevant are located within the professional sphere; that reciprocity of relation is unequal in the professional sphere where persons-to-be-reckoned-with are seen to have power over her rather than the converse; and that the most important gatekeeping function in the professional sphere involves access to her time.

It is our contention that a full explication of the functions of persons-to-be-reckoned-with within the interpersonal field will prove to be fundamental for a satisfactory understanding of the roles of such phenomena as stress and nourishment in relation to

important person-environment outcomes such as well-being, physical health, and substance abuse. Thus, we currently are emphasizing the continued development of instrumentation capable of capturing some of the dynamics of persons-to-be-reckoned-with relations and, thus, also capable of producing data which eventually can inform a full description of this interpersonal field component.

Functional relationships among interpersonal field components. This model of the interpersonal field is indebted heavily to the Lewinian notion of a dynamic life space. The full model is to be viewed as dynamic rather than static, allowing for shifts of individuals from one component to another and expansion and shrinkage of any component over time. That is, in any given time period, an individual's persons-to-be-reckoned-with component may contain more relations than does, for example, her support component, and this situation may be quite reversed during a subsequent time period. It is also quite possible that the functioning of any perceived individual may move from one component to another across time. A former support person, for example, may become a highly relevant person-to-be-reckoned-with. Issues related to the nature of the flux of the interpersonal field for specific groups of persons can be best addressed through longitudinal research. It is quite conceivable that the nature of such flux is related intimately to outcome variables that are currently of general importance within psychosocial research (e.g., mental and physical health outcomes).

As Figure 1 illustrates, it is also possible for overlap of functions to occur within any one relationship; an individual may be perceived as providing support and reference functions as well as the gatekeeping function associated with persons-to-be-reckoned-with, or some combination of two of the three. Currently, we hypothesize that the degree of overlap constitutes a measure of intimacy and will investigate this conjecture within our research programs.

The need to conceptualize function overlap has been challenging theoretically, and has resulted in a deepened appreciation for the need to construct a model that is dynamic rather than static as well as the need to produce instrumentation that is capable of being responsive to the functional shifts within any given relationship. In the case of functionally overlapped relationships, temporal emphasis may prove to be of greatest importance. We hypothesize that an overlapped relationship probably emphasizes one of the three component functions of the interpersonal field at any given moment. For example, in a relationship with a spouse that is most often functional within the support category (love, affirmative physical contact, constructive confrontation, physical task assistance, and so forth), it is still conceivable that, at some moments, an individual may perceive the spouse as exclusively a person-to-be-reckoned-with. We further hypothesize that the extent to which intimate relationships contain frequent

functioning within the reference and persons-to-be-reckoned-with components of the interpersonal field (relative to amount of functional support) may be related to important mental and physical health outcome variables.

Use of the Model

Currently, we have designed a program of research that will utilize the interpersonal field model in such a way as to maximize the possibility of identifying person-environment variables that are related to mental health and self-system outcome variables within the United States population of career-committed women. We have defined mental health outcomes in a conventional manner, and thus, are concerned with behavioral, affective, and cognitive indicators of mental health such as alcohol and substance abuse, mental health services utilization, and measures of well-being, life satisfaction, and depression. With reference to self-system outcomes, we are concerned primarily with gender role, self-valuing, career attainment and satisfaction, and personal life attainments and satisfaction.

The major hypothesis that we intend to test states that the functioning of the total interpersonal field produces identifiable patterns of stress and nourishment experiences which are significantly related to mental health and self-system outcome variables within our target population. Thus, within the conceivably large array of variables that co-vary with the functioning of the interpersonal field (such as motivational dynamics, energy levels, decision-making capabilities), we hypothesize that stressors and nourishers and the experience of these phenomena are of great importance relative to our outcome measures of interest.

Stressors are situational demands which seem to an observer to require adaptation or coping. We define nourishers as positive life events which seem to an observer to compel experiences of uplift. Stressors and nourishers are of primary importance for our research questions since there is a sizeable body of literature that has established a relationship between these variables and the mental health outcomes with which we are concerned (e.g., Dohrenwend & Dohrenwend, 1974; Kanner, Kafry, & Pines, 1978). While the relationship between stress and nourishment has not been identified clearly, there is growing evidence that they are not independent phenomena. Gurin, Veroff, and Field (1960) have shown that the absence of stress does not compel the presence of positive affect. Others have shown that positive consequences may accrue to the experience of stressful events (e.g., Antonovsky, 1979; Weiss, Ilgen, & Sharbaugh, 1978). Fulfilling experiences, or nourishment, such as self-expression and joy may emerge from efforts to cope with stress (Selye, 1974). In fact, Kanner, Coyne, Schaefer, and Lazarus (in press) have demonstrated a positive correlation between hassles and uplifts

within a general population sample. Supported by this literature, we do not consider stress and nourishment to be mutually exclusive phenomena; rather, we are interested in the role that the interpersonal field plays in creating patterns of stress and nourishment that can be shown to be related to mental health and self-system outcomes.

Important to our consideration of these issues is a reliance upon the notion that stress and nourishment are experienced phenomena. That is, conventional stressors and nourishers are not experienced as stress and nourishment by each individual. We view the terms stress and nourishment as referring to an experienced stressor and experienced nourisher. This is not, of course, an hypothesis that is original to our research program; there is a wide body of empirical literature which references individual variation in the experience of daily hassles and uplifts. However, for the most part, this body of literature suggests that the major source of variance may be located within the individual (e.g., Campbell, 1981; Rabkin & Struening, 1976). A notable exception to such interpretations can be found in literature where buffering has been used as an explanatory concept when considering the positive aspects of social support as a mitigating agent between stressors and stress experience and mental health outcomes (e.g., Gore, 1980; Liem & Liem, 1978). Such a concept is allied more closely with our own thinking because it recognizes the contribution of person-environment transactions. However, with some exceptions (Eckenrode & Gore, 1981), this literature tends to view stressors as standing apart from social support mechanisms. Indeed, the term "buffer" itself rather clearly connotes the theoretical intention behind this construct. In contrast, we believe that the experiences of stress and nourishment are phenomena which emerge from the interpersonal field (which contains social support functions in addition to references and persons-to-be-reckoned-with relations).

We believe that there are identifiable patterns of interpersonal field functioning that are related to identifiable patterns of stress and nourishment that are in turn related to the mental health and self-system outcomes that have been discussed above. It is not, for example, that the interpersonal field provides a buffer between stressors and the experience of stress (as well as mental health outcomes); rather, both emerge from the field itself. Our transactions with our life spaces produce both our joys and sorrows and we owe a major portion of our "coping" and "adjustment" behaviors to the pattern of functional relations within these interpersonal fields.

Of critical interest to us currently are such issues as the location of stressors within the interpersonal field in relation to the location of nourishers and the location of experienced stressors (stress) and experienced nourishers (nourishment) with respect to each other as well as given outcome variables. It could be that these variables require location within the same

interpersonal field component in order for an individual to successfully achieve with respect to an outcome variable such as well-being. Of course, it is equally possible that, with respect to a given outcome variable, a minimal presence of nourishers across all three components may effectively balance the exacerbated presence of stressors within one of the components.

A reconsideration of Sherif. At this point in the discussion, another look at the Sherif concept of the self-system will assist a fuller understanding of the model as well as the contribution to it of the strength of the Sherif thinking. Her concept of the self-system refers to attitudinal schemata that are formed as a result of full transaction with one's social environment. This latter concept of social environment is not too distant from the life space notion of Lewin, although Lewin emphasizes a life space that may contain phenomenologically important physical objects, while Sherif does not.

The constructs of nourishment and stress represent aspects of the Sherif self-system. These constructs denote phenomena which emerge from the transaction with the life space (or social environment) in that the life space also produces the specific events including stressors and nourishers, within which the person participates. This concept of self-system allows us to release the Lewinian notion of the natural necessity of conflict and intrapersonal tension, especially as we consult the literature which is beginning to suggest that stress and nourishment are not bi-polar anchors of a unidimensional construct, but may even positively co-vary within the life space. At no point does Sherif suggest that attitudinal schemata need be in conflict with one another.

The concept of self-system produces flexibility in the model for it allows the researcher to pull from the interpersonal field those attitudinal schemata which appear empirically and theoretically to be related to the outcomes of interest. In our use of the interpersonal field model, stress and nourishment are those aspects of the self-system that have been shown to be related to those outcomes which presently concern us. Other aspects of the self-system will be appropriate to other research issues, but in all situations where the model is employed, we believe that fruitful inquiry will rely on research questions that concern the patterns of functioning within the interpersonal field that are related to self-system variables which are in turn related to the outcome variables of interest.

It may readily be seen at this point that our interpretation of transactional theory views outcome variables to be that set of variables that can be productively isolated as person-oriented rather than transactionally-oriented. In a path modeling analysis sense, outcome variables constitute the end-point of the path. It is quite useful and theoretically sound to abdicate reliance on person-environment variables at this point; i.e., to relate the transactional model to information that is person-centered (e.g.,

substance abuse tendencies, level of self-valuing).

With a shift of emphasis, we are able to employ the concept of the self-system in such a way as to provide a second set of outcome variables that we view as important for our research interests. In addition to mental health outcomes, we are interested in a set of self-system outcomes that are person-centered (e.g., gender role, self-valuing). While it is possible to view the self-schema variables of stress and nourishment as person-centered, our use of these variables is quite different from our use of self-system outcomes in that we are theoretically linking them to interpersonal field events (stressors and nourishers) and view them as a person-environment transactional product. We view the outcome variables as variables that need to be construed in a conventional person-centered way since a major concern of psychological science involves the explanation of individual psychological dynamics and those who are interested in action research, such as ourselves, are ultimately interested in enhancing the quality of the individual life. In order to accomplish both these goals, at some point, theoretical and empirical processes must arrive at person-centered descriptions.

NOTES

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For our purposes, stress and nourishment refer to negative and positive phenomenological experiences which may result from stressors and nourishers.

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FROM SOCIAL SUPPORT TO SOCIAL NETWORK

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STUDYING SOCIAL SUPPORT IN SOCIAL NETWORKS

Where once analysts tended to treat social support as a single global concept, most now distinguish between types of supportive behaviors, e.g., emotional and material aid. Yet many studies still conceive of "support" itself as a real phenomenon, underlying and unifying all of the various supportive behaviors (see the critiques in Hall & Wellman, 1982, 1984; Wellman, 1981). Moreover, most studies have focused on the consequences of supportive behavior (perceived or unperceived), asking, for example, if social supports buffer the effects of stresses on health (see the reviews in Hammer, 1983; Thoits, 1982).

Our research group has taken a different approach, studying flows of resources among members of social networks. This approach begins with the ties and networks themselves, and leaves open questions as to what sorts of resources flow through these ties and networks and whether such resources are indeed supportive. In this paper, I present our findings about the kinds of supportive resources which are actually flowing through a naturally-occurring sample of ties and networks, and the extent to which different kinds of resources occur together as "strands" of the same ties. This work is part of our research group's longer-term goal of understanding the circumstances under which various sorts of supportive resources flow through ties and networks. In short, we hope to predict to "support" but not from it.

The East York Study

Our data come from interviews conducted in 1977-78 with a

quasi-random subsample of 29 of the 845 randomly-sampled East Yorkers we had first surveyed in 1968 (see Wellman, 1979 for the results of the first survey). East York is a predominantly British-Canadian, working and lower-middle class, residential area of central Toronto. It has a long tradition as a tranquil, cohesive community, insulated from the metropolitan hurly-burly. Although many of the "East Yorkers" had moved from the area by the time of our second study, they clearly had all been residents of a particularly good place to find supportive ties.

We interviewed the East Yorkers about their 337 active community ties: persons outside of their households with whom they were significantly "in touch." Many of the tape-recorded interviews lasted a total of ten to twelve hours over four of five sessions. The time we invested in these interviews has given us much information to complement the statistically more reliable information obtained in the original 1968 survey. For example, we obtained much more information about many more ties in the network: how the persons first met, the circumstances of their jobs and home lives, the extent to which they are companions, and the kinds of things they do for each other. We also obtained what is to our knowledge the first-ever structural descriptions of a sizable sample of personal networks. In addition, a mail questionnaire (Table 1) provided systematic information about which of fifteen types of supportive resources the East Yorkers have ever given to, or received from, each network number.¹

Composition and Structure

While the East Yorkers' eleven (median) or so active ties are only a small fraction of the 1,000 - 2,000 persons they are apt to know (Boissevain, 1974; Pool & Kochen, 1978), they are the relationships which East Yorkers actively think about and maintain (although not necessarily with frequent contact). The networks of these active ties are much more structurally complex than the densely-knit local and kinship solidarities which social support research often uses as a normative criterion. For example, the average East Yorker must deal with three otherwise unconnected pieces of his/her network: one isolate, one dyad, and one larger, internally-connected, component. Moreover, the components themselves are often composed of several clusters--densely-knit internally but only thinly-connected with each other. While only one-third of all ties are directly linked in the median network (density = .33), the components themselves have appreciably higher internal densities (median density = .67), providing significant structural bases for the coordinated provision of resources and social control.

The many links between network members means that East Yorkers must deal with network structures and not just juggle sets of disconnected ties. Most ties are "structurally embedded":

TABLE 1

East Yorker Aid Questionnaire

1. Gave help with small household jobs (such as minor repairs to house, car, cottage; small amount of help with housework)
2. Did other small services (such as driving person to doctor, occasional childcare, errands).
3. Gave help with big household chores (such as major repairs, regular help with housework).
4. Did big service that took a lot of time or effort (such as regular daycare, looking after sick person for a long time).
5. Helped out in dealing with organizations, agencies, the government (such as helping with an application for government benefits).
6. Gave or loaned household items (such as food, tools, washing machine, lawnmower).
7. Gave or loaned small amount of money.
8. Gave or loaned money for a mortgage, down payment, large home improvement.
9. Gave or loaned large amount of money (but not for a mortgage or down payment).
10. Gave advice about getting along with family members (such as marriage problems, raising children).
11. Gave other emotional support during routine or minor upset.
12. Gave emotional support during major crisis or long-lasting problem.
13. Gave information about possible job openings, promotions for other person.
14. Made important job contact for other person (such as telling an employer about him/her).
15. Gave information about house or apartment for rent or sale

Each item asked separately for each network member: "You to (Name)" and "(Name) to you." Each item circled "Yes" or "No."

conditioned by the two persons' mutual links with other network members. Indeed, most interactions themselves are not private affairs but take place in the presence of others. They are private only in that they often take place in homes, summer cottages or on the telephone--physical contexts which do not facilitate the incorporation of strangers and casual acquaintances into the interactions.

The networks usually contain a variety of ties. For example, only a minority of ties in most networks are with kin; most are with friends, neighbors and coworkers. Less than half of the ties are with socially-close intimates; these tend to be ties with immediate kin and friends. Most of the ties extend well beyond the neighborhood but within the metropolitan area. The median network member lives 10 miles away. While the median network member is in contact with the East Yorker somewhat less than once per week--using in-person contact more than telephone contact to keep in touch--the frequency of such contact ranges from several times a day (for some neighbors and coworkers) to less than once per year (for some physically-distant relatives). While most East Yorkers are in contact with three or more network members per day, a few socially-active ones are in contact with more than seven network members per day and a few isolates are only in contact with one or two in their average day. Such variation in the composition and structure of these ties and networks strongly suggests marked variation in the type, extent and breadth of social support available through them.

TAKING SUPPORT APART...

Commonly-Available Resources

Is "support" a generalized resource, ranging over a broad spectrum of interpersonal aid, or are there marked differences in the types of resources which ties and networks deliver to the East Yorkers? The data show that in the broadest possible sense, almost all of the East Yorkers' active ties are supportive: 95% convey some sort of companionship aid in one direction or the other. The few totally nonsupportive ties are usually to persons with whom the East Yorkers interact only because they are juxtaposed in the same social contexts as kin, coworkers or neighbors. (Note however, that 28% of the ties only provide either companionship or aid, but not both.) The prevalence of some sort of supportive resources in these ties and networks helps to debunk even further the old scholars' and politicians' tale of contemporary interpersonal isolation (see also Fischer, 1982; Wellman & Leighton, 1979).

Nine strands dominate the contents of these networks out of the nineteen strands of companionship and aid we have studied (see Table 2). Each of these nine strands is present in at least

TABLE 2

Percentage of Ties and Networks Providing Specific Strands of Support

	Percent of all Ties Sending a Specific Strand:		Percent of Networks in which East Yorker Receives Strand From:		
Strands	To EY'r Only	To & From EY'r	From EY'r Only	At least 1 Tie	50% + of Ties
<u>Common Strands</u>					
Sociability	a	71	a	100	90
Doing Things Together	a	62	a	93	45
Discussing Things	10	53	a	90	48
Minor Emotional Aid	10	41	7	83	45
Family Advice	10	29	10	76	28
Major Emotional Aid	10	25	8	69	31
Minor Services	15	32	7	83	38
Minor Household Aid	13	26	9	90	31
Lending House Items	7	32	5	83	45
<u>Uncommon Strands</u>					
Formal Group Act.	a	19	a	48	7
Major Household Aid	6	9	4	55	3
Major Services	4	4	3	45	0
Small \$	8	8	5	62	10
Big \$ (non housing)	2	0	4	28	0
Housing \$	1	0	3	28	0
Organizational Aid	7	6	4	38	3
Job Opening Info	5	1	5	31	3
Job Contacts	3	1	4	28	0
Housing Search Aid	5	1	3	34	3
Sample Size		336 Ties		29 networks	

^a"Companionship" variables assumed symmetric, to and from East Yorkers and network members.

one-third of the ties and three-quarters of the networks. Together these nine strands comprise 82% of all the different strands of support actually going from network members to East Yorkers. They are the kinds of resources which most East Yorkers can reasonably expect to get from many of their active ties. (For

convenience, I report data in the remainder of this paper for one direction only: the strands of aid which East Yorkers report receiving from each network member. However as the first three columns of Table 2 show, these data are quite symmetrical).

Almost by definition, the most common type of companionship is sheer sociability--the pleasure network members get from each other's company. Indeed, it underlies the other, more focused types of companionship: discussing things, sharing informal activities, mutual group participation. These shared activities--from discussing the Toronto Maple Leafs to going away on joint vacations--often are important foci (Feld, 1981) for ties and networks, bringing members together to reaffirm, readjust and sustain their ties.

Most ties in most networks provide some sort of emotional aid. This aid is usually minor, such as being a good listener during routine upsets or giving short-term advice about family problems with spouses or children. Moreover, most ties in most networks also provide some sort of small services as part of their relationship. These small services consist of providing "minor services" (e.g., driving a person to the doctor; occasional child care), helping with small household jobs (e.g., repairs to the house or car) or lending/giving household items (e.g., cups of sugar, lawnmowers). Every East Yorker is involved in at least one relationship where one or both parties provide small services for the other. For example, East Yorkers get such minor services from an average of four active ties.

Thus companionship, emotional aid and small services form the continuing basis of East Yorkers' ties and networks. They provide what East Yorkers routinely expect to get from their ties--in both daily life and in crises. Employed men, in particular, send and receive small services, as their social circles use job-related skills to help each other out. Women--either as homemakers only or in paid employment (as well as homemaking) have especially high levels of companionship and emotional aid: they are the ones who supply the time and affect to keep going most ties with kin, friends and neighbors. Yet women, too, exchange significant amounts of small services, as they work together in child care and husband care (see Luxton, 1980; Rosental, Marshall & Synge, 1981; Stack, 1974; Wellman, 1985; Wellman, Carrington & Hall, in press).

Uncommonly-Available Resources

The nine other strands are each present in less than one-fifth of the ties. Yet if the great majority of ties do not provide these sorts of resources, many networks do. Thus most East Yorkers get "major household aid" (e.g., major home repairs, regular housework) from at least one network member.

While a sizable minority of network members have lent small amounts to the East Yorkers, only a small number have lent (or given) large sums of money either to buy a home or for other

purposes. These are usually intergenerational transfers from parents or loans from siblings.

Only a minority of the East Yorkers's active ties have provided them with information about new jobs or housing. The data thus support Granovetter's (1973, 1982) argument that if such job searches are to succeed, the searcher must use a very large number of ties. Hence, even though socially-close, intimate ties are more likely to provide help on a per tie basis, the rarity of the information means that the much more numerous and socially heterogeneous, non-intimate ties are more likely to provide useful information on a per network basis.

While the commonly-available strands of ties call for mutually gratifying companionship or the provision of nontangible emotional aid or non-onerous small services, the uncommonly-available strands call for time and effort, the transfer of material wealth, or the provision of specialized information. These are resources which network members appear less likely to possess or to transfer to others. But even when such strands are available, they may not be conveyed. In some cases, East Yorkers do not want the specialized resources these strands deliver, e.g., they may not be looking for a new job. In many other cases, East Yorkers prefer to purchase such resources on the open market or get them through formal organizations. At times, East Yorkers do not even think to ask (some or all) network members for certain resources, as they do not even consider getting such support to be part of their companionate relationship. Thus a number of East Yorkers got their mortgage money from banks rather than incurring reciprocal obligations with network members.

Support clearly is a complex, multidimensional array of specific resource transfers and not a single unidimensional multipurpose tie. There are nineteen widely-available types of support, offering varieties of companionship, emotional aid and small services. In addition, while other types of support are less common they still provide resources which are important to the individual and the ongoing reproduction of the social system. To rule them out is to accept unconsciously the manner in which industrialization, capitalism, bureaucratization and urbanization have constricted the range of aid which network members provide to each other. Moreover, even, if the number of network members who give such aid is small, the availability of such aid from somewhere in the network is often crucial to the East Yorkers' well-being. Considered as a property of ties, these uncommon forms of aid are indeed rare. But considered as a property of the network—often the more appropriate unit of analysis, many East Yorkers have received aid that has been crucial in seeing them through major illnesses or getting them good jobs.

AND PUTTING IT BACK TOGETHER AGAIN

The nineteen categories of "companionship" and "support" are only a starting point and not a set typology. Although a global support construct clearly is not tenable, there is no reason to assume that all nineteen types of supportive behavior are in fact independent of one another. Indeed, many strands are significantly correlated (see Table 3).

I used variable cluster analysis to find out which strands tend to be packaged together in the same ties. The procedure uses a correlation matrix to divide a set of numeric variables into unidimensional clusters in a way that maximizes the sum of the original variables' variance across clusters that is explained by the cluster components. Unlike factor analysis, the variable cluster procedure produces hierarchical solutions, providing rich information about relationships between clusters. Moreover, variable cluster analysis is only concerned with producing packages of associated variables; it does not search for underlying factors.

A five dimensional solution provides the best balance between increasing explained variance and minimizing single-variable clusters (Table 4):²

- INFORMATION (job information, job contacts, housing search);
- COMPANIONSHIP (sociability, discussing things, doing things together, formal group activities);
- EMOTIONAL AID (minor emotional aid, family problems advice, major emotional aid, major services);
- SERVICES (minor services, lend/give household items, minor household aid, major household aid, organizational aid);
- FINANCIAL AID (small \$, big \$, housing \$);

The cluster structure shows a distinct hierarchy (Figure 1). Each division stage splits off a single dimension containing three to five specific kinds of support. The clustering reveals distinct packages of support coming from different persons in the East Yorkers' networks. While at the highest level of generality (before any clustering) almost all ties provide "support," focusing in more closely first reveals that the provision of information comes from ties which do not provide much of any other type of support. Subsequent clustering reveals that the support strands are usually packaged in substantively similar areas. Note that the strands do not cluster according to other criteria such as whether they provide large or small quantities of aid.

The clusters show that the more diffuse kinds of support--information and companionship--often come from different persons than those kinds of support which give directly targeted aid--emotional aid, services and financial aid. While almost all the emotional aid and services variables are significantly correlated with each other (as is "small \$"), there are some differences between those supporters who give emotional aid and those who give material aid in the form of money services.

TABLE 4

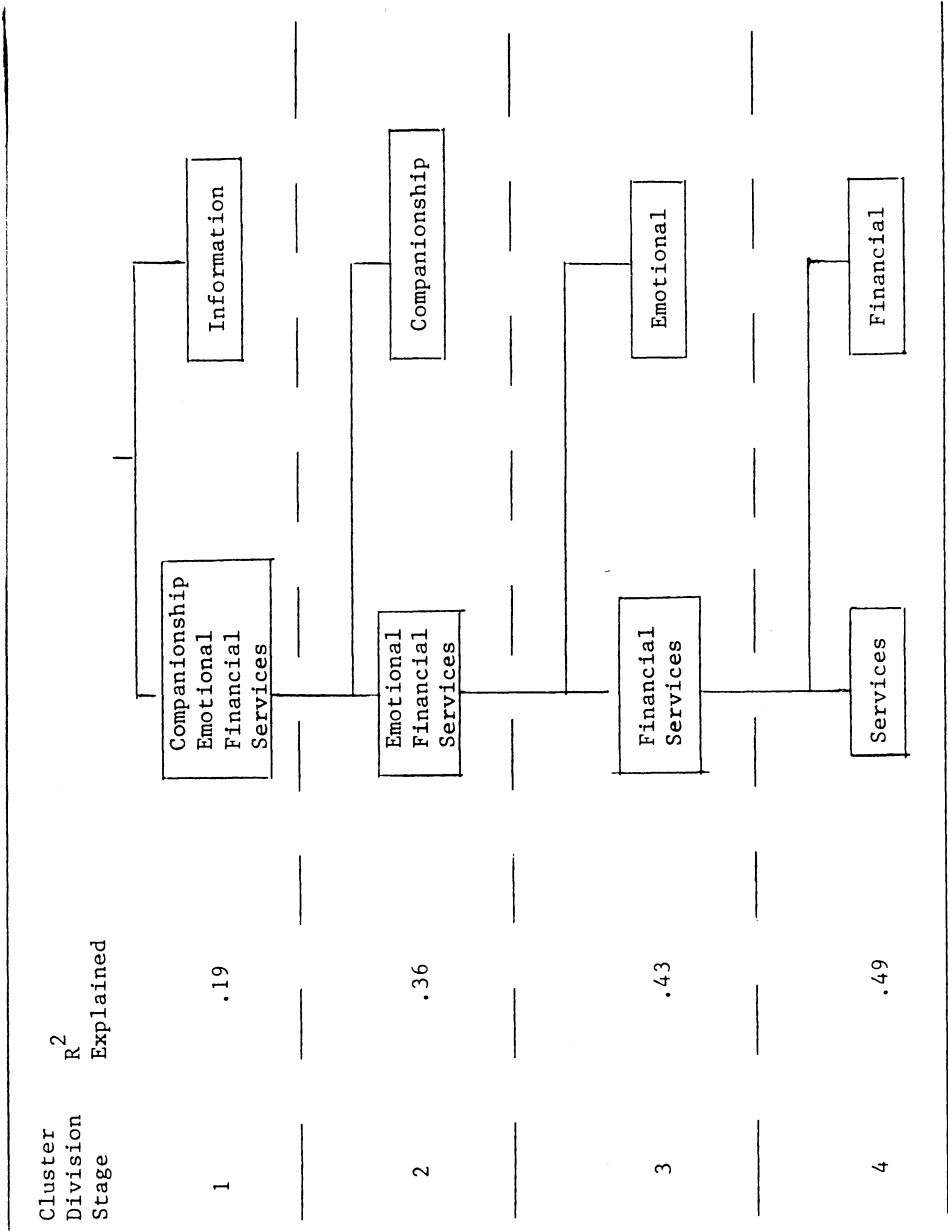
Clustering of Strands in Network Members' Ties

Cluster	Variable	R with own Cluster	R with next Highest Cluster
1	<u>Information</u>		
	Job Opening Info	.78	.02
	Job Contacts	.80	.01
	Housing Search Aids	.13	.02
2	<u>Services</u>		
	Minor Household Aid	.64	.07
	Minor Services	.45	.06
	Major Household Aid	.42	.13
	Organizational Aid	.20	.04
	Household Items	.50	.14
3	<u>Companionship</u>		
	Doing Things Together	.45	.03
	Discussing Things	.70	.01
	Sociability	.39	.02
	Formal Group Activities	.24	.01
4	<u>Emotional</u>		
	Major Services	.25	.09
	Family Advice	.65	.12
	Minor Emotional Support	.66	.06
	Major Emotional Support	.53	.12
5	<u>Financial</u>		
	Small \$	.60	.16
	Big \$	.58	.04
	Housing \$	.39	.02

Total Variation Explained = 9.38

Proportion of Variation Explained = 0.49

FIGURE 1. Cluster Structure of Support.



Moreover, major services come more from the persons who give emotional aid than from the persons who give other forms of services. Such major services are usually given to distressed persons whose families (or themselves) are suffering from acute or chronic physical or mental illness.

Companionship, emotional aid and services are the core of these networks, with at least one strand from these three dimensions of support present in most ties (Table 5). Indeed, almost all of the East Yorkers have received some form of companionship, emotional aid and services from at least one network member, and many East Yorkers can count on getting support

TABLE 5

Percentage of Ties and Networks Providing Specific Dimensions of Support

Dimensions	Percentage of All Ties Sending a Specific Dimension			Percentage of Networks in which East Yorker Receives Strand From	
	To EY'r Only	To & From EY'r	From EY'r Only	At Least 1 Tie	50% + of Ties
Information	7	3	6	45	0
Services	13	51	10	93	65
Companionship	a	78	a	100	94
Emotional Aid	13	49	12	90	52
Financial Aid	10	8	8	69	10

^a"Companionship" variables assumed symmetric to and from East Yorkers and network members

TABLE 6

Correlations Between Five Dimensions of Support

Information	1.00				
Services	.13*	1.00			
Companionship	.10	.18	1.00		
Emotional Aid	.10	.33*	.09	1.00	
Financial Aid	.13*	.27*	.05	.21*	1.00
	Info.	Serv.	Comp.	Emot.	\$

* Significant at < .05 level.

in these dimensions from at least half of the members of their networks (Table 5). In addition, a significant minority of East Yorkers have received financial aid or specialized job/housing information. Although more narrowly distributed, such resources have made significant differences to many East Yorkers' lives (see Wellman, Carrington, & Hall, in press for details).

Multistrandedness and Specialization

While the five dimensions are quite distinct, they are often significantly correlated with each other (Table 6). This suggests that the ties are not strictly specialized in the kinds of resources they carry. Yet few ties carry a wide variety of resources. Most provide three or less specific types and broad dimensions of aid. The most common packages are one or two strands of companionship intertwined with one or two strands of services and emotional aid.

It is apparent that companionship, services and emotional aid are the resources which East Yorkers want and get most frequently from their network members. Although they could purchase these services on the open market, it is unlikely they could buy the same combination of frequent, flexible, nuanced, low cost support that network members often supply. Hence the East Yorkers are more willing to invest the time and effort necessary to maintain their ties--and support--then they are to spend the money to purchase equivalent services.

Yet the specialized nature of most ties means that the East Yorkers must maintain a diversified portfolio of relationships. In order to have access to a wide range of resources, they must search among a number of ties. Rather than broadly-supportive "general stores" of aid, East Yorkers' ties are more specialized "boutiques."

The Variety of Support in Ties and Networks

The evidence suggests that the social networks from which resources come are much more than support systems: there is much variation in the frequency with which different kinds of resources occur in ties and networks, most ties convey only a few kinds of resources, and the same kinds of resources are often packaged together in the same specialized ties. Moreover, many resources only flow in one direction (see Table 2 above). Such unreciprocated aid usually involves the large-scale expenditure of money (e.g., mortgage loans), time and effort (e.g., major services) or specialized access to information (e.g., job contacts).

This picture of diverse resource flows fits quite well with what we have discovered about the variegated composition and lumpy structure of the East Yorkers' ties and networks (Wellman, Carrington & Hall, in press). The networks mix local,

metropolitan and long-distance ties; kin, friends, neighbors and coworkers; regularly seen acquaintances and rarely seen intimates; densely-knit clusters set in the midst of more sparsely-knit ramifying ties. Although the networks' unevenness and lumpiness makes for low communal solidarity and problematic access to supportive resource, the networks' heterogeneity, multiple parts and uneven internal densities provide useful structural bases for dealing with routine and extraordinary pressures, opportunities and contingencies. While the densely-knit clusters within these networks provide a solid structural basis for cooperative activity--with structurally embedded cluster members interacting whether they want to or not, the diverse, often asymmetric, ties--organized in multiple clusters and dyadic relationships--give the East Yorkers direct and indirect access to the wider range of resources available in other social circles.

On a day to day basis, the help comes mainly in the form of "band-aids": emotional aid, small services and companionship to help the East Yorkers deal with the stresses and strains of their current structural locations in large and small-scale divisions of labor. Yet over time, many of the networks provide "ladders": services, money and material goods to help the East Yorkers change their present situations (jobs, houses, spouses). Moreover, some networks give East Yorkers "levers" to change the world: several use their ties to fight for animal welfare, against the effects of food additives on hyperactive children, and for better local schools. Hence these band-aids, ladders and levers are as much structural social facts as they are individual psychological phenomena.

TOWARDS A NETWORK ANALYSIS OF SOCIAL SUPPORT

Reaching the Limits of the "Social Support" Approach

While the "social support" approach has usefully unlocked analytic attention from its fixation on doctors, hospitals and households, its inherent limitations are now seriously constraining theory and research:

--Even though analysts now recognize that "support" may be a variety of things--and not a unidimensional construct--most look only at the consequences of such "supports." By reducing ties down to their supportive elements, this approach underplays such other important elements of the relationships as information flows, normative constraints, network stresses and strains, and the ways in which ties link persons with larger social structure. It turns researchers away from investigating the circumstances under which different kinds of supportive resources may--or may not--flow through ties. It hinders them from evaluating how capitalism, industrialization, urbanization, bureaucratization and technological change have fostered the development of specialized

ties and networks.

--Treating all ties as supportive implicitly homogenizes the ties into equal-status "liking" relationships voluntarily chosen as a result of interpersonal attraction. Yet many ties are neither equal-status, "liking," nor voluntarily initiated. Network members often have little say about some of those with whom they must maintain ties. Moreover, differential access to resources makes many ties asymmetric and systems hierarchical. Indeed, patron-clientage may be as important a principle of social bonding as liking.

--Despite the often-used "support systems" rubric, analysts have tended to study disconnected sets of ties. They have rarely gone beyond these dyads to study how the structure of these can affect the flow of resources in a system.

--Treating relations of support as a system in its own right unnaturally connects such relations from the large-scale divisions of labor in which they are embedded. While all research involves some analytic isolation of domains, to isolate interpersonal resources from the larger social systems which are the sources of such resources is to confuse the local manifestation for the larger process. The supportive resources people want, have available, and transfer are intrinsically a part of their location in large (and small) scale divisions of labor.

The "Social Networks" Approach

Our research group has tried to overcome many of these limitations by studying social networks and not social support. Social network analysis starts with the simple, but powerful, notion that the primary business of sociologists is to study how the structure of social systems allocates resources. While this focus on social structure may seem obvious, it deemphasizes analyses of why people act and emphasizes the study of structural constraints on their actions. Moreover, it shifts attention away from seeing the world as composed of egalitarian, voluntarily chosen, dyadic ties and towards seeing it as composed of asymmetric ties, with varying contents, bound up on hierarchical structures (Berkowitz, 1982; Wellman, 1983).

Network analysts believe that the most direct way to study a social structure is to analyze the patterns of ties linking its members. They try to describe these patterns and use their descriptions to learn how network structures constrain social behavior. Their descriptions are based on the social network concept of ties linking nodes in a social system--ties that connect such nodes as persons, groups, organizations, and clusters of ties. They treat social systems--large and small--as networks of dependency relationships resulting from the differential possession of scarce resources at the nodes and the structured allocation of these resources at the ties. Some analyses, such as our own, record multiple types of ties between individuals in

order to study the complex ways in which these multistranded ties link specific members of a social system. Other analyses focus on a few types in order to study their overall pattern in a social system.

The network analytic approach has substantial advantages over the social support approach in the study of the implications of interpersonal relations for health. First, by defining the contents of ties as sets of "resources," it leaves open the question of which resources are being conveyed instead of assuming that all convey a unidimensional thing called "support." Second, by always seeing ties within the context of networks, it avoids interpreting behavior solely on the basis of dyadic characteristics. Third, by allowing the study of flows of resources in both directions, it facilitates the study of asymmetry and reciprocity. Fourth, by accepting that network members often have access to different kinds and quantities of resources, it handles unequal relationships routinely. Fifth, by developing concepts and techniques for studying how such unequal relations link together, it fosters the study of cumulatively hierarchical--and sometimes stratified--social systems. Sixth, by studying links between organizations, institutions and clusters of persons--i.e., by studying "networks of networks" as well as networks of individual persons--it can accommodate large-scale structural phenomena in the same analytic framework as interpersonal relations.

Network analysis has developed rapidly since the 1960s. Where its early practitioners tended to see "the social network" as a metaphor for describing webs of relationships--especially those cutting social group and category boundaries--network analysts have now developed large sets of concepts, methods and findings. Indeed, many network analysts are now working to develop a comprehensive paradigm for the interpretation of all social phenomena in terms of structured patterns of relationships. In such work, network analysis is at the center of the ways in which researchers pose questions, organize data collection, develop analytic methods and interpret substantive findings. It is time that studies of support go beyond their current metaphorical allusions to social networks and make fuller, more powerful use of the social network approach.

NOTES

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¹Wellman, 1982; Wellman, Carrington and Hall, in press provides a more detailed discussion of method, and of the composition, structure and contents of these networks. While these papers discuss the networks of all 33 interviewed respondents, the present paper reports on the 29 interviewed respondents who completed the follow-up "support" questionnaire.

²I used the Statistical Analysis System's (SAS) VARCLUS procedure. The variable clustering began with a randomized input of variables and used the SEED option to avoid an anomalous local optimum which misassigned Formal Group Activities. Solutions with more than five dimensions only detached single variables from clusters.

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PART III

LONELINESS AND PERCEIVED SUPPORT

THE PSYCHOLOGY OF LONELINESS: SOME PERSONALITY ISSUES IN THE STUDY OF SOCIAL SUPPORT

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With certain exceptions, the literature on social support has not emphasized personality issues, focusing instead on the availability of support networks, and the ways in which they contribute to health and well-being. Similarly, little attention has been paid to those factors that might influence the probability of either developing or accessing a supportive social network. Research on relationships and interpersonal behavior more broadly conceived, however, suggests that individuals vary considerably in their ability to achieve interpersonal goals as well as in the skill and confidence necessary to interact with others effectively. The purpose of the present paper then is to consider the role of personality and individual differences in the experience of loneliness and by implication in the process of social support.

The literature on the psychology of loneliness is a logical point of departure in part because it shares with the concept of social support the assumption that intimate and social relationships are essential and fundamental to human functioning. In fact, at this level of abstraction, the criticality of relationships becomes an implicit theme linking together an amazing array of ideas and data about experience. In an important way these approaches also diverge, however. The social support literature considers relevant issues from the outside inward--from the social network toward the experiencing person--finding vulnerability to stress when support is not forthcoming. Research on loneliness begins with the subjective impressions of the lonely person and then looks outward for explanations of the discomfort. Consequently, these two approaches view the social exchanges between an individual and his or her social environment from different perspectives creating the potential for

cross-fertilization.

THE PSYCHOLOGY OF LONELINESS

By definition, loneliness occurs when one's network of casual and intimate relationships is either smaller or less satisfying than one desires (cf. Peplau & Perlman, 1982). Whereas it is a common experience--26% of the respondents in one survey reported having felt lonely during the past two weeks--it is also clearly unpleasant. It is a frequent complaint of persons seeking psychotherapy, it has been implicated in a variety of social and personal problems and it is possibly related to the appearance of medical symptomatology. Whereas, loneliness does not appear to be related to gender, provided that competently developed psychometric instruments are used, it is related to age with the highest rates of loneliness occurring among adolescents and young adults, declining thereafter except among certain "at risk" populations such as divorced and widowed persons. There is a wealth of evidence indicating that measures of loneliness do, in fact, predict various features of relationships and interpersonal behavior (e.g., amount of time spent alone, marital status, dating frequency, satisfaction with relationships, etc.), and further evidence that loneliness is an empirically distinct construct. Thus, from several perspectives including psychometric, empirical and practical, it would appear that the concept of loneliness affords a useful, parsimonious and valid way of conceptualizing and measuring dissatisfaction with one's intimate and social relationships.

Research on loneliness has also begun to address important conceptual questions and to provide empirical bridges to related constructs and research areas. For example, Anne Peplau and her colleagues at UCLA not only stimulated much of the current interest in this area, but also demonstrated that causal attributions about loneliness function similarly to achievement-related cognitive processes. They found that loneliness is associated with the use of internal and stable attributions even when transient external factors would appear to be salient (Peplau, Russell, & Heim, 1979). Another innovative direction has been undertaken by Karen Rook (in press) who has compared measures of loneliness and social support directly and has identified previously neglected disruptive and problematic aspects of social networks.

Personality and Loneliness

The research on loneliness that we have conducted at the University of Tulsa has included many features, but began with the loosely conceived question of how does anyone remain lonely given the unpleasantness of the condition and the availability of a

potential social support network. This line of reasoning occurred to us because of the finding that college students are a particularly lonely group and yet college students, in the abstract at least, have a readily available pool of potential friends, lovers and companions. One of our initial assumptions was that certain personality factors were involved as had been proposed earlier by Weiss (1973). Along the way, several of our studies and those of others pointed to the utility of assessing personality in the study of loneliness.

To begin with, lonely persons consistently describe themselves differently than do not lonely persons and specifically in ways that imply at least temporary if not more stable individual differences. For example, loneliness has been found to be reliably associated with a variety of personality and attitude measures. These measures may be categorized into four groups of variables as follows: (1) inadequate social skill (e.g., shyness, introversion, low assertiveness, etc.; Horowitz & French, 1979; Jones, Freeman & Goswick, 1981); (2) emotional arousal and conflict (e.g., depression, anxiety, neurosis, etc; Russell, Peplau & Ferguson, 1978; Jones, et al., 1981); (3) poor self-regard (e.g., low self-esteem, poor social self-concept; e.g., Goswick & Jones, 1981a); and (4) negativistic attitudes (e.g., hostility, externality, pessimistic attitudes toward people, life and society; Jones, et al., 1981).

Whereas such comparisons between self-report instruments are hardly conclusive, it is important to note that these findings are quite stable and have been replicated across samples varying in age, ethnic identification, and culture. For example, we recently completed a survey involving comparisons of loneliness and other measures between college students in Puerto Rico and Oklahoma (Jones, Carpenter & Quintana, 1982). Our major findings may be summarized as follows: The Puerto Rico sample reported significantly greater loneliness; the two samples differed significantly on most of the personality variables that we assessed; these differences could not be attributed to differences in social desirability; and yet the pattern of personality variables that predicted loneliness was virtually identical across samples both in terms of the order as well as the magnitude of statistical relationships. It is, of course, commonplace to dismiss such data as mere self-reports, that is, as an imperfect and less desirable way of assessing behavior. By contrast, these data may also be viewed as self-presentations which, while not necessarily veridical in the behavioral sense, are nevertheless indications of how respondents think about themselves and how they want to be regarded by others.

Another clue that certain personality factors are ingredients in the recipe for loneliness is the common finding that loneliness is more strongly related to qualitative as opposed to quantitative indices of relationships. For example, Cutrona (1982) reported that, among college students, measures of satisfaction with one's

friendship, romantic and family relationships were better predictors of loneliness initially and again seven months later than were variables such as distance from home, frequency of contact, and so on. We have replicated this basic finding many times, most recently in a study of loneliness, social support and health habits among a group of chronic diabetics (Carpenter, Hansson, Roundtree & Jones, in press). These findings suggest that perhaps above some minimal level, the absolute amount of social contact may not be the most important feature in exchanges with one's social network, at least as far as loneliness is concerned. Instead, the type of contact would appear to be far more important as well as the degree to which one's social network lives up to one's expectations and needs. Thus, whereas we can conceptualize social networks existing as demographic realities, it is nevertheless important to recognize that how a person feels, thinks and acts with respect to that network is a subjective and interpretive process and, at least in part, subject to the influence of the kind of person who does the thinking, feeling and acting.

Furthermore, the subjective nature of loneliness may also be seen in studies that we and others have conducted in which college students were asked to keep records of their naturally occurring social interactions (e.g., Jones, 1981; Wheeler, Reis, & Nezlek, 1983). Whereas these studies indicated that loneliness was correlated with certain features of participants' interpersonal environments (e.g., the gender and intimacy of social contacts) they also indicated that lonely college students do not lack for social opportunities per se. On the contrary, loneliness was not related to the number nor the length of interactions, and in fact, lonely participants reported a greater diversity of social contacts. However, research also indicates that lonely college students believe that they have less contact with others than do not lonely persons (cf. Russell, et al, 1978). This does not prove that stable dispositional factors cause loneliness, but again, it clearly locates at least a portion of the experience of loneliness within the interpretive framework of the experiencing person rather than exclusively in the lonely person's social environment.

From another line of investigation, we know that lonely persons react to and perceive initial group and dyadic interactions differently than do persons who are not lonely. For example, in a series of studies examining dyadic and group conversations among strangers, we found that lonely as compared to not lonely participants: (1) rated their own performance more negatively on several interpersonal dimensions; (2) expected more negative ratings from others; (3) rated others more negatively, (this was particularly true for lonely men); (4) were less attracted to others; (5) expressed less interest in seeing their partners again; and (6) were less accurate judges of their partner's self-evaluations (Jones, 1982; Jones et al., 1981;

Jones, Sansone & Helm, 1983). It is, of course, possible that these results reflect the current emotional state of lonely students and nothing more. Alternatively, it is possible that the personality correlates described previously underlie not only loneliness but these patterns of interpersonal behavior as well.

In addition, several behavioral and laboratory studies suggest that lonely persons act differently in interpersonal situations and in ways that would appear to reduce the likelihood of restoring mutually satisfying relationships. For example, in one study we examined actual verbal behavior in heterosexual dyadic conversations between strangers (Jones, Hobbs & Hockenbury, 1982). We found that high as compared to low lonely participants were more self-focused and less responsive toward their partners: They talked relatively more about themselves; changed the topic more frequently; and asked fewer questions of their partners. Excessive self-focus, in turn, would seem to have implications for the lonely person's interpersonal behavior: (1) lonely persons may be attending to their own discomfort which is likely to make their discomfort worse; (2) lonely persons may fail to perceive the social overtures and social reinforcements tendered by others; and (3) lonely persons may be less rewarding or attractive as interaction partners. If so, it is not surprising that, once initiated, loneliness tends to persist. Further research has indicated that lonely as compared to not lonely participants were less accurate encoders of non-verbal communications, used anormative and less intimate self-disclosures, were lower at social risk-taking, and yielded more extreme responses to social influence attempts (Gerson & Perlman, 1979; Hansson & Jones, 1981; Moore & Sermat, 1974; Solano, Batten & Parish, 1982).

Taken together, these studies seem to show that lonely persons are apprehensive, negativistic and ineffective in their interactions with others, in short, that they lack social skills and poise. In our previous work we have emphasized the implications of these findings for the probability of restoring mutually satisfying relationships among lonely people. In addition, these results would seem to suggest that certain persons may be particularly vulnerable to loneliness by virtue of inadequate social facility and/or heightened anxiety. But a problem of interpretation exists. Even though some of the studies outlined used behavioral measures and laboratory settings, most have nevertheless been correlational in design. Consequently, it is not possible to determine if personality predispositions result in loneliness, or if once people become lonely they begin to doubt their interpersonal abilities and act accordingly.

Three recent lines of investigation provide preliminary solutions to this stalemate, however. First, longitudinal data is now available suggesting that shyness, an important individual difference variable in this regard, is a precursor rather than a consequence of loneliness among beginning college students (Goswick & Jones, 1981b). Second, experimentally induced changes

in social skill (and to that extent changes in personality) have been found to consistently result in significant reductions in self-reported loneliness suggesting a causal link between the two phenomena (cf. Jones, Hansson & Cutrona, in press).

Third, for the past few years we have been investigating external situations and events that lead to loneliness. We call our approach relational stress because it involves the assessment of life events that are associated with loneliness such as emotional conflicts with significant others, being left-out of social activities, break-ups, separation, and so on (Jones, Cavert, Snider & Bruce, in press). We have found that the situations and events that disrupt relationships and result in loneliness are multidimensional and may be characterized as unpleasant interpersonal events that are difficult to predict or control. Relational stress results from events that tend to last a long period of time when they occur, but that are also relatively infrequent. High levels of relational stress appear to result in unpleasant cognitive and emotional states such as anxiety and self-recrimination, whereas the absence of relational stress does not necessarily result in pleasant experiences and self-assurance. With regard to the work of Peplau on attributional factors in loneliness, we have found that increases in relational stress are associated with increasingly strong feelings of dissimilarity from others even in the context of situations and events that would likely make anyone feel lonely.

Of relevance to the present discussion, however, are studies in which we have crossed relational stress with either chronic loneliness or some relevant personality variable such as shyness to predict previously reported correlates of loneliness, for example, depression, avoidance of others, and so on. Our purpose was to determine the relative proportion of variance in such correlates accounted for by external, situational factors versus dispositional variables. Basically, our analyses have indicated that some loneliness related reactions (e.g., depression, anxiety, etc.) were related exclusively to relational stress, that is, to the unpleasant and disruptive interpersonal experiences that the respondent reported having recently undergone. By contrast, other correlates of loneliness (e.g., poor self-regard, avoidance of others) were predictable only from the perspective of the personality of the respondent, independent of relational stress.

In addition, we found that relational stress scores were relatively stable over time, and that at least some measures of personality assessed at one point in time were significant predictors of subsequent relational stress assessed two months later. Thus, as has been suggested by Gormley (1983) and others, one advantage of a personality perspective may be the identification of the kinds of situations that certain people habitually choose or stumble into. The main point here, however, is that at least some of the experiences associated with loneliness are related exclusively to dispositional

characteristics, that is, one's personality or the fact of having been lonely for some period of time.

To summarize, research on loneliness implicates personality and individual differences factors in its origins, development and prognosis and this may be seen in the relative discrepancy between the lonely person's perceptions and actions with respect to his or her social network versus the characteristics of the network more objectively assessed, as well as in studies demonstrating either 1) a temporal sequencing of personality and loneliness, or 2) the independence of loneliness correlates with respect to personality and situational predictors.

It is not suggested that external factors are never decisive nor important in the occurrence of loneliness. On the contrary, we have systematically demonstrated the influence of external situations and events on the experience of loneliness. Even these studies do not rule out the role of dispositional factors and internal processes, however. Instead, they suggest that the connection between external and objectively defined conditions and loneliness is filtered through considerable psychological processing. Thus, the argument in a nutshell is that properties of the person preexistent to a given interaction (e.g., attitudes, temperament, expectations, memories for previous experiences of relevance, fears, etc.) determine, in part, the style, tone, and outcome of the interaction. Over time, and in concert with other factors, personality variables would be expected to influence the breadth and quality of relationships (i.e., the social network), and hence both the probability and reality of loneliness.

Critique of the Literature

On the other hand, research on loneliness raises many unanswered questions and there are important limitations to its generalizability and interpretation. For example, we have only recently begun to make procedural distinctions on the basis of dimensions of obvious relevance to loneliness such as state vs. trait loneliness, loneliness due to immediate circumstances vs. chronic dispositional loneliness, or loneliness imbedded in the context of classical forms of psychopathology vs. loneliness that is more circumscribed. A very serious limitation is that most investigations examining loneliness and interpersonal behavior, including our own, have involved interactions among strangers in artificial situations. Whereas relating to strangers and the initial stages in the acquaintanceship process are of interest and importance in their own right, such studies do not begin to capture the breadth and complexity of a person's interactions, commitments, and feelings with respect to his or her social network. The potential differences between interactions among strangers and interactions that unfold within the context of on-going and intimate relationships is a case in point, as are variations rooted in differing social roles, for example,

parent-child, employee-supervisor, romantic partners, family members, and the like.

A related point is that our research on loneliness has proceeded in a narrow and construct-centered manner, more or less independent of broad theoretical considerations. Loneliness research has produced conceptual accounts of part-processes or mini-theories rather than the big picture of loneliness as a feature of the terrain of human nature. In other words, loneliness research has been decontextualized from the biological, historical, social and cultural context in which human behavior and experience have evolved. Some of these problems can be addressed by moderate shifts in focus, methodological refinements, and of course, additional research. The greatest need, however, is for greater theoretical development with regard to the role of personality in loneliness and, for that matter, in inadequate relationships and social support generally.

PERSONALITY AND INADEQUATE RELATIONSHIPS

Conceptually, this issue would appear to reduce to two basic questions: (1) what are the personality characteristics involved in inadequate relationships; and (2) how do they influence one's relationships and hence the occurrence of loneliness, the lack of social support, and so on. Sufficient data are now available with which to sketch an outline of the structure of relevant personality dimensions.

Personality Structure and Relational Competence

To some extent, the personality foundation on which loneliness and relational problems are built consists of deep-seated neurotic inadequacies that render the formation of reciprocal relationships continually difficult along with most other tasks of life (Duck, in press). Whereas one would not want to overemphasize the role of classical psychopathology in problematic relationships, it is a reality with which those of us interested in relationships from a non-clinical perspective have yet to adequately deal. The point here is simply that some people are habitually ineffective in relating to others for reasons of emotional disturbance that go beyond their immediate interpersonal circumstances, although there may well be a reciprocal influence between interpersonal and emotional problems. Such persons may show a life-long pattern of relational failure and loss and are undoubtedly over-represented in surveys of loneliness.

However, the literature on loneliness suggests that the dispositional vulnerability to relational failure is more often some form of deficient social skill or lack of confidence. Personality variables such as shyness, self-consciousness, introversion, externality, sociability, self-esteem,

assertiveness, and so on, have been repeatedly correlated with loneliness. Thus, it seems quite likely that some combination of these variables comprise a syndrome of characteristics that dramatically increase the likelihood of loneliness and relational failure, particularly in the face of events that disrupt normative patterns of relating (e.g., divorce, role change, widowhood, new situations, etc.). It is interesting to note these variables have also been implicated in the availability of social support (Lefcourt, Martin & Saleh, in press; Sarason, Levine, Basham & Sarason, 1983; Stokes, 1983).

But the question arises as to which variables are most strongly related to loneliness and hence are the best candidates for inclusion in the syndrome. In a preliminary assessment of this issue, we compared loneliness scores to 25 personality and attitude variables previously found to be correlated with loneliness (Jones, et al., 1982). The variables represented constructs falling into three categories: (1) measures assessing social skill and confidence in interpersonal settings (e.g., assertiveness, shyness, social anxiety, self-esteem, etc.); (2) attitudes toward relationships and human nature (e.g., attitudes toward love, jealousy, the trustworthiness of human nature, etc.); and (3) attitudes towards life and society (e.g., locus of control, alienation, purpose-in-life, etc.).

The variable pool was reduced using a principal components factor analysis. The composite variable accounting for the greatest proportion of variance in this analysis was called relational competence. Variables such as self-esteem, assertiveness and masculinity loaded positively on this factor, whereas shyness and social anxiety were negatively loaded. In a subsequent regression procedure, relational competence accounted for the largest proportion of unique variance in loneliness scores. Similarly, factor analytic studies of the structure of multivariate personality inventories consistently show that variables subsumed by the concept of relational competence (e.g., shyness, social anxiety) explain the largest proportion of variance among item responses, perhaps because of the relevance of such items to a broad spectrum of interpersonal consequences (Howarth, 1980).

We would expect then that the components of relational competence would be related to various features of the development and change of an individual's pattern of relationships and our research in this area has demonstrated this to be the case. Research on shyness, to select one example, indicates that it is inversely related to a variety of relational measures including number of friends, social satisfaction, dating frequency, closeness to family and friends, and satisfaction with one's social participation (cf. Jones & Russell, 1982). Moreover, we have found that shyness is related to reduced levels of social support and in one procedure, shy as compared to not shy participants indicated that they were less likely to have friends

who: were easy to talk to; gave good advice; did what the respondent wanted to do; looked up to the respondent; bolstered the respondent's ego, and so on. Perhaps more persuasively, we have found that both strangers and friends rate shy persons as less attractive, less friendly, less open, less warm, etc. (Jones & Briggs, 1983).

Additional research (e.g., Carpenter, et al., in press; Hansson, Carpenter, Nelson & Slade, 1983; Morgan, Hansson, Indart, Austin, Crutcher, Hampton, Opegard, O'daffer, in press) has revealed that various measures of relational competence not only predicted relational problems, but among "at risk" populations, they were related to a variety of health and adjustment variables as well. For example, among diabetic patients, assertiveness was strongly related to adjustment to diabetes, health habits, and confidence in being able to control one's illness. Specifically, diabetic patients scoring higher on the measure of assertiveness were more likely to have made the necessary changes in their lives to accommodate to their condition and to comply with their medical regimen (e.g., changes in diet, taking medication, etc.) Also they were more confident in their ability to do whatever might be necessary to control their diabetes. Interestingly, assertiveness was more strongly related to these and related variables than were various measures of social support (e.g., family cooperation, contact with family, help when ill, etc.). Similarly, among elderly persons, assertiveness was associated with greater satisfaction with primary and secondary relationships and better emotional adjustment, but also with better health and better health habits. Among permanently disabled adults (e.g., paraplegics, quadraplegics), assertiveness predicted loneliness, adjustment and social comparison following the onset of the disability. These findings have been replicated and extended for other behaviors of practical importance as well. For example, in a recently completed study, assertiveness and other measures of relational competence predicted the effort and effectiveness with which unemployed persons sought and secured jobs.

To summarize, preliminary evidence suggests that relational competence: (1) subsumes those personality variables most strongly related to loneliness; (2) is strongly related to other measures of relational success and failure including social support; (3) is related to indices of health, health behavior, and adjustment; and (4) is, in some instances, more strongly related to health and adjustment than are measures of social support. Thus, what we have conceptualized as relational competence seems to be directly involved in the status and quality of one's relationships. Moreover, similar to social support, it may be that relational competence is a moderator of the effects of stress on health and well-being, although this hypothesis has not been extensively examined. Alternatively, it may be that relational competence determines the extent to which the availability of a supportive social network buffers the health and adjustment

consequence among stressed populations.

Personality Dynamics

But prediction and explanation are not synonymous and thus the thornier but more important issue concerns how personality factors, considered under the general rubric of relational competence or one at a time, influence the process of relational formation and change. I would like to propose four possible mechanisms suggested by our recent and continuing research in this area.

First, personality factors may be involved in the process of conceptualizing one's relationships, that is, what one perceives, thinks, and expects with regard to a social network. For example, pessimistic and cynical attitudes toward other people appear to be stable features of relational incompetence and might render relationships that would ordinarily be satisfying, less so. By contrast, it may be that some dispositions result in overly positive expectations for one's relationships and specifically expectations so exaggerated that any available relationship would seem somehow lacking by comparison. It is even possible, indeed likely, that extremely positive and extremely negative expectations are held by the same individuals. For example, persons low in relational competence may, by virtue of unpleasant previous experiences or simply the lack of experience, expect the worst in their actual relationships while fantasizing idealized versions of longed-for relationships. There is indirect evidence of this effect with regard to loneliness (cf. Jones, 1982). In one study, for example, we found that lonely, as compared to not lonely unmarried college students were more cynical about the reasons that people get married as well as more pessimistic about the prospects of a long-lasting and satisfying marital relationship either for themselves or other people. In another study, however, we found that loneliness was related to highly romanticized conceptualizations of love relationships. For example, lonely as compared to not lonely respondents were more likely to endorse statements such as "I believe in love at first sight," and "for every person there is one perfect mate".

Other loneliness findings suggest analogous mechanisms. Lonely college students report having less frequent contact with intimate others, i.e., less contact with friends and family and more contact with strangers and acquaintances (Jones, 1981). At first glance this would seem to suggest that loneliness results from less intimacy and this is certainly a viable interpretation of these data. On the other hand, it is also possible that lonely persons calibrate their relationships more conservatively, for example, perhaps they are more likely to label a relationship of a given duration and intimacy as an acquaintanceship rather than a friendship.

Thus, to some extent, relationships are in the eye of the

beholder. More importantly, our data clearly show that the extent to which this is so is strongly related to the variables that comprise the construct of relational competence. For example, shyness is related to the accuracy with which one judges how one is being perceived by others; specifically, shy persons assume rejection from others even when it does not occur (Jones & Briggs, 1983). Of course the implication is that expecting the worst is a prophesy that is easily self-fulfilled with the net effect being less social support and greater loneliness.

Another type of personality mechanism has to do with constructing satisfying relationships and a supportive social network. The availability of relationships (particularly non-family relationships) might be affected by dispositional characteristics in either of two ways. On the one hand, our data clearly suggest that persons low in relational competence (i.e., persons who are shy, unassertive, etc.) fail to capitalize on the interpersonal opportunities available to them (cf. Jones & Briggs, 1983). This may occur because of compensatory disinterest, fear of failure, disruptive anxiety, interpersonal clumsiness, and so on. On the other hand, persons low in relational competence may betray their nervousness and discomfort in social interactions. Also, they tend to elicit gratuitous attributions from others beyond their shortcomings. For example, shy persons are readily perceived by observers as shy, but they are also seen as snobbish, unfriendly, untalented, etc. (Briggs, Snider, & Smith, 1983). In the first instance, personality factors interfere with relationships because the person is insufficiently skilled at initiating or maintaining relationships. In the latter case, the influence derives from the probable reactions of others to having perceived and misperceived these characteristics.

In this regard, it should be noted that our initial attempts to determine if lonely persons were differentially and perhaps negatively perceived were inconclusive. However, more recent investigations have demonstrated that lonely college students were perceived as "difficult to get to know" and "liking themselves less" (Jones, et al., in press; Solano, et al., 1982). Also, one recent study (Williams and Solano, 1983) indicated that their roommates reported less intimate relationships with lonely college students. Thus, either by virtue of their failure to initiate and nourish a supportive social network (due to their own interpersonal limitations), or as a consequence of the responses and stereotypes that their reticent behavior appears to elicit, persons low in relational competence are more likely to be lonely.

A third type of mechanism involves accessing one's relationships or social network in times of need. If a person is connected, however tentatively, to a potentially supportive network of caregivers (e.g., parents, friends, co-workers, etc.), but does not receive social support, it may signify either the failure to solicit help or ineffective solicitations. Research on help-seeking (e.g., Williams & Williams, 1983) points to a variety

of factors that inhibit or eliminate approaches to others for assistance. Asking for help may be viewed as a weakness. Similarly, embarrassment, loss of self-esteem, and reticence might constrain requests for help as might a lack of facility and persistence (e.g., obtuse, half-hearted and clumsy requests). Our research (Jones, Hansson & Cutrona, in press) indicates that lonely elderly persons often fail to avail themselves of community services (e.g., meals-on-wheels, friendly visitor programs, etc.) that would not only meet their physical needs, but also that might serve as "network builders" toward more frequent and broader interpersonal contacts and reduced isolation. Respondents who underutilized or avoided these services indicated that they were afraid of letting strangers into their homes and expressed contempt for those who did use such services. Thus, suspicion and excessive pride may inhibit the acquisition of both material and interpersonal support. Less obviously, the failure to appreciate one's own frailties and other's assets may prevent one from successfully accessing a supportive network.

Finally, personality factors may be important in one's efforts at maintaining relationships and social support. The necessity and potential difficulties of maintaining primary relationships over time are well recognized, if for no other reason than the current divorce rate. Less obvious is the need to maintain and strengthen ties among interpersonal environments that are not always supportive (Rook, in press) and particularly in conjunction with events that often result in the de facto loss of both primary and secondary relationships. For example, both divorce and widowhood typically eliminate many former relationships originated on a couple-companionate basis, especially for women (e.g., Lopata, 1969). Unemployment more or less automatically severs one's ties with co-workers, but may also interfere with non-work relationships by virtue of limited funds for socializing as well as embarrassment over having lost one's job. Similarly, conditions associated with apprehension and stereotyping on the part of others may lead to being stigmatized and ostracized, precisely at a time when social support is severely needed. Examples here include mental and physical handicaps, and to a large extent, the process of growing old.

It is proposed that stressful life events, and particularly those events that directly or indirectly threaten relationships, place a premium on the individual's coping abilities, in this instance, on their ability to maintain (or perhaps even regain) supportive relationships. This idea is similar to the environmental-docility hypothesis (Lawton & Simon, 1968) which suggests that the effects of environmental stress are greatest in the absence of personal coping resources. Relational competence is, of course, a type of coping resource and our research among "at risk" populations is beginning to suggest that such personality characteristics play a major, if not the major role in determining the consequences of such stressors.

In conclusion, our research and the research of others as well, is consistent with the interpretation that certain people are particularly vulnerable to loneliness because they are relatively deficient in relational competence. It is also clear that factors external to the individual (what we call relational stress) contribute to the probability that loneliness will occur. It is worth recalling, however, that we have some evidence to suggest that loneliness may be overdetermined--to use an old psychoanalytic term--in that persons low in relational competence may be more likely to experience relational stress. Also our research is beginning to show that relational competence is correlated with at least some measures of social support. In closing, however, the principal point is that research on loneliness would argue for the consideration of what the individual brings to the social network along with what the network does in the process of social support.

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THE FUNCTIONS OF SOCIAL BONDS: PERSPECTIVES FROM RESEARCH ON SOCIAL SUPPORT, LONELINESS AND SOCIAL ISOLATION¹

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Social bonds have long been considered essential for healthy functioning (Durkheim, 1897/1951; Faris, 1934; Murray, 1938), but this idea has been elaborated and tested most extensively in the past decade. Why this topic has captured so much attention at this point in history is an interesting question (cf., Gottlieb, 1983a). What is most clear is that researchers from quite diverse disciplines are actively working to understand how social bonds affect well-being. Interest in this topic is reflected in three different lines of work: research on social support, loneliness and social isolation. Studies of social support suggest that social relationships facilitate adjustment to stressful life circumstances and thereby decrease vulnerability to stress-related disorders. Research on loneliness emphasizes basic human needs for satisfying social ties and suggests that the absence or disruption of social bonds is in itself a cause of emotional distress. Sociological analyses of social isolation suggest that social relationships serve to inhibit deviant behavior as well as to provide support and companionship. This research has linked social isolation to psychopathology and to increased mortality due to alcoholism and suicide.

Empirical results from these different literatures thus converge on the general proposition that interpersonal ties function to sustain or enhance mental and physical health. Few efforts have been made, however, to bridge these conceptually related lines of work. Studies of social support, loneliness and social isolation have evolved independently of each other, despite their common emphasis on the health consequences of social bonds. For example, several excellent reviews of social support research (e.g., Cohen & McKay, 1984; Heller, 1979; House, 1981; Mitchell, Billings, & Moos, 1983; Turner, 1983) and loneliness research

(Peplau & Perlman, 1982) have appeared recently, yet cross-references to these complementary literatures are rare. Sociological discussions of social isolation (e.g., Hughes & Gove, 1981; Galle & Gove, 1978) similarly seldom incorporate perspectives from the logically related literature on loneliness.

There is much to be gained by efforts to integrate these different perspectives. These literatures offer different views on the basic functions of social ties, on the nature and causes of deficient social ties and on the consequences of such deficiencies. Thus, a systematic comparison of these viewpoints should stimulate theory development and suggest new hypotheses for research. In addition, an evaluation of empirical findings from these different literatures would greatly expand the knowledge base needed to guide intervention efforts.

This paper undertakes only part of the large task of integration. It contrasts the perspectives of the social support, loneliness and social isolation literatures with respect to a key theoretical issue: what functions of social relationships are important for physical and mental health? This issue seems to be at the heart of current efforts to understand how social ties affect well-being. The paper begins by examining how each research tradition has approached this issue. Three broad functions of social bonds are distinguished in the first section. The differential importance of these functions in specific situations and for specific mental health outcomes is considered next. This is followed by a discussion of the negative aspects of social bonds. This section considers how the positive and negative aspects of social bonds may covary and how they may jointly affect well-being. A final section provides an overview of the major themes developed in the paper and suggests directions for further research.

FUNCTIONS OF SOCIAL BONDS

People who are lonely, socially isolated or without social support are presumed to be vulnerable to emotional and physical problems because they lack something essential that is available only through interpersonal transactions. Our task as researchers or practitioners is to identify what it is that is missing. This focuses our attention on the content of social exchanges and the functions they serve. In keeping with their differing historical origins, the literatures on social support, loneliness and social isolation emphasize somewhat different functions of social bonds.

Perspectives of Social Support Research

Much of the current research on social support stems from parallel interests in identifying factors that reduce the effects of life stress and in evaluating the effectiveness of

paraprofessional counselors (Gottlieb, 1983a). Early epidemiological analyses indicated that characteristics of the social environment affect vulnerability to stress. People who lacked viable social ties were found to be more susceptible to disease than people who had such ties. This research prompted the hypothesis that social relationships moderate the effects of stress by providing needed assistance (Cassel, 1974). The help-providing functions of social ties were similarly highlighted in early studies indicating that most people seek help for personal problems from close associates rather than from professional mental health workers (e.g., Gurin, Veroff & Feld, 1960). Studies documenting the effectiveness of counseling provided by individuals with little or no professional training further supported the idea that significant helping resources exist naturally in people's social networks (cf., Kelly, Snowden & Munoz, 1977).

The historical roots of current social support research thus converge on an interest in the help-providing functions of social ties. Although researchers have conceptualized and operationalized social support in very different ways, a recurring theme through much of this work has been a concern with how help provided by one's social network ameliorates the effects of life stress. Indeed, use of the term support itself connotes a central concern with the provision of help.

Recently, a number of typologies of support have been developed (see reviews by Barrera & Ainlay, 1983; House, 1981). For example, researchers have distinguished tangible support (i.e., the provision of material aid and services) from emotional and informational support (i.e., empathic listening and guidance about problem solving). These efforts to distinguish various types of support are extremely useful in determining which kinds of help are most beneficial in which situations (Cohen & McKay, 1984; House, 1981).

There is a potential risk, however, that the construct of social support will become so elastic as to include all beneficial interpersonal transactions. For example, shared leisure and expressions of liking are included as types of support in some taxonomies (Barrera & Ainlay, 1983; Cobb, 1976; Cohen & Hoberman, 1983; Hirsch, 1980; Lin, Dean & Ensel, 1981). While it is likely that such exchanges do contribute to mental health, it is not clear that they should be conceptualized as social support. Rather, they represent forms of companionship that may have little to do with helping the recipient. It seems most useful to retain a somewhat narrower meaning for the term support, a meaning that refers primarily to helping behaviors. Support can then be studied as one important category of a broader set of health-sustaining functions of social ties. This is not merely a semantic issue, since grouping quite diverse types of interpersonal exchanges under the common heading of social support is likely to create conceptual and methodological confusion.

Construing social support as the provision of various types of help does not mean that the behaviors typically subsumed under the label "emotional support" (e.g., House, 1981) must be excluded. It is emotionally supportive to say "I still like you" or "I care about you a great deal" to a college student who is distraught over failing an important exam. Revealing to a forlorn friend that one has experienced a similar embarrassing problem is another example of emotional support. Both of these examples involve emotionally expressive behaviors--communication of liking and intimate self-disclosure--but they can be regarded as helping behaviors because they occur in response to learning of another's problems. The social context that gives these behaviors their meaning is the revelation of a personal problem. The behaviors serve an instrumental function rather than a purely expressive function because their purpose is to cheer or comfort the recipient. In a different context (such as a candlelight dinner) the same behaviors might have a different meaning and, therefore, different consequences (such as pleasure or excitement).

To summarize, the unifying theme of social support research is a concern with the different types of help provided through social ties to those who are experiencing life stress. A variety of behaviors, including emotionally expressive behaviors, may be construed as help-oriented because they are prompted by awareness of another's problems. The social support literature thus emphasizes the instrumental functions of social ties. The loneliness literature, in contrast, is informative about expressive as well as instrumental functions of social ties.

Perspectives of Loneliness Research

It is fairly commonplace to observe that humans are inherently social creatures and, in fact, there is a tendency to regard such a statement as so self-evident that it requires no further elaboration. Yet it is important to know what it is that motivates people to form intimate bonds and what causes such intense distress when these bonds are lost or never attained. Research on loneliness attempts to answer these questions.

Several different strands of research prompted an interest in loneliness--the subjective experience that occurs when one's existing relationships are judged to be deficient in either quantity or quality (Peplau & Perlman, 1979). Observations of the effects of infant-parent attachments and studies of grief and marital dissolution were among the earliest influences (Weiss, 1973). Early studies by Bowlby (e.g., 1960) indicated that infants who fail to develop strong bonds with an adult caretaker are at risk for developing aberrant behavior patterns. Such bonds were hypothesized to provide the child with a reliable source of security as well as a source of joy (Bowlby, 1977). This research, along with that of Ainsworth (e.g., 1964), led to the view that emotional attachments to others are necessary for

healthy human development. Theorists extrapolated from this research to posit the existence of a similar need for interpersonal attachments among adults (Bowlby, 1977; Henderson, 1977; Kahn & Antonucci, 1980; Weiss, 1974). Research on reactions to the death of a spouse confirmed the importance for adults of close emotional bonds (e.g., Marris, 1958; Parkes, 1972). The prolonged and intensely painful grief reactions observed in these studies indicated that the loss of an intimate social bond markedly disrupts psychological functioning. Studies of the stressful and often ambivalent emotions commonly experienced during marital dissolution further emphasized the difficulty with which attachment figures are relinquished (Weiss, 1975).

Drawing on attachment theory, loneliness theorists argue that people seek to establish close bonds with others who can reliably provide comfort and security (e.g., Weiss, 1974). People are believed to function most effectively "when they are confident that, standing behind them, there are one or more trusted persons who will come to their aid should difficulties arise" (Bowlby, 1973, p. 359). This idea overlaps considerably with the importance attributed to help-giving by social support theorists.

Beyond this emphasis on the instrumental value of social bonds, loneliness theorists emphasize that social bonds are sought in and of themselves because they provide opportunities for pleasurable companionship and intimacy. From this perspective, social interaction does not serve an extrinsic purpose but, instead, is desired for its expressive and hedonic aspects (cf., Lin et al., 1981). Pleasurable companionship refers to such activities as shared leisure and recreation, discussion of common interests and spontaneous play. These represent activities that are engaged in for their own sake. Intimate exchanges with another person are also often sought for their own sake. Self-disclosure, for example, occurs not only out of a need to seek help for personal problems but also out "of a simple desire to be known or understood" (Fromm-Reichmann, 1959). Discussion of personal aspirations and fantasies, expressions of affection and enactment of private jokes or shared rituals are further examples of intimate interactions that may be initiated for purely intrinsic reasons.

While sociable and intimate interactions may at times serve to provide distraction from stressful problems, they more typically serve to enhance mood and feelings of self-worth directly. Weiss (1974) commented in this regard that friendship ties "offer a base for social events and happenings" and "in the absence of such ties life becomes dull, perhaps painfully so" (p. 23). Thus companionship and intimate interaction represent centrally important contexts in which people reward themselves and each other by arranging mutually gratifying activities. In addition to providing opportunities for mutual reward, companionship and intimacy may provide novelty, excitement and

challenge. Lawton and Nahemow (1973) proposed that the optimal physical environment for human adaptation is one that provides a sufficient level of challenge and stimulation. Presumably this idea would apply to the social environment as well.

Such pleasurable interpersonal experiences are thought to be highly desired by most people; indeed, some researchers regard them as basic social needs (e.g., Sullivan, 1953; Weiss, 1973, 1974). Accordingly, individuals who lack opportunities for companionship and intimacy suffer emotional distress, or loneliness. Most loneliness researchers view loneliness as an acutely painful condition (Peplau & Perlman, 1982) that may lead to depression and other psychological problems over time (Brennan & Auslander, 1979; Perlman & Peplau, in press; Rook, in press). This distress is compounded by the fact that cultural stereotypes portray lonely people as losers in the social marketplace (Gordon, 1976; Parmelee & Werner, 1978). Thus, loneliness is a stigmatized condition.

Social support researchers, in contrast, have rarely described lack of social support as a condition that in itself is aversive, although some researchers have observed that the disruption or loss of relationships is a source of stress (Thoits, 1982). Moreover, lacking others to turn to for help has rarely been characterized as a stigmatized condition. In fact, cultural values regarding self-reliance might serve to cast nonhelp-seekers in a favorable light. Thus, loneliness and perceived lack of support would appear to differ in their emotional impact, in their public meaning and in the type of social deficits involved.

In sum, the loneliness literature and the social support literature offer complementary perspectives on the beneficial functions of social ties. Social support research emphasizes the contribution to healthy functioning of help provided through social relationships. Loneliness research calls attention to the value of pleasurable companionship and intimacy in enhancing mental health. A third emphasis is provided by theory and research on social isolation.

Perspectives of Social Isolation Research

Analyses of the effects of social isolation have been conducted primarily by sociologists and are derived historically from Durkheim's (1897/1951) theory of social integration. Within this research tradition, social integration serves two purposes that are relevant to this paper. First, by embedding the individual in a web of interpersonal influences, social integration helps achieve compliance with group norms. That is, social ties serve to maintain social order by restraining deviant behavior. Second, social relationships impart meaning to one's existence. Membership in a socially cohesive group gives the individual's life meaning and purpose. Absence of strong interpersonal connections accordingly contributes to despair, and

in extreme cases, to suicide. This emphasis on the meaning imparted to existence by social bonds, apart from any explicitly instrumental function, parallels the emphasis in the loneliness literature on the intrinsic satisfactions provided by social bonds. The regulatory functions of social ties, however, are emphasized in neither the loneliness literature nor the social support literature and, therefore, warrant further consideration here.

Although sociologists were originally most interested in understanding how social regulation affected society by helping to maintain order and stability, more recent analyses have examined how regulation affects individuals. An interesting example of this type of individual-level analysis is Hughes and Gove's (1981) research on the effects of living alone versus living with others. They conceptualized those who live alone as being vulnerable to psychological disorders for a number of reasons, including lack of social regulation. Those who live alone are considered more likely to engage in risky and self-destructive acts because they lack others to restrain them. Those who live alone are also at a disadvantage because they lack "structure and input into daily affairs." Hughes and Gove did not elaborate this interesting idea further, but it might refer to the stability and structure imposed by social role obligations. The parent of a young child or the minister in a small town, for example, might be forced to resist feelings of depression and fatigue because they are obligated to function effectively in the roles they occupy. "Structure and input into daily affairs" might also include prompting of stable health-sustaining behavior, such as proper diet, exercise, adequate sleep and periods of relaxation. Thus, the regulatory functions of social ties may include not only deterrence from deviant acts but also promotion of healthy behavior.

The sociological emphasis on interpersonal regulation suggests an alternative interpretation of studies that have linked social ties to health and well-being. Respondents who were conceptualized as lacking social support may have been more vulnerable to health threats not because they lacked others to provide specific types of assistance but because they lacked others to deter them from ill-advised behavior. Similarly, people identified as lonely may have been at greater risk for psychological problems not because they lacked opportunities for pleasurable companionship but because they lacked others to prompt them to function more effectively.

Social support theorists might argue that this idea of regulation is represented in taxonomies of supportive behaviors that include feedback to the individual (e.g., Barrera, 1981; Caplan, 1974; Gottlieb, 1978; House, 1981; Schaefer, Coyne & Lazarus, 1981). While feedback and regulation are somewhat similar, important differences also exist. In the social support literature, feedback is often construed as information that helps the individual to better evaluate a problem or possible coping

responses (e.g., Schaefer et al., 1981). This suggests that feedback would often be solicited by the individual specifically to improve his or her problem solving. The sociological idea of regulation, in contrast, connotes feedback that need not be focused on problem solving but, rather, may be offered for a wide range of behaviors. For example, a woman might be advised by friends to change aspects of her job performance even though she had not defined this as a problem area and had not sought advice about it from anyone. Another difference between these two perspectives is that social support theorists often imply that feedback must be positive or affirming to be helpful. Kahn and his colleagues (Kahn, 1979; Kahn & Antonucci, 1980), for example, define as a key element of social support "expressions of agreement or acknowledgement of the appropriateness or rightness of some act or statement of another person" (p. 85). While such affirmation is undoubtedly beneficial in many cases, the sociological orientation suggests that feedback from others which is not affirming may also be beneficial. Intervening to stop someone from a personally desired but reckless course of action, such as plotting revenge against an unfair boss, is very likely beneficial even though it is distinctly not affirming.

In this vein, Hughes and Gove (1981) hypothesized that the regulatory actions of others may provoke psychological distress even though they lead to less deviance and more stable functioning. "Constraint may be the source of considerable frustration; at the same time, it tends to reduce the probability of problematic or maladaptive behaviors" (p. 71). From their discussion of the dual effects of interpersonal regulation, they derive the interesting prediction that in areas where social integration is strong enough to reduce the suicide rate, levels of psychological distress will simultaneously be elevated.

Summary

In keeping with their different historical origins, the literatures on social support, loneliness and social isolation provide somewhat different answers to the question of which functions of social ties are important for health and well-being. These literatures point to three broad functions as being most important. Social support theorists emphasize the importance of different types of help provided by others in ameliorating the effects of stressful events. Loneliness theorists conceptualize social bonds as satisfying inherent needs for companionship and intimacy. Social isolation theorists stress that social relations contribute to the regulation of behavior, including inhibition of deviant behavior and prompting of role performance and healthful habits. Each of these broad functions could be further disaggregated into its constituent elements (cf., Novaco & Vaux, 1983). Social support researchers have already distinguished several different types of help (e.g., House, 1981). It would be

useful if the general ideas of companionship, intimacy and regulation were similarly refined. This would allow determination of which specific interpersonal exchanges subsumed by these general concepts have the greatest impact on functioning. For example, expressions of affection may be more beneficial than the novelty provided by shared recreation.

DIFFERENTIAL IMPORTANCE OF FUNCTIONS ACROSS SITUATIONS AND OUTCOMES

An implicit assumption in the preceding discussion is that the three major functions identified--help, companionship/intimacy and regulation--have unique effects on well-being. In fact, the rationale for developing taxonomies of basic functions is that these functions have distinguishable effects (cf., Cohen & McKay, 1984). From this perspective, deficits of particular types of social exchanges cannot be compensated for by an abundance of other exchanges (Weiss, 1973). If this assumption is correct, it suggests that different functions may vary in importance across situations and health outcomes. The research implications of this idea are examined below.

Importance of Functions in Different Situations: Buffering and Main Effects.

As conceptualized in this paper, the provision of help by others is most useful in reducing threats to well-being by facilitating adaptation to stressful events. That is, help is most important when the individual's equilibrium has been disrupted by some event or set of events. Companionship and intimate interaction, in contrast, are most useful in providing positive inputs to well-being, such as recreation and affection. That is, these pleasurable interpersonal events are important not so much for restoring the individual to a prior level of functioning as for elevating the current level of contentment. If we imagine psychological functioning to have a true neutral point, then companionship and intimate interaction serve to promote positive mental health by raising functioning above the neutral point, while receiving help with pressing problems serves to prevent impaired mental health by restoring disrupted functioning to the neutral point. Main effects of social contact would therefore be expected when that contact involves companionship, while buffering effects of social contact (interactions of stress and social contact) would be expected when that contact involves help.

This analysis helps to resolve the controversy surrounding main effects versus buffering effects by providing a theoretical basis for predicting main effects as well as buffering effects. It may also provide a basis for reconciling inconsistent results

from studies that have used very different measures of social support. Studies that report main effects may have operationalized support in ways that tend to capture companionship rather than help-giving, while the reverse may be true among studies that report buffering effects. In this regard, Cohen (this volume) concluded from a review of empirical studies that main effects appear primarily when global "structural measures" of support are used, while buffering effects appear primarily when specific "functional measures" of support are used. Structural measures assess such characteristics of social involvement as how many friends an individual has, how often these friends are seen, whether or not the individual is married, and so forth. Functional measures assess the extent to which the individual receives specific types of help from others (e.g., material aid, emotional support, advice).

One interpretation of the pattern noted by Cohen is that structural measures have little to do with support per se (i.e., provision of help) and, instead, actually assess sociable interaction. That is, people who report that they have many friends or get together with friends often are, in effect, telling us that they are actively involved in shared leisure and other forms of pleasurable social interaction. Consistent with this, Fischer (1981) found in a large survey of people's social networks that use of the term "friend" to describe a relationship with another person was more strongly related to the frequency of socializing than to the amount of emotional support received. If structural measures implicitly assess the amount of enjoyable social interaction, then main effects would be expected since everyone presumably would benefit from such interaction. If functional measures, in contrast, assess the amount of help received from others, then buffering effects would be expected since only those respondents whose well-being is threatened by specific stresses would benefit from a high level of help.

This discussion highlights the confusion that can result when measures that assess quite different types of social exchanges are grouped under the common label of "social support." More important, it points to a need for researchers to be explicit about the functions that their measures are capturing and about the situations in which these functions are most important.

Thus far the discussion has focused on only two of the three major functions identified. It is less clear in which situations the regulatory function of social bonds would be most important. At a societal level, sociologists have argued that regulation is particularly important during periods of rapid social change, such as the large-scale migration from rural to urban areas that occurred at the turn of the century (Meier, 1982). Extrapolation from this argument would suggest that at the individual level, social regulation is most useful in maintaining stable functioning during periods of rapid personal change, such as the often abrupt transition from employment to retirement or the transition from

high school to college. Regulation would also seem to be particularly useful in supplementing internal restraints on deviance during periods when such restraints are likely to be weak. During times of intense anger toward someone or exposure to antisocial values and practices, countervailing restraining influences may be critical.

Importance of Functions for Different Outcomes.

If it is correct to assume that different functions of social bonds have relatively unique effects, then careful consideration must be given to the measures of well-being used in research. For example, if the primary value of companionship is to promote positive mental health rather than to alleviate psychological distress, then exclusive use of measures of psychiatric symptoms may be inappropriate. The benefits of companionship may be most apparent through use of measures that assess positive dimensions of functioning, such as happiness or life satisfaction. That is, people with differing levels of companionship might not differ on scales of psychiatric symptomatology, but they might differ on measures of happiness or life satisfaction. Failure to include such measures might obscure the benefits of companionship. Conversely, if help provided by others serves primarily to reduce threats to well-being, then measures that reflect relief from distress would seem most appropriate. For example, receiving advice about how to resolve a pressing problem may not increase happiness but might alleviate anxiety or feelings of incompetence and thereby restore equilibrium.

Similarly, it may be possible to detect the unique effects of social regulation only by including measures that tap the occurrence of deviant or risky behavior and the quality and stability of role performance. Hughes and Gove (1981), for example, sought to investigate the effects of social regulation through use of self-report measures of alcohol and tranquilizer consumption, seat belt use, carelessness leading to household accidents and other indicators of risky behavior.

Empirical support is beginning to emerge for the assumption that the different functions of social bonds have distinguishable effects (Cohen & Hoberman, 1983; Schaefer et al., 1981). Nevertheless, researchers have noted that conceptually distinct social exchanges sometimes overlap in their effects (e.g., House, 1981; Cohen & McKay, 1984). For example, depression may result from a prolonged deficit of satisfying activities (Lewinsohn, 1974), as well as from exhaustion due to prolonged stress (Wolpe, 1971; Selye, 1952). Thus, insufficient opportunities for satisfying companionship as well as insufficient help from others may result in the same outcome--depression. Similarly, stress caused by unresolved problems may be alleviated not only by problem-focused help but also by the distraction provided by socializing with friends (Barrera, 1981). These examples suggest

that different functions of social bonds may at times produce similar outcomes, even though they do so through rather different intervening processes. If this is so, then the important questions to be addressed are: 1) Do the functions differ in how consistently they achieve desired effects? 2) Do they differ in the magnitude of their effects? 3) Are the gains achieved through these different processes equally durable? For example, socializing with friends may provide relief from stress less consistently, less powerfully or with less lasting impact than problem-focused help.

Summary

The various ways that the functions identified may map onto specific situations and outcomes are summarized in Table 1. Whether the hypothesized distinctions and relationships presented in this table are valid is an empirical question. For example, the different categories of exchanges identified may prove to have overlapping effects. Moreover, it is possible that these functions are highly intercorrelated, such that people who receive support also tend to have sociable and intimate interactions and to receive regulatory feedback. This would increase the difficulty of determining which types of interpersonal events have which effects. Reis (in press) suggested that factor analytic procedures will be required to demonstrate the discriminant validity of hypothesized functions of social contact and that procedures for partialling (e.g., partial correlation analysis, structural equation analysis) will be required to estimate the unique variance attributable to each function.

NEGATIVE ASPECTS OF SOCIAL BONDS

The preceding sections considered only the beneficial functions of social bonds. Identification of such beneficial functions is an important priority, yet researchers have increasingly argued that social relationships may detract from as well as enhance well-being (Wortman & Dunkel-Schetter, 1979; Gottlieb, 1979; Heller, 1979; Rook, 1984; Schaefer et al., 1981; Wellman, 1981). In fact, a basic premise of social exchange theory is that relationships entail costs as well as rewards (Homans, 1974; Thibaut & Kelley, 1959). Abundant evidence of the harm that people are capable of inflicting on each other is provided by statistics on such social problems as divorce, child abuse, abuse of elderly parents, and crimes of passion. A sizable psychiatric literature suggests that disturbed family relationships play a role in the onset of psychopathology (Jacob, 1975; Liem, 1980). Events involving interpersonal conflict or dissolution of relationships are among the most stressful entries in many popular scales of stressful life events

TABLE 1. Mapping of Functions of Social Bonds onto Situations and Outcomes

Functions	Type of Social Exchange	Major Benefits	Salient Situations	Salient Outcomes
I. Help e.g.: Emotional support Informational support Appraisal support Instrumental support	Examples: Empathic listening Expression of caring/ concern Reassurance of worth Advice Provision of material aid/services	Reduce threats to well-being (restore equilibrium, prevent pathology)	Disruption of functioning due to stressful events	Stress-related physical illness Psychiatric symptoms (particularly worry, anxiety, depression) Quality of coping responses Sense of self-efficacy
II. Companionship/ Intimate Interaction e.g.: Pleasurable social activity Stimulation/challenge Self-revelation	Examples: Shared Leisure/ recreation Discussion of common interests Humor Affection Disclosure of hopes fantasies, beliefs	Provide positive inputs to well- being (increase content- ment, promote positive mental health)	No specific situations— presumed to be beneficial in most situations	Quality of life/positive mental health (e.g., happiness, life satis- faction, sense of self- worth) Mood (absence of positive inputs leads to depression)
III. Regulation e.g.: Deterrence from deviant acts Prompting of stable role performance Prompting of health- sustaining behavior	Examples: Feedback about inappropriate behavior Direct intervention to prevent self- injurious acts Reminder of role obligations Threats for ineffective role performance Requests for better self-care Modeling of effective self-care or role performance	Prevent deviance (supplement internal restraints) Maintain stable functioning (provide structure to daily functioning)	Periods of rapid personal/social change Periods when internal restraints are weak (e.g., when intense anger is aroused) Exposure to groups espousing antisocial or deviant values	Deviant and self-injurious behavior Risk-taking (e.g., alcohol and substance abuse, reckless driving) Quality of role performance (e.g., work attendance, pro- ductivity, performance of household and parenting duties) Quality of self-care (e.g., diet, exercise, compliance with medical regimen)

(Thoits, 1982). Beyond such dramatic examples, it is intuitively obvious that social relationships can be a source of everyday hassles and demands.

For each of the three beneficial functions discussed earlier, there may be a parallel detrimental function. This section briefly considers social exchanges that exacerbate threats to well-being, that provide negative inputs to well-being and that promote unstable, deviant functioning. This section also considers how these negative exchanges may interact with positive exchanges to affect well-being.

Exacerbation of threats to well-being. Help provided by others may inadvertently reinforce sick role behavior or dependence (DiMatteo & Hays, 1981). Having to rely on others for help may cause feelings of guilt or shame, particularly if the individual is not able to reciprocate the help received (cf., Fisher, Nadler & Whitcher-Alagna, 1982; Walster, Walster & Berscheid, 1978). In addition, ineptly provided support may exacerbate rather than alleviate the recipient's feelings of distress. Wortman and her colleagues (Wortman and Dunkel-Schetter, 1979; Wortman, this volume) offer a cogent analysis of the psychological mechanisms that often make the informal support offered to cancer patients and other victims more harmful than helpful. Thus, it is unwise to assume that lay helping responses are beneficial simply because they are well-intended.

Provision of negative inputs to well-being. Social relationships may be a source of negative inputs that directly detract from well-being. Criticism, rejection, exhausting demands, unwillingness to provide help, violation of privacy or confidentiality, and exploitation are all examples of such negative inputs. Interestingly, evidence on the developmental course of close relationships suggests that the balance of positive to negative exchanges shifts over time, such that negative exchanges become increasingly common (Rands & Levinger, 1979). Thus, relationships that initially provide the greatest opportunities for pleasurable companionship and intimacy may be quite vulnerable to the development of aversive exchange patterns (Altman & Taylor, 1973).

Promotion of unstable, deviant functioning. The processes of interpersonal influence within specific subcultures may encourage unconventional values and practices (Fischer, 1976), as illustrated in extreme form in the bizarre behavior of members of such religious cults as the People's Temple in Jonestown. Indeed, a central tenet of both functionalist theory (e.g., Durkheim, 1893/1964) and family systems theory (e.g., Minuchin, 1974) is that groups often reinforce the deviant behavior of a member because such deviance benefits the group, for example by enhancing group solidarity. At a more mundane level, it is clear that others frequently prompt behavior that is unhealthful rather than healthful by encouraging us to eat too much, drink too much, drive

too fast or break rules. Kaplan (this volume) found, for example, that teen-age diabetics with an extensive peer network were less likely to comply with their treatment regimens than relatively isolated teen-age diabetics.

Combined Effects of Positive and Negative Social Exchanges on Well-Being

A potential paradox seems to exist here. On the one hand, considerable research appears to support the conclusion that social ties are essential for health and well-being. On the other hand, a separate body of evidence appears to point to the opposite conclusion--that social ties often detract from well-being. How can these divergent views be reconciled?

Theoretical perspectives emphasizing the role of choice in the construction of social networks may help to provide a resolution. Social exchange theorists argue that we choose our social ties on the basis of their capacity to provide rewards relative to costs (Homans 1974; Thibaut & Kelley, 1959). Choice, therefore, should lead most people to construct social networks composed predominantly of rewarding social ties. Nevertheless, it is obvious that many constraints operate to limit choice as the major determinant of those with whom we interact (Fischer et al., 1977). While the majority of one's social contacts are likely to be positive, it is plausible that at least some contacts are unwanted and aversive in nature.

Given this general backdrop of largely positive social encounters, it might be expected that negative encounters would have disproportionate impact precisely because they are rarer and therefore more salient (cf., Kanouse & Hanson, 1972). That is, negative social encounters may be particularly potent because they are more unusual and more unexpected. Alternatively, it might be hypothesized that positive encounters offset, or "buffer," the effects of negative encounters. A third hypothesis is that negative and positive social experiences affect well-being in a simple additive fashion.

Relatively few studies have measured both negative and positive social exchanges. As a result, little basis exists currently for choosing among these hypotheses, although preliminary evidence favors the view that negative events exert a disproportionate impact on well-being. A study of elderly widows' social networks found that negative exchanges detracted from well-being to a greater extent than positive exchanges enhanced well-being (Rook, 1984). While this finding requires replication in other studies, it suggests that researchers who fail to assess negative social encounters may overlook a particularly important source of variation in psychological functioning.

In addition, findings from investigations of the stress-buffering hypothesis are vulnerable to an important alternative interpretation if negative interpersonal encounters are not measured. Differences in well-being between those with

varying levels of social support are typically attributed to limited opportunities for low-support individuals to obtain help with pressing problems. Yet it is plausible that those who lack supportive social ties might additionally be burdened with troublesome ties that exacerbate the problems experienced. Thus, an apparent effect of low support would actually be due to aversive social exchanges that compound the effects of stressful events. This stress-exacerbation hypothesis is rarely considered but could readily be tested if negative as well as positive social encounters were assessed.

More research is needed to determine how positive and negative social encounters covary. For example, they may be inversely correlated if those who have a strong network of supportive social ties tend to be adept at avoiding troublesome social interactions. Alternatively, those who are most active socially may have the highest risk of troublesome encounters simply by virtue of their greater contact with others. In this case, supportive and problematic social interactions would be positively correlated. Yet another possibility is that these domains of experience are independent.

CONCLUSION

The question of how social bonds affect health and well-being is exceedingly complex and warrants the considerable attention it has received from a variety of disciplines. Yet efforts to develop sophisticated answers to this question have been hampered by the fact that researchers working within a particular research tradition rarely incorporate the perspectives of those working within a different tradition. This paper sought to contribute to an integration of these different perspectives by examining how social support, loneliness and social isolation researchers conceptualize the functions of social bonds.

Through this review, three complementary functions were identified--help, companionship/intimacy, and regulation. These functions were hypothesized to affect well-being through different intervening processes --by reducing threats to well-being, by providing positive inputs to well-being and by promoting stable, nondeviant functioning. Moreover, these functions were seen as being important in somewhat different situations and for different outcomes. Synthesizing the perspectives of these three literatures thus provides a much broader view of how social ties affect well-being than is possible by considering any one literature alone.

This broader view helps to balance the current preoccupation with how social relationships protect us from threats by also emphasizing how social relationships increase the quality of our lives. It is not necessary to posit the existence of threatening events to predict that those who lack social ties are vulnerable

to impaired health and well-being. The absence of opportunities for pleasurable companionship and intimacy may in itself cause psychological disorder, yet this idea runs the risk of being overshadowed by the popular focus on support. Moreover, simply including pleasurable social interaction in taxonomies of social support does not eliminate this risk, since such inclusion is usually based on the idea that pleasurable interaction helps alleviate the distress caused by some problem (e.g., Barrera, 1981; Cohen & McKay, 1984). What is needed are analyses that grant companionship and intimacy a conceptual status that goes beyond stress alleviation—one that recognizes their role in enhancing rather than protecting mental health. Lawton (1978) commented in this regard that we tend to settle for bland definitions of well-being based on the absence of pathology. We need greater attention to positive interpersonal experiences that enrich the quality of life.

Another contribution of the broad framework outlined in this paper is that it suggests alternative interpretations of previous studies that have documented benefits of social interaction. These benefits may have been due to help, companionship/intimacy, regulation or some combination of these three functions. Recognition of these alternative explanations is important not only for our basic understanding of how social relationships affect well-being but also for how we conceptualize interventions. Implicit assumptions about the functions of social bonds are likely to influence the goals and content of social network interventions. For example, what is the appropriate balance of affirmation and esteem-enhancement to confrontation and persuasion in social support groups? If we regard the major function of support groups to be providing affirmation, what safeguards are needed to ensure that inappropriate behaviors are not inadvertently reinforced? On the other hand, if we conceptualize actions that challenge people's views as beneficial, then what measures are needed to prevent attrition by participants whose most salient personal needs are for affirmation by others? Similarly, how should we design interventions to aid those who lack companionate as well as support-providing relationships? Such individuals may benefit most from support groups that are structured to allow time for informal socializing as well as for problem-focused discussion.

This paper also considered how negative social exchanges may affect well-being through intervening processes that parallel those posited for positive exchanges--by exacerbating threats to well-being, by providing negative inputs to well-being and by promoting unstable, deviant functioning. Different models were presented for how positive and negative social exchanges may combine to affect well-being, and a stress-exacerbation hypothesis was suggested as an alternative to the stress-buffering hypothesis. Unless researchers assess both negative and positive interpersonal experiences, it will not be possible to choose among

these alternative hypotheses. Moreover, these issues have implications for interventions as well as for research. For example, while most discussions of social network interventions focus on the need to facilitate formation of positive relationships, it may also be important to consider interventions to help people alter or disengage from troublesome relationships (1984).

The ideas developed in this paper provide a useful framework for analyses of the effects of social bonds on health and well-being. Yet many questions remain about how to operationalize such a framework and about how to incorporate other theoretical distinctions that were intentionally ignored here for the sake of simplicity. Three issues seem particularly important in efforts to advance theory and research.

First, what major functions of social bonds are missing from the set described in this paper? One likely candidate is the provision of help to others (as opposed to the receipt of help from others). It is widely believed that we need to feel important in the lives of others and that we derive significant psychological benefits from engaging in prosocial behavior (Reissman, 1965; Weiss, 1974). More generally, what is the best strategy for deriving taxonomies of the functions of social bonds (Peplau, this volume)? This paper adopted the strategy of synthesizing ideas from compatible research literatures, but other strategies might be equally useful.

Second, how should individual differences be conceptualized in analyses of the functions of social bonds? On the one hand, there is much appeal to the idea that research might enable us to identify "basic" functions of social bonds, or functions that are universally important for well-being. On the other hand, personality characteristics undoubtedly affect the readiness to seek help from others, the need for sociable and intimate interaction, and the responsiveness to feedback from others (cf., Hansson, Jones & Carpenter, in press). There is a risk of tautology if we must posit multiple individual needs on a post-hoc basis to explain empirical results (Reis, in press). How should we resolve this tension between the search for basic (universal) functions and recognition of the importance of individual differences?

Third, what links exist between the structure and functions of social ties? Specifically, are certain types of relationships or patterns of relationships best suited to performing certain functions? Weiss (1973, 1974) argued, for example, that intimacy is best attained through an attachment relationship (such as a relationship with a spouse or romantic partner), while sociable companionship is best attained through ties to a social group. Kin ties have been characterized as better suited to provide instrumental support than emotional support or companionship (e.g., Novaco & Vaux, 1983). If certain types of relationships do indeed become "specialized" to perform particular functions

(Weiss, 1973, 1974), then a single relationship will not suffice to perform the diverse functions that contribute to well-being. Whether a single relationship is sufficient to sustain well-being or whether a set of relationships with certain structural characteristics is needed remains an important question with implications for mental health policy and intervention (Kiesler, 1983).

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LONELINESS RESEARCH: BASIC CONCEPTS AND FINDINGS

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Nothing, the old adage says, is as powerful as an idea whose time has come. In the social science community, the recognition that social relationships are essential to personal health and happiness, that "friends are good medicine," is such a timely idea. Indeed, so fundamental are social ties that several independent research traditions have developed in this area, each with a somewhat different focus. Thus the co-existence of work on social support, loneliness and social isolation attests to the vital importance of social relations. The independence of these research traditions is understandable, given their historical and disciplinary origins. But the time is now right for researchers in these separate fields to become acquainted with each others' work.

My goal in this paper is to outline in broad strokes current research on loneliness, summarizing key findings and identifying conceptual and methodological issues. I think that those who study social support will find work on loneliness both comforting, for the similarities it displays to their own work, and instructive, for the new perspective that it offers (see also Rook, in press).

Loneliness, the distressing feeling that one's social relationships are deficient in some important way, is all too common a human experience. It has been estimated that roughly one American in four has felt lonely in the past few weeks (Bradburn, 1969). Perhaps 10% of the population suffers from severe and persistent loneliness (Peplau & Perlman, 1982). Although some people are at greater risk for loneliness than others, no segment of society is totally immune.

A BRIEF HISTORY OF WORK ON LONELINESS

Although Freud himself did not directly address the problem of loneliness, the earliest psychological discussions of loneliness were influenced by the psychoanalytic tradition. Perhaps the first work on loneliness published in English is a paper by Zilboorg that appeared in 1938. He linked loneliness to personality traits of narcissism, egocentrism, and hostility, which he believed had their origins in faulty parenting during infancy. Fifteen years later, Sullivan (1953) gave loneliness a place of prominence in his theory of personality development. He wrote that in preadolescence, a powerful human need for intimacy first emerges, making teenagers especially vulnerable to the driving force of loneliness. About the same time, Fromm-Reichmann (1959) published an influential paper on loneliness, based on her clinical work with schizophrenics. She, too, emphasized that loneliness is a distressing and powerful experience, often traceable to childhood experiences in the family. A common theme in the works of this period was that loneliness is a painful subjective experience, distinct from the objective state of being alone.

In the 1960s, 64 new English-language publications on loneliness appeared. Some works, such as those by Carl Rogers (1961, 1973), continued to draw primarily on clinical observations. Others, such as The Lonely Crowd by Riesman, Glazer, & Denny (1961), called popular attention to the possible impact of social changes on personal relations and loneliness. Also evident in the 1960s was the beginning of empirical research on loneliness. For example, several sociological surveys investigated loneliness and social isolation among older adults both in the United States and in Europe (e.g., Blau, 1961; Donson & Georges, 1967; Lopata, 1969; Lowenthal, 1964; Shanas et al., 1968; Tunstall, 1967).

In the 1970s, work on loneliness expanded rapidly, spurred by Robert Weiss's book, Loneliness: The Experience of Emotional and Social Isolation (1973). Weiss offered an interactionist view of loneliness as stemming both from personal vulnerabilities and situational constraints on relationships. More recently, loneliness research has taken many directions. Several researchers have developed and validated instruments to assess loneliness (see review by Russell, 1982). In part because of the availability of these loneliness scales, numerous studies have begun to examine the personal characteristics and social behaviors of lonely people (see chapter by Jones in this volume), and to investigate the links between loneliness and personal well-being (see Perlman & Peplau, in press). One indication that work on loneliness has joined "the establishment" is that NIMH has sponsored two conferences on loneliness, one in 1979 (see Peplau & Perlman, 1982) and a second in 1982 (see Peplau & Goldston, in press). Today, research on loneliness is well-established.

A DEFINITION OF LONELINESS

Loneliness has been defined in a variety of ways (Table 1). However, virtually all definitions share three main points of agreement. First, loneliness is a subjective experience and is not synonymous with objective social isolation. As Paul Tillich (1952) observed in The Courage to Be, "Our language has wisely sensed the two sides of...being alone. It has created the word "loneliness" to express the pain of being alone, and it has created the word "solitude" to express the glory of being alone."

TABLE 1

Definitions of Loneliness

Loneliness...is the exceedingly unpleasant and driving experience connected with inadequate discharge of the need for human intimacy, for interpersonal intimacy. (Sullivan, 1953, p. 290).

Loneliness is a sentiment felt by a person....(experiencing) a wish for a form or level of interaction different from one presently experienced. (Lopata, 1969, pp 249-250).

Loneliness is caused not by being alone but by being without some definite needed relationship or set of relationships....Loneliness appears always to be a response to the absence of some particular type of relationship or, more accurately, a response to the absence of some particular relational provision. (Weiss, 1973, p.17).

Loneliness (is) the experiencing of a lag between realized and desired interpersonal relationships as disagreeable or unacceptable, particularly when the person perceives a personal inability to realize the desired interpersonal relationships within a reasonable period of time. (de Jong-Gierveld, 1978, p.221).

Loneliness is the unpleasant experience that occurs when a person's network of social relations is deficient in some important way, either quantitatively or qualitatively. (Perlman & Peplau, 1981, p. 31).

Loneliness (is) the absence or perceived absence of satisfying social relationships, accompanied by symptoms of psychological distress that are related to the actual or perceived absence....I propose that social relationships can be treated as a particular class of reinforcement....Therefore, loneliness can be viewed in part as a response to the absence of important social reinforcements. (Young, 1982, p. 380, italics deleted).

Making this distinction clearly has permitted researchers to investigate empirically the relationship between loneliness and aloneness. As expected, there is a positive, but only moderately strong association between the two.

On the average, lonely people report having fewer friends and less contact with other people (e.g., Jones, 1982; Perlman, Gerson, & Spinner, 1978; Russell, Peplau, & Cutrona, 1980). Both teenagers and adults are significantly more likely to report feelings of loneliness when they are alone than when they are with other people (Larson et al., 1982).

Second, loneliness results from a deficiency in a person's social relationships. This deficiency has been variously described. Some (e.g., Sullivan, 1953; Weiss, 1973) emphasize the notion that basic human needs for intimacy are not being met. Others (e.g., Perlman & Peplau, 1981) take a more cognitive view that there is a discrepancy between the type, quality, or quantity of relationships that a person wants and those that the person perceives himself or herself as having. All agree, however, that some sort of relational deficit is a defining feature of loneliness.

Third, the experience of loneliness is aversive. Although some (e.g., Moustakas, 1975) have encouraged lonely people to emphasize the opportunities loneliness provides for personal growth and insight, lonely people seldom view their experience as pleasant. Loneliness is typically associated with such feelings as depression, emptiness, anxiety, boredom, helplessness, and desperation (e.g., Rubenstein & Shaver, 1982; Russell, Peplau, & Cutrona, 1980).

TYPES OF LONELINESS

Two ways of classifying loneliness have proved useful, one based on the duration of loneliness and a second based on the nature of the relational deficit involved.

Chronicity.

Loneliness can range from fleeting twinges of discomfort to severe and persistent feelings of intense misery. Researchers and clinicians have largely ignored transient feelings of loneliness, and focused instead on more enduring loneliness. Young (1982) recently proposed a distinction among three types of loneliness. Transient or everyday loneliness refers to brief and occasional lonely moods. Situational loneliness occurs when a person has had satisfying relationships until some specific change occurs, such as moving to a new town or getting divorced. Situational loneliness can be severely distressing, but does not invariably last for long time periods. When a person has lacked satisfying social relationships for a period of two years or more, Young

classifies them as chronically lonely. These distinctions in chronicity seem intuitively sensible, and research (e.g., Hojat, 1983) has recently begun to demonstrate differences between situationally and chronically lonely people. Whether acute situational loneliness or chronic loneliness is more harmful to well-being is an important question for further investigation.

Nature of social deficits.

Types of loneliness can also be identified in terms of the specific social deficit involved. Probably the most popular loneliness typology is Weiss's (1973, 1974) distinction between the loneliness of social isolation and the loneliness of emotional isolation. In his view, emotional loneliness is based on the absence of an intimate attachment figure, such as might be provided for children by their parents or for adults by a spouse or intimate friend. Social loneliness occurs when a person lacks a sense of social connectedness or community that might be provided by having a network of friends and associates at work or school. Weiss believes that emotional loneliness is the more serious condition, and that the two types of loneliness are associated with distinctive affects (see Rubenstein & Shaver, 1982).

A consideration of types of loneliness points to one of the most pressing needs for our understanding of social ties, namely the development of a comprehensive analysis of the basic functions of social relationships (see also Rook & Peplau, 1982). Weiss's typology of loneliness grew out of his own analysis of what he called six basic "provisions" of social relationships: attachment, social integration, opportunity for nurturance, reassurance of worth, a sense of reliable alliance, and the obtaining of guidance. Weiss (1974) suggested that no single relationship can provide all of these essential ingredients, and so a satisfactory social life requires a network of different types of relationships. Social support researchers have also proposed taxonomies of types of social support (e.g., House, 1981; Kahn & Antonucci, 1980). For example, Cohen and Hoberman (1983) distinguish among social support that provides belonging, self-esteem, appraisal and tangible aid. In her chapter in this volume, Karen Rook identifies three basic classes of relationship functions: help or assistance; companionship and intimacy; and the social regulations of individual behavior.

Although there is no dearth of proposals about the functions of social relationships, most are based on intuition and common sense, not on theory or data. Nor does there seem to be much consensus about what strategy might best be used to develop a systematic taxonomy of relationship functions and their corresponding relational deficits. Such an undertaking seems essential, however. A taxonomy of relationship functions would permit us to ask, for instance, whether all relational deficits

lead to loneliness. My hunch is that they do not. Does a person who can't get a lift to the airport or who can't find a reliable mechanic feel lonely -- or merely frustrated? A taxonomy of relationship functions might also help to clarify the difference (if there is one) between loneliness and perceived social support, since the two might be linked to different relationship functions. The task of mapping the major classes of relationship functions, of identifying the types of relationships in which these functions can and typically do get met, and linking these to subjective feelings of loneliness and social support seems essential.

HOW TO MEASURE LONELINESS

Given that loneliness is a subjective experience, its measurement must ultimately rely in one way or another on verbal self-report. (For a review of measures assessing loneliness, see Russell, 1982). One common approach has been to ask people one or more direct questions about their feelings of loneliness. A typical survey item comes from Bradburn's (1969) Affect Balance Scale, and asks if the person had felt "very lonely or remote from other people" during the past few weeks. A cross-national survey of older adults (Shanas et al., 1968) simply asked respondents in general how often they were lonely.

In recent years, considerable effort has gone to developing and validating multiple-item loneliness scales, both in the United States (e.g., Rubenstein & Shaver, 1982; Young, 1982), and in Europe (de Jong-Gierveld, 1982). Illustrative of these measures is the UCLA Loneliness Scale (Russell, Peplau, & Cutrona, 1980; Russell, 1982). This scale has twenty items, ten worded in a positive or socially-satisfied direction (e.g., "There are people who really understand me") and ten in a negative or lonely direction (e.g., "There is no one I can turn to"). Respondents indicate how often (never, rarely, sometimes, often) each statement describes them. This scale, like most loneliness measures, is quite global and does not identify the specific type of relationship, such as marriage or friendship, that is missing. (For a new scale that does distinguish types of relationships, see Schmidt & Sermat, 1983). The UCLA Scale performs well on traditional psychometric criteria. It has a coefficient alpha of .94 and a test-retest reliability of .73 over a 2-month period. It has been shown that volunteers for a loneliness clinic and members of such at-risk groups as people seeking help with social skills, divorced adults, and prison inmates score high on the measure. We have also provided evidence for the discriminant validity of the scale, showing that it measures loneliness per se, not merely related concepts such as depression, anxiety, or social inhibition.

The development of reliable and convenient loneliness scales has provided a useful impetus to research, leading to much new

information about the personal and situational correlates of loneliness (e.g., Jones in this volume). These scales have also been useful in assessing the effectiveness of interventions to alleviate loneliness (e.g., Jones, Hobbs, & Hockenbury, 1982).

An important but unanswered question is whether scores on measures of loneliness and of perceived social support are interrelated. The general content of at least some social support measures, such as the Social Support Questionnaire (SSQ) developed by Sarason, Levine, Basham, and Sarason (1983) seems somewhat similar to the content of loneliness scales, although the response format is quite different. Indeed, a recent study (Sarason, Sarason, Hacker, & Basham, in press) found a significant correlation between the SSQ and the UCLA Loneliness Scale. It is also noteworthy that many of the reported correlates of the SSQ, such as anxiety, depression, and introversion are also correlates of the UCLA Loneliness Scale (Russell, Peplau, & Cutrona, 1980). A more systematic comparison of measures of loneliness and social support seems a fruitful direction for future research. One goal would be to determine in which populations and under what circumstances measures of loneliness and social support identify the same individuals as experiencing relational deficits.

THE CAUSES OF LONELINESS

A concern with etiology has characterized much of the research on loneliness. In understanding the causes of loneliness, it is useful to distinguish predisposing factors that heighten a person's vulnerability to loneliness and precipitating events that trigger the onset of loneliness.

Factors that Predispose a Person to Loneliness

Characteristics of the person. It is clear that some people are more susceptible to loneliness than others. A large number of studies (reviewed in Peplau & Perlman, 1982; Perlman & Peplau, in press), conducted mainly but not exclusively with college students, have examined the personality correlates of loneliness. Fairly good evidence links loneliness to shyness, introversion, lower affiliative tendencies, a lack of assertiveness, external locus of control, greater self-consciousness and lower self-esteem. There is also evidence that some lonely people may have faulty social skills. Jones (1982), for example, suggests that lonely people are often self-focused and nonresponsive in their interactions with others. Presumably, these factors affect loneliness because they make it difficult for people to sustain satisfying relationships, or to initiate new relationships to replace ones that have been lost (e.g., by moving or divorce).

Childhood experiences may also influence the likelihood that a person will become lonely. In particular, lonely people have

(or at least believe they have) parents who were colder and less nurturant. In one study (Brennan & Auslander, 1979), for example, lonely teenagers reported greater parental rejection and less encouragement to strive for popularity than did nonlonely teenagers. Another finding is that lonely people are more likely to be the children of divorce (Shaver & Rubenstein, 1980). The younger the person was when the divorce occurred, the greater the probability that they will report loneliness in adulthood. These childhood experiences may affect later loneliness in many ways, such as depriving a person of opportunities to gain social skills, fostering feelings of low self-esteem, or creating feelings of interpersonal mistrust.

Loneliness has also been linked to several demographic factors (see review by Perlman & Peplau, in press). There is a well-documented association of loneliness and age, but counter to stereotype, loneliness is greatest among adolescents and declines with increasing age. Whether this is a developmental pattern associated with aging or a cohort effect due, perhaps, to generational differences in willingness to reveal feelings of loneliness cannot be determined from existing cross-sectional data. There is also good evidence that married people are less likely to be lonely than others. It should be noted, however, that some married people (18% in one large survey, cited in Perlman & Peplau, in press) do report loneliness. Variations in the extent of loneliness among the never-married, divorced, and widowed have not been consistent across studies. Loneliness is common among the widowed but, as might be expected, seems to decline over time (e.g., Lopata, Heinemann, & Baum, 1982).

The issue of whether men or women are more vulnerable to loneliness has not been resolved. In general, no sex differences have been found on multi-item loneliness scales, such as the UCLA Loneliness Scale. On single-item questions, however, such as those used in surveys, women are more likely to describe themselves as lonely than are men. Whether this is due to a real gender difference in loneliness, or to a gender bias in self-disclosure about loneliness is not known. Finally, there is good evidence that loneliness varies by socio-economic status; loneliness is more common among lower income groups. Data on possible racial and ethnic differences in loneliness are not currently available.

Characteristics of the environment. Some social situations are undoubtedly more conducive to loneliness than others, although researchers are only beginning to examine this topic systematically (Jones, Cavert, Snider, & Bruce, in press). Social psychological theory suggests several features of situations that may increase the risk of loneliness. For example, life situations vary in the opportunities they provide for social contact and the initiation of new relationships. Some constraints such as time, distance, and money, are fundamental. The single parent on a tight budget may not be able to afford the babysitter who would

permit time for adult social activities. Constraints may also limit the availability of potential friends and lovers. People who are "different" from those around them--the only black family in the neighborhood, the only single person in the apartment complex--may have fewer opportunities to form relationships. For example, because women live considerably longer than men, older widowed women have fewer prospects for remarriage and are significantly less likely to remarry than are older widowed men. Finally, it is also likely that some social settings foster superficial interactions that may not satisfy intimacy needs. For instance, in a workplace that creates competition or hostility among co-workers, friendly relations are unlikely to develop.

In summary, a variety of personal and environmental characteristics can predispose an individual to loneliness. But predisposing factors do not invariably lead to loneliness. The shy teenager who has been in the same school system since kindergarten may, over the years, have developed a satisfying social life despite being shy. Only when the teenager is faced with making new friends, perhaps when going away to college, will shyness be a problem. The experience of loneliness is triggered by some change in a person's life.

Events that Precipitate Loneliness

Precipitating events are factors such as moving to a new community or separating from a spouse that change a person's social life in some significant way. Precipitating events create a mismatch between the person's actual social relations and the person's social needs or desires. A change in either part of the equation without a corresponding change in the other can create loneliness.

Perhaps most often, loneliness results from a change in the person's actual social relations, such as the loss of an important relationship or separation from a loved one. Divorce, bereavement, and geographic mobility are common causes of loneliness. When Cutrona (1982) asked college students what initiated their experiences of loneliness, the three most frequent answers were leaving family and friends to begin college, the breakup of a dating relationship, and problems with a friend or relative. A decrease in the quantity or quality of social ties is a typical precursor to loneliness.

Loneliness can also be triggered by events that change a person's social needs or desires. Our needs for human contact are not constant. When we are working feverishly on an overdue report, we may crave solitude and try to avoid people as much as possible. When we are sexually aroused, we may ardently desire the company of a particular partner. One implication is that feelings of loneliness are probably not constant through out the day or week, but rather vary depending on our goals and activities (Larson et al., 1982). In addition to these short-term

fluctuations in social needs, more major changes can also occur. Ten-year-old children seldom complain of loneliness because they lack a boyfriend or girlfriend. But sometime during adolescence, through a combination of maturation and changing social expectations, the desire for a dating partner becomes intense. If the right partner does not materialize, the teenager's feelings of loneliness may become equally intense. The general point is that changed social needs and wants that are not accompanied by appropriate changes in actual social relations can precipitate loneliness.

COGNITIVE FACTORS IN LONELINESS

The possible importance of cognitive factors in the experience of loneliness has frequently been discussed, although empirical research is limited. One factor concerns the personal standards that people use in evaluating their social relationships (Peplau, Miceli, & Morasch, 1982). Subjective assessments of the quantity and quality of interpersonal ties are comparative, and involve judging oneself against a variety of standards. These standards may not be consciously-articulated, but they are evident in the complaints of the lonely: "I don't have enough friends" or "No one really cares about me." Both our own past experience and social comparisons with others influence our judgments of the adequacy of our social ties. For example, one study (Perlman & Goldenberg, 1981) found that students who believe they have fewer friends than their peers are likely to be lonely. Another study (Cutrona, 1982) demonstrated that satisfaction with current relationships is affected not only by comparisons with peers, but also with one's own previous relationships. We need to know more about these subjective standards for evaluating the adequacy -- or inadequacy -- of social ties.

Once people decide that their social life is inadequate, they are typically motivated to try to understand the causes of their plight. Discovering the reasons for one's loneliness helps to make sense of a difficult situation and may suggest possible remedies. Our research at UCLA (Peplau, Russell, & Heim, 1979; Michela, Peplau, & Weeks, 1982) indicates that people give varied explanations for loneliness, varying along dimensions of internality (blaming the self versus external causes) and stability (citing unchangeable causes versus changeable ones) (Michela, Peplau, & Weeks, 1982). There is some evidence that depression, a common correlate of loneliness, may be most likely when a person attributes their loneliness to internal, stable causes, such as being physically unattractive or having an unpleasant personality. Coming to blame oneself for loneliness may also contribute to the frequent link between loneliness and low self-esteem (Peplau, Miceli, & Morasch, 1982). Finally, although good data on this point do not exist, it has been

suggested that when loneliness persists over time, there may be a typical attributional shift toward more internal and stable causes, with a related increase in depression and decrease in self-esteem (Peplau, Russell, & Heim, 1979).

LONELINESS AND PSYCHOLOGICAL WELL-BEING

Loneliness researchers have usually started with the premise that loneliness is itself a distressing and harmful experience, worthy of study in its own right. Hence it has not necessarily been considered important to demonstrate that loneliness leads to other forms of mental disturbance. Nonetheless, in recent years, there has been growing interest in identifying the harmful mental health consequences of severe and persistent loneliness (Peplau & Goldston, in press).

The most firmly established link between loneliness and psychological well-being is the correlation of loneliness and depression. Studies using short self-reports of depression find that people who say they are lonely also say they feel depressed (e.g., Perlman, Gerson, & Spinner, 1978; Russell, Peplau, & Ferguson, 1978). Studies using longer depression scales such as the Beck Depression Inventory also find a strong association between loneliness and depression (e.g., Bragg, 1979; Russell et al., 1980; Weeks, Michela, Peplau, & Bragg, 1980; Young, 1982). At the same time, it is important to recognize that loneliness and depression are distinct although partially overlapping phenomena (Russell, Peplau, & Cutrona, 1980). Not all lonely people are depressed, and not all depressed people are lonely. This finding led Bragg (1979) to propose a distinction between "depressed loneliness" and "nondepressed loneliness." In a study of college students, Bragg found that depressed loneliness was associated with fairly global negativity, seen in dissatisfaction not only with social relations, but also with school, work, and many facets of life. In contrast, nondepressed lonely people expressed dissatisfaction only with their social relations; they were not necessarily unhappy about other aspects of their lives. More recently, Young (in press) has proposed that "lonely depression" be considered a major subtype of depression in which social deficits are of central importance.

Evidence about loneliness and other aspects of psychological adjustment is more limited. Diamant and Windholz (1981) found a strong relationship between loneliness and scores on Zung's Clinical Index of Suicide Potential. Lonely people have been found to score higher than the nonlonely on measures of neuroticism (Berg et al., 1981; Diamant & Windholz, 1981; Hojat, 1982). In structured psychiatric examinations, lonely old people were more often judged to have mental symptoms needing treatment (Berg et al., 1981). Among college students, loneliness has been linked to indices of poor personality integration and general

maladjustment from the Tennessee Self-Concept Scale (Goswick & Jones, 1981). Among adolescents, loneliness is associated with poor grades, expulsion from school, running away from home, theft and vandalism (Brennan & Auslander, 1979). One survey (Rubenstein & Shaver, 1982) found that lonely people report more psychosomatic symptoms such as headaches, poor appetite, and tiredness. In summary, studies of loneliness are consistent with studies of social support in showing that perceived relational deficits can be detrimental to mental health.

It should be noted, however, that in studies of loneliness and psychological adjustment, most issues of causality are currently uncertain. Studies of loneliness and psychological well-being have typically been correlational and have used cross-sectional data. In some cases, the direction of causality is clear. For example, it seems improbable that loneliness causes bereavement. In other cases, however, it is a matter of judgement whether loneliness is considered the cause or the effect, or whether causality is considered to be bidirectional. For instance, it is likely that chronic loneliness can lead to depression, but it is also possible that depression may itself lead to disruptions in a person's social relationships that result in loneliness.

LONELINESS AND SOCIAL SUPPORT

In reviewing loneliness research with an eye to work on social support, three unresolved issues are salient.

First, what is the relationship between loneliness and perceived social support? In a very general way, the concepts of loneliness and social support can be viewed as opposites: loneliness refers to the experience of deficits in social relations; social support refers to the availability of interpersonal resources. But a closer examination reveals differences of emphasis between the two areas (see Rook's analysis elsewhere in this volume). For instance, social support research has given prominence to instrumental assistance, a theme largely absent from studies of loneliness. It is not clear whether researchers should be encouraged to use the concepts of loneliness and perceived social support in more restricted and differentiated ways, or to treat them essentially as synonyms. A key empirical question is whether scores on measures of loneliness and perceived social support are consistently highly correlated. In other words, are both measures identifying the same individuals as having problematic social ties?

Second, what are the basic functions of social relations, and which of these are most vital to personal well-being? Both loneliness and social support research make assumptions, seen most clearly in the content of measuring instruments, about the important features or functions of social bonds. We need a more

systematic effort to analyze the diverse content of social exchanges, and to develop taxonomies of key functions. Only then will we be in a position to ask which kinds of social deficits are most detrimental to mental health.

Finally, what are the links between objective characteristics of social relations and perceptions of loneliness and/or lack of social support? Both research traditions distinguish between objective features of social relations (e.g., frequency of contact, amount of time together, type of relationship, reciprocity of exchange) and subjective perceptions of social ties. We are only beginning, however, to understand the links between these objectives and subjective experiences, and the processes by which an individual translates one into the other. This is an important direction for future research, and one where loneliness and social support researchers might profitably share ideas and methodologies.

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PERCEIVED SUPPORT AND SOCIAL INTERACTION AMONG FRIENDS AND CONFIDANTS

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In this paper we will review the research we have been doing over the last few years attempting to differentiate among the various facets of the social support construct. We will cite both correlational questionnaire studies as well as behavioral studies, and will conclude with the opinion that the effects of support are much more complicated than the original proponents of the concept had imagined. In all probability, we will raise more questions than we answer, and will offer the hypothesis that the effectiveness of support may not reside in any particular behavior of significant others, but in how that behavior and ensuing relationships are perceived. We also will suggest that individuals are not simply passive recipients of support, but in part, determine the nature of the supportive behavior they receive from others.

The original epidemiological research that sparked the current wave of interest in social support demonstrated that psychosocial assets, broadly defined to include both personal and environmental characteristics, had a role in positive health outcomes (de Araujo, van Arsdell, Holmes & Dudley, 1973; Nuckolls, Cassel, & Kaplan, 1972). This message was important in order to counter the then dominant medical belief that disease and symptomatology were exclusively programmed by internal biological variables (Cassel, 1976; Cobb, 1976). Unfortunately, the field did not go beyond the demonstration that environmental and life style variables influence health outcomes. Little attention was given to the fact that the predictors used were extremely heterogeneous and were confounded with other psychosocial variables, such as personal competence (Heller, 1979). Researchers worked toward repeated demonstrations of predictive validity (that support influenced health) and paid little attention to construct validity (Heller,

Swindle, & Fodacaro, 1982). The result is that we still have little idea about effective, ineffective, or noxious modes of support. Despite a growing literature on social support, the exact mechanisms by which social support may exert an influence on health and well-being remain unknown.

Until recently, the field was marked by conceptual ambiguity (Heller, 1979; Thoits, 1982). Previous studies often used only global measures of social support and did not clearly distinguish its different facets. The failure to explicate the relationship between the various facets of support and other aspects of the coping process probably accounts, in part, for the equivocal results of past studies of stress buffering. Attempting to better operationalize the social support construct, Heller and Swindle (1983) distinguished between four aspects of social support: the structure and function of social networks, the personal attributes and competencies of individuals that make it likely that social networks will be accessed and maintained, the cognitive appraisal that support is available, or that further help is needed and finally support seeking behaviors that result from the appraisal process (see figure 1). Specifically, whether an individual uses

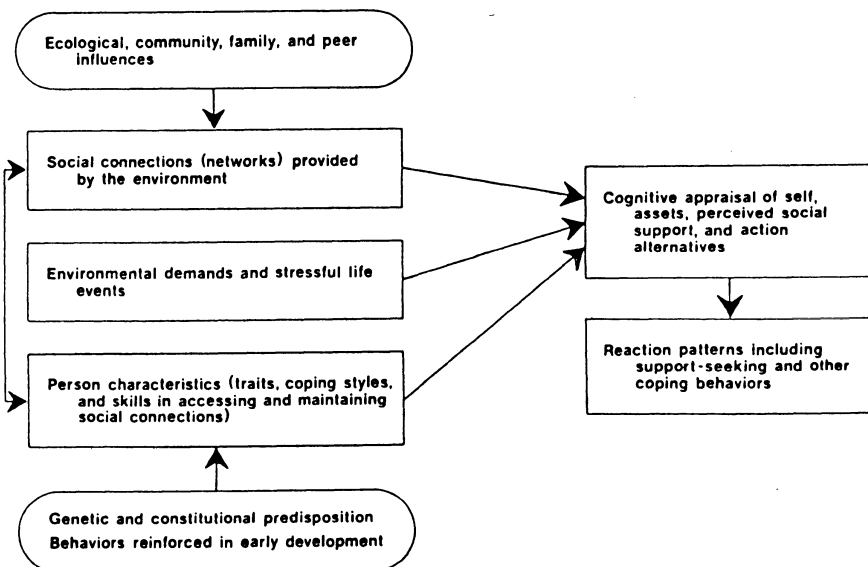


Figure 1. A Model of Social Support and the Coping Process.

support effectively during stressful events depends upon support availability, personal competencies in accessing supportive others, and the reaction patterns developed over time in which seeking support from others becomes part of an effective coping strategy.

Factor analyses which we conducted in a number of studies revealed distinct facets of support which were related to symptomatology and other outcome measures in different ways. Two studies can serve as illustration.

THE IDENTIFICATION OF FACETS OF SUPPORT

In a doctoral dissertation by Ralph Swindle, 113 college students (63 males and 50 females) were asked to complete a battery of support and competency measures. Factor analyses of subject responses revealed separate interpretable factors for support and competence variables. Within the support domain, Swindle was able to differentiate between:

Support seeking - the number of persons contacted for stressful events in which support was sought, and the rated helpfulness of that support.

Perceived support from friends and family - the Procidano and Heller Perceived Support Scales and the Cohesiveness subscale of the Moos Family Environment Scale.

Support transactions - the percentage of events in which emotional support or problem solving help was received when support was sought.

Network structure - the percentage of events for which peer support or parent support was considered most helpful.

Not only were facets of support clearly distinguishable, but they were independent of competence measures that were used in this study (The Haan scales of coping and defense, Joffe & Naditch, 1977; and Levenson & Gottman, 1978 social competence measures). The finding that facets of support could be distinguished from measures of social competence was particularly gratifying because we had been concerned that some of the findings in the literature reporting positive effects of support might have been due to the operation of social competence (Heller, 1979). That social support and social competence are distinct constructs that can be measured independently should encourage researchers to include measures of both constructs in future work so that any differential association with health and well-being can be determined. Indeed, in the Swindle study, measures of social competence were more highly related to reported symptomatology (MMPI-D; the brief Symptom Inventory, Derogatis, Lipman, & Covi, 1973) than were any of the support indices. Among the support measures, only Perceived Family Support (1983) was consistently

related to reported symptomology.

A second study by Fondacaro and Heller (1983) expands upon and confirms these findings. In this study, social network and perceived support measures were administered to 137 male college students along with measures of alcohol use and psychological symptomatology. As Table 1 indicates, a factor analysis of the support and competence scales reveals independent dimensions of network functions, perceived intimacy/support and social competence. This is basically a replication of the Swindle finding of separate facets of support, which can be differentiated from measures of social competence. In addition, Fondacaro and Heller found that the best predictors of self-reported drinking were the number of network members who drink, network density and the amount of social contact with network members. Drinking was

TABLE 1

Social Network, Perceived Support, and Social
Competence Variables and Factor Loadings

Factor and Variance Accounted For		
1. Network Functions (37.1%)	Multidimensionality	(.75)
	Emotional Support	(.74)
	Informational Support	(.72)
	Problem Support	(.72)
	Material Support	(.65)
	Companionship Support	(.46)
2. Perceived Intimacy/Support (20.6%)	Network Intimacy	(.70)
	Very Intimates	(.64)
	Intimacy with Father	(.61)
	Intimacy with Mother	(.53)
	Perceived Support-Fr	(.49)
	Perceived Support-Fa	(.45)
3. Residual (16.1)	Proportion of Males	(.96)
	Proportion of Females	(-.96)
4. Social Competence (11.4%)	Dating Competence	(.86)
	Assertiveness	(.58)
	Heterosexual Intimacy	(.39)

(from Fondacaro & Heller, 1983)

more likely to occur when an individual had high contact with a dense network of drinking buddies. Here was an instance in which social ties were associated with an increase in problem behavior (moderate to heavy drinking) rather than its diminution.

Overall then, we would conclude that previous research has treated the social support construct in a too global and undifferentiated manner. Facets of social support can be identified, and their relationship to outcome measures depends upon the context and group studied. Not only is there a greater need for precision in measuring social support, but greater care must be paid to the outcome measure as well. General measures of life satisfaction and morale, psychological symptomatology and specific problem behaviors are not equivalent; their links to facets of social support must be separately established.

RESEARCH ON PERCEIVED SOCIAL SUPPORT

Our work on perceived support came from the recognition that a distinction should be made between the objective characteristics of social networks and the impact that such relationships have on the individual. Thus, a large portion of our work has been concerned with the Perceived Support construct. As a first step, Procidano and Heller (1983) developed a Perceived Social Support Inventory which was designed to measure the extent to which an individual perceives that his/her needs for support are fulfilled by friends (PSS-Fr) or family members (PSS-Fa). The research using this instrument can be summarized as follows:

1. Among college students, perceived family support was found to be inversely related to the report of neurotic symptoms (Procidano & Heller, 1983; Swindle, 1983). Perceived friend support was inversely related to reported feeling of loneliness but was unrelated to amount of social contact (i.e., time spent with friends; Corty & Young, 1980). Loneliness and perceived support are both summary appraisal judgments, which can be made independent of specific behaviors, such as time spent with friends.

2. Among former mental patients, perceived support from friends and family was lower for recovered schizophrenics than for patients recovering from an affective disorder. Perceived support reported by both patient groups was lower than support reported by normals (Frame, 1981). Among adult male alcoholics, nonrelapsers were higher in perceived support than were relapsers (Rosenberg, 1979).

3. Daughters of alcoholics and mental patients reported perceiving less family support during their teen years than did daughters of normal fathers (Benson, 1980). Furthermore, the best predictors of the current positive adjustment of daughters of alcoholic fathers did not include the father's reported drinking level. Rather, daughters' adjustment was related to reported

absence of conflict in the home, the ability to relate to the alcoholic parent, and the level of perceived family support.

The above represents a summary of research findings for studies of perceived support conducted at Indiana University. The Procidano and Heller (1983) scales are being used by others as well and we can report on the results of two such studies. Steinmetz, Lewinsohn, and Antonuccio (1983) found that perceived family support was one of seven significant predictors of successful outcome in a psychoeducational group treatment program for depression. This study was conducted with a community sample of 75 individuals who met the Research Diagnostic Criteria (Spitzer, Endicott, & Robins, 1978) for unipolar depression. In a second study, Frazier (1982) found that among low-income, elderly community residents living in New York City, perceived support, as measured by the Procidano and Heller scales, were better predictors of both self-esteem and depression than was frequency of contact with friends or relatives.

EXPERIMENTAL STUDIES OF SUPPORTIVE INTERACTIONS

While the studies reported thus far seemed promising, we were troubled by the correlational nature of the evidence we had been accumulating. Could we not go beyond self-report measures and questionnaire studies, the type of research that had become the norm in the social support field? A major problem with this body of research is its vulnerability to alternative explanations. For example, it may be said that social support is more a reflection of personal characteristics than social networks. Individuals who are more socially competent are more likely to show better adjustment and higher levels of social support than are less socially competent persons (see the chapter by Sarason & Sarason in the volume). We hoped to obtain causal evidence concerning the effects of social support by conducting experimental studies. While subjects might report a stress buffering effect for social support, would individuals placed in a stressful situation display greater competency and less emotional arousal if given an opportunity to interact with a close friend?

In general, the results of our experimental studies were not uniformly positive in that supportive interactions with others did not aid performance during exposure to laboratory stressors. We suspect that the link between support perception and supportive behaviors is more complicated than most of us initially imagined.

In a study by Deborah King (1979), male and female college students were placed in an experimental setting in which they were led to anticipate receiving an electric shock either in the presence of a friend, a stranger or alone. While previous studies concerning social affiliation under stress have produced equivocal results (Bond & Titus, 1983; Epley, 1974) most of this research has been conducted with subjects and companions who did not know

one another. It seemed reasonable that during stress, affiliation among friends should have beneficial effects (see positive findings by Davidson & Kelley, 1973; and Kissel, 1965). King found that women, but not men, who anticipated shock in the presence of a stranger showed higher levels of physiological arousal than women anticipating shock in the presence of a friend or who were by themselves. Thus, strangers were found to increase arousal under stress. However, a friend did not produce a calming effect. Anticipating shock in the presence of a friend did not reduce arousal levels for either men or women compared with anticipating shock alone.

We had followed the custom in the social affiliation literature, in not allowing friends and companions to talk to one another, thus testing the effects of "mere presence". Perhaps reducing the fear of shock in a strange psychology laboratory was too much to ask of any friend under these conditions, so we began searching for a more natural stressor.

A second experimental study was conducted by Ralph Swindle (1983) as part of his dissertation. College student males were asked to participate with their closest male friend. Subjects took several support and personality questionnaires and then were assigned to listen to and respond to a stressful problem solving tape. Midway through the tape, there was a break, and subjects were allowed to think and take stock of the situation either alone, in the presence of another student whom they did not know (stranger), or in the presence of a close friend. The dependent variables in this study were measures of problem solving effectiveness and ratings of emotional arousal. We wanted to know whether the opportunity to talk over a problem with a friend would lead to greater problem solving effectiveness and less arousal than would dealing with that problem alone or in the presence of a stranger.

The problem situation asked the subject to role play a discussion with his girl friend who had just announced that she may be pregnant and who asks the subject to accompany her to the student health center for a pregnancy test. The story line that unfolds in the role play provides the subject with information that could lead him to question the girl friend's fidelity. The pause in the tape at midpoint allowed the subject to discuss the tape (or the entire experiment) with his friend, with a stranger or allows the opportunity to plan subsequent responses alone. The problem of a potentially pregnant girlfriend was chosen for the role play stressor because male college students had previously reported that it was a frequent and common worry or concern.

Results indicated that high perceived support (PPS) subjects in interacting with their friends spent more time talking (less silence) and more time discussing the task than did low PPS subjects. They spent less time in irrelevant (non-task) talk and less time talking about the experiment in general. When talking to a stranger, high PSS subjects also spent more time talking than

did low PSS subjects--but their discussion with a stranger was not task relevant, that is, there was more non-task talk with the stranger.

Also, an affiliation preference was demonstrated in this experiment. After the initial set of problem situations was presented, but before the interaction, subjects were asked whether they wanted to talk to their friend about the problem situation. Seventy percent of the subjects said that they wanted to do so, and indeed, their talk with friends was task relevant. Of all the pre-test variables, the best predictor of the desire to talk about the situation was perceived friend support ($r=.36$). However, the interaction with the friend had no effect on later problem solving competence or rated emotionality in responding to subsequent role-play problem situations. Being with a friend and talking about the problem situation, even for subjects who wanted to do so, did not affect later task performance.

We can speculate about any number of reasons why interactions between male friends might not have improved problem solving or reduce emotionality in this situation. For example, companions may have found it too difficult to provide effective support in this situation. Also possible is that male friends in talking about a possible pregnant girlfriend suspected of infidelity, might offer advice that would be rated as noncompetent by peer judges (e.g., "dump her"). However, what should be emphasized is not our ability to explain away nonsignificant findings, but their possible validity. What this experiment may be telling us is that social support may affect motivation for interaction, and level of interaction behavior (task talk)-- but not coping behavior in stressful situation.

BEHAVIORAL OBSERVATION OF SUPPORTIVE DYADS

In this series of studies we were interested in distinguishing between the behavior of intimate and less intimate dyads. In the first study, Procidano and Heller (1983) studied interaction rates between companions. Subjects reported for an experiment with either their older sibling or a close friend. Subjects were told that they would be left in a room with their companions while we supposedly were readying the equipment. In actuality, their verbal interaction during the five minute wait period was being recorded. It was found that individuals who reported high levels of perceived support from their friends, spoke longer in interaction with friends or sibs, and spoke about themselves more than did individuals of low perceived friend support. Individuals with high perceived support from friends were more open and disclosing to both friends and sibs. Perceived support from family members affected interaction rates with companions in a more specific manner. Perceived family support only affected interactions with sibs, in that individuals who reported low

family support when interacting with their sibs showed a marked verbal inhibition compared with members of the other groups.

This study suggested that interaction rates could distinguish between high and low support dyads. The next step was to determine whether the content of support provided by high and low support dyads also could be differentiated. In a second study, the current authors observed the behavior of high and low supportive friends and content analyzed their conversation. High and low perceived support subjects were asked to participate with a close friend. Subjects were asked to tell their friend about a current worry or concern in a study to determine "how friends help one another with the hassles and problems of daily living". Companions were told to "behave as you normally do when your friend tells you about a personal difficulty". Again the subjects were told that the experimenter was not ready and that they could "sit and chat for about five minutes". After the five minute anticipation period, subjects were told that the experiment would begin. They spent fifteen minutes talking about a worry or concern of the designated subject and then waited together during a post-experiment five minute period. At the beginning of this third period subjects were told that the experiment was over, but in fact the tape recorder was kept running.

The content of the conversations between high and low perceived support subjects could not be distinguished during any of the three experimental periods. Subjects displayed remarkable consistency in their friendship behaviors. Companions, for the most part, acted in the role behaviors they thought friends should adopt--asking questions, offering minimal encouragers or describing their own personal experiences. Only in the third period, when the subjects thought that the experiment was over did we begin to see some of the interaction level findings reported by Procidano and Heller. Perceived support, a summative rating measure of support experienced from friends as a group, was not predictive of subject and companion talk, but rating of intimacy with the specific companion brought to the experiment was. Intimate friends continued on task, talking about the subject's problem in the third period even though they were no longer required to do so.

In addition, there was an interesting interaction between the subject's support preference and the companion's behavior during the third period. Brian Lakey had previously developed a questionnaire to measure support seeking strategies. As used in this experiment, we found that in the third period, problem oriented support seekers (those who said they would "seek out a knowledgeable friend who would offer suggestions about how you could solve a problem") actually received more advice from their companions ($r=.40$; $p < .001$). On the other hand, emotional support seekers (those who said they would seek out a friend who could make them "feel better about the problem" or who could "take their mind off the problem") received less advice ($r= -.27$;

$p < .03$) and fewer interpretations ($r = -.25$; $p < .05$) from their companions, but received more frequent descriptions of others' personal experiences ($r = .38$; $p < .002$). These admittedly tentative findings offer the possibility that individuals either choose friends who are adept at meeting their specific support preferences, or that close friends know what their friends want and are willing to provide it.

DISCUSSION AND CONCLUSIONS

The literature on social support has treated the construct in a much too global fashion. It seems that the concept "social support" has come to stand for any type of social interaction that is associated with health and well-being. Clearly, progress will not be made until we have a clearer idea of the specific mechanisms by which support may operate. How can we hope to develop intervention programs to "increase social support" without first understanding these mechanisms?

In our own research, we have found that distinct facets of support can be distinguished, and that the various facets have different relationships to various mental health outcome measures. For example, there is a closer link between perceived support and perceived mental health status than there is for network measures of support, support-seeking behavior, or support transactions. When one considers the multifaceted nature of social support, perhaps it should not be surprising that we were unable to identify the specific behaviors that distinguished between high and low perceived support dyads. Even though intimate dyads continued talking about problems when they were no longer required to do so, high support dyads did not show differential amounts of advice, agreement, interpretations or reflections of feeling.

There were differences between the behavior of close friends and less intimate dyads, but our behavioral findings occurred in unanticipated directions. Our subjects seemed to know the cultural expectations for friendship, so when we asked them to role-play a discussion of a personal problem, they showed the role behaviors associated with friendship--asking questions about their friends' problem, offering minimal encouragers and describing personal experiences of others that are related to the topic under discussion. These behaviors were demonstrated by the majority of friend dyads regardless of their closeness. It was only when they thought they were no longer being observed did differential behavior of intimate dyads appear. Close friends continued talking about the problem they had previously initiated, while more distant dyads switched to non-problem related topics. Even so, close friends did not engage differentially in behaviors we usually associate with "support"--that is, they did not show differential amounts of advice, agreements, interpretations or reflections of feeling. We received the distinct impression that

while these behaviors are what psychologists think of as "support", college students do not talk or think like psychologists. If they behave supportively with one another, it is in other ways. One intriguing and unanticipated finding was the synchrony between the style of support seeking that subjects had endorsed in a testing session two weeks to two months before the experiment, and the later observed behavior of close friends. Problem-oriented support seekers received advice from their close friends while emotional support seekers were told about similar problems of others. In other words, the behavior of close friends matched the subjects' support seeking style. We wondered to what extent subjects "shaped" their environments, choosing friends who met their needs. We also wondered whether friends shaped their behaviors toward what they perceived to be the subjects' style of support seeking.

Is it likely that our research results, as well as similar findings of others, are confounded by common methodological biases? This is a real possibility when support and outcome measures are assessed only from the perspective of a single individual. The same biases that lead to the perception of high and low support also can lead to expansive or constricted network descriptions or to positive or negative reports of well-being. What are needed are outcome measures that are "outside the head" of the subject. This can be accomplished either by obtaining confirmation of support status by significant others who know the subject well, or by conducting behavioral observation or experimental studies similar to those described in this paper.

We did several experiments in which we looked at the behavior of close friends and were unable to demonstrate that friends improve performance under stress. To be sure, we did find affiliation preferences--subjects reported that they would rather experience stress in the company of a friend than alone--but we did not find any increments in their problem-solving ability or any decrements in physiological arousal associated with the presence of a close friend. Perhaps our methodology was inappropriate in that the help friends provide may be more subtle than we were able to observe. Friends may find it difficult to provide effective support in laboratory situations, or positive effects may not appear until some time after the person has experienced the stressor. But there is an alternative interpretation. The positive effects of support, when they occur, may not be in the specific behavior of close friends, but in how that behavior is perceived and interpreted. Believing that you have friends who could help you in time of need may be more important than the help they actually provide.

While our research has emphasized perceived support, the factors that contribute to the perception of support have yet to be clarified. Blazer (1982) comes to a similar conclusion in his longitudinal study of social support among the elderly. Blazer found that elderly persons with low perceived social support had a

mortality rate that was 3 1/2 times higher at a 30-month follow-up period than did elderly with adequate to good perceived support. Yet he cautions that "those aspects of perceived support that are important predictors of mortality have yet to be determined" (p.693). Believing that one is part of a group, or that someone else "cares" have been suggested as important ingredients in support. However, more prosaic mechanisms also are possible. For example, supportive others may increase the likelihood that health enhancing behavior will occur, such as encouraging visits to physicians when symptoms are reported, or insisting that proper diet is maintained.

In conclusion, we hope that future research moves beyond the demonstration phase (i.e., Does social support have an effect?) to a more careful explication of the processes underlying the support phenomenon. The structure, function and quality of relationships with others in one's network; individual competency levels; the nature of environmental demands and whether they can be modified; and how these variables are perceived and appraised; all contribute to eventual coping and adjustment, and should be studied.

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PART IV

STRESS, COPING AND MALADAPTION

LIFE STRESS AND HUMAN DISORDER: CONCEPTUALIZATION AND MEASUREMENT
OF THE DISORDERED GROUP

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Since the publication of the Dohrenwends' classic book a decade ago (Dohrenwend & Dohrenwend, 1974), substantial improvement in the conceptualization and measurement of life events and social support has occurred. New, improved inventories appear regularly, more complex models are proposed, and advances in statistical sophistication are particularly noteworthy. In spite of these efforts, Rabkin and Struening's (1976) conclusion, in reviewing the pre-1976 research, that life events account at the very best for 9% of the explained variance in predicting human disorder(2) still holds true. Although the evidence on social support is less extensive, recent studies which control for other input variables suggest that support is also a weak predictor of outcome status (e.g., Monroe, 1983; Williams, Ware, & Donald, 1981). In almost reflexive style, social scientists have viewed the poor predictive power of these input variables as due to inadequacies in measurement, research design, and theory. Such inadequacies may indeed prove to be limiting factors, but the possibility that other factors may account for the limited predictive power of event and support variables seems worth entertaining.

In proportion to the attention afforded input variables in the stress-disorder equation, there are two factors that investigators have largely ignored. The first is the conceptualization and measurement of disorder itself, or, in other words, of the variable which is being predicted. The second is the mechanism or pathway by which stress effects a pathological response. As Hinkle (1977) noted some time ago, stress-disorder relationships will only be meaningfully understood when specific

mechanisms of the relationship are studied. But, of course, in order to study specific mechanisms, it is first necessary to accurately conceptualize and measure human disorder. Therefore, we will focus on this latter issue.

Current Conceptions and Measurement of Disorder

The majority of studies in the stress-disorder area, particularly those in recent years, have treated disorder in a totally undifferentiated manner. Unless a study is concerned with a particular disease, specific disorders are rarely distinguished. Disorder is viewed as a unified concept, and life stress is apparently assumed to act uniformly on the heterogeneous array of conditions that must exist in the disordered group.

This conception of disorder has dictated the type of measurement of disorder applied in this research. Disorder is measured in a very nonspecific manner: that is, the predicted entity is not generally measured as a specific type of disorder. Rather, a heterogeneous mixture of either physical complaints or psychological affective states or both are assessed. Usually, the list of physical complaints is not comprised of symptom complexes of specific disease categories; instead, somatic changes generally associated with stress reactions are assessed, such as headaches, acid stomach or indigestion, insomnia, heart beating hard or pounding, and poor appetite (Holahan & Moos, 1981; LaRocco, House, & French, 1980). The same form of assessment is applied to so-called psychological disorder. Almost all of the scales are loaded heavily with negative affect items (mainly anxiety, irritability, and depression), while a few incorporate general indicators of emotional disturbance, such as life dissatisfaction and a lack of positive well-being (Williams, Ware, & Donald, 1981). In a critical review of these types of scales, Goldberg (1972) demonstrated that they assess mainly the negative affect associated with general emotional disturbance; no legitimate statements about specific psychopathologies may be made. Even many scales with nominal specificity (such as the Center for Epidemiological Studies-Depression Scale) identify general emotional disturbance as opposed to specific primary disorders (Boyd, Weissman, Thompson, & Myers, 1982).

It seems then, that the predicted entity in most current research represents a host of conditions characterized by general negative affectivity and stress-responsive somatic complaints. Under these circumstances, it is obviously impossible to reach any conclusions concerning the relation of stress to a specific disorder. More disturbingly, this form of assessment has the unstated underlying assumption that the nature of the relationship between life stress and human disorder is similar across disorders. This assumption is almost certainly wrong. Surely the effect of stress on disorder will vary in quantity and kind depending on the psychobiology and stress-reactivity of the

disorder in question. It seems to us that only the broadest, most undifferentiated principles of stress-disorder relationships may be derived from such a measurement approach.

In order to make an inroad into understanding the nature of what is currently being predicted, it is useful to examine the best predictor of disorder. In practically every case where it has been assessed in a longitudinal design with at least two measurement occasions (Time 1, Time 2), the level of prior disorder (Time 1) is by far the most powerful predictor of subsequent disorder (Time 2) (e.g., Monroe, 1983; Thoits, 1982; Williams et al., 1981) accounting for as much as 30% - 40% of the total variance in disorder one year later. In comparison, events account for 1% - 9% of the variance, generally around 3%. Furthermore, in the few instances where the effects of prior disorder have been controlled statistically, the social support variable contributed 0% - 1% of the variance in psychological disorder (Monroe, 1983; Williams et al., 1981).

The meaning of these findings is clear: the bulk of high-scoring respondents on the current measures of disorder in general population-based samples are characterized by psychobiologic traits that result in a chronic, though perhaps fluctuating, state of physical and/or emotional disturbance. Moreover, environmental events and supports, as currently assessed, do not appear to be significant, independent predictors of the level or course of their disorder. As to the size of this group, there are several suggestions that it represents up to a quarter of general population samples. For instance, on the basis of their massive epidemiologic work, Link and Dohrenwend (1980) estimated that 25% of the community population is psychologically distressed at any point in time, and our discussion above suggests that this distress is best predicted by prior distress. Thus, the majority of this group is probably characterized by chronic conditions. More directly to the point, both Hinkle (1974) and Thurlow (1967) have demonstrated that approximately one-quarter of community samples exhibit chronic patterns of psychological distress, life and job dissatisfaction, and a wide variety of medical disorders. These individuals show a persistent, general susceptibility to psychological and physical disturbance over many years of follow-up.

In terms of more specific psychopathologies, the rate of chronic conditions varies with the disorder. In affective disorders, however, Brown (1979) found that over 50% of depressive cases in a large community sample exhibited a chronic pattern, and this figure is close to the rate found for both chronic subsyndromal (Depue, Slater, Wolfstetter-Kausch, Klein, Goplerud, & Farr, 1981) and full syndromal depressions (Depue & Monroe, 1978).

Thus, across various types of populations, a chronic pattern of disorder exists at a rate of about 25% or higher, and so the suggestion that individuals with this pattern comprise the bulk of

high-scoring respondents on measures of general disorder in most life stress research seems highly plausible.

Composition of the Chronically Disordered Group

Certainly, among the high-scoring respondents on scales of disorder are individuals who are displaying an acute episodic pattern of disturbance: from a generally normal baseline, they will show a sharp, relatively rapid elevation in disorder, only to return to normal baseline levels as the episode or stressor subsides. In view of the weak predictive power of events and support variables, it is either the case that acute episodic patterns are not stress-initiated in any regular fashion, or that they are stress-initiated but constitute too small a subgroup relative to chronic conditions to significantly affect the stress-disorder correlation. It may be that the most fruitful approach will be to examine acute episodic and chronic patterns of disorder separately; if so, then the composition of both the chronic and acute groups should be defined so that they may be more precisely measured.

Outside of the acute episodic forms of disorder, there are at least four subgroups that comprise the disordered group. All are characterized by a chronic, although perhaps fluctuating or intermittent, pattern of disorder.

Personality Traits and Disorders

Because of the heavy loading of negative affect items on scales of psychological disorder, one can expect a group of high-scoring individuals on these scales who have personality traits, or even diagnosable personality disorders, that result in the persistent subjective experience of negative affect, life dissatisfaction, fatalism, and alienation. That such a group exists is suggested by the fact that almost all personality inventories, when factor analyzed, yield two major factors or personality dimensions. In past years, one of these factors was usually labelled neuroticism, which is characterized especially by high trait anxiety. However, Watson and Tellegen (1984) have shown that the neuroticism factor, when defined by subjective experience, may be more broadly conceived as a dimension of Negative Affectivity. Ranging from a low point defined as a calm, relaxed state, this dimension is characterized at higher levels by negative affect (anxiety, depression, irritability), alienation, and heightened, persistent stress-reactivity to daily life circumstances. This pattern is viewed as a stable life pattern and, concordantly, Lykken (1982) has shown that the dimension is strongly influenced by genetic factors. This latter finding is consistent with the fact that neuroticism has consistently shown one of the highest heritability estimates of any personality dimension (Fuller & Thompson, 1978).

Thus, in view of the fact that most scales of psychological disorder assess general negative affect, it is likely that a substantial portion of the chronically disordered group will be characterized by the personality trait of Negative Affectivity. Furthermore, most of the more "specific" personality disorders (although few are very specifically defined) are also strongly characterized by negative affect and, hence, they may contribute to the heterogeneity of the chronically disordered group.

Psychopathological Disorders

The chronic-intermittent subtypes of psychopathological disorders probably represent a significant proportion of high-scoring respondents on scales of general psychological distress. Because these scales assess mainly negative affect, affectively disturbed psychopathologies, in particular, may be overrepresented in the disordered group. This would include the depressive and anxiety disorders, both of which can have a substantial combination of anxiety, irritability, and depressed affect (Depue & Monroe, 1978; Gersh & Fowles, 1979). Both of these categories of disorder have substantial subgroups, ranging from 20% - 50%, who are characterized by a chronic, fluctuating course (Akiskal, 1983; Gersh & Fowles, 1979). For the depressive disorders, this not only appears to be true for the group of individuals who reach full syndromal status (Depue & Monroe, 1978), but also for less severe, subsyndromal conditions identified in the community (Brown, 1979) or in a nonpatient university student population (Depue et al., 1981; Depue & Evans, 1981; Depue & Monroe, 1983). Moreover, in the latter populations, the subsyndromal forms of disorder far exceed the rate of full syndromal disorder (Akiskal, 1983; Depue et al., 1981; Depue & Evans, 1981). For instance, in university populations, subsyndromal cyclothymia and dysthymia may account for as much as 7% - 10% of the population.

Taken together, these facts are of special relevance to life stress researchers since most studies use either community or university populations as opposed to patient samples. Therefore, the disordered group identified in these studies is likely to be well represented by chronic forms of subsyndromal affective disorders.

It is crucial to realize that distinguishing broader categories of psychopathology within the larger disordered group is only an initial, yet insufficient, step. If psychopathologists have learned anything about these disorders in the past two decades, it is that tremendous genetic, and hence biologic, heterogeneity exists within the broader categories (Depue & Monroe, 1983). A sobering example is provided by mental retardation, for which over 200 genetic causes have now been identified (Buchsbaum & Rieder, 1979). Similarly, because the pathophysiology of many psychopathologies is associated with

disturbances in neurotransmitter metabolism (Depue & Monroe, 1983), the magnitude of potential heterogeneity is revealed when it is recognized that there are more than a hundred neurotransmitter synthetic and degradative enzymes known.

An example from our own research may help to illustrate this important point. The group of individuals experiencing subsyndromal chronic depression, or dysthymia, is composed of at least five subgroups (Akiskal, 1983). Outside of their common core of depressed behaviors, they are biologically, and probably etiologically, quite different. The first subgroup is comprised of individuals suffering from chronic medical or psychiatric disorders which result in secondary dysphoria. The second subgroup represents patients who have had a major depressive episode, but who have since shown chronic dysthymia. The final three subgroups are amply found in university populations as well as in the community. All three have chronic fluctuating dysthymia, but their features differ widely. One is a psychomotorally retarded-hypersomnic group that is possibly related to bipolar affective disorder (Akiskal, 1983; Depue et al., 1981). A second is also endogenomorphic in form and course, but is agitated, hypsomnic, and stress reactive (Depue, 1984). A third subgroup, which is biologically and familially different from the latter two subgroups (Akiskal, 1983), is chronically dysphoric with only slight fluctuations in level of depression. The form and course of disorder is nonendogenous and highly cognitive. These individuals are characterized mainly by severe personality disorder and chronic limitations in personal and role functioning. They are highly obsessive and highly reactive to stress, even of minimal intensity. They view themselves as incapable of coping with life's many crises and as at the mercy of fate (Akiskal, 1983; Depue, 1984).

It is clear from our own research that the latter three subgroups of dysthymia will comprise a significant portion of disordered groups in the community or university on measures of general negative affect. It is reasonable to doubt that a single stress-disorder model will fit these groups equally well. For example, the last subgroup described above may require a model that relies heavily on personality and coping behavior in order to account for the large impact of minor events or daily hassles. On the other hand, our experience thus far does not suggest that the dysthymic subgroup that may be related to bipolar disorder is inordinately stress reactive, nor do consistent personality or coping patterns appear to play a dominant role in episode onsets. It may be that more relatively "pure" stress initiation of episodes or exacerbations of disorder occur in this form of dysthymia. Whether the exact nature of our speculations eventually proves to be correct is not the important point, however; it is that differential stress models may be a necessity in view of the heterogeneity of psychopathological disorders.

Secondary Chronic Dysphorias

Particularly in community samples, a portion of high-scoring respondents on both somatic complaint and negative affect scales will represent persons with chronic secondary dysphorias resulting from a primary, chronic, life threatening or life-altering, medical disorder or physical handicap. These individuals may suffer serious limitations in self-care, mobility, and social and physical activities, as well as the threat of sudden or certain premature death. They are often subject to persistent, unalterable distress. This group may also exhibit elevated reactivity to minor stressors in view of their possibly altered ability to cope, both on a physiological and psychological level, and to elicit material and social resources. The issue for stress-disorder modeling in this instance would seem to be much more stress maintenance of chronic psychological distress than a matter of stress initiation. Moreover, the variables of coping and social support may play key roles in this model in view of the deficits these people suffer.

Negative Affect Arising From Persistent Difficulties

Dysphoria, hopelessness, anxiety, and pessimism are commonly associated with the presence of persistent stressors that, at least currently, appear unalterable. The negative affectivity accompanying the stress generally lasts as long as the stressor persists (Akiskal, 1979). Persistent life difficulties, such as chronic unemployment, chronic marital discord, and unsafe neighborhoods, may have a continual, prolonged effect on affective state. Indeed, Brown (1979) has suggested that severe life difficulties can contribute independently to the onset of depressive episodes in both patient and community samples. It is likely, then, that individuals undergoing chronic, severe difficulties will score high on scales of general disorder.

The chronic difficulties group raises problems for life stress investigations. Should the life difficulty exceed the temporal boundaries of a study's measurement range, the difficulty may not be reported as a newly occurring event. Thus, a lack of events will be associated with high disorder, and an effect of stress will appear to be minimal. Chronic difficulties should not suffer from extreme recall error; therefore, they ought to be assessed for at least one year prior to the current study's time frame. It would also seem particularly relevant for the difficulties group to assess the degree of congruence between the demands of the difficulty and the form of support available (e.g., emotional, material), as suggested recently by Cohen and McKay (in press). The degree of incongruency may yield a more predictive index than a simpler social support dimension, such as amount or satisfaction of support.

Implications of Chronicity for Stress-Disorder Research

There are at least five implications for stress-disorder modeling and prediction arising from the heavy representation of chronic cases in disordered groups.

The Role of Stress Maintenance in Chronic Disorder

The implicit assumption underlying most studies in the stress area is that stress acts by initiating an acute episode of disorder. That is, events are viewed as having valid effects when they occur during times of relatively normal functioning and when they subsequently are followed by a rapid elevation in disorder. Indeed, researchers go to great lengths to establish that events are independent of disorder (that is, that they clearly precede onset of the disorder), and that events are independent of the negative consequences of a disorder's effect on behavior (such as arguments with spouse resulting from the increased dependency accompanying depression). In cases where event independence cannot be proved, evidence is quickly dismissed as too grossly confounded with disorder effects to interpret. Interpretation in this case, however, is only confounded if one assumes that the only meaningful effect of stress is to initiate acute episodic disorder. With chronic disorder, the occurrence of stressors and disorder are necessarily concurrent. Thus, the maintenance of chronic disorder by life stress represents an important issue, especially in light of the relatively large proportion that chronic cases comprise of the disordered group. The issue of maintenance will be discussed further below.

Nature of the Response to Stress

If stress-maintenance of disorder serves as a better model for chronic conditions, then we may have to broaden our conception of the nature of stress responses. One generally assumes that life stress will result in large increases in disorder as an acute episode is initiated. In chronic disorders, this model may not always hold. Significant portions of chronic dysthymias never or infrequently achieve the proportions of major episodes (Akiskal, 1983; Depue et al., 1981). Indeed, that chronic subsyndromal forms often lack the occurrence of major episodes suggests that their biology is vulnerable to low-grade dysfunction but not the exaggerated disturbance found in major episodes of disorder. Similarly, individuals characterized by high scores on the Negative Affectivity personality dimension generally maintain a heightened level of disorder over time; distinct "episodes" of increased disorder scores are difficult to define in the data (Watson & Tellegen, 1984). Chronic conditions, therefore, may not reflect stress effects as episodic elevations in disorder.

Measurement of other forms of response may be more sensitive

in detecting the effects of stress on chronic conditions. Our endocrine research on cyclothymia provides one example. We have found that the defining feature of hormonal response (serum free cortisol) to stress in chronic cyclothymic subjects is not an elevated stress response. Rather, the cyclothymes were characterized by extensive variation in cortisol level after stress, and by a very slow, unstable recovery of cortisol to baseline values. In other words, chronic cyclothymia is characterized by a persistently inadequate modulation system of cortisol secretion (Depue, Kleiman, Davis, Hutchinson, & Krauss, in press). This inadequacy is reflected most clearly by high variation in cortisol level after stress as opposed to episodically excessive elevations in level. Interestingly, we have found a similar pattern of intraindividual variation using self-reported daily behavior ratings in chronic cyclothymia (Depue et al., 1981).

In sum, chronic subsyndromal conditions are characterized by persistent, mild biological dysfunction. Apparently their biological systems are poorly modulated such that disequilibrium frequently occurs. Measurement of the resulting intraindividual variation at the level of self-report inventories of disorder may prove to be the most sensitive assessment of the effects of stress on these conditions.

Stress Reactivity and Minor Events

One of the strongest factors loading on the personality dimension of Negative Affectivity is Stress Reactivity (Tellegen, 1982). Individuals high on the Negative Affectivity dimension report that they are emotionally and physically overresponsive to even minor, daily hassles. Life's daily demands are essentially a series of crises for these people. For the chronic cases characterized by these personality traits, minor events, which would not affect most individuals adversely, may play a significant role in the maintenance of persistent disorder. In this light, it is interesting that minor daily hassles have been found to be better predictors of concurrent and subsequent psychological symptoms than major life events, even once initial symptom status was controlled statistically. When the effect of life event scores were removed, hassles and symptoms remained significantly correlated (Kanner et al., 1981; Monroe, 1983). In view of the fact that these studies measured psychological disorder in the usual nonspecific manner, the results may relate well to the relation between minor stressors and the maintenance and exacerbation of chronic disorder.

Minor events or daily hassles would seem important to measure because they are not assessed by major life events inventories. Their importance may stem from the possibility that major life events could operate by affecting the person's pattern of daily hassles. Separate from the impact of life events, however, many

hassles have their origin in the person's characteristic style, routine environment, or their interaction. As such, minor events may have stressful effects independently of life events. However, because stress-reactive individuals probably generate many daily hassles, and overreact to independently occurring minor events, the impact of minor events is not independent of personality factors. It is probably the case that global perceptions of one's overall level of demands, and of one's resources to meet those demands, influence specific responses in any assessment of hassles. From such a perspective, the details of which minor events are cited by the person are less important than the overall level of hassles and the subjective distress they generate (Monroe, 1983). Dependence on personality does not diminish the importance of minor events within the framework of stress-maintenance of chronic disorder. Indeed, because of their greater retest stability, relevant personality traits may prove to be the more reliable predictor of chronic disorder than either major or minor events.

The possibility that minor events may play a maintenance role in chronic disorder raises a potential problem. Research has often relied on the number of events as a measure of life stress. In the case of major life events, this may be a sensitive measure in relation to acute onsets because the frequency of major events is low and their impact is high relative to minor events. That is, because of low frequency, only a small portion of the sample will experience a large number of major events, and if these events do initiate disorder, disorder scores are likely to be elevated in at least some of these persons relative to the remainder of the sample. Thus, at least a small to moderate correlation between major events and disorder is possible. However, the frequency of minor events may be quite similar, or show only small variation, across the range of disorder scores. Nondisordered and disordered individuals may not differ significantly in frequency of minor events, although they will differ greatly in magnitude of disorder. Correlations between minor events and disorder will accordingly be minimal. Note that Brown (1979) found no difference between depressed and normal community subjects in the frequency of minor events; he dismissed them as insignificant. This lack of difference may only exist, however, on the frequency variable. For stress-reactive persons, the impact of minor events (and "impact" will need careful definition and study) may be much more sensitive a measure than their frequency.

The Genotype-Environment Covariance Problem

Researchers interested in the independent and interactive contribution of life events and social support to disorder often categorize subjects in their samples by constructing a fourfold table, as shown in Table 1. Here, the main effects of events and social support on disorder scores may be assessed separately, and

TABLE 1

HYPOTHETICAL LEVELS OF DISORDER AS A
FUNCTION OF LIFE EVENTS AND SOCIAL SUPPORT

Social Support		
	<u>Low</u>	<u>High</u>
<u>Low</u>	Slight-to-Moderate Disorders	No Disorders (Normals)
<u>High</u>	Highest Disorder (Chronic Conditions)	Slight-To-Moderate Disorder

their interaction may be used to analyze the buffering hypothesis of social support. In other words, on the bases of their event and support data, subjects are assigned to one of the design cells, where the effects of environment on disorder can be analyzed.

There is a very basic flaw in this practice. Such analyses implicitly assume, as one may in true experimental designs, that the subjects have been randomly assigned to the cells. Obviously no one in stress research assumes this to have occurred, but what is assumed is the essence of random assignment: that the subjects across the cells do not differ in meaningful, significant ways other than on socioenvironmental (events, support) scores. The socioenvironmental factors are essentially taken as independent variables in that they are assumed to have happened to the person. Thus, it is said that high levels of events or low social support cause an increase in disorder. This assumption may be stated differently: genetic homogeneity across the subjects in the design cells is assumed. Thus, any differences in the distribution of phenotypes (i.e., disordered behavior) across cells must be due to environment. This is the well-known assumption underlying genotype-environment interaction models (Fuller & Thompson, 1978; Vogel & Motulsky, 1982).

Particularly in the case of chronic conditions, these assumptions may not be warranted. There are two broad reasons for this. First, as we noted above, individuals with personality traits (such as stress reactivity), chronic life-threatening medical disorders, or chronic subsyndromal affective disorder, may generate many of the events or chronic difficulties which occur to them. We know that to be the case in chronic cyclothymia (Depue

et al., 1981). The events typically are generated by poorly reasoned or impulsive decisions; irresponsible behavior; and being incapable of functioning and, therefore, not maintaining work or school activities. Second, a major category of generated events is interpersonal: these individuals are often in perpetual conflict with close associates or vacillate between extreme clinging dependency, abuse and neglect (Goplerud & Depue, 1984). In all of these instances, the close associates become alienated and often withdraw emotionally or physically. This process has been noted for depressed patients as well. Both Brown (1979) and Paykel (1979) have noted that actual or threatened loss of social support is the main event category that is associated with the onset of depression. This state of affairs may occur either as the direct result of disorder or it may be that the same factors that cause the disorder also influence other traits, such as social competence (Heller, 1979), that then affect interpersonal relations directly. The latter example is suggested because chronically disordered individuals often have histories of poor social competence and lack current social supports. This would be congruent with the findings of Miller and Ingham (1976) that persons with low social support also had an absence of "many casual. . .superficial. . .less intimate friends" (p. 57). This suggests the possibility of a general lack of social competence (Heller, 1979).

If this is an accurate characterization of the corollaries or sequelae of chronic disorder, then categorizing subjects on the basis of their event and social support scores may represent a unique taxonomy based on socioenvironmental diagnostic characters. Persons positive on both socioenvironmental indicators (i.e., high events, low support) will represent the most severely disordered subgroup; these will be the highly chronic conditions. Those individuals negative on the two indicators (low events, high supports) will be nondisordered (see Table 1). Of course, false positives and negatives will occur, as they do in any diagnostic scheme. The most important point is that such a taxonomy leads to dissimilar people across cells. To put it differently, it leads to an uneven distribution of genotypes across cells. Therefore, an unequal distribution of phenotypes (disorder) across cells, usually taken as evidence of a causal effect of life stress, may be no more than a reflection of unevenly distributed genotypes that resulted from the socioenvironmental taxonomic procedure. No causative statements would be legitimate. This of course, is the familiar problem of genotype-environment covariation (Fuller & Thompson, 1978; Vogel & Motulsky, 1982). It occurs whenever an environmental variable is assumed to be independent of a genetically influenced dependent variable turns out to covary for some reason with that variable.

It is possible, even likely, that both interaction and covariance occur in this case. With chronic conditions, it will be very difficult to rule out covariance, and it is probably

unnecessary to do so. Events and loss of supports, even if generated by the chronic disorder, may play a significant role in maintaining or exacerbating disorder. In order to study a more "pure" stress initiation model of disorder without the covariance confound, attempts to establish "genetic" similarity of subjects will be important. This is done routinely in psychopathology research (Depue & Monroe, 1983). This would involve, for example, an assessment of history and chronicity of disorder, and of personality traits that may contribute to chronic negative affectivity and high stress reactivity. In other words, it would be important to exclude the individuals characterized by chronic disorder or personality traits, and to statistically control course and history of disorder variables in analyses.

One line of research suggested by this discussion is to define what are the relevant socioenvironmental indicators of the chronic group. If this group of individuals is seriously disordered, they may exhibit several types of impairment. We have already mentioned social impairment, which would result in low social support, and coping impairment, which may increase the frequency and/or impact of events, even minor ones. But chronic cases also may be occupationally impaired; have less well paying jobs and, hence, lower family income; and generally be of lower socioeconomic class. Interestingly, Lin, Dean, and Ensel (1981) demonstrated that 25% of the explained variance in negative affect scores was accounted for by family income. Moreover, Brown (1979) found that those individuals with low social support and high frequency of severe events were of lower class, and chronicity of disorder was associated with all three variables. Thus, family income and socioeconomic class may reflect, in part, the presence of the chronically disordered group.

Psychometric Ceiling Effects and Disorder Scores

Even if major and minor events have a significant effect on the level of disorder, the already high levels of disorder associated with chronic conditions may approach the inventory's ceiling. Further increases in disorder scores would be limited and, accordingly, reduce the correlation between events and disorder. This problem may be particularly problematic in life stress research because often the inventories have very few items. It is important to have a score range that will be sensitive a) to both moderate increases and decreases, so that both intraindividual variation in mild chronic cases and the effects of minor stresses can be assessed, as well as b) to large elevations that occur with the onset of an acute episode of disorder.

Concluding Remarks

There are several main points that we have attempted to emphasize. Perhaps most important is that level of disorder and

person factors are the best currently available predictors of human disorder. Environmental factors have proven to be either weak predictors (e.g., major life events) or may be the result of person factors (e.g., low social support). The most notable person factors are a) personality traits that are characterized by a chronic sense of negative affect and heightened stress reactivity, rendering even minor daily events stressful; b) psychopathological disorders, especially those with affective symptoms, of either full or subsyndromal status; c) chronic life-altering medical disorders or physical handicaps; and d) perhaps a general susceptibility to medical and psychological disorders (Hinkle, 1974; Thurlow, 1967). In all of these cases, the conditions are persistent or chronic, and each involves issues concerning limitations in coping, the alienation of social supports, and increased impact of minor life events. In view of the heterogeneity of person factors contributing to scores on a general scale of disorder, it seems imperative to begin the process of refining our measurement of these factors.

The very existence of the groups just mentioned does not conform to the implicit assumptions of current life stress research that a) disorder represents a uniform concept, and b) that a similar stress model holds for all disorders. We tried to show that different groups may require different types of stress models. These models need to differentiate initiation, relapse, and maintenance processes; to consider the effects of both major and minor life events; to examine the role of personality and coping behavior; to treat more seriously the effects of chronic disorder in modifying the psychosocial environment; and to be aware of the various forms of stress response in different conditions of disorder (e.g., acute and chronic).

Life stress research is at a highly primitive level of analysis: although the conceptualization and measurement of environmental and psychosocial variables is becoming more sophisticated, most of the studies still generally ask the simplest of questions (is stress correlated with disorder?) in the context of poorly conceptualized and measured human disorder, and within a design that measures the input variables so infrequently that little can be concluded about causation. Questions concerning the more precise nature of the process involved in the interaction between an environmental event and psychobiologic functioning are seldomly entertained and more rarely investigated. By focusing on the nature of the disordered group, its measurement, and implications for stress models, we hope to raise some of these questions. But what we wish to convey most is that our knowledge about the nature of the process underlying a stress-disorder relationship will not be greatly advanced by sole reliance on the study of nonspecifically-defined disorder in large epidemiologic designs. Although such studies have been useful for exploring points of correlation, it is now time to complement the epidemiologic approach with one involving more intensive,

frequently repeated measurements of individuals who have known traits and specific disorders.

NOTES

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We will use the term "disorder" in a general way to refer to the physical and psychological states which are currently assessed in life stress research.

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LIFE EVENTS, SOCIAL SUPPORT AND CLINICAL PSYCHIATRIC DISORDER

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LIFE EVENT METHODOLOGY

My aim in this paper is to review the current state of findings on recent life events and, to a lesser extent, social support in clinical psychiatric disorder. Life events are highly relevant to the discussion of social support, since social support is postulated to have major protective effects against them and they are often studied together with it. The life event field is now well developed and in the last few years has produced its own reaction in the form of a vigorously critical lobby. This will also enable me to make some methodological points which I believe are equally important for social support but are sometimes disregarded.

The main methodological problems in collection of life event data are well known (Brown, Sklair, Harris, & Birley, 1973a; Paykel, 1983). Most prominent in that of retrospective reporting. The inaccuracies of recall that afflict anyone may be compounded by attempts to explain illness in terms of a real or imagined life stress and by misperceptions due to factors such as depression.

A second problem is that events do not occur in a vacuum: we all to some extent create the events that afflict us. In particular illness itself creates events, such as loss of job.

The well known pioneering technique of Holmes and Rahe (1967) has in recent years come in for a good deal of criticism (Brown, 1981; Dohrenwend & Dohrenwend, 1981). In evaluating this it is important to keep different aspects separate. Holmes and Rahe produced several advances: a relatively comprehensive list of events, a method of data collection using a self-report questionnaire, the Schedule of Recent Experience (SRE); a method of quantification based on scaling and adding the event scores.

The life events list has deficiencies, since some of the events seem more like symptoms (e. g. change in sleep pattern) (Hudgens, 1974), but these are potentially remediable.

The main criticism concerns the self-report method. There is a crucial dichotomy in data collection methods, between self-report checklists and interviews. The self-report checklist is appealing, as an economical method of data collection, but it presents major problems. It can be quite difficult to define an event. For instance, in assessing an argument with someone close, one needs to specify the relationship, the magnitude and the persistence of a dispute which is to be considered worth recording, rather than trivial. Adequate definitions are too cumbersome to be incorporated in a short questionnaire. Added to this is the difficulty in recalling time of occurrence. Our own experience is that it takes frequent reminders, and anchoring by clear dates such as Christmas, to avoid a tendency of subjects to report events occurring well outside the time period specified in a study.

The better approach is to use an interview. Both Brown's work (Brown et al, 1973a; Brown and Harris, 1978) and our own studies, are based on semi-structured interview methods. Brown's interview is the most detailed available. Our own, described fully elsewhere (Paykel, 1983) is shorter and less probing but easier to use where there are time limitations. It nevertheless takes between a half and one and a quarter hours, covering in considerable detail 64 events with, in its more recent version, judgements of independence and objective negative impact of each event which has occurred. All of my own studies used this method, except for studies of event scaling (Paykel, Prusoff, & Uhlenhuth, 1971; Paykel, McGuiness, & Gomez, 1976) and one study of self reported life events against self reported symptoms (Uhlenhuth & Paykel, 1973).

Reliability and validity

There have been a fair number of studies of reliability and validity which are reviewed more fully elsewhere (Paykel, 1983). The controversy can mostly be resolved by looking at the method employed.

Table 1 lists findings from studies of test-retest or inter-rater reliability, mainly the former. In this and subsequent tables, where possible, I have indicated percent concordance, based on proportion of events reported on either occasion which are reported on both, but in many of the studies only a correlation is reported. I have divided studies into those using a self-report method, most commonly the Schedule of Recent Experience, and those using an interview method. A few studies used intermediate methods such as interview only for events listed as present on a self-report questionnaire, or a very standardized interview with no probing. Some simplification of complex

TABLE 1

Studies of reliability of life event reporting

	Test-retest Interval	Concordance		
<u>Self-report</u>				
Casey, Masuda, & Homes (1967)	Nine months	.74*		
Thurlow (1971)	Two weeks	.78*		
	Two years	.07	-	.34*
McDonald, Pugh, Gunderson, Rahe (1972)	Six months	.48	-	.60*
Jenkins, Hurst, Rose (1979)	Nine months	.38	-	.45*
Horowitz, Schaefer, Hiroto, Willner & Levin (1977)	Six weeks	.71	-	.90*
<u>Intermediate</u>				
Steele, Henderson, Duncan-Jones (1980)	Ten days			
	Total score	.89	-	.94*
	Specific events		70%	
<u>Interview</u>				
Paykel (1983)	Inter-rater			
	Specific event	95%		
	Month of Occurrence	85%		

* Correlation

findings has been necessary. As can be seen from Table 1, although there is some variation, self-report methods tend to have low reliabilities: interview or intermediate methods acceptably high reliabilities.

Another test is to examine the fall-off in mean number of events reported in the general population as time periods extend back into the past. In ill subjects some recent peaking of events is to be expected, if events cluster before illness. In the general population events should be randomly distributed in time.

Table 2 shows the results of some studies. The self-report studies have found substantial fall-off of 4-5% per month. The interview studies have found acceptably low rates of fall-off of 1-3% per month. The exception is a study by Schmid, Scharfetter, Binder, (1981) using Brown's interview, but without training (Brown & Harris, 1982).

An approach to validity is comparison of information provided by the subject with that provided by another informant. Studies are summarized in Table 3. Most have reported percent agreement. Again concordances have been low with self-report methods, reasonably high with interview methods, particularly in Brown's studies. There are some exceptions. The study of Hudgens et al. was an early one using only briefly and broadly specified life events. The report by Neugebauer (1983) needs to be taken seriously. The interview method does appear to have been adequate. The subjects were schizophrenics with at least two previous hospitalizations, and the report does not make clear whether they were currently ill. These subjects would be expected

TABLE 2

Studies of fall-off of event recall in general population

	<u>Time Period</u>	<u>Fall-off</u>	<u>Fall-off per month</u>
<u>Self-report</u>			
Jenkins et al (1979)	9 months	34%-46%	4%-5%
Uhlenhuth, Hamberman, Balter & Lipman (1977)	18 months	66%	4%
Monroe (1982)	8 months	36%	5%
<u>Intermediate</u>			
Henderson et al. (1981)	12 months	32%	3%
<u>Interview</u>			
Paykel (1980)	6 months	9%	1%
Schmid et al. (1981)	6 months	62%	10%
Brown & Harris (1978)	6 months	8%	1%
Brown & Harris (1982)	1 year	34%	3%

TABLE 3

Studies of patient-informant concordance for life events

	Agreement for individual events
<u>Self-report</u>	
Rahe (1974)	.07 - .75*
Yager, Grant Sweetwood & Gerst (1981)	35%
<u>Intermediate</u>	
Schless, Mendels (1978)	43%
<u>Interview</u>	
Hudgens, Robins & DeLong (1970)	57%
Brown et al (1973a)	
Schizophrenics 3 month	81%
Depressives 1 year	79%
Brown and Harris (1982)	78%
Neugebauer (1983)	22%

*Correlations of total scores

to be at the lower end of reliability although Brown's method has been found reliable with them.

As a general conclusion, careful interviews tend to produce data of adequate reliability and validity. Methods which are intermediate between interview and self-report produce intermediate results. When self-report checklists are used, there are serious deficiencies in data collection. The self-report technique has additional disadvantages in that it does not permit assessment of independence of events from effects of illness. Self-report checklists have generated much interesting research and 15 years ago they appeared quite reasonable. However, their time is past. Future studies of recent life events will need to use interview methods.

Events due to illness

A second major methodological issue is elimination of events which may be consequences of illness. One approach is to confine attention to time periods clearly antedating illness. For depression it is usually possible to date onset; for schizophrenia it is more difficult, although an attempt needs to be made if events are to be related to development or relapse of symptoms.

The second way of tackling the problem is the concept of the "independent" event introduced by Brown and colleagues (Brown et al., 1973a; Brown & Harris, 1978); the idea that on detailed scrutiny certain events can be isolated which would be unlikely to have been brought about by illness, if it were present. Although some events, such as illnesses and deaths of other persons, are almost always independent of the subject, there are many events which depend on particular circumstances, so that the judgement is better an individual one.

Exclusion of events which are consequences of illness is very important in most studies, whether of patients or of general populations. Very similar issues arise in relation to social support, and I will return to these later.

Quantification

In the quantification of stress three approaches have been used; additive summation of consensus weights (Holmes and Rahe, 1967), categorization of events into different types such as undesirable or exits (Paykel, Myers, Dienes, Klerman, Lindenthal, & Pepper, 1969), rating of threat in each event based on its detailed context (Brown & Harris, 1978). Consensus weights and categorization are less prone to bias based on knowledge that the subject has become depressed: individual judgements of threat are more sensitive. Although the merits of the different approaches have been vigorously argued I do not know of evidence from empirical studies indicating that the results they produce are different: we certainly found overlap for undesirability categorization based on the event label and for individualized judgements of contextual threat (Paykel, 1983).

The one method which has considerable problems and is best avoided, particularly in retrospective studies, is the subjective judgement of stress made by the subject who recently experienced the event. The judgement is likely to be circular: the subject who develops psychiatric disorder, or milder upset, after the event is likely to regard the event as stressful. Brown's rating of contextual threat (Brown & Harris, 1978) is careful not to use this but the surrounding circumstances of the event.

CONTROLLED STUDIES OF DEPRESSED PATIENTS

In presenting findings I will largely concentrate on patients with clinical disorder treated by psychiatrists, although this is a little out of keeping with the major emphasis of this meeting. I do so mainly because the argument in psychiatry as to what quality and severity of disorder in the community comprise a "true" case still goes on. Some have argued that disorder identified in community surveys corresponds to "distress" rather than psychiatric illness and that the latter is less related to social stress (Bebbington, Tennant, & Hurry, 1981). This criticism can most easily be dealt with by looking at treated psychiatric samples.

Depression has been the disorder most studied. Table 4 summarizes findings of 20 published retrospective comparisons of psychiatrically treated depressed patients with control groups, reviewed elsewhere (Paykel, 1982). Eight studies from USA, England, Italy, Kenya and India, employed general population controls, including one study of elderly patients (Murphy, 1982). All found more events reported prior to depressive onset although in one study, with small numbers, the difference did not reach significance. Two studies compared depressives with medical patient controls. Both found more events reported by depressives but the differences were not very striking or clearly attributable to causes rather than effects of depression. However, events may also cluster before onset of medical illness or admission to hospital.

Comparisons of depressives and other psychiatric patients are summarized in the tables. Depressives have been found to report more events than schizophrenics, indicating in the retrospective frame greater causative effects. In one study (Leff & Vaughn, 1980) differences for independent events were suggestive but not significant and there was no difference for undesirable events. Two additional studies not shown in the table failed to find differences between depressives and schizophrenics, but life event methodologies were limited (Eisler & Polak, 1971; Lahniers & White, 1976).

Some comparisons with mixed psychiatric patients have also suggested greater effects in depressives, but not fully consistently. On the other hand two comparisons of suicide attempters with depressives found more events in suicide attempters.

Some other studies using different designs or samples are not shown in the tables. In a small study (Paykel, 1974) using the patients as their own controls, we found that event rates in depressives dropped on follow-up but not fully to general population level. In another study (Paykel and Tanner, 1976) there was an excess of events in the three months prior to relapse, in relapsing depressives compared with non-relapsing controls. One study has shown heroin addicts with secondary

TABLE 4

Nature of controls	Author	Excess any events	Excess separations	Excess other types of events
General population	Paykel et al (1969)	Yes	Yes	Various especially undesirable events.
	Thomson & Hendrie (1972)	Yes	Not reported	More stress overall.
	Cadoret, et al (1972)	Suggestive	Suggestive	Not reported
	Brown et al (1973b)	Yes	Not reported	Markedly and moderately threatening events.
	Fava et al (1981)	Yes	Yes	Undesirable, negative impact.
	Chatterjee, et al (1981)	Yes	Yes	Health, inter-personal
Medical patients	Murphy (1982)	Yes	Suggestive	Health
	Forrest, Et al (1965)	Yes, weak	No	Social factors
	Hudgens, Et al (1967)	Yes, weak	No	Moves, inter-personal discord
Other psychiatric patients	<u>Schizophrenics</u>			
	Beck & Worthen (1972)	Yes	Suggestive	Events of higher rated hazard.
	Brown et al (1973b)	Yes	Not reported	Events/moderate and marked threat over longer time.
	Jacobs, (1974) Et al	Yes	Yes	Undesirable, health financial, inter-personal discord.
	Leff & Vaughn (1980)	Suggestive	Not reported	Not for undesirable events.
	<u>Suicide attempters</u>			
	Paykel et al (1975)	Fewer events in depressives	No	Fewer events in depressives, especially undesirable upsetting.
	Slater & Depue (1981)		No. Fewer exits	Fewer, independent events.
	<u>Mixed psychiatric patients</u>			
	Sethi (1964)	Yes	Yes	Not reported.
	Levi, Et al (1966)	Yes	Yes	Not reported
	Malmquist (1970)	No	No	No
	Uhlenhuth & Paykel (1973)	No	No	No

depression to have more stressful events than those without depression (Prusoff, Thompson, Sholomskas & Riordan, 1977) while another study obtained similar findings for depressives compared with non-depressed schizophrenics (Roy, Thompson, & Kennedy, 1983), as did a questionnaire study of secondarily depressed and non-depressed alcoholics (Fowler, Liskow, & Tanna, 1980). A follow-up study of addicts found events to be associated with continuing depression and failure of recovery, as measured by the Beck Depression Inventory, although the direction of the association was not clear (Kosten, Rounsaville, & Kleber, 1983).

A specific hypothesis would suggest that depression and only depression is induced by certain types of events. Most prominent in the literature is the role of loss.

Interpersonal separations of various kinds have received the most study. Findings from the studies are summarized in table 4. Sixteen studies reported specifically on recent separations. In ten, depressives reported more separations than the control groups, which included both general population and other psychiatric patients, suggesting some specificity. There was, however, no excess over medical patients. Two studies not only found exit events related to depression but also found that their converse, entrance events, were not (Paykel et al., 1969; Fava, Munari, Pasvan, & Kellner, 1981). However, one study (Slater & Depue, 1981) found that primary depressives making a suicide attempt had experienced more exits than those that did not, indicating a greater relationship to the attempt. Some additional studies, not shown in the tables, have found a relationship between depression and recent deaths (Paykel, 1982).

Also common in the studies are arguments and discord with various key interpersonal figures. They may involve threat of separation. Some of these events may be consequences rather than causes of depression, for the judgement of independence is difficult to disentangle here. These events also figure in studies of other patient groups such as suicide attempters described later.

As can be seen from Table 4, a wide variety of other events is also involved. In general the studies suggest only weak specificity. There is some relationship between depression and interpersonal losses, but these also precede other disorders, and many depressions are not preceded by them. The strongest relationship appears when events are categorized in rather broad terms such as "threatening" or undesirable. This extends well beyond interpersonal loss.

It should be emphasized again that these are all studies of disorder sufficiently severe to present to psychiatrists. In the case of depression the link with recent events is obviously strong.

Endogenous depression

A topical issue involves the distinction between endogenous and neurotic depression. A proportion of depressive episodes are not preceded by life events: perhaps about 20% in the controlled studies. However, as the term is usually employed endogenous depressions are also regarded as showing a specific symptom pattern including greater severity, psychomotor retardation or agitation, sometimes depressive delusions, early morning wakening, diurnal variation with morning worsening. In factor-analytic studies (Mendels & Cochrane, 1968) precipitant stress has generally been found to load on neurotic or reactive factors as expected, but life event precipitation has usually depended on global judgements based on limited information, and with knowledge of the symptom pattern so that bias can intrude.

Several recent studies using more rigorous life event methods have shown that life events and symptom pattern are only weakly related. In our first New Haven study (Paykel et al., 1969) symptoms were rated by one rater and life event information collected blind by another. Absence of life stress showed only a low correlation, although in the predicted direction, with an endogenous symptom factor. In a subsequent study in London (Paykel, 1979) using a more crude clinical judgement as to whether the depression was precipitated, there was no relationship with symptom pattern.

Brown, Bhrolchain, & Harris, (1979) found that depressives characterized as psychotic or neurotic on the basis of their symptoms showed only a very small difference in the proportions whose illnesses had been preceded by a severe event or major difficulty. When the depressions were divided into those with and without such an event or difficulty relatively few individual symptoms distinguished the groups. Benjaminsen (1981) compared neurotic and non-neurotic depressives and found that almost equal proportions had experienced a stressful event.

One group not yet adequately studied are bipolar manic-depressives. Two controlled comparisons (Ambelas, 1979; Kennedy, Thompson, Stancer, Roy, & Persad, 1983) have suggested that life events are related to the development of mania. This may particularly apply to first attacks: subsequent mood fluctuations in manic depressives appear less stress-related (Hall, Dunner, Zeller, & Fieve, 1977). One study (Glassner & Haldispur, 1983) found more evidence of stressful life events in bipolars with onset over the age of 20 than under, suggesting that constitutional factors might be more important in the latter.

OTHER DISORDERS

Studies of life events in other disorders have been much less frequent than those for depression. Table 5 summarizes studies comparing other specific psychiatric patient groups against

TABLE 5

Controlled studies of life events in other disorders

			<u>Findings in index patient group</u>
<u>Disorder</u>	<u>Controls</u>	<u>Author</u>	<u>Excess events</u>
Schizophrenia	General population	Brown & Birley (1968)	Yes, 3 weeks before onset
		Jacobs & Meyers (1976)	Yes, relatively weak
		Schwartz & Myers	Yes. Minor symptoms in community.
	Non-relapsing schizophrenics	Leff et al (1973)	Yes. Relapses on placebo.
Suicide attempts	General population	Hendrie et al (1975)	No. More male schizophrenics in low stress category
		Paykel et al (1975)	Marked excess.
		Cochrane & Robertson (1975)	Especially shortly before attempt.
	Other psychiatric patients	Isherwood et al (1982)	More events, especially unpleasant, disrupted interpersonal relations. Males only.
		Stein et al (1974)	Higher stress scores than general population and accident control group
		Luscomb et al (1980)	Yes. Only separations studied
	Normals	O'Brien & Farmer (1980)	Yes. Only found in older subjects.
			Yes. General practice consultant controls.
Suicide independent.	Natural deaths	Hagnell & Rorsman (1980)	More events than either control group. Not all
	Non-suicidal patients	Pokorny & Kaplan (1976)	More events in psychiatric patient suicides.

general population and other controls, excluding the comparisons with depressives already shown in Table 4.

There have been several studies of schizophrenics. Brown and Birley (1968) found more independent events than in general population controls but only in the three weeks before onset or relapse. The differences were considerably less than in Brown's later studies of depression. Jacobs and Myers (1976), using our methods, found more life events in the year before first onset than in the general population, but here too differences were relatively weak and for a small group of events categorized as likely to be independent of the subjects's control they did not reach significance. The studies by Schwartz and Myers (1977a,b) involved schizophrenics in the community not undergoing major relapse. Life events were more common in those showing minor symptoms, particularly of depression and anxiety, than in the general population.

Among studies not involving general population controls Leff, Hirsch, Gaing, Rohde, (1973) found life events more common in schizophrenics who relapsed than in those who did not, but only in placebo rather than active drug groups from two maintenance drug trials. In extension of these findings Leff and Vaughn (1980) later found that life events tended to occur in relapsing schizophrenics whose families did not show expressed emotion. Hendrie et al. (1975) compared schizophrenics with neurotics and personality disorder patients using the Holmes-Rahe questionnaire. Among males but not females, schizophrenics were predominantly characterized by low stress, neurotics by high stress. Comparisons with depressives have already been shown in Table 4. They tend to show fewer events, or fewer subjects having a major event, in the schizophrenics.

One other study used different methods. Steinberg and Durell (1968) in an epidemiological study, found the inception rate for schizophrenia significantly raised in the early months of military service. Overall life events do appear to contribute to schizophrenic onset and relapse, but to a much lesser extent than in depression.

Suicide attempters are a special group of patients often not showing features of persistent depression. Three studies have made comparisons with general population controls. In a representative sample we found a great excess of events of many stressful classes compared with general population and depressive controls, particularly in the month and often the week or less before the attempt (Paykel, Prusoff, & Myers, 1975). Cochrane and Robertson (1975) studied only the relatively infrequent male attempters and found, on a self-report checklist, a considerable excess of events in the prior year, involving particularly unpleasant events and disrupted interpersonal relationships. Isherwood, Adams, & Hornblow, (1982) also used a modified Holmes-Rahe methodology. Suicide attempters showed much higher stress scores than general population controls or a second control

group of drivers presenting after car accidents.

Several studies have made comparisons to patient control groups. Two studies are shown in Table 4 and have already been referred to. We found suicide attempters to have experienced more events than depressives, particularly for events which were undesirable, outside the subjects' control, or likely to be upsetting. Slater and Depue (1981) found that primary depressives who made a suicide attempt had more independent and exit events spread over the year before the attempt than other primary depressives.

Other studies are shown in Table 5. Stein, Levy, & Glasberg, (1974) studied recent separations and found these commoner in suicide attempters than other psychiatric patients. Luscomb, Clum & Patsiokas, (1980) obtained similar findings for a range of events but only in older age groups. O'Brien and Farmer (1980) used a different control group - young people consulting general practitioners for various reasons, who might themselves show elevated life event rates. The Holmes-Rahe event list was used. Much higher event rates were found in suicide attempters. One year, but not 3 months, after the attempt there was a decrease in the number of events reported.

Actual completed suicide is difficult to study in this way since the main protagonist cannot be interviewed. Hagnell and Rorsman (1980) found more undesirable events such as object loss, blows to self-esteem, occupational problems and moves of home in suicides from a psychiatric cohort study than in normals or natural death controls, although not all the events would appear to have been independent. Pokorny and Kaplan (1976) found more life events, particularly where scores had also been high on a measure of "defenselessness" mainly reflecting depressive content, in subjects from a follow-up of psychiatric patients who committed suicide than in controls from the same follow-up who did not. Murphy, Armstrong, Hermele, Fischer, & Clendenin, (1979) in two uncontrolled studies also found that alcoholics who committed suicide tended to have high rates of recent loss of close interpersonal relationships.

An unresolved issue is the relationship of anxiety disorders to life events. Comparisons have mainly been with depressives, rather than general population controls. We (Uhlenhuth & Paykel, 1973) found no differences between events or overall stress reported in the two disorders in a study by questionnaire where most patients had mixed disorders (see Table 4). Two other studies did not involve psychiatric patients. Barrett (1979) found in symptomatic volunteers that undesirable events and exits were more frequent before depression than anxiety, a pattern suggesting a non-specific excess of negative events. Finlay-Jones and Brown (1981) in general practice consultants, found that "loss" events tended to precede depression while "danger" events preceded anxiety.

Some other studies of mixed neurotic patients are not

included in the tables as they were not retrospective controlled comparisons of psychiatric patients. Cooper and Sylph (1973) used Brown's interview in a careful study of patients with new episodes of neurotic illness consulting general practitioners, comparing them with non-psychiatric consultants. The neurotic group had experienced more independent events in the three months before onset, particularly severely threatening events, unexpected crises and failures to achieve life goals. Miller, Ingham, & Davidson, (1976) found that general practice consultants experienced more threatening life events than non-consultants and that number of threatening events was strongly related to severity of psychiatric symptoms. Davies, Rose, & Cross, (1983) found worse outcome in general practice consultants with psychiatric symptoms where there were intervening major life events.

In a longitudinal study of mixed psychiatric outpatients and non-patients employing the Holmes-Rahe questionnaire, patients reported more undesirable and uncontrollable and fewer desirable events than non-patients, but the time period was concurrent rather than related to onset (Grant, Sweetwood, Yager & Gerst, 1981). As subjects were followed longitudinally the same classes of events were related to symptom fluctuations.

A few comparisons of mixed inpatients with the general population, without diagnostic breakdown, have been published but the heterogeneity of subjects limits possible conclusions.

MAGNITUDE OF EFFECT

It has often been pointed out that the life events implicated in psychiatric disorder, although stressful, usually fall short of major catastrophes. Marital arguments, separations and divorce, are not uncommon in the general population. Case-control studies ignore base rates. In circumstances where the causative event is common and the disorder uncommon it is obvious that most event occurrences are not followed by disorder.

In an attempt to quantify the causative effect Brown, Harris, & Peto (1973b) used the "brought forward" time - an estimate of the average time by which a hypothetical spontaneous onset was advanced by life events. They obtained values of about two years for depression both in their patient study and in our own study. For schizophrenia the figure was much smaller, 10 weeks.

We used a modification of an established epidemiological concept, the relative risk (Paykel, 1978). This is the ratio of the rate of disease among those exposed to a causative factor to the rate among those not exposed. Applying this to our own data we obtained figures of around 6:1 for the risk of developing depression in six months after the more stressful classes of events. Other studies gave similar findings. Risks fell off rapidly with time. Causative effects appeared considerably weaker for schizophrenia with relative risks of only 2-3 over 6 months, but were higher for suicide attempts.

Relative risks of this magnitude indicate an effect which is important, but not overwhelming (Paykel, 1979). They are consistent with follow-up findings of subjects in the community undergoing single major events, such as bereavement. In these circumstances, although distress is usual and help from family physicians and community agencies may be common, overt presentation of major disorder to the psychiatrist is relatively rare.

These findings suggest disorders with multi-factorial causation, in which any single factor may account for only a relatively small proportion of the variance. Although events are important, a large part in determining whether an event is followed by disorder must be attributed to other modifying factors. There may be a whole host of these, both genetic and environmental, ranging from biochemical, through personality and coping mechanisms, to social. There is certainly much room for social support and its absence as a modifying factor or independent causative factor, after the variance due to life events is taken out.

SOCIAL SUPPORT METHODOLOGY

To consider social support, I would first like to compare the problems of methodology with those for recent life events. Assessment of social support, whether currently or retrospective, carries similar problems.

The first problem is that of information gathering. In some respects this may be easier. Information regarding life events has to be collected retrospectively in most cases, in the sense of being after the event occurred: information regarding social support can be truly concurrent. However, the same issues of reliability of different methods arise. There is a surprising wealth of reliability data, favorable and unfavorable, on life event assessment. Although some data certainly exist, I am not aware of a similar critical evaluation of the adequacy of different methods for social support assessment, particularly comparisons between self-reports, standardized and more probing interviews.

A second issue concerns perception versus reality. Its nearest parallel lies in methods used to quantify life event stress, where I earlier emphasized the need to ignore any report of the subjective effect of the event, and to depend instead on the detailed circumstances and surrounding context. This careful sorting of actual situation from its perception may be harder to achieve for social support. Some aspects such as contacts with friends may be easy to quantify objectively; others, such as being understood by one's spouse, are more difficult. Most of the difficulties could be overcome. What is worrying is the frequency with which perceived lack of support, or assessments likely mainly to reflect it, is treated in published papers as though it

automatically measured an actual non-supportive situation. The problem is worsened by the strong possibility that such perceptions may reflect depressed mood.

A third problem has its parallel in the concept that some life events are independent of illness and others are not. How independent is social support? This seems to me the most difficult problem. It is easier to make the judgement for a point occurrence event than for a persisting situation. Even for events the problem is not fully solved since the judgement of independence usually involves the immediate circumstances of the event rather than long-term antecedents. It does not fully take account of the possibility that some personalities may, via life style, put themselves into circumstances where events, even if initiated by others rather than the self, happen more often. For social support it seems probable that such crucial matters as the number and supportiveness of friends, confiding relationships with spouse and others, family, housing and economic resources will very much be determined in the longer term by personality assets and deficits and by coping strategies.

The issue has received only a limited amount of empirical study, and that mainly of symptoms rather than personality. Blazer, (1983) in a community study of the elderly found that social support improved on follow-up in those who had been depressed, which would suggest that it might have been worsened by depression. In our New Haven studies of depressed women (Weissman & Paykel, 1974) we found a variety of aspects of social maladjustment, including social contacts and marital relations appeared to be secondary consequences of depression and improved as did symptoms. In outpatients Flaherty, Gaviria, Black, Altman, & Mitchell, (1983) recently found measures of social support and social functioning to be correlated.

I raise these issues because I discern in the psychiatric literature the beginnings of a similar reaction to that which has already appeared against life events. Among others Henderson, having started very much from the standpoint of social support has concluded that perception rather than lack of social relationships is important in neurosis, and that personality, particularly neuroticism, is the crucial underlying variable (Henderson, Byrne & Duncan-Jones, 1981; Henderson, 1983). These questions will need to be tackled, and will require further elaboration of methodology.

SOCIAL SUPPORT AND TREATED PSYCHIATRIC DISORDER

The crucial work of Brown and Harris (1978) on depressed women in the community in Camberwell, and later in the Hebrides, has been central in recent psychiatric discussions. I will not discuss it in detail because it is dealt with much more appropriately elsewhere in this volume. The main findings of four vulnerability factors - not working outside the home, presence of

several young children, absence of a confidant and early loss, are well known. The opposing reaction has taken several directions, including the suggestion that the vulnerability factors, rather than modifying susceptibility to life events, have independent effects in producing disorder (Tennant and Bebbington, 1978), and more recently, the argument that cases in the community correspond to distress rather than major illness.

We have already seen quite strong life event findings for treated major psychiatric illness. Does this also apply to social support? It is important to ask the question, since Brown and Harris (1978) did not find vulnerability factors acting to the same extent in psychiatrically treated depression.

Studies of early loss in depressed patients will not be reviewed in detail here. Careful attention to selection of controls is required. Findings are much less clear-cut than for recent life events (Paykel, 1982). They suggest an effect but one which is relatively weak and not very specific to depression. In any case early loss is less germane to current social support.

The literature on social support in patients is diffuse and not easy to review. It overlaps with other areas such as demographic risk factors for psychiatric illness. Controlled trials showing efficacy for such interventions as social work, family crisis therapy or psychotherapy may also reflect a social support element. I shall confine myself to a sketchy review of a few recent studies looking more directly at the relevant variables.

Roy has published a series of studies in patients directly related to and generally confirming Brown's findings. British depressed women showed a higher incidence of all four of Brown's original vulnerability factors than did gynecology patients (Roy, 1978). In a similar comparison of Canadian women (Roy, 1981a) and men (Roy, 1981b) with orthopedic controls, all Brown's vulnerability factors except presence of young children were replicated. In a comparison of depressives with personality disorder patients (Roy, 1981c), the factors were not fully specific; early loss and unemployment were related to depression in women, while in men only early loss was. The findings for early loss were also confirmed against other psychiatric patients (Roy, 1981d). None of these studies included recent life events, so that it was not clear whether the factors were acting primarily to modify event effects or independently.

Murphy (1982) examined some social factors in depressed elderly patients and general population controls, in addition to recent life events. Unfortunately, although intimacy of relationships was rated, it was only reported in relation to depressed subjects identified in the community, where it did increase vulnerability to effects of life events. Early loss of parent did not differ between patients and controls. In a follow-up of elderly depressed patients (Murphy, 1983) absence of social support as measured by marital status, living alone, or

intimacy of relationships did not predict poor outcome, although major life events did.

In a follow-up study of younger depressed patients, Surtees (1980) found that poor social support, on a composite measure of contacts, living group and confiding relationships over the follow-up period, was associated with poorer outcome. The effect, which was weak, was partly independent of life events but a little more marked in their presence.

Slater and Depue (1981) found that suicide attempters were more often living without a confidant than were primary depressive controls. However, in the majority of cases this was because of an earlier exit event; events and social circumstances may be confounded. In actual suicide the extensive sociological, epidemiological and clinical literature on social isolation is consistent with marked effects of social support although inter-relationships with effects of life events have not been examined. Social support, as contrasted with other social factors, requires more explicit study in schizophrenia.

In studies using less diagnostic specificity Henderson, Duncan-Jones, McAuley, & Ritchie, (1978) and Brugha, Conroy, Walsh, Delaney, O'Hanlon, Dondero, Daly, Hickey, & Bourke, (1982) found non-psychotic outpatients to show deficiencies in social interactions and close relationships compared with general population controls. The second of these studies found the difference in neurotic depressives but not retarded depressives on the PSE/CATEGO system. Neither study reported life events. Our own study of social adjustment in depressed women (Weissman and Paykel, 1974) found much impairment but a good deal of it appeared to be a secondary consequence of the depression.

In studies outside the psychiatrically treated range Miller et al (1976) found in general practice consultants that absence of friendship was associated with higher symptoms, even without threatening events. Davies et al. (1983) found general practice consultants with psychiatric symptoms to show less improvement on follow-up if social support was less, but mainly in the absence rather than presence of negative life events. In a study of mild post-partum depression, mostly untreated (Paykel, Emms, Fletcher, & Rassaby, 1980), we found poor marital relationships and absence of support associated with depression; this only occurred in the presence of stressful life events, which were themselves strongly associated with depression.

Overall the findings do suggest an association between absence of social support and clinical depression. However there is a surprising dearth of studies explicitly investigating a causative role for social support in psychiatrically treated samples. Most studies have been in milder depression. More studies are needed of severe unipolar and bipolar affective disorder and of schizophrenia and other disorders. The findings for depression do not necessarily establish the association as causal or make it clear what comes first.

CONCLUSIONS

The study of recent life events and psychiatric disorder has now proceeded with modern methods for something over 15 years. The methodological problems are at least clear, if not solved. Adequate reliability can be achieved with careful interview procedures, but not by self-report. It is possible to exclude events subsequent to onset or in other ways potentially due to illness.

Findings in samples of clinical major disorder presenting to psychiatric facilities show conclusively that life events precede depression at considerably greater than control rates. Although interpersonal separations are particularly involved, the effects are not very specific and spread across most threatening or undesirable events. The presence or absence of life event stress relates only weakly to the endogenous symptom pattern.

Findings in schizophrenia are less impressive. Although life events do precede schizophrenic episodes at greater than control rates, the effect is a weak one. On the other hand, suicide attempts have a high relationship to recent events. Completed suicide is less well studied, but probably has some association with events. The position regarding severe anxiety disorders is less clear, as most samples studied have not been receiving psychiatric treatment.

For social support, similar issues of methodology require attention, particularly the separation of objective situations from perception and whether it is the personality or social environment which is the primary deficit. Psychiatrically treated depressive disorder does appear to be associated with absence of social support. More studies are required of severe and bipolar depression, of other disorders and of the nature of any causal links.

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SOCIAL SUPPORT, LIFE EVENTS AND DEPRESSION

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There has been an intimidating growth of interest in the relationship of social support and health. Cobb and Jones (1984) cite 34 recent reviews and puzzle about this explosion of interest for, as they note, there is hardly enough empirical material to justify it. Instead of adding to them we will deal with the implications of one program of research which for the last 15 years has concerned itself with the onset and course of affective disorders - largely of depression in women. Attention has been concentrated on one sex and one condition; a common core of measures concerning stressors has been employed and these have been utilized by many other research centers. It is this intense and protracted concentration of effort that we trust justifies such a parochial review.

One reason for the choice arises from the problem of replicability. It is now clear that, for depression at least, almost any result is possible if the closest attention is not given to the definition of both support and depression. For support it is essential to take into account both its lack as well as its presence, and also the range of phenomena that can be encompassed by either state. Equally essential is the distinction between support in terms of a few core relationships, particularly with husband and a close confidant, and support seen in terms of a much wider network (Surtees, 1980; Gottlieb, 1981). Some examples may be useful. When a high threshold of severity is taken to define depression there is evidence that the amount of general support in terms of the presence of a network of ties has apparently little correlation with risk of onset of severe depression (what we call "cases") once core supportive relationships have been taken into account (Brown & Harris 1978; Surtees, 1980). At the same time it is hardly surprising that

general support is linked with depression when this is considered in terms of a change of mood reported by patients in general practice (Miller, Ingham & Davidson, 1976). But there is yet a third band of severity of more mild syndromes of depression (what we call "borderline cases") that are more than a change of mood but less than a fully fledged syndrome ("cases"); these again behave differently and their onset shows no correlation with the kind of support that relates to onset of more severe conditions (Brown & Harris, 1978a). This appears to be due to the fact that those that are protected in terms of social support tend to develop mild conditions rather than a full syndrome following a major loss or disappointment. There has also been considerable controversy about the claim that for severe depressive conditions such support behaves interactively in the sense that it tends only to raise risk in the presence of a major loss or disappointment. But again, clarity of definition is essential. For example, if, as is frequently the case, chronic longterm depressive conditions antedating the stressors being examined are not excluded, such an interactive effect is bound to be attenuated. This will be so since, in contrast to conditions developing during the study period, such chronic conditions will more often have low support without a stressor (because the latter will tend to have occurred before onset which is earlier than the period of study) (Brown, 1981).

Such conceptual infelicities have been the rule rather than the exception and, as already noted, it is possible to obtain practically any result concerning stressor, support and depression depending on how depression and support have been defined. Controversy based on such shifting sands is on the whole tiresome and unproductive. We wish to go beyond this and the many disputes resulting from shortcomings in measurement and research design, to consider the equally vital question of the need to create effective models taking into account a range of factors involved in etiology and course, and accompanying theory that can both interpret the model and point the way forward. Neither task can possibly be accomplished in the short-term; their development needs to go hand in hand and for each to continually influence the other. Without such intermeshing of model and theory our findings will almost certainly remain puzzlingly inconsistent and our theoretical excursions anarchic. (See Gore (1981) and Broadhead, Kaplan, Wagner, Schuenbach, Grimson, Heyden, Tibblin & Gihlback, 1983) for much the same point). One of the reasons why we need such models is that so many of the factors apparently influencing depression or other conditions for that matter, are highly correlated even when we have ruled out measurement artifacts. To take a simple example: practical and emotional support received by a woman at the time of a major crisis is highly correlated with support received on a much longer term basis and this with various "personality" measures such as her ongoing feelings of self esteem. In turn this may well even be found to correlate highly

with support received many years before - even as long ago as childhood and adolescence. It is unlikely that simple causal links will adequately describe the influences underlying such associations. It is essential if we are to gain insight into, say, the role of support at the time of a crisis, to take into account the possible role of such antecedent experiences. Faced with such a daunting task some may urge that we bypass it by providing support at the time of a crisis on an experimental basis. However, knowledge achieved in this way is almost bound to be insufficient without the background provided by an effective model and theory to help plan for the intervention itself and to interpret results. How, for instance, is the acceptance of such support related to wider factors? How far could the results be generalized to other populations?

Before proceeding a comment is necessary about terminology. Given the high correlation between so many of the measures dealing with social support, and indeed their correlation with other apparently quite distant variables, it is difficult at present to be confident about what measures represent even if we have no doubts about their sensitivity and accuracy. We face, in a particularly serious form, the classic indicator-concept problem--Blalock's two languages (Blalock, 1968). Because of this, we will begin by using the term social support somewhat cavalierly to apply to any phenomena that at all loosely relates to the first OED definition: "an act of preventing a person from giving way, backing him up, or taking his part: assistance, countenance, backing". It is only as we insert our measures into quite complex causal models that we can hope to gain some insight into what they in fact signify in terms of the etiology of depression; and it is only when we come later in this essay to deal with such models that we will try to specify what it is about support that may be involved.

The model developed on the basis of the first stage of the research program in Camberwell in South London suggested that certain provoking agents occurring not long before onset determine when an episode of depression takes place. Most significant are severe events which usually involve an important loss or disappointment, if this is understood to encompass not only loss of a person but loss of a role or even an idea. A second less important provoking agent concerns ongoing major difficulties such as might be brought about by overcrowded and poor housing or by a husband who drinks heavily. At a conservative estimate the original study concluded that such provoking agents probably bring about two thirds of all episodes of depression among psychiatric patients (excluding certain groups such as those with bipolar disorder and depression secondary to other conditions) and about 85 per cent among women developing depression in the general population (Brown & Harris, 1978a). However, in both instances the chance of such experience bringing about depression is highly related to the presence of vulnerability factors and the most

important of these apparently involves lack of social support, particularly marital intimacy, as reflected in ability to confide and to trust in a core relationship. Such vulnerability factors relate to a higher risk only in the presence of the provoking agent, but have little or no association without them.

THE ISLINGTON SURVEY

A central issue stemming from the original Camberwell research is the apparently critical role that a husband plays in protecting from psychiatric disorder women aged between 18 and 65. The measure of intimacy was crude. The scale was a four-point one: The high point "a" was used largely for women considered to have a close, intimate and confiding relationship with their husband or boyfriend. A relationship with a husband or boyfriend was rated "a" if the woman mentioned him as a confidant either in response to a direct question or spontaneously, provided there was nothing in the interview as a whole that contradicted this. We occasionally, for example, placed in an "a" relationship a husband not named as a confidant in this way but where the overall account made it clear he was treated as such. The second point "b" was kept for women without such an intimate tie but who nevertheless had a confiding relationship with someone else such as a mother, sister or friend seen at least weekly. The third and fourth points "c" and "d" covered other women with a confidant seen less often and those who mentioned no confidant at all.

The rationale for our emphasis on a husband was derived from Robert Weiss's well known discussion of various kinds of "provisions" offered by relationships and the existence of certain lack of interchangeability between them (Weiss, 1969). We did not want to prejudge the necessity of a sexual relationship to reach the highest point of intimacy and did include occasionally relationships with other women with whom they lived - in subsequent work in the Hebrides, for example, we included a number of mothers in this category. (Brown & Prudo, 1981; Prudo, Brown & Harris, 1984; Prudo, Brown, Harris & Dowland, 1981). None the less the "a" category in practice has been most solely based on the quality of tie with a husband. Nor should it be seen as necessarily reflecting the predominant role of confiding - indeed, given the questioning involved, it is probably best seen at present as a general index of the quality of the marriage.

The scale turned out to be a powerful mediator between provoking agent and depression: a type "a" relationship showing by far the lowest risk, but with some hint that a type "b" relationship might also reduce it (Brown and Harris, 1978). Despite the controversy that resulted from this set of findings (e.g. Tennant & Bebbington, 1978; Brown & Harris, 1978b) it is noteworthy that seven of the eight studies which have kept closely to the original study in terms of measurement and analysis have

produced comparable results (Campbell, 1982; Cope, 1983; Brown & Prudo, 1981; Parry, 1983; Finlay-Jones, 1983; Costello, 1982; Martin, 1982; Bebbington et al., 1983). In addition a related study by Paykel (1980) produced similar findings, and one by Murphy (1982) with the elderly fits the model if the criteria for intimacy are extended to cover all sources of confiding (see Brown and Harris, 1984).

However, such replications have been based, like the original study, on a cross-sectional design. While there is now very good evidence (reviewed by Paykel, 1983) that a life-event instrument based on an intensive interview can achieve excellent accuracy and validity when used retrospectively, there is an obvious risk of a woman's current depression influencing reporting and thereby biasing results when we deal with "softer" measures concerning support. And such doubt must be particularly strong for the measures concerning the self which we wished to introduce. With this in mind we spent several years developing a new interview-based instrument, the Self Evaluation and Social Support Schedule (or SESS) for use in a new prospective enquiry (O'Connor and Brown, 1984). The interview takes from 3 to 5 hours to administer, usually spread over two visits, and involves the interviewer in making several hundred ratings concerning all aspects of support, and relationships and activities in the area of marriage, children, housework, participation in the labor market and contacts outside the home. The measures dealing with self esteem focus on the respondents' evaluation of characteristics or dispositions such as physical attractiveness, intelligence, and success in different roles such as motherhood, and general self acceptance. Emphasis is placed on those attributes to which they themselves give particular importance, and to spontaneous comments they make about themselves in the course of the lengthy interview.

These measures have been used in a new prospective enquiry in Islington in North London. One of its basic purposes was to try to sort out how much the apparent protective function of social support derives from emotional and practical help at the time of a crisis, and how much arises from the impact of a woman's sense of closeness to others and from her continuing emotional concern well prior to any such provoking crisis. In our interpretation of the Camberwell results we have emphasized the latter: how feelings of self esteem, partly arising from such ongoing support, provided the critical intervening link. In brief, that clinical depression is a consequence of feelings of hopelessness about some key plan, aspiration or commitment and that while this is initiated by the provoking agent, it is the ongoing feeling of low self esteem that is critical since it is this that serves to generalize the hopelessness to the future and the woman's life as a whole (Brown and Harris, 1978). But even given the broad applicability of this interpretation, there are a number of other more complex, or alternative, possibilities. Low self esteem would be expected to

play a role in increasing the risk of depression irrespective of whether it derived from lack of ongoing support or largely derived from early experience of loss and rejection in a woman with a good deal of current support. A more complex possibility is that high self esteem may influence the perception and reporting of current support; a person with high self esteem may tend to gloss over potential shortcomings in relationships and support. On the other hand, such a person may also be more skilled at creating adequate networks of relationships and support in itself may not play a critical etiological role. There are a number of such complex possibilities and it is probably not helpful to dwell too much on them at this point. Our strategy is to develop a fairly straightforward causal model and once this is established, to explore more complex possibilities by adding further variables. At present we are only able to present the results of an initial foray into the data of our current prospective study in Islington, focussing on the support obtained a good deal prior to the onset of depression rather than at a point of crisis in the follow-up period.

We interviewed just over four hundred, largely working-class, women with children living at home. We used the SESS, together with clinical measures based on the PSE. When the women were seen a year later the clinical measures were repeated, and the "harder" Life Event and Difficulty Schedule (LEDS) was used to cover the one year period between the two interviews. We also questioned in detail about coping with crises and difficulties occurring in the follow-up period. As in the original Camberwell survey the rate of depression was high: 18% experienced caseness of depression at some time in the year prior to first interview and approximately half the conditions were chronic in the sense of lasting continuously for at least one year. This is, of course, a disturbingly high rate, bearing in mind the severity of the conditions involved. (Inclusion of "borderline case" conditions roughly doubles these rates). However, it must be recalled that we are dealing with an inner London borough where social problems are common. Twenty-one percent of the random sample, for example, were single parents at the time of the first interview - somewhat more than double the national average (HMSO 1983). But while the rate of caseness among such women is double that of married women - 35% versus 17% - the rate among the remaining women is still high.

The central role of severe events and major difficulties in provoking onset of caseness of depression has again been confirmed and also the role of lack of support as a vulnerability factor. For a married woman it is the quality of her tie with her husband that is critical in this vulnerability. In our first attempt to measure "marital support" we considered it to be low if either confiding in husband or emotional support received from him was low or where we had recorded evidence of negative interaction between the couple - a scale which took account of the extent of

arguing, strain, violence, indifference, and coldness. (As with all such rating scales, judgements were based on answers to detailed questioning about actual incidents and behavior and did not rely merely on general statements, and at times differed quite markedly from the subject's assessment). However this initial index ignored two other measures dealing with the marriage obtained in a similar way which were also clearly related to risk of developing an onset of depression - the "identity destructive" and the "security diminishing character" of the marriage. They were both based on information we had collected about the husband; while largely based on the present, the rating of "security diminishing character" did at times take account of the recent past - for example, that two years before the interview the husband had taken a year off work for no good reason. After considering the pattern of interrelationship of all five measures we formed a second, alternative, index based only on "negative interaction" and "security diminishing characteristics". In practice this second index was very highly correlated with the first and gave practically identical results when related to provoking agents and depression. However, we believe it to be superior in a theoretical sense because lack of confiding and lack of emotional support do not relate to a raised risk when the presence of "security diminishing character" or "negative quality of marriage" are absent. (The measure of "identity destructive character" of the marriage was not included in this second index as it did not increase its predictive value).

We also considered the quality of other ties in terms of the support they provided. Women were requested to describe what they meant by "someone to whom they felt very close" and asked whether they had anyone like that at present. They were questioned in detail about all such relationships, focussing on the quality of the interaction, and the degree of confiding that took place on intimate matters. Four-fifths of the women mentioned someone other than their husband and children living at home as "very close" - but, for those naming a person, on average only two were mentioned. Despite this, there were still considerable differences in the quality of such relationships and we have dealt with this by defining as "truly close" those relationships involving contact at least every 2 to 3 weeks and a reasonably high level of intimate confiding. Unlike marital support, our analysis of the various scales dealing with "very close" relationships indicated that confiding might well be one of the factors of critical theoretical significance. (We return to this point later). For married women experiencing a provoking agent in the follow-up period, lack of a very close relationship (irrespective of whether it is "truly close") was unassociated with risk of depression: here the quality of the link with husband was of overwhelming significance (Table 1, right-hand two columns). For women living alone something like the reverse occurred. The existence of a lover (for just over a third of the

TABLE 1

Onset of depression in follow-up period among those with a provoking agent by marital status, marital support and type of "external" very close relationship. (Excluding women cases of depression at first interview.)

		<u>Married</u>	
		<u>Poor marital support</u>	<u>Good marital support</u>
<u>Type of very close relationship</u>	<u>Single parent (at beginning follow-up period)</u>		
	%	%	%
None	44 (4/9)	30 (3/10)	14 (1/7)
Not "truly close"	27 (4/15)	20 (4/20)	0 (0/13)
"Truly close"	4 (1/23)	25 (6/24)	17 (3/18)
Totals	19 (9/47)	24 (13/54)	11 (4/38)

"Good Support" 8% (5/61) (Col 1 row 3 & col 3)
 "Poor Support" 27% (21/78) (Col 1 rows 1 & 2, & col 2)

$p < .02$, 1 df.

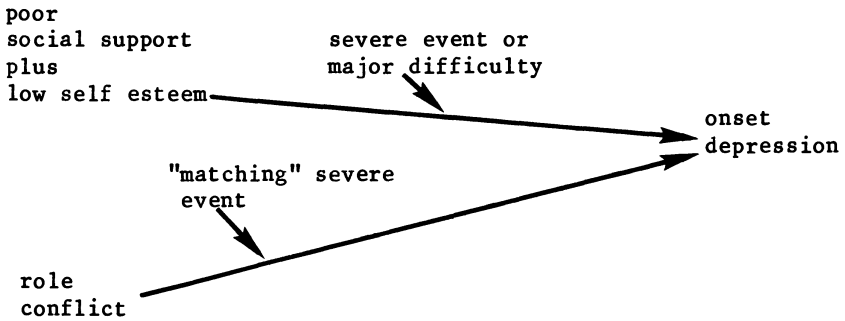
women), even taking into account the quality of such a relationship, provided no protection. (Lovers appeared to be important in the sense that depression may well be provoked when something went wrong). However, for those living alone, the presence of at least one very close relationship with someone

other than a lover, and whether or not this was "truly close", was highly related to the development of depression (left-hand column, Table 1). These results quite closely replicate those from the cross-sectional studies reviewed earlier.

We have yet to deal with our new measures of self. An index of low self esteem based on three of our fourteen measures of self was highly associated with onset of depression ("low self acceptance", "negative evaluation in terms of roles" and "negative evaluation of personal attributes"). Among women experiencing a provoking agent and who were not already cases of depression, 31% (16/51) of those with low self esteem experienced an onset of caseness of depression in this follow-up period compared with 11% (10/88) of other women ($p < .01$). When the two types of support are considered for the sample as a whole results for low self esteem very closely parallel results for onset of depression among those with a provoking agent. That is, among the married, low self esteem relates to marital support and among single parents to type of support from "very close relationships". (Women who were cases of depression at the first interview were excluded from both tabulations.)

There was one unexpected result. At the first interview women were rated in terms of the amount of role conflict that was apparent from what they said about their lives. Conflict arising from diverging obligations, particularly between domestic and external spheres, was related to risk of depression despite being unrelated to low self esteem. The conflict was directly reflected in the woman's actual situation (e.g. full-time work versus time spent with children), although its rating was in each instance based on the reporting of the woman. Among those with a provoking agent a third of those with role conflict developed depression compared with half this rate for those with no such conflict. Further analysis indicated that this increased risk was almost always explained by the "match" between the life event preceding onset and the conflict present at the first interview. For instance, a single parent whose conflict arose from the clash between the obligation to work and the obligation to be a good mother developed depression when she found out that her school-age daughter was pregnant, thus underlining her failure in one of her obligations. It is important to note that unlike low self esteem, role conflict was entirely unrelated to the two types of social support.

Since role conflict when occurring with a "matching" severe event increased risk irrespective of levels of self esteem and support, we have the beginnings of a causal model that explains most instances of depression occurring in the follow-up period:



A QUESTION OF INTERPRETATION

In coming to terms with such diverse material, some may still question whether social support is of any importance in reducing risk of depression. After all there is now excellent evidence for the role of personality factors - particularly feelings of low self esteem and perhaps this is all that is involved? Although the results from the Islington prospective study seem to indicate that social support plays an important role in some way, the evidence still might not convince a resolute critic.

For example, in the only other major prospective study concerned with depression, Henderson and his colleagues (1981) in Australia claim that they found no evidence for such a link. Their instrument utilizes sophisticated notions concerning forms of social support, particularly influenced by the writings of Robert Weiss, who has delineated five major "provisions of social relationships" (Weiss, 1969). An extensive structured Interview Schedule for Social Interaction (ISSI) has been developed to reflect such provisions. But results of the prospective study are nevertheless puzzling. The indices of support based on measures of the type of social contact, its presence or absence and its frequency (called Availability measures) do not relate to the development of psychiatric disorder in the presence of life events; it is only support based on the person's dissatisfaction with the support he or she is receiving that relates to increased risk of psychiatric disorder (op. cit.: 137-155). Interpreting this result in conjunction with correlations found with certain features of personality made at some of the follow-up interviews (measured by the Eysenck Personality Inventory - EPI), they suggest that social support in itself may be unimportant. They argue that because the Adequacy measure of support is entirely based on the respondent's evaluation of his or her relationships, there is the possibility that personality factors that have nothing to do with support are the reason for the measure's ability to predict the occurrence of psychiatric disorder, since they relate separately both to satisfaction and disorder. In a

similar way, it might be argued that our results may be due to a tendency for "neurotic" women to underplay the support they receive and that the link we have uncovered between low support and depression is spurious, the real association lying in the reporting bias produced by depression prone personality characteristics. Our whole approach to the measurement of "support" emphasizes obtaining actual instances of behavior to substantiate ratings and has been designed to preclude just such an interpretation; and studies in which husband and wife have been independently interviewed have confirmed the overall validity of the approach (e.g. Brown & Rutter, 1966; Rutter & Brown, 1966). Platt has recently argued that we might have been too optimistic in the weight we have placed on such agreement between the reports of husband and wife (Platt, 1980), and others have criticized the assumption that perceived lack of support automatically reflects an actual non-supportive situation (e.g. Vachon, Sheldon, Lancee, Lyall, Rogers & Freeman, 1982). But, we know of no evidence that would suggest that the ratings we have used are likely to be significantly biased and there is a good deal of work to suggest that they are not. For example, Quinton and his colleagues in a prospective study of women who had spent time in care have predicted current social adjustment equally well by using a measure of the husband's own account of his behavior as with the woman's reports about him and the marriage (Quinton, Rutter & Liddle, 1984).

But even if we feel reasonably confident in our own results, the Australian ones obviously present a challenge to our theoretical model. We have already made clear that we do not rule out the role of personality - the results are puzzling because even if personality factors play the kind of role envisaged they would still be expected to have at times a deleterious impact on quality of relationships and thereby on measures of support. Using the same terms as Paykel (see chapter this volume), although we have just ruled out implications concerning reporting bias, there is still the problem of independence raised by the Australian findings. That is, given their perspective, the prior personality of the subject would have been expected to create a lower availability of support. Why therefore is there no association between their index of Availability of Support and depression? It is therefore somewhat of a relief to realize that there are a number of technical shortcomings that probably explain this failure. For instance, although it is confiding and the general relationship with a husband that has consistently been shown to play a critical role in the development of depression only one item of the eight used in their main Availability index relates to confiding (see also Brown & Harris, 1984). Nor do they devote much attention to contrasting husband with other relationships. However, what may be as fundamental is the way the notion of attachment has been handled. It is not, for instance, distinguished from the provision of practical help and the

emotional support and advice accorded by close relationships. A good case can be made that attachment purely and simply in the sense of persons-to-whom-one-feels-close does not protect against depression (O'Connor and Brown, 1984). With our own instrument, SESS, the rating of "felt attachment" was made for each core tie with others living outside the immediate family. We took into account the importance for the women of the other person just being there, the feeling that they would always be there and always willing and able to help out, and the extent to which the woman could imagine the other person not being there. In short, we feel there are too many limitations in conceptualization, measurement and design of the Australian project for it to shake seriously the widespread evidence for the critical role of support. We therefore feel justified in assuming that our measures represent genuine differences in what is actually provided by social support, not just differences in satisfaction; and from this it follows that our results reflect an important protective role for social support, not just a spurious association produced by its link with personality factors.

But the issue of what our results reflect still remains. Provoking agents usually involve a loss or disappointment. One immediate complication is that such loss may involve withdrawal of support and the critical distinction made in our model between provoking agent and vulnerability factor would then become confused. Surprisingly, there is little to suggest that this occurs. For those developing depression in the follow-up period, the Islington index of support (with its different versions for single parents and married women) is changed in only one instance by the event preceding onset. However, it might still be argued that change is still possible within what is after all a broad index of support: a woman with two "truly" very close relationships would not change in terms of the index of support used for single parents if one died. But change in the index is still negligible when this kind of loss is taken into account.

What, then, is lost? We are not in a position to give a final answer, but it is clear that provoking events as a whole are much better characterized in terms of loss of a valued idea than loss of a person - although it should be added that actual loss of a person sometimes follows after the onset. In schematic outline a woman might have had long-term difficulties in her marriage, including suspicions about her husband's fidelity; then an event conveying unequivocally that he is unfaithful, followed by the onset of depression, and some time later a marital separation. As an example it conveys another significant point - in many instances the valued idea, say of a viable marriage and a notion of oneself as attractive enough to retain the loyalty of one's husband, has been gravely jeopardized well before the occurrence of the provoking agent. The provoking event merely serves to confirm the loss of something that in one sense can be said to have already been lost.

But this confirming role of the event may not always be so straightforward. In the Camberwell research we showed that often minor events occurred just before onset where a major ongoing difficulty brought about onset and it appeared that in these circumstances something quite small could cause a woman to reassess her life as a whole (Brown & Harris, 1978a). A minor incident, perhaps even with some positive features such as a sister's engagement to be married, may in this way lead to profound feelings of hopelessness by pointing a contrast to the subject's own situation. We now realize that severe events too may play such a reassessment-forcing or, as we now prefer to say, confirming role. For example, one woman developed depression directly upon hearing the news that her brother, age 14, living with her parents in Scotland had been admitted to hospital seriously ill with cancer. (He died soon after the onset of her depression). We believe it likely in such instances that the event confirms and provokes feelings of hopelessness about one's own life - in this instance we had established on an earlier visit that she had a difficult marriage and had touched upon her fantasy of returning to her family in Scotland. Financial and other circumstances made it impossible for her to get away to be with him at such a crucial time and this probably brought home to her just how hopelessly trapped she was in her current life situation. In a similar way we had seen how news of quite a positive event (a daughter's first pregnancy on the Scottish mainland) had brought home to a middle-aged woman on a council estate in the Hebrides how trapped she was with her irritable unemployed husband: "I became so depressed then" she told us weeping "because I knew I couldn't be with her during the pregnancy" (Brown & Prudo, 1981). However, whatever credence is placed on a particular mechanism, it seems clear that for most of the women with onset of depression the event involved loss of a valued and cherished idea, and that, for a significant proportion, the valued idea and any support that in her mind should have been linked to it had already been seriously threatened, if not largely lost, by the time of the provoking crisis. This is also consistent with our view that it is not merely loss itself that is critical in depression. It is a sense of hopelessness about retaining, retrieving or replacing an object of value, or indeed of being worthy enough to be linked with such a source of value. The experience of failure and inadequacy may therefore be closer than loss and separation to the depressive experience - loss and separation perhaps only appear to be important as they so often are the precursors of interpretations of failure and inadequacy.

Given that the provoking event rarely in itself involves a direct loss of support, how should we interpret the impressive association between our measures of lack of support and depression following a severe event? While any answer at this point must be provisional it is already possible to discern the broad outlines of a plausible interpretation. We have focussed on the quality of

support at about the point of first interview and its association with the development of depression in the following year. We have not yet analyzed the material we collected at follow-up about support during crises during the follow-up year and which may have preceded onset. However, emphasis on support around the time of first interview is probably in any case justified for the following reasons. In most instances onset of depression seems to have followed quickly in the wake of the provoking event - almost half the women described it as occurring within one week, and three-quarters within four weeks of the event. Furthermore, in hardly any of the provoking events was it possible to conceive of practical intervention that would lead to some kind of quick resolution of the situation resulting from the crisis - a point already implicit in our argument concerning the way the event often confirms some ongoing sense of failure or disappointment. It follows that effective intervention would in many instances need to concern itself with combatting the woman's developing judgement of hopelessness. Given that the object or aspiration that is the initial focus of her concern is likely to be one to which she has long been committed, it may well be that, if ongoing support had not already been present well before crisis, little can usually be done at the point of the crisis itself to prevent a depressive disorder occurring. That is a disorder in which there is in any case a marked lack of capacity for initiative and planning - unlike a "borderline case" condition where such capacity for coping may be fully retained. But for this very reason it does not follow that support of the kind envisaged by those interested in "crisis intervention" in terms of the here and now handling of a problem will always be of little avail. As already made clear many of the severe events leading to onset were part of an ongoing and worsening situation that had sometimes begun years before. "Support" during this earlier period may well reduce risk by helping the subject to prevent the final crisis. Second, and probably more important, support of this practical kind may speed up recovery; improvement in an acute or chronic depressive condition in Islington was frequently preceded by an event leading to a resolution or significant improvement in a major problem, or to an event bringing about a reassessment of a woman's belief in her ability to deal with her life. A woman for instance, moving to a new house and thereby escaping an intolerable housing problem or, in terms of the latter, a woman recovered from a serious chronic depression soon after a surgeon told her she was very lucky to be alive following an emergency operation for a perforated ulcer. The potential here for intervention is obvious. Thus it may well be that "support", in the sense conceived by workers in the "crisis-theory" tradition, can play an important role either some time before or some time after a critical crisis (Killilea, 1982). Onset itself, at least for the women in Islington, often appeared, once the crisis had occurred, to have an inevitable quality about it - and

the likely importance of intervention seems confirmed by the swiftness with which the depression follows the provoking event.

If for the sake of the present argument we ignore the possible importance of social support at the time of the provoking event how are we to interpret our two indices? These clearly indicate that something of importance is going on. For the married woman it appears to be shortcomings in the marital relationship or simply in the behavior of the husband that are critical. The latter were on the whole quite serious, involving unemployment, a serious physical disability or heavy drinking. But since the rating of "diminishing security character" of the marriage takes some account of past unreliability, lack of support has on occasions been rated when there was no overt problem at the time of our interview. None the less, three-quarters of women low on the marital support index had a longterm difficulty rated "moderate" or "marked" associated with their husband compared with just over 40% of other women. However, although it seems likely to be the presence of negative experience within marriage that acts in low marital support as a vulnerability factor, it does not appear to be confounded with our notion of provoking agent - only 29% of all provoking agents centered on a husband, even when events such as husband's ill health or unemployment due to redundancy are included. All but one of the women with such provoking agents are low on our index of marital support; but since only 38% of the total women with onset of depression had such a provoking agent, the two components of the etiological model are clearly largely distinct. But to return to the interpretation of our indices of support: marital support relates to low self esteem and a link between support and self esteem has been shown before (e.g. Hirsch, 1980). It therefore does not appear unreasonable to assume that on the whole low self esteem is the result of the poor marital situation, although some kind of snowball effect may well be present and, as will be discussed later, there is bound to be on occasions a reversal of the direction of the effect. Therefore as a tentative model we suggest that low marital support increases the chance of a provoking agent occurring (e.g. husband's unfaithfulness, crisis over money), having already tended to reduce self esteem. Given that ties with persons named as "very close" are unrelated to risk of depression among married women, it is necessary to posit that they find it difficult to use relationships with friends and relatives to compensate for the failure, disappointment and low self esteem resulting from the marital situation: the impact of support from "very close" relationships outside the home tends to be blocked, at least as far as reducing risk of depression is concerned.

The situation of single parents is quite different. They do not have to cope with a husband's inadequacies - at least within the home - and self esteem can certainly at times grow from the very fact of coping alone. One woman with an extremely low income

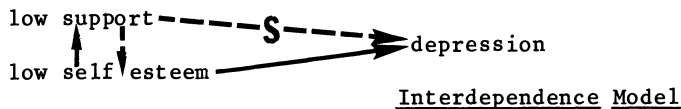
and two children (one of whom was severely handicapped) recounted with great pride how well she coped financially - for example, by her prowess and energy in shopping (buying in bulk, looking for "special offers"); and she vividly contrasted her current independence with her former frustration and hopelessness some years previously with her feckless husband and the fluctuation and insecure income he provided. Therefore we do not see lack of a husband in itself as necessarily, or even often, leading to low self esteem, although a number of the women were in terms of self esteem almost certainly still suffering from the impact of a recent marital separation. However, lack of a husband undoubtedly contributed to a markedly increased rate of provoking agents and thus increased the risk of depression. Role conflict needs also to be included for both the married and the single parents as increasing risk of depression independently of either form of support. But for women living as single parents, as we have seen, the presence of persons named as very close (as long as they were seen often and used as confidants), was highly related to a reduced risk of depression. The simplest explanation is that such outside contacts relate to self esteem in this group and that this explains the link with risk of depression. However, we cannot rule out that coping well when living alone can "unblock" constraints in forming relationships and that high self esteem relates to such support rather than vice versa.

For those rated high on the marital support index there is the same kind of possibility: the judgement of effective role performance in the marriage may relate to high self esteem (and also a lower rate of provoking agents) and this may make the impact of support from other relationships superfluous as far as protection from depression is concerned.

Before proceeding it should be noted that the correlation between marital support index and the rate of provoking agents concerning the husband has quite radical implications for our Camberwell model. It suggests that provoking agent and vulnerability factor are more often of one piece than we have so far allowed. If we take the example of the woman discovering unequivocal evidence of her husband's infidelity: the longterm difficulties in her marriage, to say nothing of her suspicions about his conduct, may well have slowly sapped her feelings of self worth and encouraged the negative ideas about herself picked up in our first contact with her. And when the final revelation occurred the resulting feelings of hopelessness could be seen as the result of an ongoing situation of lack of support (from husband), low self esteem, revelation, and hopelessness about her marriage and life as a whole forming part of one ideo-effective posture (Tomkins, 1966) or emotional schema (Leventhal, 1980). Therefore while provoking agent and vulnerability as exemplified by our support indices often appear to be quite "distinct", we have probably underplayed how at the time of onset the judgement of hopelessness can stem from a mutually reinforcing complex of

ideas and emotions. This conclusion is, of course, also implied in our discussion of the way events "confirm" lack of support that in a real sense has been lost for some time. But this makes no difference at all to the clear cut conclusions we have drawn about support interpreted broadly in terms of the quality of core relationships. These are highly correlated with risk of depression and, given their link with self esteem, may well play an important etiological role.

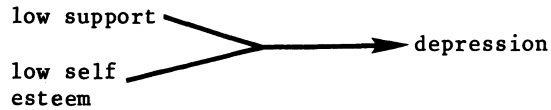
There will, of course, be the difficulty in settling the direction of influence between self esteem and support. Because of this it is of considerable interest that for women with low support but high self esteem, a combination occurring in a third of those with a provoking agent, there was a relatively low risk of depression (see column 4, Table 2). Risk was also low in the small group (about a tenth) who had high support but low self esteem. It will be essential to explore the "exceptions" in detail in terms of the qualitative descriptions on the written protocols as well as correlated variables, especially concerning other sources of self esteem such as employment, practical coping and cognitive defenses. For in this way we shall be able to confront the nascent disenchantment with social support to which Paykel refers in his contribution when he discusses independence. The implication of the criticism on grounds of independence is that the link between low support and depression is spurious: low self esteem determines both and belief in pathway S is therefore erroneous:



But there is more than one version of this perspective emphasizing the lack of independence. In one the arrow goes only one way, from low self esteem to low support, and in such an instance to brand pathway S 'spurious' will be justified. In another the lack of independence is more of an intradependence: the arrow goes in both directions, and even if pathway S does not exist, there is a direct link from low social support to depression via low self esteem. However the fact that risk was low in the group of women with low self esteem with high support suggests that the most suitable model is the interactive, not the interdependent one. That is both variables contribute to a final common pathway to depression:

TABLE 2. Social support, provoking agent, self esteem, normative role conflict, and onset of depression in follow-up period.

1st INTERVIEW		FOLLOW-UP		ONSET OF DEPRESSION AMONG THOSE WITH PROVOKING AGENT	
Support	Provoking agent	Self esteem	By self esteem	By normative conflict at 1st interview	
POOR	A) 77/115 (67%)	C) Low 49/115 -43%	15/37 (41%)	G) Conflict 2/7 (29%)	
		D) High 66/115 -57%	5/40 (13%)	H) No conflict 13/30 (43%)	
		E) Low 35/173 -20%	1/14 (7%)	I) Conflict 6/22 (27%)	
		F) High 138/173 -80%	5/48 (10%)	J) No Conflict 5/80 (6%)	
GOOD	B) 62/173 (36%)				
A x B, p <.05	C x E, p <.001	C x E + F, p <.001	C x E, p <.05	G x H, ns	I x J p <.01

Interactive Model

The key point at present is that support and self esteem are closely associated and sorting out their respective contributions will not be easy. It will of, of course, be further complicated when we take into account actual support received during crisis in the follow-up year. However, the relative contributions of support and self esteem may largely prove to be a pseudoproblem in the sense that they are seen as different sides of the same coin, mutually sustaining each other. However, if we are to advance theoretically it is essential to try to separate them.

In reviewing this preliminary material we have convinced ourselves of the importance of seeing the dauntingly large range of possible influences in terms of causal models and more speculative theory. We believe we have already achieved a certain amount of success in this venture. But as we proceed we are made increasingly aware of something we have known all along but have chosen to underplay: that the elements in our models can influence each other over time and that this is at times so marked as to make them in some sense one. This is, of course, to some degree implicit in our view of depression as a cognitive disorder: that there has to be a coming together of the various factors in the model in the mind of the individual and the degree to which this takes place over a lengthy period of time means that the elements of the model are bound to influence each other and in some sense fuse. However, we as yet remain unrepentant in our search for dominant lines of influence and we have hope that as our analysis of the Islington material proceeds, new insights and new evidence will emerge. But, although this will take several years, we already look ahead to new studies - and these seem likely to include both more intensive enquiries of the individual as well as experimentally-based intervention studies in which we attempt to manipulate support - we trust guided by a not entirely erroneous theoretical perspective.

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SOCIAL SUPPORT AND CHILDREN OF DIVORCE

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This paper examines childrens' social support in the years immediately succeeding the dissolution of their parents' marriage. The situation of parental divorce is a frequent one in our society, one that will be experienced by approximately one third of all children before they reach their eighteenth birthday (Glick, 1979). The negative consequences of this situation for children have been demonstrated to include extreme immediate distress, anxiety and aggression problems, school difficulties and impaired cognitive and sex role development (Hetherington, 1979; Kurdek, 1981).

There is much to be gained from studying support in a specified situation such as family transition following parental divorce. An in depth understanding of stress, support, and coping in a circumscribed situation should enable investigators to disentangle the complex processes by which support is obtained and subjectively processed and how it affects adjustment to the situation. In order to achieve an in depth understanding, however, it is necessary to confront some basic assessment and outcome issues in social support research.

The issue of how to assess social support has received considerable recent attention (Barrera, Sandler & Ramsay, 1981; Cohen & Hoberman, 1983; Procidano & Heller, 1983). Historically the empirical literature on social support has been marked by the use of a wide range of measures including the existence of a salient social tie (Eaton, 1978; Sandler, 1980), social network measures (Hirsch, 1979; Wellman, 1981), specific helping transactions received (Barrera, Sandler, & Ramsay, 1981), and perceived satisfaction with help received (Cohen & Hoberman, 1983; Procidano & Heller, 1983). While single measures of social support have been utilized in most studies, recently investigators

have advocated the use of several conceptually distinct measures (Sandler & Barrera, 1984; Thoits, 1982). We will argue that understanding social support in a situation requires an assessment of the sources of support, the actual helping transactions exchanged and how these transactions are evaluated by the focal subject.

The outcome issues refer both to how support affects stress and coping over time and how the different aspects of social support are changed as a consequence of the stressors of the situation. Salient issues here include whether support protects people against the negative impact of stressors and whether specific kinds of support are particularly helpful buffers of specific stressors (Cohen & McKay, 1984). On the other hand social ties can be changed as a function of the frequency and reciprocity of helping transactions. Ties can be strengthened by the receipt of support in times of need or they can be weakened by the failure to receive expected support. Similarly the occurrence of a stressful event (e.g., divorce, death) may effect the composition of social networks and how network members interact and are evaluated.

We propose that a model to study social support in a stressful situation needs to include a) a multi-dimensional assessment of social support, b) attention to how the stress situation affects each facet of the social support construct and c) examination of how each facet of the support construct affects adaptation to the stressors in the situation. In this paper we will describe this model and illustrate the utility of the model by a) applying it to the literature on children's adjustment to divorce and b) presenting early findings from a study we are conducting on stress and support in children of divorce.

SOCIAL SUPPORT AS A MULTI-DIMENSIONAL CONSTRUCT

Social support is conceptualized as including the social ties that constitute the support network, the exchange of helping transactions with network members and the individual's evaluation of the help received.

Social network properties (e.g., size, multiplexity, density, intensity, etc.) have been enumerated elsewhere (e.g., Mitchell & Trickett, 1980; Wellman, 1981). The social network is significant as the sources from which support is derived and the structure of the social network determines the availability and content of support. Cochran and Brassard (1979) for example, argue that characteristics of the parents' social support network are important for children. They note that parents' network members influence performance of the parental role in several ways. Support may help parents deal with their own life stressors which interfere with attending to the parental role. Network members may also directly encourage or discourage some child

rearing behaviors, or may do so indirectly by modeling. It is also important to note that the relationship of the supporter with the focal person often includes the exchange of other non-supportive transactions that may affect what support is offered and how the support is evaluated. For example, the recipient's assessment of his relationship with the supporter may lead to differential evaluations of the supporters' motives which may in turn effect how the support is evaluated (Fischer, Nadler & Whitcher-Alagna, 1982).

The nature of the helping transactions that are actually received has become the focus of increasing research attention. Recently, Barrera and Ainlay (1983) have presented a conceptual analysis of the content of social support. This analysis yielded six categories: material aid, behavioral assistance, intimate interaction, guidance, feedback and positive social interaction. Barrera and Ainlay (1983) factor analyzed the self-reports of supportive transactions received by 370 undergraduate students using a 40 item scale of supportive transactions received, (Barrera et al., 1981). They extracted four conceptually unique dimensions; directive guidance, non-directive support, positive social interaction and tangible assistance.

The subjective evaluation of support is likely to be the consequence of the content of support received, the context within which it is given, personal characteristics of the recipient (e.g., age, personality, prior history of support received) and the overall nature of the social ties with the supporter. Cohen and Hoberman, (1983) have developed four scales to measure different aspects of perceived availability of support, tangible (material aid), appraisal (someone to talk to about one's problems), self esteem (positive evaluation of oneself by others), and belonging (availability of people to do things with) support. They report that perceived availability of esteem and appraisal support are the most effective in protecting people against the negative effects of stress. Procidano and Heller (1983) have developed a measure to assess subjective evaluation of being supported by friends and family. Others (e.g., Barrera, 1981) have integrated a measure of subjective evaluation of support within their social support network measure by asking subjects to report their satisfaction with each type of support received.

One central issue in the social support literature is to identify the mechanisms by which support affects adjustment. For example, the hypothesis that support affects adjustment by reducing the negative impact of stress has received considerable attention (Cohen & McKay, 1984; Heller & Swindle, 1983). What are the implications of a multi-dimensional concept of support for this issue? We propose that each of the support concepts, networks, helping transactions and subjective evaluations of support, are involved in the processes by which support impacts on adjustment. For example, it may be that subjectively evaluated support yields the most consistent stress buffering effects (Cohen

& McKay, 1984). However, whether these subjective evaluations derive from the receipt of particular kinds of helping transactions, embeddedness within a particular type of social network, some personality characteristic or early childhood experience of the individual is critical to understanding the processes by which support affects coping in a stress situation. Furthermore, it may be that because of the stressors in a particular situation some support network structures may be more adaptive (e.g., low boundary density in role transitions, Hirsch, 1981). It may also be that some specific kinds of assistance may match the demands of specific stress situations (Cohen & McKay, 1984). Thus, using a multi-dimensional concept of support, we are interested both in the determinants and interrelationships between each facet of the support construct as well as in the relationship of each facet of support with adjustment.

It needs also to be acknowledged that loss of support is often a source of stress. Many of the most significant stressors in our culture involve interpersonal losses (e.g., death, divorce, moving). Other stressors (e.g., unemployment, illness) which are not themselves interpersonal losses may strain interpersonal ties and result in loss of support. The loss of support may involve the breaking of relationships, change in interpersonal transactions and/or a change in evaluation of relationships. Thus a multi-dimensional model is also seen as appropriate for assessing how support is changed in a stress situation.

SOCIAL SUPPORT AND PARENTAL DIVORCE

Let us now examine the situation of parental divorce to consider a) how children's social support is affected by this situation and b) how social support is utilized as a resource that moderates the stressful effects of this situation.

Recent reviews of the literature on the effects of divorce on children have advocated that divorce not be treated as a single event, but as a series of events that occur as the family structure changes (Hetherington, 1979; Kurdek, 1981). Wallerstein (1983) has described several coping tasks for children brought about by these events (e.g., physical separation of parents, parental arguments, parents beginning to date) which occur after parental divorce. These tasks include accepting the reality and permanence of the change, disengaging from the continuing conflicts between the parents, and resolving feelings of anger, blame and fear of abandonment.

When we consider the role of social support throughout this situation it is apparent that it is both the object of change (thus often being a stressor) and that it is utilized as a resource that moderates the stressful effects of changes. For example, while interpersonal loss experiences are a salient stressor, reassurance about parental love and information about

other children's experiences may facilitate coping with these losses. Thus, our description of the role of support in this situation includes both how support is affected by this transition and how support is utilized in coping.

Effects of divorce on social support of children

Utilizing our multidimensional model we will describe the effects of divorce on the supportive network, supportive transactions received and on childrens' evaluation of being supported.

The effects of divorce on childrens' social network will be described in terms of changes in the nature of salient relationships in the network (e.g., father, mother, friends) and effects on the structure of the network. There is a dramatic effect of divorce on the child's relationship with both the custodial and non-custodial parent. While initially there is often an increase or maintenance of the predivorce level of contacts between father and children, over time the amount of contact tends to decrease. For example, 25% of the preschool children in the Hetherington, Cox and Cox (1978) sample had more contact with their fathers two months after the divorce than before the divorce. However, most of the children experienced a gradual tapering off of contact and a sizeable number of children completely lost contact with their fathers. Hetherington et al. (1978) reported that two years after the divorce 40% of their sample saw their fathers at least once a week, while about 30% had contact once every three weeks or less. Studying a much larger sample, Fulton (1979) reported that two years after divorce only 20% of fathers continued a steady pattern of visits while approximately 30% had ceased visits with their children. The decreased contact with fathers post-divorce appears to be more marked for girls than for boys (Hess & Camara, 1979). It is also interesting to note that children in paternal custody arrangements have more frequent contact with the noncustodial parent than children in maternal custody arrangements (Santrock & Warshak, 1979).

Considerably less attention has been paid to the effects of divorce on children's network ties other than with parents. Tietjen (1982) reported that children from father-absent homes had more limited peer networks than did children from two parent homes. Hetherington, Cox & Cox (1981) described pre-school age male children of divorce as "incompetent bullies" who are excluded from interaction with their same sex peers and later increase their play with younger children. Ahrons and Bowman (in press) found that non-custodial grandparents report decreased contact with their grandchildren. One prominent new relationship which often develops after a divorce is between the child and the parent's dating partners or new spouse. Hetherington et al. (1981) report that although some step parents did not become

heavily involved with the children, others took an active parenting role. She also reported that remarriage of the non-custodial parent is often followed by decreased contact with the children.

One salient feature of the child's network is the relationship between the parents. This relationship is usually marked by hostility although it is not unusual for the parents also to provide positive assistance to each other (Colleta, 1979; Hetherington et al., 1981). Although a high degree of conflict between the parents usually precedes the divorce, conflict may escalate in the months immediately succeeding the divorce and involve attempts by the parents to form alliances with the children against the other parent (Hetherington et al., 1981; Wallerstein & Kelly, 1980). A second salient feature of the effects of divorce on children's networks is that over time two separate clusters of interrelated but minimally overlapping kin groups, the bi-nuclear family, develop (Ahrons, 1980).

Although it is apparent that children receive support from a range of sources (e.g., grandparents, friends, parents, etc.), very little is known about the content of this support, how it is distributed across supporters and how it changes from the pre-divorce to post-divorce periods. Instead, research has focused on how divorce changes children's more general relationship with members of their social network. The affective quality of the father-child relationship is marked by a strong sense of loss and desire to maintain the relationship by the child (Hetherington et al., 1981; Warshak & Santrock, 1983) as well as considerable blame of the father for the divorce (Young, 1983). The father also is often described as having a strong sense of loss which initially is manifested in overindulgence of the children. This "everyday is Christmas" behavior declines, however, and a more balanced relationship develops over the first two years (Hetherington et al., 1981; Jacobs, 1982).

Frequently, the custodial mother's contact with the child includes many more negative interactions after the divorce (e.g., often over issues of control) as well as fewer positive interactions. In the second year post-divorce mothers establish more effective parenting practices, and employ more nurturance, consistency and effective discipline in their interactions with their children (Hetherington et al., 1981). Particularly with adolescent children mothers may develop more reciprocal relationships, in which advice and intimate interaction are exchanged (Weiss, 1979). Hetherington et al., (1981) also report that teachers develop negative interactions with children of divorce, largely in reaction to the acting out behavior of these children.

While there is no direct evidence on this point, there is indirect evidence that custodial parents have relatively little understanding of children's feelings about the divorce. For example, Young (1983) assessed the relationship between children's

ratings of their divorce related feelings and parent ratings of how they believed their children would respond. On nine of thirteen scales no significant relations were obtained. Similarly, Hingst (1981) found large discrepancies between mother and child reports about the child's feelings about the divorce. There is also evidence that parents do not provide their children with very timely information about the divorce. In two studies, a significant proportion of children report never being told directly about the divorce (25%, Warshak & Santrock, 1983) or not being told until after the divorce had occurred (35%, Hingst, 1981).

Parental divorce has a negative effect on children's evaluations of their support network. Children report considerable blame of both parents for the divorce although there is more blame of the father than of the mother (Young, 1983). They also continue to strongly value their relationship with both the custodial and non-custodial parent (Warshak & Santrock, 1983; Hetherington et al., 1978). Warshak and Santrock (1983) found that children were dissatisfied with the frequency of visits with the non-custodial parent, wanting more contact than they had. Feelings of loss of a parent and perceptions of the parents being sad, upset and vulnerable are salient (Hetherington et al., 1978). Sandler, Wolchik and Braver (1983) found that children rated experiences in which friends, relatives and neighbors negatively evaluated their parents as being among the most stressful events which occur after separation.

There is currently no empirical literature which addresses how children of divorce evaluate the actual support they receive. However, we can speculate about some possible effects of children's conflicted feelings on children's evaluation of support received from their parents. For example, they may see the support as having ulterior motives and may devalue it, or they may see it as a reaffirmation of their relationship and value it highly. It is reasonable to believe that children may be sensitive to being devalued by others or stigmatized as being different after the divorce. If so, these children (particularly adolescents) may react negatively to supportive attempts in order to protect their self-esteem (Fisher, et al., 1982; Eisenberg, in press).

Effects of support on divorce adjustment of children.

The most consistent evidence about the effects of support on children of divorce is that the continuing relationship with the custodial parent (usually the mother) is a critical determinant of children's adjustment (Hess & Camara, 1979; Hetherington et al., 1978; Wallerstein & Kelly, 1980). Hetherington et al. (1981) reported that because of the mother's distress, task overload and employment, her parenting deteriorates in the year after divorce and this poor parenting correlates with increased child adjustment

problems. Hess and Camara (1979) reported that the quality of the parent child relationship was the best predictor of children's problems with aggression and social relationships with peers and of their work effectiveness in school. Guidubaldi, Clemenishaw, Perry, and McLoughlin (1983) also reported that the quality of the relationship between the custodial parent and child was positively correlated with the child's classroom behavior, popularity, school achievement and general adjustment.

There is also evidence that a continuing relationship with the non-custodial parent (usually the father) contributes to better child adjustment after divorce (Guidubaldi et al., 1983; Hetherington et al., 1978). Hess and Camara (1979) found that the affective quality of this relationship was related to child adjustment. Kurdek and Berg (1983) reported that while the frequency or regularity of visits did not relate to child adjustment, more time spent alone with the father was related to better child adjustment.

A second important feature of the child's network concerns the relationship between the parents. In fact, Hetherington et al. (1978) view the interparental relationship as the social system that is most important in determining the effectiveness of the divorced mother's interactions with her children. Evidence is consistent that a continuing high level of conflict between the parents correlates with poor child adjustment after divorce (Hetherington et al., 1978; Hess & Camara, 1979; Jacobsen, 1978; Kurdek & Blisk, 1983). Whether the effects of this continuing conflict are mediated through high parental distress, poor parenting or as another stressor in itself has not been established.

A final interesting issue relating to the child parent relationship is whether a good relationship with one parent protects the child from a bad relationship with the other. The evidence is mixed on this issue. Hess and Camara (1979) found that a positive relationship with either parent buffered the effects of a bad relationship with the other while Hetherington et al. (1981) found that only a good relationship with the mother yielded this effect. Hetherington et al. (1981) also report that a good relationship with the mother also partially protected the child from the negative effects of interparental conflict.

In terms of the wider social network, Santrock and Warshak (1979) found that child contact with additional adult caretakers was related to better child adjustment. Similarly, Guidubaldi and Clemenishaw (1983) report that when the child's custodial parent received help on household tasks or child care from his or her parents, the child performed better academically. Also, more contact between the child and the non-custodial parent's relatives was related to better child academic and social adjustment (Guidubaldi and Clemenishaw, 1983). We might speculate that these effects could be due either to children receiving some valued support from these adults or to the relief from task overload that

these adults provided for the custodial parent. As noted previously, there is evidence that mothers are able to use support from their extended network to provide relief from stress, facilitating their effectiveness in the parental role (Cochran, Campbell, & Henderson, 1982; Colletta, 1979; Hetherington et al., 1978). Hetherington et al. (1981) also found that child involvement in a well structured, predictable and nurturant school environment correlated with better child adjustment. Similarly, Guidubaldi et al. (1983) reported that school climates characterized as "safe, orderly, predictable" and that provided high achievement expectations were related to more positive school adjustment. While it is reasonable to expect that children (particularly adolescents) utilize peers as a source of support, there is no evidence about the effects of such assistance on the adjustment of children of divorce.

There is relatively little evidence about the specific kinds of supportive transactions which have a positive impact on children of divorce. Several studies (Hetherington et al. 1981; Santrock & Warshach, 1979) point to the authoritative relationship with parents, which is nurturant, well organized and consistent, as being important. Guidubaldi and Cleminshaw (1983) reported better academic and school adjustment for children who told their non-custodial parents about nice things which occurred at school. Jacobson (1978) investigated the relationship between parent discussion of the divorce with the child and child adjustment. She found that parent discussion of the divorce with the child before and after the event was related to less child psychological symptomatology after the divorce. There is no direct evidence about the effects of children's evaluations of support on their post-divorce adjustment.

PRELIMINARY RESULTS FROM A CROSS-SECTIONAL STUDY ON SOCIAL SUPPORT AND CHILDREN OF DIVORCE

Using the conceptual model outlined above it should be apparent that multiple questions need to be asked in order to understand the contribution of social support to the processes that occur in a stressful situation such as parental divorce. In this section, the following questions are addressed: 1. From what social ties do children derive different kinds of social support? The intention here is to assess the resources children have for each support function. 2. How is the structure of the support network related to the subjective evaluation of the support received? Several previous studies with adolescents and adults have assessed this issue and found that network measures such as density, multiplexity and intimacy, correlate with satisfaction with support (Barrera, 1981; Hirsch, 1979, 1980; Stokes, 1983). As noted previously, the issue is an important part of understanding the interrelations between the components of the

multi-dimensional construct of support.

Subjects

Subjects were 89 children from families where there had been a parental separation within the past three years. The sample included 59% boys and 41% girls. The age range of the children was between eight and fifteen, with a mean age of 11.6 years. The mean time period since the parental separation was 1.4 years, with 57% being within one year, 27% within two years and 16% within three years of separation. It should be noted that this sample is approximately 3/4 of a sample that is being generated to study children of divorce.

Measures of social support

Social support was assessed using an instrument adapted from the Arizona Social Support Inventory Scale (Barrera, 1981). The adaptation was designed to make the scale appropriate for younger children, while maintaining the scale's ability to assess support network, specific support functions and children's evaluation of support. Five different kinds of supportive functions were identified based on previous conceptual work on the content of support: recreation/play, advice/information, goods/services, emotional, and positive feedback (Barrera & Ainlay, 1983; Cochran & Brassard, 1979). Each kind of support was defined and children were asked to list all the people inside and outside their family who had provided this kind of support during the previous couple of months.

Network size was assessed as the total number of people mentioned for the total network as well as the family and non-family network. Size of network was also calculated within the family and non-family network. Size of network was also calculated within each support function (e.g., number who provided each type of support). Multiplexity was assessed as the total number of relations that included more than one function, and was also assessed separately for family and non-family supporters. Children also rated how satisfied they were with the support received for each function from family and non-family (using a four point scale) and how they felt about each supporter named (using a seven point scale from very good to very bad). Satisfaction with support was calculated separately for family and non-family as the satisfaction across each of the five support functions. A total support satisfaction score was also calculated as the mean across family and non-family ratings. A network satisfaction score was assessed as the mean satisfaction rating of all supporters.

All supporters were listed and subjects were asked to indicate those family members and non-family members who interacted with each other outside of their interactions with the

child. Family-non-family boundary density was assessed as the total number of connections between family and non-family supporters divided by the possible number of connections. In addition children were asked to list all those people who sometimes made them feel "angry, bad or upset". The sum total of people who were the source of such negative feelings was calculated.

Procedure

Subjects were obtained from two sources, court records of requests for marital separation and responses to newspaper articles. Letters describing the study were mailed to several thousand people who had requested a marital separation with the prior two years. A follow-up contact was made by telephone in which we requested an interview with the custodial parent and a child between the ages of eight and fifteen. The child interview included measures of stressful life events, perceptions of divorce, psychological symptomatology and competence as well as social support and lasted 1 1/2 hours. The parents completed a questionnaire battery about their own divorce adjustment as well as their children's adjustment. A workshop about children's adjustment to divorce was provided for the parents as compensation for participating in the study. It should be noted that the collection of data for this study is still in progress and that the analysis to be presented represents an early report on this data set.

Results

Providers of support. The mean number of people in each of seven categories who were mentioned as the source of each of the five support functions (and of negative feelings) is presented in Table 1. Two aspects of the descriptive information about the support network are worthy of note. First is the overall breadth of the support network. Children derive support from a variety of people beyond their immediate or even extended family. The average total size of the support network was 16.00 (9.76 non-family and 6.22 family members). Secondly, the contrast between maternal and paternal support is interesting (but not surprising). Mothers are more frequently seen as providers of each type of support than are fathers.

The significance of the differences between the proportions of mothers and fathers mentioned for each of the five support functions was tested using McNemar's test for correlated proportions (Ferguson, 1959). The proportion of mothers mentioned was higher than fathers for each of the functions. These results are not surprising in view of Fulton's (1979) finding that by two years after the divorce approximately 30% of fathers have ceased visiting their children. Thus we divided our sample to look at

TABLE 1

Mean Number of Supporters for Five Support Functions

Total Sample (N=89)

Providers of Support	Play	Ad- vice	Goods/ Services	Emo- tional	Positive Feedback	Negative Feelings	Total Support
Dad	.78	.53	.67	.39	.62	.28	2.99
Mom	.91	.74	.89	.76	.89	.20	4.19
Siblings	.96	.45	.60	.39	.74	.64	3.14
Extended Family	1.87	1.11	1.06	.58	1.17	.15	5.79
Friends	5.61	1.54	1.71	1.53	2.45	.88	12.84
Other Adults	.48	.76	.49	.30	.53	.19	2.56
Other	.69	.28	.58	.27	.53	.81	2.35
Total	11.30	5.41	6.00	4.22	6.93	3.16	33.86

Higher Contact Dads (N=54)

Dad	.91	.59	.80	.46	.80	.20	3.56
Mom	.94	.74	.89	.76	.91	.14	4.24

Lower Contact Dads (N=30)

Dad	.53	.40	.43	.27	.30	.43	1.93
Mom	.83	.70	.87	.77	.83	.30	4.00

The two-groups do not sum to the total group because of missing data for five subjects.

two sub-groups (based on custodial parent's report), one where the father had occasional or no contact with the child (36% of sample) and one where the father had regular contact (64% of sample). When only the fathers with a higher level of contact with the child are considered the difference in frequency with which mothers and fathers are mentioned as supporters narrows but continues to be significant for three areas (advice, emotional and positive feedback). It is also interesting to note that in families with higher father contact with the child the number of fathers mentioned as a source of serious negative feelings is significantly lower ($\chi^2 = 5.21$, $p < .05$) and the number of mothers mentioned as a source of negative feelings is marginally lower ($\chi^2 = 2.89$, $p < .10$) than where fathers have less contact with the children.

A one-way repeated measures analysis of variance (using a multivariate approach) was performed to compare the number of people reported as a source of each support function across the total network. A significant overall effect was obtained for support function (multivariate $F = 26.57$, df approximation 4,85, $p < .001$). Post hoc comparisons indicated that children had fewest resources for emotional support, followed by advice and goods/services (not different from each other), followed by positive feedback and had most resources for play.

Relationship between support network structure and support satisfaction. Two types of network variables (size and multiplexity) were correlated with measures of satisfaction with support. As shown in Table 2, the results indicated that multiplex relationships were related to greater satisfaction with both support received and with the supporters. Ten of these twelve intercorrelations between these measures were significant ($p < .05$). There was also a tendency for larger network size to be related to greater satisfaction with support and with supporters (six out of twelve correlations were significant).

The issue was raised about whether the relative paucity of available sources of emotional support (see Table 1) was problematic. It may be that children simply rely on fewer, more intimate relations to obtain emotional support, but that these ties are sufficient. Alternatively it may be that children are not receiving support that is needed to cope constructively with their feelings about the divorce. In order to investigate this

TABLE 2

Intercorrelation Between Measures of Support Network and Support Satisfaction (N=89)

	Support Satisfaction	Satis- faction	Non-family Satisfaction	Network Satisfaction
Network Size	.18*	.10	.20*	.17*
Family Size	.20*	.18*	.14	.24**
Non-family Size	.12	.03	.18*	.08
Boundary-density	.06	.06	.02	.03
Multiplexity	.26**	.18*	.28**	.28**
Family Multiplexity	.19*	.18*	.17*	.36**
Non-family Multiplexity	.21*	.11	.25**	.11

* $p < .05$

** $p < .01$

issue, children's reports of satisfaction with each of the five support functions were compared using a one-way repeated measures analysis of variance. The results indicated a significant overall difference in satisfaction across functions (multivariate $F = 14.10$, df approximation 4,42, $p < .001$) with the order of satisfaction (from highest to lowest) being play, positive feedback and goods/services (not different from each other), advice and emotional support (not different from each other).

Discussion and research directions. This paper presents a preliminary analysis of social support received by children of divorce. Several findings are particularly informative. The higher levels of mother as compared with father support are not surprising, although this gap does narrow where fathers have more frequent contact with their children. Even here, however, father's are less frequently mentioned as a source of the more personal forms of support (i.e., advice and emotional support). It is also notable that where fathers have less contact with their children, the fathers (and to some extent the mothers also) are more frequently reported as being a source of negative feelings. In view of the strong evidence of the importance of both the maternal and paternal relationship with the child after divorce, more research is needed on factors (e.g., custody arrangements, mediation, etc.) which affect the level of support provided by the parents.

The relatively low number of resources for and low satisfaction with emotional support received by the child is consistent with prior evidence that parents are not very cognizant of their children's feelings about the divorce (e.g., Hingst, 1981). It is also interesting to note that children continue to have very negative and distressing feelings about the divorce up to six years later (Kurdek, Blisk & Siesky, 1981), and that children's attitudes and feelings about the divorce correlate with their overall post-divorce adjustment (Kurdek & Berg, 1983). Thus, while we know very little about how children actually resolve the emotional issues aroused by divorce (Wallerstein, 1983), emotional support does appear to be a significant area where children often do not receive sufficient aide from their natural support network.

The positive relationship of support satisfaction with multiplex network relationships is similar to previous findings (Hirsch, 1979). It may be that such relationships, which are probably closer and more personal are particularly reassuring during a stress such as divorce which threatens the very integrity of the support network. These relationships should be the subject of study from several viewpoints; how children develop such relationships with friends and family, what kind of support is exchanged within these relationships, how the relationships change developmentally, etc.

In a more general sense this paper advocates the careful

study of social support in stressful situations. As an illustration of such research the present data indicate potential problem areas for the networks of children of divorce (i.e., emotional support, paternal support) and possible directions for strengthening the support network (i.e., fostering of multiplex relationships).

This evidence adds to prior research about the support network of children of divorce. The processes by which people receive assistance from their social network are complex and multiple studies are needed to identify how support matches the coping tasks of the stress situation and how the supportive ties and helping transactions change over time and effect adaptation.

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LIMITATIONS OF SOCIAL SUPPORT IN THE STRESS PROCESS

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INTRODUCTION

Numerous studies have now found that social support acts to mitigate some of the undesirable consequences of the stress process. Thus, whereas life events and daily hassles have been shown to result in increases in strain as measured in terms of physical and emotional health (Dohrenwend & Dohrenwend, 1974; 1981; Holmes & Masuda, 1974; Meyers, Lindenthal & Pepper, 1971; Rabkin & Streuning, 1976; Stewart & Salt, 1981) social support may act as a buffer, limiting the negative effects of stressors on strain (Caplan, 1974; Cobb, 1976; Dean & Lin, 1977; Gottlieb, 1978; Hirsch, 1980; Johnson and Sarason, 1979; Wilcox, 1981a). Other studies have suggested a direct effect, rather than a buffering effect, in which social support is seen as positively related to health, but in which it does not mediate the stress-strain link (Andrews, Tennant, Hewson & Valiant, 1978; LaRocco, House & French, 1980; Williams, Ware & Donald, 1981).

While this research has opened a whole new area of investigation, this chapter will argue that the use of correlational studies and an atheoretical approach has led to a simplistic model which future research will need to expand, broaden, and in some respects confront. First the epidemiological model used in these investigations will be examined. Second, types of life events in which social support would be expected to have limited or contradictory effects will be discussed. Following this discussion a number of alternative research strategies will be suggested within a model of the support process.

THE EPIDEMIOLOGICAL MODEL RECONSIDERED

Most research on social support has been correlational in nature (e.g. Hirsch, 1980; Holohan & Moos, 1981; Sandler & Lakey, 1982). This has led to certain misconceptions related to a simple and at times inappropriate epidemiological model when these results have been interpreted. In the next few paragraphs these misconceptions will be discussed along with examples which illustrate a number of points. It is not suggested here that these correlational studies have not been valuable. On the contrary, they have been essential and they have been heuristic. Nor is it suggested that they have suffered from the same phenomena as does the "correlation of number of storks and the birth rate in Holland," that is the trends of the correlations are accepted as indicative of phenomena which are causally related.

Epidemiological research has successfully employed correlational research on both disease causing agents and health factors. This model may be related back to studies in England by John Snow (1854, republished 1936) which found that persons in neighborhoods supplied with water from "down river" of London on the Thames or using certain wells were more likely to contract cholera than persons supplied with water upstream of London or using cholera-free wells. Something in the water or passed via the excretions of victims was inferred to be responsible for the deadly disease, even though the tools which would later enable researchers to identify the bacterial agents responsible for cholera were twenty to thirty years in the future. In one of the early scientifically based public health measures Snow convinced the authorities to remove the handle on the contaminated Broad Street Well to prevent its further use. This classic epidemiological model has been applied in hundreds of cases and more recently has been adapted to investigation of mental health (e.g. Midtown Manhattan study, Srole, Langner, Michael, Opler & Rennie, 1962).

This epidemiological model lends itself to mental health less well than to physical health due to a fundamental qualitative difference between mental health and physical health variables. On one hand, disease agents in the physical health chain are almost consistently unidirectionally negative in their effect. Bacteria and viruses that result in illness, result almost exclusively in illness. A little bit of typhus or cholera is not a good thing. On the other hand, "disease agents" in "mental illness" are in fact bi-directional or multi-directional.

Stressors are one of the most clear examples of this bi-directionality. While high stress may result in negative health consequences, so too, high levels of stressors have been found to result in higher motivation, improved stress management, a heightened sense of mastery and an improved self concept (Hinkle, 1974; Ruch, Chandler & Harter, 1980). Everyone who has mastered a challenge has felt the exhilaration and heightened

sense of self efficacy as a result of such events. Those who have not confronted and overcome the challenges of stressors may feel a sense of inadequacy (Bandura, 1982). On the other hand, Hobfoll and Walfisch (1983) have found prior number of recent stressor events to be linearly related to depressive emotionality experienced during an acute crisis. It is possible that stressors have a linear effect on mood, but a more complex effect on ultimate psychological and physical health. In any case, stress has both positive and negative effects.

The second point is that even for physical health the simple correlational epidemiological model suffers from another basic fault. Extreme cases which do not necessarily represent a process evidenced across cases bias the results and extrapolation of such correlational effects leads to a dangerous misrepresentation of the data (Feldt, 1961; McNemar, 1960). Such interpretation has plagued research on social support.

An example from classic epidemiology will clarify this discussion. A study of caloric intake would show either a negative, positive, or zero order effect on physical health depending on the nature of the sampling in regard to extreme cases on either side of the caloric intake of the population. If a very low income neighborhood was sampled, the correlation between caloric intake and health would be over-weighted by those individuals having a critically low intake of calories, resulting in malnutrition and possible rickets, scurvy, or lead poisoning (from eating paint off the walls). Contradictory evidence would be gleaned from a study in a middle class neighborhood where very high caloric intake would over-weight the correlation as the small percentage of obese persons would be found to have greater heart disease, hypertension, etc. A healthy population of soldiers among whom there was a limited range of caloric intake might, in turn, show a zero-order correlation between caloric intake and health. The zero-order effect would also be found in samples in which both obese and malnourished extreme groups are included. The correct conclusion is not that amount of calories is negatively or positively related to health, but rather that a critically high or critically low caloric intake is related to poor physical health. Moreover, the type of calories is a critical factor, and this would only be identified if analysis of extreme groups was undertaken, as its effect is not apparent in the intermediate range.

Returning to research on social support it may be seen that correlational studies may well have been affected by extreme groups. So, for example studies have shown that having more individuals in one's social network is related to better health (Aneshensel & Frerichs, 1982; Pearlin, Lieberman, Menaghan & Mullan, 1981; Wilcox, 1981a). However, Caplan (1974) and others have suggested, and research has confirmed, that having one or two intimate relationships is what is critical, and only those with no intimates may be at risk (Brown, Bhrolchain & Harris, 1975;

Conner, Powers & Bultena, 1979; Lowenthal & Haven, 1968; Stokes, 1983). The size effect may be a product of a threshold effect and it would be inaccurate to interpret it as a true size effect. The same may be true for density, multiplexity, and other quantitative measures of social support.

Such misinterpretation may result in two undesirable consequences. The first is the obvious possibility of being over-influenced by the effects of extreme groups. The second negative consequence ironically is an extreme underestimation of the effect of crossing between those who have and do not have the critically minimum level of social support. This is the case as the non-extreme groups moderate the statistical effect of the extreme groups. So, for example where 50% of individuals who have no intimate relations may experience extreme psychological distress following a series of major life events, the resultant correlation of number of intimates with psychological distress would be low due to the lack of difference of having from two to ten intimates. From past research we may thus both be misrepresenting the effect of social support across cases, and minimizing its effect between cases at some critical point. Social support may have a large effect on coping, but one that is reflected in whether or not individuals have some minimum amount of support and not in the degree of support over and above this level. To the extent the effect of size is binomial rather than continuous the correlation will be an underestimate by as much as a factor of $(r - r^2)$ (Rosenthal & Rubin, 1982) as for a binary distribution the predictive increase is equal to the correlation not the correlation squared. At this point we simply do not have the answers to this critical question.

A third confounding is also inherent in the epidemiological model that has been applied to research on social support. While extreme groups along the continuum of the independent variable will affect correlations, subgroups who are affected differently by the independent variable will also cause a distorted picture where non-parametric analyses are employed. In this regard, general community samples have been used in order to arrive at generalizable results (Folkman & Lazarus, 1980; Holohan & Moos, 1981; Wilcox, 1981a; Williams, Ware & Donald, 1981), but the effect may be the opposite. Community samples by their nature include various subsamples. While a low correlation is generally assumed to be representative of a weak trend across the entire sample, it may in fact indicate a strong relationship within a subsample. No study to this author's knowledge has delved into a breakdown into subsamples by a theoretical or empirically based schema.

So for example, the widows and divorcees in a given sample may benefit greatly from emotional and instrumental support in order to work, parent, pay the bills, and fix a clogged sink. Married women may benefit only minimally from social support. Unless the sample is subdivided into these two subsamples the

research would conclude that social support has a moderate effect on psychological distress. It does not. It has a great effect on certain subgroups and a minimal effect on others. Other trends may disperse fairly evenly across subsamples, but this is an empirical question. Unless one looks at those individuals receiving water "upriver and downriver" the causative and contributing factors will be masked or minimized. In fact, John Snow (1854) showed that the English tended not to get cholera in the winter because they drank boiled tea; cholera was epidemic in Scotland winter and summer because Scots drank spirits and water regardless of the season.

A final methodological point is related to recent studies investigating the "buffering" versus the "direct" effect question, which have examined samples of individuals who differ as to the extent to which they have experienced recent life events. Some studies have found support for the "buffering effect" whereby social support buffers the negative consequences of high stress, but has a negligible effect during low stress periods (Andrews, et al., 1981; LaRocco, et al., 1980; Sandler & Lakey, 1982; Wilcox, 1981a). Other studies support a "direct effect", whereby social support has a positive effect on health independent of stress level (La Rocco, et al., 1980; Andrews et al., 1978; William et al., 1981). As cross-sectional studies, however, these investigations cannot control for the possible effect of selection (Campbell & Stanley, 1963), such that persons high on recent life events may differ from persons low on recent life events (Dohrenwend, 1973; Monroe, 1982), even when these events may appear to a greater extent to be outside the individual's control (Antonovsky, 1972). These differences may effect or be related to individuals' tendency or ability to exploit potential resources.

For example, persons high on recent life events may tend to be more depressed prior to their confrontation with life stressors and may call on social support more than less depressed individuals (Coyne, Aldwin & Lazarus, 1981), while persons low on recent life events may be less adventuresome (Zuckerman, 1979), or more well insulated from potential negative events (Dohrenwend, 1973). Consequently, while it has been concluded by some that social support has a mediating effect, this may be particular to the "type" of persons who experience or report experiencing many stressful life events. It does not necessarily follow that persons who experience fewer stressful life events would be aided by greater social support if they were to experience a greater number of recent life stressors.

While true experimental design is not feasible in field research, examining the same individual during periods of high stress versus during periods of low stress and use of longitudinal designs in general would add to the validity of consequent results and as such would shed clearer light on the "buffering" - "direct" effect question (Holohan & Moos, 1981; Lazarus, 1980). A "buffering effect" would result in less psychological distress

among persons with greater resources in comparison to persons with lesser resources during high stress periods, but not during low stress periods. In contrast, a "direct effect" would result in persons with greater resources experiencing less distress both during periods of high and low stress (Hobfoll & Walfisch, 1984).

WHO MIGHT NOT BENEFIT FROM SOCIAL SUPPORT

One conclusion that may be drawn from the previous methodological discussion is that certain subsamples may not be benefited at all or may even be harmed by social support. It will be argued in this section of this paper that certain groups when studied may show surprising results in terms of social support if their needs, skills, situation, and other relevant characteristics are not carefully considered.

A better understanding of the effect of social support may be provided by an evaluation of the "ecological congruence" of the given situation. Hobfoll, Kelso and Peterson (in press) have coined the term "ecological congruence" which is defined as the fit of a given resource or set of resources to meet the emotional and task requirements of a given stress situation, for a given group, at a given point in their lives, and at a given time in relationship to the occurrence of a crisis event. It is seen as a heuristic aid to guide researchers. So, for example, married women have different needs than divorced women, and divorced women have different needs during the divorce, the separation period, and the year thereafter. Younger divorced women have different needs than older divorced women. These differences need to be considered a priori and hypotheses should be based on these considerations.

Limiting Situational Factors: Structural Determinants

The ecology of the situation is especially important when individuals in particular situations have limited access to social support. Crisis situations are one often overlooked example of this case (Hobfoll & Walfisch, 1984). During the immediate crisis period individuals are often separated from potential supportive agents. The time element involved due to the sudden onset of many crises limits the individuals' abilities to recruit friends and family who often have no way of knowing of the tragedy (Hobfoll & Walfisch, 1983). At such times individuals experiencing stress in its most intense doses must rely on internal resources such as self concept, or other resources that may be immediately summoned (Hobfoll & London, 1983; Hobfoll & Walfisch, 1984). Professional support and organized self-help groups are examples of immediately available external resources.

Another aspect of crises that limit the potential effect of social support is that they often tend to involve all of the

intimates in a given support system. For example, the serious illness of a child (Friedman, Chodoff, Mason & Hamburg, 1977) affects the parents, immediate family and close friends. Such events may render potential support ineffective as everyone involved is needing of support and may not be able to provide support. Support agents less involved in the crisis may offer support, but as intimates have been seen as the most effective supportive agents, the support of these less involved others would be less valuable (Caplan, 1974; Brown et al., 1975). Involved individuals have even been shown to have a negative effect on the health of a loved one who is ill, an effect which has been attributed to their extreme worry and over-protectiveness (Garrity, 1973; Lewis, 1966).

While private disasters affect families, public disasters affect large numbers of people in a given location (Green, 1982). When large number of lives are lost, persons injured, or when people lose their homes and property, the ability to provide support to others may be limited. Certainly, help-giving of all varieties emerges, and affected families and friends provide mutual support. However, the most deeply affected families would be least likely to be able to provide such mutual support. In recognition of these factors, some mental health authorities have activated publicly organized lay support and professional support networks following disasters (Lindy, Grace & Green, 1981).

War is one man-made disaster that has drawn research attention. As in other disasters self-help groups and professional support have been organized to aid soldiers on the front lines, families at home, those having lost a loved one, and returning veterans (Hobfoll & London, 1983; Teichman, 1975). One study has found that family support of women whose loved ones have been mobilized into the military during war has no effect on emotional distress (Hobfoll & London, 1983). The same investigation found support from friends to have a negative effect on emotional distress. Women who reported more intimate relations with friends and more use of social support during the first week of a war of sudden onset, were more likely to experience state anxiety than women who reported less intimacy and received less support. This finding was interpreted as reflecting a "pressure cooker" effect, whereby women with a common crisis shared their fears, anxieties, and rumors as to the progress of the war. Upon debriefing many of these women reported that they could find nothing else to talk about, as one reminded the other of their common plight. Still, there may be other long term positive effects of the sense of sharing that was experienced.

Life crises often occur when persons are physically isolated from their social networks. This is very common during transition periods. Recruits in the military, college freshman, new graduates, families that move their residence, new immigrants, those entering or leaving an institution (e.g., prison, psychiatric hospital) and refugees are examples of groups of

individuals who are susceptible to life crises, in a potentially taxing transition period, and who may be cut-off from social support (see Felner, Farber & Primavera, 1983).

Thoits (1982) has also illustrated how many life events are themselves events which directly or indirectly separate persons experiencing crisis from their most intimate relationships. She lists deaths of friend or spouse, divorce, separation, change in relations with spouse and family member leaving household among such support-limiting events contained in common life events surveys.

Such individuals will be at especially high risk. On one hand, they will be experiencing a stressful life event, on the other hand they will be experiencing the event at a time when their resource pool is weakened. This may even render them more susceptible to psychological stress than are loners, because they are used to utilizing social support, whereas the loner may be used to exploiting resources other than those based on social ties (Perlman & Peplau, 1981).

Limiting Personality Factors

Persons with certain personality styles also may have varying needs of social support. Persons with high self-esteem and sense of mastery have been shown to have less immediate need for social support than those with lower self esteem (Hobfoll & Walfisch, 1984; Pearlin, et al., 1981). They rely on this high self-esteem as an internal resource when confronted with stressors. If subjected to chronic stressors, however, self-esteem begins to decline as the persons' resources are increasingly taxed. At this later stage social support acts to buoy sinking self regard (Pearlin et al., 1981). This differential effect over time is consistent with models of social support which proposed that a central product of social support is the cognitive appraisal that if supported we are loved, valued, and part of a network of others who need us as we do them (Caplan, 1974; Cobb, 1976).

Persons who are typified by a trait called "hardiness" have been shown to be not only resistant to the negative effects of stressors, but are purported as seeing such events as a positive stimulating challenge (Kobasa, 1979). Because they do not feel that life events are particularly stressful and as they tend to have high internal locus of control, hardy persons might feel less need for social support, or may only require it when under very high stress. It would be interesting to investigate if such individuals even saw requesting help as a sign of weakness.

Even when lacking such personal resources and given good social support, some individuals are unable or unwilling to profit from the help they received. Coyne, et al. (1981) found that depressed individuals discredit social support, although they received more of it than nondepressed individuals. Individuals with psychiatric problems have also been found to be ineffective

in their exploitation of social support (Tolsdorf, 1976).

The effects of supportive interactions may also be limited by the personal characteristics of those offering the support. The term social support itself is misleading, as it implies that all social network interactions are supportive (Hobfoll, Kelso & Peterson, in press). Research and case studies in clinical psychology have long recognized the detrimental effects of bad advice, scapegoating, enmeshed, passive-aggressive or dependent relationships. (Oxford & O'Reilly, 1981). Yet, in social support literature attention to the negative contributions of many social interactions has been generally lacking (Hobfoll, et al., in press; Hobfoll & London, 1983).

Stigmatized Events as a Limiting Factor: Individual-Event Interaction

Events in which a stigma is attached may also effect the willingness of potential supportive agents to offer social support. In addition, an individual who is undergoing a stigmatizing event may be too embarrassed to recruit support and may even hide the fact that they are in distress.

It is the nature of many illnesses that individuals feel embarrassed or self conscious about being ill and perhaps a concomittent sense of failure (Moos, 1977). In the case of chronic illness persons may feel that they are being a burden and so wish to isolate themselves or refrain from requesting help. If an individual becomes depressed, despair may also lead to isolation or a negative response to offers of support. Chronic illness may be accompanied by an angry phase (Kubler-Ross, 1969) and the anger may be projected on potential helpers and loved ones, unintentionally pushing them away.

Intimates have also been reported to have difficulty interacting with a loved one who has a serious or terminal illness (Kubler-Ross, 1969; Meyerowitz, 1981). Individuals report that they feel they have nothing to offer the very sick or dying person, or that they feel too emotionally involved to visit. Fears of the irrational sort (e.g., "I may get breast cancer too so I can't face her") or the rational sort (e.g. "I may catch this infectious disease") would also act to limit supportive interactions with the sick individuals. Fear of or difficulty in interacting with the "mentally ill" is a special case in point (Mitchell, 1982; Perrucci & Targ, 1982).

Alcoholics are a large group of persons whose problem requires special types of social support, and for whom some kinds of support are harmful. The stigma attached to alcoholism that leads families to cover up for the alcoholic often leads to personal denial on the part of alcoholics themselves. These responses limit potential efforts by significant others to help the alcoholic. Severe alcoholics were found in one study to be negatively affected by supportive friends because the friends were

drinking companions and contact with them could not be expected to elicit much more than encouragement for drinking (Hobfoll, et al., in press). Other studies have found families to contribute to the alcoholic's drinking through a complex relationship which is probably more interactive than causative (Futterman, 1953; Kalashian, 1959; Edwards, Harvey & Whitehead, 1973). Alcoholics Anonymous, on the other hand, is a kind of directed social support that has a beneficial effect on many alcoholics.

Divorce is also a stigmatized event in many cultures and countries, even if in Western society it is a more accepted event than it was twenty or thirty years ago. Families often have difficulty accepting divorce and support may be mixed with messages of blame (Goode, 1956; Weiss, 1975).

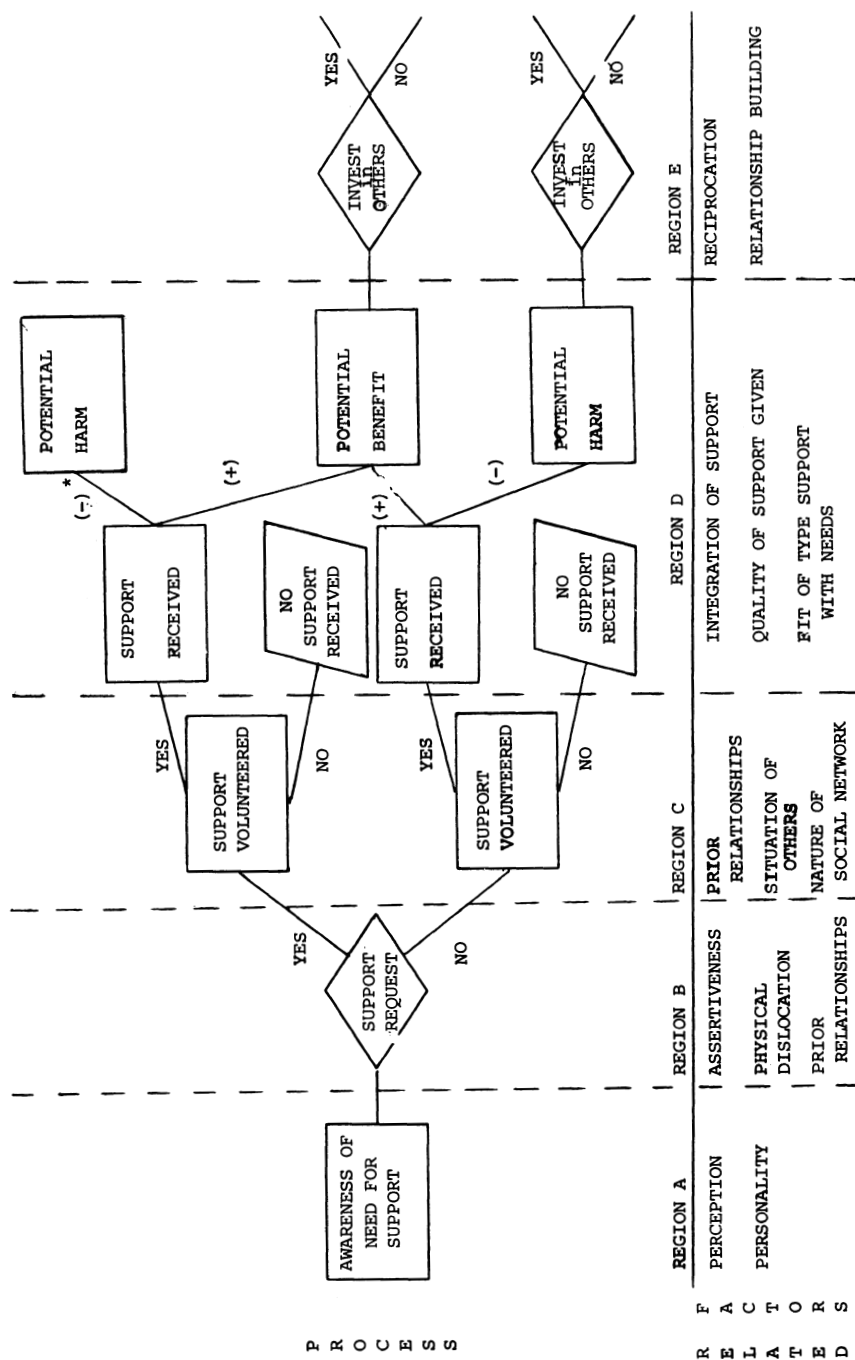
Friends and family have difficulty deciding where their allegiance lies and support systems tend to erupt and fracture during divorce, thus limiting their potential effectiveness (Wilcox, 1981b). When individuals are blamed in a divorce, they may be intentionally isolated, and singles may not fit in well with old friends who mingle in couples and whose relationships may have been based on what the couple had to offer, not its individual parts. Jealousy toward the single in social circles where couples are the norm or threat that "my spouse may be attracted by the single life too" may also contribute to isolation of divorced individuals from their support systems.

Application of Theory and Model Building

The recent explosion of research in the area of social support has provided a wealth of information about the nature of the relationship between social support, stress, and health consequences. Reviewing these studies and critical reviews of research (Cobb, 1976; Dean & Linn, 1977; Thoits, 1982) it can be seen that this area of investigation has been observational and generally atheoretical and for the most part devoid of attempts to create unifying models. This is not to be criticized, as observation is a necessary stage of scientific advancement (Kuhn, 1970). Future attempts might be focused on combining observed relationships of social support with theoretical directions derived from clinical, social, and community psychology and sociology in order to build a preliminary model of the set of behaviors related to social support. The purpose of any such exercise is to logically unify current knowledge and to challenge predictive ability of current assumptions. Research will then be able to test aspects of the model so as to create increasingly accurate explanations of the social support process. This, in turn, is a necessary step toward clinical application.

A diagram of the stages of the social support process is depicted in Figure 1 below. This diagram proceeds chronologically across five necessary stages of the support process: 1) the perception of need, 2) recruitment of support, 3) support

FIGURE 1. A Process Model of Social Support.



*Note: (+) signifies "good" support and (-) signifies "bad" support

response, 4) integration of support, and 5) reciprocation of support phases. These five phases are depicted in regions A through E, respectively, of the diagram.

A number of general corollaries for the study of social support may also be derived from this process model. The first corollary is that breakdown in any stage of the process might lead to a diminished provision of support. A second corollary is that even if support is provided it may not be effective and may even be harmful in nature. A third corollary is that the personality, social, cognitive, and demographic characteristics of the individual and of potential supportive agents (corollary four), will affect the quantity, provision and acceptance of help giving. A final general corollary of this process model is that the behavior of individuals between crises and active support periods will effect the availability and quality of support during crises and active support periods.

Beginning in Region A of Figure 1 it may be seen that the first stage in this model of social support is the awareness of the need for support. Research on loners, for example, suggests that such individuals become accustomed to their status and do not perceive a need for social relationships as a mechanism to help them solve life's problems (Lowenthal, 1964; Weiss, 1973). Such individuals probably develop the perception that they need to "go it" alone. Differences between individuals also may account for the degree a particular event is perceived as stressful or threatening and therefore would be coupled with an awareness of a need for social support (Lazarus, 1977; Meichenbaum & Cameron, 1978).

Spielberger's (1966) model of state-trait anxiety has been applied in this framework and results of a carefully devised series of studies by Teichman (1978) have shown how ego-threat and situational conditions account for individuals' desire to affiliate or isolate themselves from others. Embarrassing situations tend to result in preference for isolation rather than affiliation. As high trait anxious persons are more likely than low trait anxious individuals to react with elevations in state anxiety to ego threatening events, it is not surprising that high trait anxious individuals would be most likely to isolate rather than affiliate when confronted with potentially embarrassing life events. As discussed earlier in this paper, such embarrassment may be self-perceived in such cases as divorce, illness, work problems, and other events where stigma or feelings of failure are attached. A reaction to isolate may cut off any possibility of receiving support, even that support which may be volunteered without it being requested.

If an event or situation is perceived as requiring the support the individual must then make efforts to request support, depicted in Region B of Figure 2. Prior research and thinking on self-disclosure may be instructive in the study of this point in the model (Jourard, 1964; Mowrer, 1964). Research on

self-disclosure has indicated that individuals differ as to the degree they are willing to share intimate feelings and personal thought with others, and consequently in communicating information that they are in need. Individuals who are unable to share personal problems and feelings may give mixed messages as to their need for support and may provide an abstruse picture of just what their needs may be. Self esteem has been considered a key personality determinant of self-disclosure.

Recent attempts to enhance self esteem in assertiveness training workshops have built on similar conceptions of the personality prerequisites involved in making one's needs known (Flowers & Boomraem, 1980). Thus, while high self esteem individuals may have less immediate needs for social support during crisis (Pearlin, et al., 1981) they may be more equipped at obtaining support when they feel the need than persons low in self concept.

Work on social skills training among children and adults has also highlighted that the ability to problem solve is, in part, dependent on certain basic social competencies (Shure & Spivack, 1982; Weissberg, et al., 1981). Thus, individuals who perceive the need for social support must feel enough self esteem to assert their needs and feel that they are deserving of social support (e.g., love, affection, help). Following this they must have the social skills to translate this set of perceptions about their environment and themselves into effective behavior. So for example, loners with poor social skills have been seen as poor candidates for friendship and as actually frightening away potential support (Weiss, 1973; Fromm-Reichmann, 1959)..

Friendships also have different histories and fill a variety of social and instrumental needs. In this regard, the extent to which help has been requested and provided in the past may lead to feeling of social debt, attachment, and social ease. Research in the area of exchange theory and ingratiation may be applied to the social support process (Vanfossen, 1981; Walster, Walster & Berscheid, 1978). In general this research has shown that future social interactions may be explained by the adjudged balance of past social interactions and their outcome, as made by participants in the interactions. To the extent that social exchange theory is applicable to the support process it may be predicted that artificial attempts to activate social support where there is an absence of prior social commitment (exchange) will have dubious effects (Huesmann & Levinger, 1976). Self help groups may appear to be an exception to this, but in fact such groups may be seen as being based on a restricted sense of group belongingness and a willingness to establish social exchange in a defined framework and often along defined activities. New individuals "borrow" on the support reserve of the active members, and are expected at some future time to themselves contribute.

Following the stage at which individuals may request support there exists a stage as depicted in Figure 1, Region C where the

process of social support is out of the individual's direct control. At this point the social network needs to respond. Support may be volunteered in reaction either to a request for help or to a supporting agent's perception that there is a requirement of help even if not solicited.

While it has not been presented within this model, the work on the qualities of social networks has identified a number of parameters that are related to this phase of the support process and therefore to positive outcome during stress. Among these characteristics are density, multiplexity, size and domain (e.g., family, friends) of the social network (Hirsch, 1980; Wilcox, 1981). Qualitative characteristics of the social network -- intimacy of social relations in particular -- have also received attention (Billings & Moos, 1982; Conner, et al., 1979).

Again, research on exchange theory could be heuristic if applied to this stage of the support process. Persons who have received support in the past would be more likely to give support in the future. Such mutually supportive networks might be most effective as they would be based on an already rehearsed repertoire of behaviors. Role theory (Merton, 1957; Parson, 1951) may lead to the prediction, however, that those in giving roles in the past (e.g., mothers) may neither know how to request help from others, nor may others be accustomed to offering support to them. In this author's own research on medical distress even older mothers reported upon debriefing that they did not notify grown children of their surgery "so as not to worry them."

While personal characteristics of network members, make-up of the social network and past relationships with significant others are important, geographical factors also play a key role in this stage. Even change in an hours driving time has been shown to radically alter the nature of friendships, although it had less an effect on family ties (Lee, 1980). Since the number of intimates is usually not more than a chosen few and given the mobility of modern society, the social network is quite fragile.

When at this stage help has been provided there still exists a significant question as to whether the supportees can utilize what is offered them (See Region D). One or two studies have touched on the ability of the supportee to utilize support (Coyne, et al., 1981; Tolsdorf, 1976), but this aspect has not been well developed. While some authors have subcategorized types of social support into emotional, instrumental, advice giving components and further into subcomponents of these (Gottlieb, 1978), there has been little or no mention of the quality or appropriateness of the help given to the category of need. So, while support may be provided it may not be appropriate or the supportee may not be open to receive it.

A second aspect of the way in which support is received may be linked to personality theory. Even the best of support may be interpreted otherwise by the supportee. Referring back to research already cited, depressed persons for example were found

to belittle the support they received, despite the fact that they received more of it than non-depressed individuals (Coyne, et al., 1981). The personality interprets events and incorporates them adding meaning. So, for example, whereas social support has been seen as effective to the extent that it provides information that we are loved, needed, and important (Caplan, 1974; Cobb, 1976), individuals with low self esteem, problems trusting others or fear of rejection, may be resistant to accept such messages. This is the stuff of which transference is made, yet it has not been well addressed in the social support literature. There is strong indication from what few studies which have examined the personality of the supportee that this is a determinant factor in the effectiveness of social support (Hobfoll & Walfisch, in press; Mitchell, 1982; Sandler & Lakey, 1982; Tolsdorf, 1976), and it is at this point in the process when it would show its influence.

Where resources are available and accepted by the individual some may fit situational needs better than others (i.e., may be more ecologically congruent). In a study reported earlier by Hobfoll & London, 1983 it may be recalled that among women whose loved ones were mobilized into the military during the June 1982 "Shalom HaGalil" Israel-Lebanon conflict, those who reported more intimate friends and who reported more support during the crisis period experienced more, not less, state anxiety than less well supported women. High self esteem and sense of mastery, in contrast, were negatively related to psychological distress. Upon debriefing it became apparent that women who met and supported one another generated "rumor mills" and reminded one another of their common plight. Family support had no such negative effect, but nor did it have a positive effect on emotional distress. One interpretation of this data is that the providing of a resource was not consistent with the emotional demands of the situation. In contrast, internal resources were quite effective in the coping process (see also Pearlin & Schooler, 1979).

Following periods of crisis or high need for social support there exists a period typified by reciprocation of support, relationship building, and maintenance of current relationships (Region E). To the extent the individuals gives to others, establishes new or more intimate relations, and invests in relationships with loved ones a strong basis will be built for future demand. Such thinking is based on exchange theory (Walster, Walster & Berscheid, 1972), and findings in the social support literature that have shown size, density, multiplexity and more important, intimacy, to be critical elements of effective social networks (Brown, et al., 1975; Hirsch, 1980; Wilcox, 1981a, 1981b).

If interventionists are interested in affecting social support systems for the benefit of individuals this relative period of calm can be exploited to build these aspects of social networks. Such intervention will require application of knowledge and theory in regard to social anxiety, social competence and

problem-solving. Techniques of systematic desensitization, assertiveness training, cognitive restructuring, and instruction in social skills are potential strategies. A preventive approach will be critical because persons who do not have these skills and competencies will not be able to instantaneously obtain them during crisis. These characteristics require time and significant investment to develop. It also follows from earlier discussion that having a supportive social network is not enough. Because crises so often occur during transition periods individuals also require the ability to construct new social networks and be open to developing intimate relations outside of family and life-long friends (Felner, Farber & Primavera, 1983). Professionals may aid in his process by promoting self help groups during predictable crisis periods, even before the crisis occurs (e.g., impending divorce, entering university, impending retirement, pre-location, post-relocation, and when chronic illness is suspected (Bloom, 1979).

CONCLUSIONS

This paper has attempted to outline methodological limitations in the study of social support, to present personal and structural-environmental characteristics that may limit the effectiveness of social support, and to outline a possible process model in which social support may be studied in a more wholistic fashion. To the extent that future research is applied to this or other models, attempts at clinical application using social support strategies can be attempted with greater confidence.

The employment of the concept of "ecological congruence" will be a key factor in this regard. The time has arrived when just as researchers now regard as naive the notion that all stress is bad, likewise all social support is not necessarily good, is not appropriate in all situations, nor to everyone, nor is it always "supportive." To some, on the other hand, it is more vital than most research would have us believe. Certainly research will show the models and concepts presented here as naive. To the extent that they do so, however, they will need to borrow from areas of psychology and sociology that have until this time remained isolated from the current body of research.

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PART V

HELPING AND THE COST OF CARING

THEORY INTO PRACTICE: ISSUES THAT SURFACE IN PLANNING INTERVENTIONS WHICH MOBILIZE SUPPORT

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This paper addresses two sets of issues that have cropped up in the planning of intervention programs that mobilize or improve the social support available to vulnerable populations in the community. The first set of issues emerged in the process of planning the design of a supportive intervention for adolescents whose parents had recently separated or divorced. I was faced with the practical matter of deciding exactly what kinds and sources of support would most appropriately meet the needs of youth facing this family disruption, a decision which ultimately entailed an analysis of the best match between the several stressful demands surrounding this life event and the supportive provisions that different people could marshal on behalf of these youth. In particular, I debated the virtues of an intervention involving their participation in a peer group context as opposed to one emphasizing their involvement with a single close associate or confidant. The considerations surrounding this practical decision proved highly instructive because they spotlight issues that practitioners must confront in planning appropriate support-mobilizing interventions among other client populations. They also reveal how wide the gap still is between basic research on social support and the information needed by community mental health workers and clinicians. The second set of issues also were prompted by practical concerns. The Ministry of Health in Ontario is eager to find ways of preventing the institutionalization in nursing homes and in other costly government settings of elderly persons who are in frail health but still capable of living in the community, with extra support. Recognizing the critical role of family caregivers in diverting the elderly from institutions, Ministry officials want to know what can be done to shore up these intergenerational caregiving relationships. In planning a study

of these relationships, I gained some insights into a set of social-psychological factors that have an important influence on people's perceptions of the supportive character of their interpersonal ties, and in the process, I came to appreciate some of the complexity underlying many of the subjective measures of social support that are presently being used. Equally important, by identifying these social-psychological variables, it may become possible to alter aspects of the parties' helping transactions, both to produce more favorable perceptions of their supportive character, and to increase the likelihood that these caregiving relationships will endure.

CONFIDANT OR PEER GROUP SUPPORT

Recent reviews of the literature concerning the nature and measurement of social support reflect a growing consensus about its multidimensional character. Specifically, researchers have operationalized the construct in three ways, one focussing on the extent and quality of the individual's ties to a network of close associates, the second probing the degree of affective solidarity between the individual and a single close associate, and the third concentrating on the individual's perceived sense of social support, which is a subjective estimate of the adequacy of the support that can be garnered from the social field. This last approach to measurement is largely phenomenological in nature, not transactional. Indeed, as Gore (in press) points out, the psychological sense of support is fairly stable over time and may be more properly conceived as a personal characteristic that colors self-perception than as an environmental resource. In contrast, the first two approaches tap experienced support but differ in the unit that is regarded as the most critical source of support, namely, the more diffuse ties of the social network and the more intimate tie afforded by a confidant. Those interested in the social network's role in mitigating stress (e.g. Hirsch (1980); Wellman (1981)) are involved in the study of support systems while those who spotlight the confidant's protective influence (e.g. Brown, Bhrolchain & Harris, (1975); Miller & Ingham (1976); Lowenthal & Haven (1968)) are involved in the study of close relationships or social intimacy.

Although there is disagreement about whether health protection is predicated upon the existence of a social intimate or upon participation in a supportive social orbit, there is broad consensus among researchers about the range of supportive provisions that both sources can render. The categories of my original classification scheme of informal helping behaviors (Gottlieb, 1978) have been collapsed and reorganized in a number of ways but generally, have been distilled to 4 classes of social support: emotional support, tangible aid, cognitive guidance (appraisal), and socializing and companionship. A fifth class of

support also appeared in the classification scheme but has received much less attention. Originally, I called it "indirect personal influence", but now refer to it as, "milieu reliability" because this phrase captures the idea that support is conveyed through the individual's faith that, should the need arise, specific people or resources they control will be quickly mobilized. O'Connor and Brown (1984) had the same idea in mind when they recently coined the phrase, "felt attachment", signifying "feelings of inner security or safety felt to reside in or derive from the relationship irrespective of whether they were reflected in behavior: for example, the importance of the other person just being there, the feeling that they would always be there and always willing and able to help out, and the extent to which the woman could imagine the person not being there" (p. 7). Milieu reliability differs from the more global psychological sense of support because it is an appraisal of a specific feature of the individual's relationships, not an evaluation of the overall adequacy of the available support.

In sum, while empirical work has illuminated the substance of support, it has not informed our knowledge about whether a single dyadic bond or a network of close ties is most strongly implicated in reducing vulnerability to different stressors. Practically, should we attempt to design interventions that increase relational intimacy, examples being Lewis' (1978) intimacy therapy and Miell's (1983) strategic self-disclosure, or should we concentrate on ways of mobilizing or restructuring the social field in which people are embedded so as to optimize the support it provides? This is the sort of question that mental health practitioners are left with when they read the conflicting conclusions reported in the literature. For example, in their recent review of 17 studies examining the buffering effect of social support, Mitchell, Billings & Moos (1982) point out that in some instances network measures of support condition the impact of certain stressors on certain aspects of psychological functioning, and in some instances interaction with a single confidant has a health-protective impact. Moreover, there is little consistency in the results of studies that pit the two types of support sources against one another. For instance, Lowenthal and Haven (1968) found that elderly persons who maintain contact with at least one confidant, even in the face of reduced social contact with others over time, report more positive mood states, greater life satisfaction, and better health status than those without such a close tie. Similarly, Henderson and his colleagues (Henderson, Byrne, Duncan-Jones, Scott & Adcock, 1980) found that the support indexes that tapped social integration buffered the effect of stress on psychiatric symptoms for men, but that the indexes tapping close affectional ties did so for women. And Miller and Ingham (1976) report a lower level of symptomatology among both men and women with diffuse support (knowing many people in the neighborhood and at work), but find that this type of

support is particularly important for men while the presence of a confidant proved of most importance for women. Can we conclude on this basis that social support interventions designed on behalf of men should feature manipulation of their broader field of contacts while those planned for women should concentrate on strengthening close attachments? Does the male sex role confer a diminished need for emotional intimacy during periods of adversity, while the female role heightens this supportive requirement?

Furthermore, when we examine more closely certain intervention programs that mobilize social support by creating peer support groups it is not clear whether their beneficial consequences stem from exposure to the group's supportive milieu or to a single supportive companion. For example, in Vachon, Lyall, Rogers, Freedman-Letofsky & Freeman's (1980) study of support groups for widows, those in the intervention group were purported to have adapted more quickly than the controls, both socially and psychologically, because of their involvement in a self-help group. Yet, a closer reading of the actual intervention reveals that each of the widows was contacted by a woman who was herself a widow and had resolved her own bereavement reactions. Moreover, the authors report that these "widow contacts" were available to the widows for as long as they were needed. Initially, they offered one-to-one support as needed, including practical help in locating community resources, supportive telephone calls, and face-to-face interviews" (p.1382). Only later did the widows become involved in small group meetings. Are the results then attributable to the group exposure, the more intimate dialogue between the widow and her "widow contact", or a combination of the two? Similarly, in Minde, Shosenberg, Martin, Thompson, Ripley, & Burns (1980) widely cited study of support groups for the parents of premature infants the same confounding of dyadic and group support may have occurred. Here, a nurse was designated the group coordinator but in addition, "talked to virtually all the mothers and fathers in between group meetings" (p.5). In fact, the authors report that in three cases the parents "attended their first group meeting only after they had four to six individual sessions with the group coordinator" (p.7). Finally, in the only study that ostensibly compared confidant support and self-help group support, Barrett (1978) found that widows improved equally in both conditions, neither group showing greater gains than a control group composed of widows awaiting participation. However, even here, the intervention involving the creation of confidant pairs was not a pure test of the impact of support from a single intimate tie because group discussions were held following each of a series of intimacy tasks in which the confidant pairs engaged. But Barrett (1978) offer an important lead about the planning of interventions when she states that: "The confidant group strategy might be more effective if limited to widows who have no confidant prior to treatment" (p.29). Her words signal the fact that the type of supportive intervention

that is called for should be determined by first examining people's unmet needs for different kinds of support from different sources.

To summarize, the empirical evidence both from studies of the stress-buffering role of social support and from intervention programs involving the mobilization of social support suggests that both a confidant and a network can provide a measure of health-protection in the face of adversity. But the intervention studies have tended to confound the two while alerting us to the need to assess in advance whether clients' deficits lie in the area of relational intimacy or peer group ties. Similarly, the buffering studies point to the fact that confidant support and network support are called for in response to different types of stressors, suggesting the need to examine more closely the fit between the demands associated with particular types of life events and transitions and the provisions available from these two sources of support.

As Lowenthal and Haven (1968) point out, even a confidant was not capable of moderating the depression experienced by those elderly who fell seriously ill during the two-year period prior to follow-up. Succinctly, they state "a social support--such as an intimate relationship--may serve as a mediating, palliative, or alleviating factor in the face of social losses, but one should not expect it to cross system boundaries and serve a similar role in the face of physical losses" (p.27). In short, the choice between interventions aimed to improve relational intimacy and those that optimize peer group support, should be made on the basis of knowledge about the "system boundaries" that Lowenthal and Haven refer to and which signify different classes of stressors. Moreover, there are certain life changes and transitions that pose a series of adaptive challenges that call for a succession of different types and sources of support as the individual grapples with shifting demands.

One way to investigate the correspondence between types of stressful demands and the unmet supportive provisions that can be rendered by a confidant or a peer group, is to devise programs or evaluate existing interventions that systematically vary the two elements. In what follows, I outline a set of considerations that entered into the decision to mobilize peer group ties rather than a confidant to meet the supportive needs of young adolescents whose parents had recently separated. I hope that these considerations will be instructive both to mental health practitioners who are faced with decisions about the occasions when these two sources of support are called for, and to researchers who are interested in discerning the process whereby different sources of support ameliorate stress.

Application to Intervention with Youth whose Parents Have Separated

The intervention program which involved the creation of support/discussion groups for young adolescents whose parents had recently separated or divorced was predicated on research showing that marital disruption predisposes the children to adverse health and social consequences (Hetherington, 1979; Kurdek, 1981). In fact, in the very first group session we convened, the participants, who ranged in age from 11 to 14 years old, were asked to enumerate the issues they wished to discuss in subsequent meetings. In doing so, they touched on most of the themes that Wallerstein and Kelly (1980) address in their book, Surviving the Breakup. They talked about feeling angry with their parents about the breakup, feeling sad a lot of the time, blaming themselves for the burdens they had placed on their parents, and feeling torn between allegiance to one or the other parent. They also talked about the fact that they had seen their grades in school plummet because they seemed to have a hard time concentrating. On the social side, they gave voice to feelings of embarrassment and most notably, to feelings of estrangement from the peer group in which they had formerly been active participants. In fact, one of the participants caricatured the tension she experienced in her contacts with former close friends by drawing a cartoon in which she is holding a bag called "My secret" while her friend is rambling on and on about an irrelevant subject which the artist epitomized with the words "Blah,blah,blah." The cartoon depicts the thwarted need to confide about her parents' separation. Another participant simply wrote the words, "Am I different?" above a cartooned self-portrait. Wallerstein and Kelly also observed that the children feel stigmatized regardless of how visible and widespread divorce is in the local community. Finally, evidence that the process of marital disruption entails exposure to a cluster of secondary stressors was provided by the participants' reports of having to take a job, moving from one home to another, and being assigned a variety of new household responsibilities necessitated by their single-parent situation.

The critical question for planning interventions on behalf of these youth is whether a confidant, a network of peer ties, or a combination of the two can most effectively render the supportive provisions required for meeting the many stressful demands triggered by the separation. Our experience suggests that there is a hierarchy of needs of which the first and foremost is to mitigate the catastrophic sense of loss and the attendant anxiety about abandonment and social rejection. Indeed, while the severing of the marital bond catalyzes these anxieties, the parents' preoccupation with their own post-separation problems intensifies the childrens' attachment-related insecurities. Illustratively, one of our group participants drew a picture of his mother waving goodbye from a car as she set off on a weekend

jaunt with her boyfriend. Wallerstein and Kelly's (1980) data also document the fact that the parents are simply not available to their children, noting that one third of their sample of children "experienced moderate to severe deterioration in their mothers' day-to-day care for them" and that "only 10% of the youngsters felt strongly that the father was sensitive and understanding" (p.41-42). In short, in the aftermath of parental separation parental support is chimerical.

A second demand is related to the first but has more to do with the childrens' need to come to terms with the meaning of the event for their own identity, self-concept, and self-trust. Their worries about being different from others not only shook their relations with others but also shook their sense of themselves. Moreover, the stigma they felt carried over to anxieties about being different later in life as well--seeing themselves as forever handicapped and vulnerable.. Their parents' separation undermined their self confidence and self-trust largely because it raised questions about the extent to which they were personally implicated in its genesis and presently culpable for experiencing conflicting loyalties to the two parties. In short, to the degree that their parents' separation made so much of their present life discontinuous with their past life, and to the extent that it supplanted peer ties that reinforced their former social identities (Hirsch, 1981), it fundamentally undermined a stable sense of self.

A third demand placed on the children stems from their ongoing relationships with their social networks. Once the separation has actually occurred, their relations, especially with parents, are conflictual, and sibling relations are also tense. Our participants talked openly about the discomfort they experience when one parent asks them for sensitive information about the other and when one parent confides in the child about matters that exceed his/her maturational level. Further, when custody, visitation, and other domestic arrangements are made, the children want a part in decision-making but do not assert themselves, fearing they will add to their parents' distress or provoke rejection. Beyond parental relations, interactions with other relatives and especially with the parents' boyfriends or girlfriends are fraught with difficulty. Breaking the news of their parents' divorce to peers and teachers can be an insurmountable challenge when they recall their own reaction to its mishandling by a parent.

Keeping in mind these details of the stressful demands placed on the children of separation--their feelings of loss, abandonment and rejection, their threatened sense of identity, and their ongoing interpersonal difficulties--what supportive provisions can a confidant as opposed to a network of peer ties provide? And where is the deficiency in the two types of support most acute? In practical terms, would a support group composed of similar peers or a confiding relationship with a peer or adult

supply better support to meet the three types of adaptive challenges they face? Historically, the latter option has been chosen in North America, taking the form of the Big Brothers and Big Sisters organization. More recently, another voluntary organization called Parents Without Partners offered children from separated homes a chance to socialize with one another (Parks, 1977). In contrast, a self-help style support group offers an altogether different forum for addressing the childrens' concerns.

In designing the support group program we were mainly concerned with compensating for the loss or depletion of their peer network and fostering a sense of reliable alliance with those who were also undergoing the same family crisis. Moreover, we opted for the more diffuse ties of a peer network rather than introducing or strengthening a single close tie because we found that their parents' separation had fragmented their wider ties to family members and friends while simultaneously drawing them into a closer relationship with one or two social intimates. The data we gathered from 10 adolescents before we launched the support group meetings revealed that, with only one exception, each of the children nominated one person to whom he/she felt especially attached and whom, above all, he/she trusted and relied on in special ways. For three children it was a family member, for two a neighbor, two others had a close friend in this privileged category, one nominated his father's girlfriend, and one the Big Sister to whom she had been matched. We did not want to risk the possibility of undermining these close relationships by grafting a new close tie onto their social field. It follows that the Big Brothers and Big Sisters organization may be unintentionally supplanting the close relationships of children from single parent homes by martialing a new outside relationship instead of cultivating close ties inside their networks or reinforcing those that exist. Before a match is made, the organization should first assess the availability of a confidant in the child's existing social orbit, and explore the child's feelings about whether there is someone with whom he/she would like to develop a close relationship. This observation simply underscores an earlier point, namely, that intervention must be guided by prior assessment of the presence and sufficiency of diffuse and intimate support.

By opting for the support group format, precedence was given to redressing what Weiss (1973) has called, "the loneliness of social isolation" over "the loneliness of emotional isolation". The literature on mutual-aid groups shows quite clearly that the opportunity to meet others with similar problems is the single most important element that moderates the feelings of undesired uniqueness that members bring to the group (Gottlieb, 1982). For adolescents in particular, the feedback of the peer group can bring a measure of stability to their shaky self-conceptions, helping to normalize their feelings and counteract the stigma attached to their identities as children from separated homes. As

noted earlier, many of the children are upset by their feelings of conflicted loyalty to their parents, which leads them to trust themselves less. By first recognizing that other children are experiencing the same emotional turmoil, they come to see it as a natural bi-product of the situation to which they have been exposed rather than as a reflection of their own emotional instability. In short, the process of social comparison helps them to reattribute the cause of their distress to the situation and their feelings are thus less threatening because they are not causally linked to some deficiency in their own character. This is a unique advantage of the peer network relative to the intimate bond; while a close relationship can provide feedback that stabilizes feelings and self-views (Swann & Giuliano, 1982), it provides only a single point of reference for judgments about the causes of these feelings. In contrast, the peer network not only contains multiple sources of feedback, but also offers the extra consensus information that leads to situational as opposed to dispositional attributions about the causes of those feelings (Kelley, 1967). Thus, in trying to come to grips with their mixed emotions about their parents' separation, emotions that de-stabilize their own self-views, group members see others feeling much the same way and conclude that their feelings arise from a compelling situation, not from their personal deficiencies. In this way, their peer group interactions can help them to maintain the emotional equilibrium that is necessary to shore up their coping efforts.

The group experience also prevents the participants from blaming themselves for being unable to effect certain changes in their family situations. For example, some of the youth in our program had attempted to entice the noncustodial parent to return home. When they learned that other group members had also made unsuccessful efforts to convince their parents to try to make a go of it again, they were less likely to make internal attributions for failure, reassigning the cause of their lack of control to situational forces beyond their influence.

The support group addresses the third demand facing these children--the management of their ongoing network relationships--by directly examining ways of dealing more effectively with difficult social situations. The two group convenors, senior university students in their late twenties, both of whom have lived through a family separation of their own, asked the group members to jot down tricky scenarios that they had trouble handling, and then suggested they role-play and critique alternative ways of responding. There is nothing novel about this dimension of the group's work, consisting as it does of problem solving and behavioral rehearsal, except that once again, it draws the members' attention to the commonalities in their predicament and underscores the value of modeling and personal testimony in the mutual aid process. The toughest vignettes for the children to handle were those that called for them to be assertive toward

their parents about their rights and needs in the domestic context because much of the style of assertiveness they had seen at home led to relationship collapse.

A few additional comments about the provisions of peer group versus confidant support merit attention. First, what may appear to be a defense of the superiority of peer group ties over a close relationship in responding to the demands of parental separation should be more appropriately seen as a preliminary exercise in matching types and sources of supportive provisions to types of stressful demands. To this extent, I am pursuing a line of analysis similar to Cohen and McKay's (in press) in which they elaborate a "stressor-support specificity model" except that I have concentrated on the kind of support rendered by a peer network as opposed to a confidant, and I have emphasized the need to assess people's access to both types of support prior to planning interventions. The thrust of both analyses underscores Pearlin and his colleagues' (1981) general principle that the coping variable under investigation must be matched with the stress situation it is presumed to affect. No doubt, the analysis will become more complex (others might say refined) as dispositional variables are integrated within this situationally-specific perspective. For example, our support group intervention might be of benefit only to children with an internal locus of control who can make better use of the support they receive (Sandler & Lakey, 1982), or it might be better suited to children who have a modicum of social skills or who are more willing to disclose personal information in a group setting (Graham & Gottlieb, 1983). Shy, more withdrawn children might profit more from intimacy enhancement, in which case they might have a Big Brother or Sister assigned to them. But practitioners might become paralyzed by all these details, and national organizations like Big Brothers/Sisters are not about to shut down operations until all the data are in. However, they may be interested in conducting evaluative research that addresses the question of who benefits most and least from their interventions. To my knowledge, no study has systematically examined the personal characteristics of children who weather a family breakup better or worse, however measured, upon being assigned a Big Brother or Big Sister. Once we have enrolled a sufficient number of participants in our support group intervention we hope to undertake just such an analysis.

Second, even within a peer support group intervention, the more intimate interaction offered by a confidant can occur. For example, our support group members requested that we plan several social activities such as an afternoon swim party, a dance, and a trip to a local conservation area. On these occasions, the two convenors observed certain children pairing off while others initiated an intimate dialogue with one or the other convenor. This natural phenomenon not only reveals that the more diffuse support of the group is insufficient for some participants, but

also that there is a good deal of flux in some of the childrens' supportive requirements. It is not clear whether the participants brought a need for intimate dialogue to the group or whether the group process aroused this need by including companions who were especially drawn to one another and then sought one another out when the opportunity for a private exchange arose. Moreover, the tendency toward "buddying" has been noted in many reports of mutual aid and support group interventions. Certain members contact one another between group sessions, individuals become attached to their (self-help) group "sponsor", and these dyads endure long after the last group meeting. In short, the group ties afford access to more intimate bonds which are integrated within the individuals' everyday lives. In network-analytic terms, a unidimensional tie to a co-member of the support group is transformed into the more intimate multidimensional tie when that co-member also becomes a close friend.

FACTORS CONDITIONING PERCEPTIONS OF RELATIONAL SUPPORT

A second issue in the study of social support also deserves more detailed analysis in order to inform intervention strategies. It concerns the factors affecting people's perceptions of relationships as supportive or not, that is, the factors influencing people's experience of being supported by others. Many of the current measures of support seem to probe this phenomenological dimension, reflecting the researcher's belief, that "...social support is likely to be effective only to the extent perceived" (House, 1981, p.27). However, no one has inquired into the factors that influence perceptions of relational support. We know precious little about the mental calculus involved in these judgments. No doubt, they are conditioned in part by the respondent's affective state at the time of reporting, and in part, by stable personality traits. But, in addition, social-psychological factors and interactional processes between helpers and help recipients must lie anterior to the perception of relationships as supportive or not. This latter set of factors is more amenable to change than the former two and constitute the subject of the discussion that follows. Before identifying these factors, I will again briefly describe the practical situation that has drawn my attention to them.

There has been much interest among gerontological researchers in the factors that distinguish between the elderly who are institutionalized in nursing homes and those who remain in the community. Studies comparing these two groups' health status, income levels, and former proximity to institutional facilities have ruled out these potential explanatory factors, instead converging on differences in their access to social ties and particularly, in their access to a "family caring unit" (Brody,

Poulshock & Masciocchi, 1978; Smyer, 1980). The latter term disguises the fact that it is the younger generation of daughters and daughters-in-law who are the principal sources of support for their elderly relatives (Brody, 1981). Recognizing the importance of their role in preventing the institutionalization of the elderly, several investigators have examined the burdens this role incurs for the helpers, as well as ways of lightening their load. For example, in New York, the Community Services Society offers both support groups and direct services to alleviate some of the emotional and practical demands placed on those attempting to meet the supportive needs of elderly relatives. It seems evident that those attending this program do not perceive their relationship with their elderly family member as mutually supportive, nor can they garner sufficient support to sustain their helping role from other parties in their natural networks. Equally important, the elderly recipients of their support no doubt also sense that the helping relationship is jeopardized.

So long as both parties to this helping relationship perceive it as supportive, it will endure; perceptions of the relationship as unsupportive threaten its stability and may lead to its collapse, an outcome that could trigger institutionalization of the elderly family member. What are the interactional and social-psychological processes that condition the parties' perceptions of their relationship as supportive or not? A complicated calculus is entailed in this overall judgment, including at least four factors:

- . the extent to which the parties originally entered the relationship voluntarily;
- . the extent to which it constrains the freedom of action and decision-making of the parties;
- . the extent to which it creates feelings of indebtedness that cannot be dissipated; and
- . the extent to which the aid actually rendered communicates pejorative self-perceptions on the recipient's part.

The first factor draws attention to the historical context in which supportive relationships are spawned. The parties' views of whether their relationship was chosen of their own will or imposed upon them will certainly color their feelings about all subsequent transactions involving the provision of aid. Fundamentally, the issue revolves around the parties' views of whether there was a felt commitment to support one another or an obligation to do so. To the extent that the relationship grew out of a sense of affective solidarity, the aid and support rendered will be interpreted as signs of caring and concern. On the other hand, if situational demands, societal norms, or family pressures thrust the parties together, they may begrudge the aid they give and receive. From an attributional perspective, when the parties assign the genesis of their helping relationship to external situational causes rather than internal dispositional causes, they will prize the relationship less and accept the aid less readily.

In the case of the elderly, when failing health or fateful events require an intensification of support, it will be accepted more readily from helpers who are perceived as acting of their own volition rather than out of force of circumstance. In light of this, we should reconsider recent policy recommendations involving the provision of financial incentives to families that take in their elderly relatives instead of institutionalizing them (Sussman, 1976). This proposal can backfire because the motives of families spurred into caregiving only by the promise of such monetary benefits would be transparent to their elderly relatives and negatively color their perceptions of the help they render.

A second factor undermining perceptions of support in relationships is suggested by reactance theory (Brehm, 1966) which predicts that aid that limits the individual's freedom of choice will be negatively colored. More simply, when the strings attached to being helped constrain freedom of choice or action, the help or the helper is resented. A major cost of being placed in the role of help recipient is the loss of independence for the sake of preserving the interdependence with the helper. The elderly, in particular, worry about the loss of control over their lives that may result from moving into a family member's home, perhaps because retirement and other age-related constrictions of their field of activities have already narrowed many options. Equally, when they voice their attitude of "not wanting to be a burden on my family" the elderly are expressing their consciousness that they may infringe on their relatives' space of free movement and engender reactance on the latter's part. Hence, when helping is perceived to limit either party's freedom of choice or when it hems in their movement, their relationship will be viewed as less supportive.

To some degree, every occasion that calls for outside help signals the recipient's dependency on the helper. But it is the interpretation of the nature of the dependency that is most important in determining the acceptability of the help rendered. The more legitimate the reasons for needing help, the more it will be accepted and appreciated. Here again, an attributional perspective suggests that needs for support arising from the force of circumstance are viewed as more legitimate than needs for support that reflect personal deficiencies or skill deficits. Indeed, support is less likely to materialize when helpers believe that the need was brought on by the helpee's own doing or by personal flaws than when it was brought on by fateful events or by factors beyond the helpee's control. It follows that from the recipient's point of view, support that communicates pejorative causal attributions will do damage to his or her self-regard. One apt illustration comes from Vaughn and Leff's (1981) study of differences in the beliefs of high and low EE (Expressed Emotions) relatives about the causes of schizophrenia. They show that high EE relatives saw the patient's behavior as deliberately difficult, levelling the majority of their critical remarks at longstanding

personality traits of the patient, whereas low EE relatives viewed the behavior as resulting from a disease over which the patient had little control. The schizophrenics who relapsed may not have been able to tolerate the overstimulation of their relatives' critical remarks and hostility, and they may have suffered psychologically from the implicit message that they were to blame for their own victimization. Returning to the elderly, when help is interpreted by both parties as a response to needs brought about by advancing years and declining health, rather than by personal failings, it will be extended and accepted more readily.

The fourth and final factor affecting the parties' sentiments about the helping relationship has been recognized in several studies comparing the network ties of clinical and general population samples. Studies by Tolsdorf (1976), Hammer and her colleagues (1978), and Sokolovsky et al (1978) have observed that patients tend to maintain asymmetric helping relationships, failing to reciprocate the support they receive from others. An imbalance in helping exchanges makes interaction less satisfying for both parties because the helper is drained and the recipient feels uncomfortably indebted, suffering also a decline in good feelings about him/herself. However, these studies do not tell us whether the recipient's failure to reciprocate support stems from social skill deficits, from the tendency of network members to induce dependency on the part of the distressed individual, or from such low levels of self-esteem that the individual comes to believe that he/she is incapable of extending the most rudimentary of supportive provisions. In helping relationships among kin, considerations of reciprocity may not be limited to ongoing interactions but may take into account past and anticipated exchanges, as well as more general norms about family and intergenerational obligations. For example, an elderly relative may interpret the abundance of help presently received from her niece as due repayment for past help or even as a debt that will be cancelled by future gifts. To the extent that family norms concerning intergenerational relations dictate that her niece has a rightful obligation to support her in her dotage, equity considerations do not enter at all. But both parties must accept these norms! More generally, unmet expectations of equity in helping may be due either to an inability to repay help or to an insensitivity to the need to do so. In either case, the supportive character of the relationship will be undermined.

To summarize, in a study we are presently conducting, we are examining the helping relationships between elderly persons and their younger generation caregiving relatives, attempting to predict their satisfaction with the supportive character of these relationships from information gathered about the following four questions:

1. Why are the role partners prompted to help one another? Is it out of necessity or role prescriptions on the one hand, or out of personal volition based on affective solidarity on the other?
2. Why do the role partners need help? Is it due to the force of circumstance, thus making it more acceptable and legitimate, or due to some personal failings or deficiencies?
3. Does the process of accepting and extending help entail a felt constriction of the parties' space of free movement, thus engendering psychological reactance, or are the strings and constraints minimal?
4. Does acceptance of the help bring about uncomfortable feelings of indebtedness? Do the two parties have the skills and resources to repay one another's aid?

We have written 20 items tapping each of these social-psychological domains, one set for the caregiver and one for the elderly relative. We will administer them separately to each party and then examine their independent and additive contribution to the prediction of the parties' satisfaction with the supportive dimension of their relationships. Ultimately, we hope that the knowledge we gain about the relative importance of each of the four social-psychological factors affecting the parties' satisfaction with the support they exchange will allow us to prepare guidelines for strengthening intergenerational caregiving relationships.

CONCLUSION

The ideas expressed in this chapter result from the blending of theory and practice. The idea that certain types of stressors call for specific types of supportive provisions represents a recent theoretical advance that can give more direction to the planning of clinical and preventive interventions involving the mobilization of social support. However, when interventions are designed, they call for knowledge that is also capable of informing decisions about the sources of support who are likely to have most influence, and then they must be predicated on an assessment of the extent to which these sources already exist in the target population's ongoing networks or whether they should be grafted onto their networks. Additionally, intervention programs must take into consideration whether the consensus information provided by similar peers is more significant to the target population, given their developmental stage, the demands posed by the stressors they face, and their present access to such peers, or whether the attachment and emotional intimacy afforded by a

confidant is more significant. Moreover, theoretical notions regarding the match between stressors and supportive resources pale in the face of the knowledge that is gained from observations of supportive requirements and processes that emerge in the course of actual supportive transactions. Here, we observe a great deal of flux in supportive requirements as demands and priorities for their resolution shift, and as a result of actual exposure to supportive exchanges. The desire for greater intimacy with a peer or with an adult convenor that we observed during the course of our work with teens may have surfaced only because the participants had an opportunity to sample the support expressed in the group context, and to discern which peers were most similar to themselves along certain valued personal and experiential dimensions. Thus, the receipt of social support has reactive effects of its own, triggering needs for more, different kinds, or different sources of support. Experience from practice enriches and elaborates theory, affording a more dynamic, processual perspective on the field of inquiry.

Similarly, if we fail to inquire about how perceptions of relational support are refracted by historical exchanges between the parties as well as by the norms, meanings, and motives that lie beneath their ongoing exchanges, we have little chance of discovering the ways in which interactional and social-psychological processes condition the conclusions people draw about relational support. More important, by ignoring these anterior phenomena, we close off avenues to intervention that can restore or promote the psychological sense of support in relationships.

NOTES

There are other threats to the stability of the helping relationship, including external pressures such as stressors directly affecting the caregiver or his/her spouse and children, as well as lack of support from these nuclear family members to sustain the caregiving role.

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SOCIAL SUPPORT AND THE ALLEVIATION OF LOSS

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The earliest treatise on grief and mourning to which researchers working in the area of bereavement invariably refer is Freud's 1917 paper "Mourning and Melancholia". The ideas expressed in this monograph became not only the basis for the psychoanalytic theory of depression, but also greatly influenced later conceptions of the emotion of grief. Central to Freud's approach was the notion of the "work of mourning" or "grief work", whereby he meant the process through which the grieving survivor becomes free of his or her ties to the deceased, achieving a gradual detachment by means of reviewing the past and dwelling on memories of the deceased.

It is interesting to note that while Freud was preoccupied with explaining emotional aspects of "grief work" and its function in the process of recovery, paying little or no attention to social aspects, in his private life we have a classic example of the provision of social support during the crisis of bereavement, and a clear illustration of grief work taking place through discourse with a friend. This is documented in Freud's correspondence with Ludwig Binswanger (1957, see also Blau, 1975), with whom he was closely acquainted. Over a period of years both Freud and Binswanger lost beloved family members for whom they grieved deeply. As these tragic events occurred, the two friends communicated their losses to each other, expressing their anguish and despair. In turn, they received expressions of consolation and deep understanding, borne of their own grief and similar experience, and warm assurances of friendship. Significant is the fact that throughout the correspondence each made constant reference to his own losses, telling of his own pain or suffering, while at the same time showing a full awareness of this need to communicate his own previously dormant or at least longer-standing

losses. Clearly, the exchange was mutually beneficial and was recognized as such.

This personal experience had little impact on Freud's theoretical analysis and it is largely due to his influence that the study of grief concentrated for many years on intrapersonal aspects of the emotional process. Only during the last two decades have grief researchers turned their attention towards studying the role of social support in alleviating the impact of partner loss (e.g. Lopata, 1973, 1979; Maddison & Walker, 1967; Vachon, Formo, Freedman, Lyall, Rogers, & Freeman, 1976). This long neglect of the role of social support is somewhat surprising. After all, bereavement is not only one of the most intensely painful experiences that a person is likely to endure but an experience which is caused by the termination of an intimate relationship and the abrupt loss of social support that had been exchanged with the spouse. It would seem obvious, therefore, that the widowed should be particularly in need of social support to substitute for the deficits left by the death of their partner.

The purpose of this paper is to examine the hypothesis that social support furthers the recovery from grief and reduces the risk of poor mental and physical health outcomes among bereaved persons. The first section briefly outlines a deficit model of partner loss which emphasizes the importance of social support for coping with bereavement. The second section then reviews evidence that links the availability of social support to the amelioration of health outcomes following bereavement. Although this work is generally consistent with the notion that social support buffers individuals against the negative impact of partner loss, such purely correlational findings are notoriously ambiguous with regard to causal interpretations. In a third section we therefore discuss bereavement intervention studies as a rare example of research in which the availability of social support has been systematically manipulated. The final section then reviews the implication of these findings for the fields of bereavement as well as social support.

A DEFICIT MODEL OF BEREAVEMENT

The deficit model of partner loss combines assumptions from stress and social support theory to account for the health consequences of bereavement. The basic notion of stress theory is that stressful life events play an important role in the etiology of various somatic and psychiatric disorders. This assumption has been supported by a large body of empirical research which related stressful life events to increases in various physical and mental illnesses (e.g. Dohrenwend & Dohrenwend, 1974; Filipp, 1981; Rahe, 1979) and identified some of the neurochemical and hormonal processes assumed to mediate these relationships (e.g. Ader, 1981; Glass, 1977). Conjugal bereavement is the highest ranking

stressful life event in the Social Readjustment Scale (Holmes & Rahe, 1967) and has been linked to significant deterioration in mental and physical health as well as to increases in the risk of suicide and mortality from natural causes (e.g. Lynch, 1977; Parkes, 1972; Stroebe & Stroebe, 1983).

A central assumption of social support theory (Cobb, 1976; Cohen & McKay, 1983; House, 1981) is that close interpersonal relationships buffer individuals against the negative impact of stressful life events. In an analysis of potential causes of the debilitating effects of partner loss, Stroebe, Stroebe, Gergen and Gergen, (1980, 1982) argued that marriage provides individuals with support to an extent that it fosters a mutual reliance of spouses as their major, often exclusive, source of social support. Marital partners spend more time with each other than other adults, their relationship is more intimate and they provide a range of instrumental, emotional, and social functions for each other that would be hard to equal in other small groups. Since bereavement signifies the termination of this close tie, the loss of a partner should lead to deficits in a number of areas which can be broadly characterized as instrumental support, validation support, and emotional support. It is these deficits which result in, or at least contribute to, the negative health consequences of bereavement.

Instrumental support refers to the provision of material aid as well as to giving advice. Due to the specialization of roles in marriage, the bereaved survivor is likely to be confronted with a substantial loss of material and task support. The more specialized the marital roles, the more drastic will be the effects of partner loss on instrumental support and the more the bereaved person will need others in the network as substitutes. If a widow has never dealt with financial matters or a widower with the demands of child care, the sudden necessity to cope with these tasks, if others are not available to take over for them, will be highly stressful.

Validation support refers to the role others play in helping individuals to evaluate themselves as well as to assess and structure their environment. Effective behavior requires an assessment of reality and of one's own abilities that may take place under ambiguous circumstances. Such evaluations are therefore largely dependent on social comparison processes (Festinger, 1954) and the loss of a partner (that is, of someone who fulfilled a central role in comparison processes) may lead to drastic instability of such judgements. Social comparison processes may also be instrumental in judging the appropriateness of one's own emotional responses (Schachter, 1959), which are likely to be in a state of particular turmoil following the death of the loved person. Many of the bereaved are fearful that they are "going crazy" (Glick, Weiss & Parkes, 1974), and without the

support of someone to reassure them that such emotional responses are the result of the temporary strains of grief and not indicative of a deep-seated mental illness, the fear of going out of one's mind may become an all-engrossing interpretation of reality.

Emotional support is usually taken to mean (cf. House, 1981) the "unconditional positive regard" of the supporter. One does not have to hide weaknesses or put on an act. Such a total trusting relationship may be rare even in marriages, but some approximation to this type of emotional support seems to be on most people's minds when they think of others as "supportive"; Gottlieb (1978) found that more specific acts of social support reported by a group of respondents fell into this category than in any of the others. Whatever the effects of emotional support on ameliorating the bereavement outcome, its effects on health for other stresses seems clearer and more positive than for the other forms of support (House, 1981).

It follows from this perspective that the health consequences of bereavement should be moderated by the availability of alternative sources of social support. Since the problems confronting the bereaved have been analyzed extensively (e.g. Parkes, 1972), it would even be possible to differentially predict the types of social support that should be particularly important during different phases of bereavement. Unfortunately, however, the evidence available to date does not allow us to evaluate such specific hypotheses, as most studies have used only global measures of social support. We will thus have to content ourselves with addressing the more general question of whether the provision of alternative social support lessens the deleterious effects of bereavement.

SOCIAL NETWORK AND SOCIAL SUPPORT

The Availability of Social Support.

The hypothesis that social support reduces health risk would be corroborated if it could be shown that the health status of widowed persons who have others available to them for support is better than for those who do not. One source of information is provided by studies of living arrangements of the widowed; those who live alone lack the potential support of a cohabiter. However, since it will become evident from the review of these studies that confounding factors prevent one from drawing causal conclusions, this research will only be described rather briefly.

The impact of living alone has been examined for a variety of health and well-being measures. Clayton and her colleagues (e.g. Clayton, Halikas & Maurice, 1972; Clayton, 1975; Bornstein, Clayton, Halikas, Maurice, & Robbins, 1973) looked at the effects

of living alone on depressive symptoms among a sample of recently bereaved. While Clayton (1975) did not find that living alone influenced the depressive symptomatology of very newly bereaved (at one month), there were some indications from the analyses of the data reported in this particular publication that with time (after a year) those living with others did better. More clearcut conclusions were reached by Bornstein, et al. (1973), reporting on data from the same sample. The widowed living with their families had lower depression rates at 13 months than those who did not. In fact, only two of the 16 widowed in their depressed group lived with their families, in contrast 35 of the 76 who were not depressed lived in a family situation. Bornstein, et al. (1973), concluded that lack of support could be regarded as a contributing factor to the depression of widowhood. Similarly, Lopata (1973) reported that widows whose adult children lived in the same city did much better in their recovery from bereavement than those who had no children living locally.

A number of studies have found that social ties and interactions with others are related to mortality risk. While most of these do not separate the widowed from the other non-married status groups (e.g. Berkman & Syme, 1979; Kobrin & Hendershot, 1977), specific information of the impact of social ties on mortality risk among the bereaved is available from a study by Bunch (1972). Bunch examined the relationship of recent bereavement of a parent or spouse to suicide in bereaved persons. While her main interest was with the excessive rates of suicide among the recently bereaved (it was five times greater than in the general population), she also reported that the bereaved who committed suicide seemed to be receiving less support from their relatives, that their bereavement caused them more social disruption, and that they were more likely to have been living alone. Previous psychiatric breakdown was also more likely among bereaved who committed suicide compared with non-suicide bereaved. Bunch (1972) concluded that suicide was most likely when the bereaved person had shown a previous tendency to breakdown and was not, during bereavement, closely supported by a family group. Similar findings and interpretations are reported in two further studies of suicide among the bereaved (Wenz, 1977; Bock & Webber, 1972).

However, living with children or family is not always a straightforward help, as the above studies might imply. Vachon, Formo, Freedman et al. (1976), found that widows living alone with dependent children reported more stress than others who were living alone or with children and someone else. The presence of another adult seemed here to mediate the stress of dealing with children while bereaved. It appears then that the presence of dependent children added to, rather than alleviated, the stress. Vachon et al. (1976) further reported that widows who had no children or only children living outside the city where they lived said they were under less stress than did those who either lived

with their children or whose children lived in the city. This seems to be in contradiction to Lopata's (1973) findings, but Vachon et al. (1976) pointed out that the widows in her sample were frequently having trouble with their children. Thus, while living with family might protect the bereaved from the extreme detriments to health, the added problems involved in running and being part of a larger household might have some adverse effects on the adjustment and well-being of the bereaved person.

While studies of the relative effectiveness of different sources of support on extreme mental or physical debility or mortality are not available in the literature, a few studies have looked more closely at the quality and sources of social support and their mediating effects on well-being or depression. For example, Lowenthal and Haven (1968), who examined the impact of having an intimate relationship or confidant, included some data on the widowed in their study. They reported that 45% of the widowed who said they had a confidant were depressed, compared with 73% who said they had no confidant. Unfortunately, the interpretation of this finding remains unclear. Although it might be indicative of the relative importance of emotional social support in times of stress, it could as well be interpreted as showing that depressive individuals are less likely to encourage (or acknowledge) this type of close relationship.

Bankoff (1983), who explored the utilization of social networks in ameliorating psychological distress, found that at different phases of bereavement different sources of social support were linked to improved well-being. Initially interactions of the widows (who were relatively young) with their parents were of most help in reducing distress. Later in bereavement the most helpful were single or other widowed friends. The source of help was found to be more significant for outcome than the type of help given.

In two other studies relationships with family members emerged as critical to outcome. Maddison (1968) reported that support offered by the widow's mother was crucial: widows who complained of the mother's nonsupportiveness, or as being unable to meet her needs, went on to become bad outcome bereaved. Glick et al. (1974) pointed to the benefits of having a supportive family: 70% of their younger widows said that they had received more support from their families than their friends, and benefitted from this.

While other studies confirm that contact with relatives and friends is positively associated with level of life satisfaction and morale among widows (Bahr & Harvey, 1979, 1980; Morgan, 1976), it is likely that friends and family have differential impacts in ameliorating the effects of bereavement. It could be that family were particularly important for the younger widows of the Glick et al. (1974) and the Bankoff (1983) studies because their spouses had died relatively young, and their parents were more likely to be alive than those of the average widow. Further

information on this is provided by Arling (1976) who compared various sources of social support (including family members, friends and neighbors) on the morale of the elderly widowed. Arling found that "contact with family members, especially children, does little to elevate morale, while friendship-neighbouring is clearly related to less loneliness and worry" (p. 757). Similar findings were reported by Pihlblad and Adams (1972), who found that interaction with friends was more closely associated with life satisfaction (measured by the Havighurst Life Satisfaction Scale) than was interaction with children and relatives. Thus, when the comparison is between children and friends, rather than parents and friends, the latter are clearly more helpful. This could be due partly to the additional stresses which children bring to the widowed as we discussed above. It could also be due to the positive effects of social comparison available from interaction with peers (especially other single or widowed persons) and parents (who are likely in many cases to have been widowed too).

Satisfaction with Social Support.

That bereavement outcome is also related to satisfaction with social support has been demonstrated in a few studies. Maddison and Walker (1967) obtained questionnaire data from 132 widows, who were asked to give subjective reports on their health during the previous year. From this data an illness score was derived. Twenty good and twenty bad outcome widows were selected from the total sample on the basis of these scores. Interviews were then conducted with the 40 widows to obtain information about the people with whom they had contact during the bereavement crisis (the first 3 months after death) and their perception of the helpfulness of these people. Helpfulness was rated for the following forms of interaction: expression of affect, review of the past, orientation towards the present and future, and provision of concomitant needs. While there was no overall difference in perceived "helpful" responses, significantly more bad outcome subjects gave the response "unhelpful" than did good outcome subjects for the first three forms of support (expression of affect, revision of past, and orientation to the present and future). On the fourth (provision of concomitant needs), bad outcome subjects gave more "needed" responses than good outcome, indicating that no such interaction had occurred, and that they perceived their environment as failing to meet their needs during the crisis. Bad outcome subjects thus perceived deficits in areas which approximate roughly with emotional (expression of affect) validation (orientation) and instrumental (provisions of needs) support.

Although these findings are consistent with the assumption that a supportive social environment provides some protection against the impact of bereavement, a number of alternative

interpretations are equally plausible: First, the perception of the environment as unhelpful could have been a misperception. Second, their perception could have been valid, but the lack of support could have resulted from their own behavior. Third, since perceived social support was assessed retrospectively at the time at which health outcomes were measured, the perception of social support could have been colored by the poor health of the poor outcome widows. Although the authors present additional data to counter some of these alternative explanations, any causal interpretation of their results must remain tentative.

More recently, a prospective study by Vachon, Sheldon, Lancee et al. (1982), in Toronto, Canada examined, among other variables, perceived deficits in social support as a correlate of persistent high distress among a sample of widows. The study from which Vachon and her colleagues drew this data was of 162 widows under 70 years of age, who were followed for the first two years of bereavement. Vachon et al. first identified variables at one month after bereavement which predicted high distress at two years. The most important variable out of a total of ten was initial score on the General Health Questionnaire (GHQ), a test developed to screen for nonpsychotic psychiatric illness. Three further variables related to social network, either to utilization of, or satisfaction with, resources that it provided. They then isolated two subgroups of widows, "enduring high distress" (who had high distress scores on the GHQ at one and 24 months) and "enduring low distress". Twenty-nine of the 99 widows who completed the two year study had enduring high distress and 30 enduring low. Patterns of distress were related to perceived deficits in social support. Sixty-nine percent of the high and 34% of the low (either enduring low or became low) distress reported a deficit in social support. Patterns of distress were also related to perceived health and financial problems. Thirty-five percent of the enduring high distress group had health problems, whereas only 4% of the low distress group did. Forty-two percent of the former had financial problems, whereas only 10% of the latter did. It seems therefore that financial difficulties might have been a source of additional stress early on in bereavement, and might explain why some of these widows were in the high distress group.

Vachon et al. concluded that while the situation variables (health and finances) were important correlates of high distress, a very common deficit was social support. Although the causal direction remains unclear in this type of study, one plausible interpretation of these findings suggests that, due to poor health and financial difficulties, these widows strained their social support networks to an extent that the other groups did not. Thus their social support network would in fact be less supportive in terms of the needs of the widows, but may have provided a similar level of help to that received by the low distress group. However, in this case of extreme need, it was not enough.

Conclusions

The studies reviewed in this section have shown that high risk of poor outcome among the bereaved is associated with living alone, with poor contact with family and friends, and with perceptions of non-supportiveness from the social network. They have not demonstrated, however, a causal relationship between lack of perceived or actual support and poor health outcome among the bereaved. There are a number of reasons why such a causal interpretation would be premature. First, living alone is likely to affect patterns of illness behavior quite irrespective of actual health status. For example, it seems plausible that bereaved individuals who live alone would be more likely to seek medical attention as an excuse simply to receive social support from the doctor than would widowed who have other sources of social support. A similar bias might operate with regard to admissions to hospitals. Those living alone would be more likely to be admitted to inpatient care, not because their health status is worse than those living with others, but because they lack the presence of someone in the home to take care of them when sick. Thus, they would tend to be overrepresented in statistics of in-patient care, which could lead to the spurious conclusion that living by oneself has such drastic effects that hospital inpatient treatment is necessary.

A second issue is more critical for establishing the cause-effect sequence, and is more difficult to control for or test empirically, namely, that those who live alone and who actually lack social support, or who perceive their friends and relatives as non-supportive, may actually be socially isolated because of their own mental or physical health status. In other words, the interpretation would be that selective processes operate to determine which bereaved are in the "non-support" versus "support" samples, rather than that the protective features of social support per se operate positively on the health of those who receive it (Heller, 1979).

Lopata's (1973, 1979) work points to the possibility that personality differences determine the availability of intimates. She noted that those who develop satisfactory relationships and weather bereavement well, typically had a higher education, a comfortable income and the "physical and psychic" energy needed to initiate change. It seems plausible that individuals who are severely depressed or ailing in some way are less likely to develop or be able to maintain a friendship. Suggestive data on this point are provided by Parkes (e.g. 1972) from his London study of 22 widows. Those who expressed most anger after bereavement became more socially isolated than those who were not so angry (they also reported their health as being worse, although they did not consult their doctors more than the other widows). As Parkes (1972) commented:

"Whether they drove their friends and relatives away or whether they dealt with their angry feelings by shutting themselves up at home, the result was loneliness and insecurity" (p. 105).

A conclusive elimination of these alternative interpretations would require experimental studies, i.e. investigations in which conditions are created that differ systematically in level of social support and in which subjects are randomly assigned to these conditions. Fortunately, such studies have been conducted in the context of evaluation of the impact of therapeutic intervention programs. We will therefore discuss this research in the next section.

THERAPEUTIC INTERVENTION PROGRAMS AS SOCIAL SUPPORT

In a recent handbook of grief counseling, Worden (1982) regards the role of mental health professionals in grief therapy as substituting for religious and family institutions, which have ceased in recent years to give the necessary support to the bereaved person. In many western cultures, organized support has become a replacement for the decline in informal support from the social support network. Similarly, Raphael (1980) stated,

"The background to all bereavement counseling is general support, support that offers human comfort and care that accepts and encourages appropriate grief and mourning".
(p. 62)

Such therapy, then, covers the different aspects of social support that were outlined in the deficit model. Offering human comfort provides emotional support, the provision of care implies instrumental support, the encouragement of appropriate grief supplies one type of validation support. Thus, corroboration of the hypothesis that social support reduces risk would be provided if it could be shown that counseling enhances well-being and reduces mental and physical symptomatology.

Unfortunately, assessment of the effectiveness of intervention programs is frequently judged by the organizers or participants of these programs themselves (cf. Hiltz, 1975: McCourt, Barnett, Brennen & Becker, 1976: Silverman & Cooperband, 1975) or in terms of the need for and use made of a service by the bereaved (e.g. Abrahams, 1972). However, for an examination of the impact of the various types of intervention of health outcome a comparison of the outcome of participants with those of similar bereaved who have been excluded from the program or therapy in question is essential. A number of biases are introduced if this is not done. For example, if participation is voluntary, it is likely that those bereaved who enter such programs and stay in them long enough for assessment (a) feel they need and want this type of support and (b) find it effective or helpful in some way. Differences in outcomes of those who receive help through such

intervention and those who chose not to participate could then be due to selection factors. As Vachon, Lyall, Rogers, Freedman, Letofsky, and Freeman (1980), pointed out, most newly bereaved are somewhat passive and unable themselves to reach out for help, so that programs which rely on the bereaved responding to announcements of the service in the media are likely to obtain those who are less passive and more receptive to an offer of support. There are some indicators (see, e.g. Silverman & Cooperband, 1975) that those bereaved lacking in social support may be the ones who choose to enter such programs. Other research (Vachon et al., 1980) suggests that those who drop out of the intervention may actually be those with fewer social supports available to them, and the worse off generally. Thus there are critical differences between participants and non-participants in the need for social support, the perceived availability of social support and in actual social support, which must be taken into consideration when evaluating the effectiveness of a program actually designed to provide support for bereaved persons.

Although it is sometimes impossible, even when outcome assessment is a key question, to allocate the bereaved to participation versus non-participation in an intervention condition (cf. Lieberman & Videka-Sherman, 1983), to examine the effect of a social support program on health outcome it is necessary to randomly assign the bereaved to either an intervention condition or to a control group which receives no counseling and to compare these groups by quantitative measures of state of health or change in health, ideally before, during and after the intervention and, to be truly diligent and exhaustive at a follow-up some time later as well.

The majority of studies of intervention among the bereaved which comply with the above specifications confirm the general hypothesis that providing support for the bereaved improves their outcome (Parkes, 1980). Thus, an examination of investigations of the different types of counseling should give further insight into the nature of the support given, and the conditions under which it is effective.

Voluntary Aid

One type of voluntary aid for bereaved persons is the so-called "self help" aid, of which widow-to-widow programs are exemplary. The principle behind self-help groups, that grieving persons may be best helped by others who have been through and mastered the traumatic event themselves, is expressed by Silverman and Cooperband (1975):

"The evidence points to another widow who has coped and accommodated as the best caregiver. Very often the first question a widow helper is asked is, "How am I going to manage?" The second question is, "How did it happen to you?" The new widow seems to be seeking a role model,

someone with whom to identify. This other widow can be a friend, a neighbour, or a relative. She offers an opportunity to talk with someone who indeed really understands. She can provide perspective on feelings; she provides a role model; she can reach out as a friend and neighbour - not someone defined as concerned with abnormal or deviant behavior." (p. 11)

Empathic and instrumentally constructive social support is thus provided by a similar other in a non-directive manner. In particular, one distinctive feature of this type of intervention is that it offers, to a greater degree than in other programs, appropriate validation support: counseling is conducted by someone who has been through the same experience and can validate the bereaved person's interpretation of the event, its impact on the bereaved and the various changes that necessitate adjustments.

One study of self-help intervention which conformed to the methodological requirements outlined above was conducted as part of the larger investigation undertaken by Vachon and her colleagues (e.g. Vachon, 1979; Vachon et al. 1982), which was described in the previous section. Vachon et al. (1980) examined the efficacy of a widow-to-widow program by assigning participants either to an experimental intervention condition ($N = 68$) or to a non-intervention control group ($N = 94$). All were interviewed at home at 1, 6, and 24 months after the husband's death, and were given the Goldberg General Health Questionnaire (GHQ), to evaluate overall disturbance. On this measure the intervention and the control subjects were similar at one month (i.e., before assignment to these groups had been made) with regard to general distress level.

The intervention was conducted by a widow of longer standing, who had overcome her own bereavement, and who had attended training seminars examining bereavement problems, the provision of supportive counseling and the availability of community resources likely to be helpful to the new widows. Vachon et al. (1980) described the intervention as providing emotional, cognitive and practical support to the individuals, categories which are similar to those of our deficit model. Intervention in this study was not limited in duration or to set times, and the widow being counseled was free to contact the intervener when she wished. Thus, in this respect, it was comparable with informal support systems.

Only on certain specific items were differences between the intervention and control subjects apparent at 6 and 12 months. At 6 months intrapersonal items were the ones that differed between the groups (for example, intervention subjects were more likely to feel better than they had), whereas at 12 months interpersonal items such as the initiation and resumption of social ties distinguished between them, in favour of the intervention group. Finally, at 24 months, intervention subjects had fewer symptoms on the GHQ, the only time at which a difference in overall disturbance between the two groups became apparent. Following

this analysis of the nature of the items on which the groups differed at the different time points, Vachon et al. (1980) suggested that the intervention group were "proceeding along a hypothetical pathway of adaptation faster than controls" (p. 382). There was some evidence that participation not only accelerated recovery but also reduced risk. After two years significantly fewer women in the intervention group who had high distress scores on the GHQ at 1 month were still in this high risk category, as compared with non-intervention controls.

Vachon et al. (1980) are cautious in interpreting their data in view of a number of shortcomings, and urge replication. Parkes (1980) also pointed out that the "self-help" provided by the widow counselor was backed by the advice and guidance of a skilled and experienced professional, and that the very positive results of this study may not be generalizable to more typical self-help widow-to-widow groups which function without such backing. Nevertheless, the design seems adequate to support the hypothesis that social support both reduces the risk of bad outcome among the widows and shortens the duration of their grief.

An intermediary step between using nonprofessional persons who rely on their personal experience to help them counsel others and professional mental health workers to provide social support for the bereaved is to employ the help of trained volunteers. Parkes' (1979) study is the best example that we have found to date of an assessment of voluntary help services, that is, those in which support is offered to the bereaved by sympathetic, well-informed persons who have been selected and trained by professionals. Parkes' study was part of a much larger provision of care extended to dying patients and their families at St. Christophers' Hospice in London, with which Parkes has been closely involved as consultant psychiatrist since its inception.

High risk bereaved were selected according to a number of criteria (see also Parkes & Weiss, 1983), including lack of supportive families, low socioeconomic status, young age and an evaluation by the nursing staff (with whom, due to the nature of hospice care, they would have become much better acquainted than would be the case in a general hospital). A small "imperative need" group of bereaved was excluded from the study as their need for support was so great that it was felt it would be unethical to withhold this had they been randomly assigned to the control condition, as were the bereaved who were included in the study. The intervention group (N = 32) were given the help of the voluntary service, controls (N = 35) received no such help. Support for the intervention group was provided in the bereaved's home, and besides the emotional and instrumental help typically offered, included "befriending" the person. Thus the volunteer would become more of a confidant than would be possible under professional counseling.

Parkes (1979) examined changes in health in the four years following bereavement. During the first year of bereavement

differences between the intervention and control groups were not apparent, but in the following three years overall scores on health outcome were better for the intervention group than for the control group, and two out of three measures of change in health favored the intervention group. High risk persons who had been assigned to the control group had significantly worse health scores than the low risk bereaved (none of whom had received the support). High risk persons who had received the support had similar health scores to the low risk group. Parkes concluded that voluntary intervention is effective in reducing the health risk in potentially bad outcome subjects.

Professional Help

In a well-controlled study conducted in Australia, Raphael (1977) provided professional support for a sample of conjugally bereaved widows, comparing health outcomes in this group with those of a sample which did not receive crisis intervention. The study was designed specifically to test the effectiveness of an intervention technique in improving mental health of the widows. Two hundred recently bereaved (within seven weeks of the death) widows, who had agreed to a request to participate in the study, were interviewed in-depth and examined for criteria of high risk. On the basis of previous research the following risk factors were devised to delineate high risk: (1) Perceived non-supportiveness of the social network during the crisis of bereavement, as measured by the questionnaire developed by Maddison and Walker (1967); (2) "Traumatic" death of the partner; (3) Ambivalent marital relationship with the deceased; (4) The presence of additional concurrent life crisis.

High risk individuals (those showing one or more of the above risk factors) were randomly assigned either to a treatment group (N = 31) which received intervention, or to a control group (N = 33) which did not. Following her own model (Raphael, 1971) of "selective ego support for ego processes stressed by the crisis experience" (1977, p. 1451), Raphael's technique of intervention was to provide support for the "expression of grieving affects" and facilitation of the review of positive and negative aspects of lost relationships. Thus, it is probably true to say that more emphasis was placed on emotional and validation, rather than on instrumental, support in this particular study. Support was, however, tailored to the specific needs of the particular individual. Thus, for example, if the bereaved felt that the social network had failed to provide adequate support, she would be encouraged to express anger or guilt.

Intervention took place in the widow's home, each session lasting at least two hours. The average number of sessions was 4. Intervention was terminated by three months after the death. Outcome was assessed thirteen months after the death by means of a general health questionnaire, which was sent to the subjects

apparently independently through the university medical school, thus minimizing the possibility of any cues to responses that the intervenor might have given. Unfortunately, no pretest health measures were taken and compared with the post test scores, but this is not critical to the design, as assignment to intervention and control groups was random.

Of the treatment group for whom complete data were available ($N = 27$) 21 became good and 6 bad outcome. Of the controls ($N = 29$) 12 became good and 17 bad outcome ($p < .02$). Comparison was then made between these high risk groups with those who had not reached the high risk criteria described above. There were no significant differences between the high risk intervention group and this low risk group on the health measures after thirteen months, whereas the high risk control group who had received no support had significantly poorer health outcomes than the low risk group. This implies that intervention contributed to making the high risk group more similar to the low risk.

Raphael also analyzed results for the subgroups selected into the high risk category by each risk factor (perceived non-support, traumatic death, ambivalent relationship and additional crises). Of interest here is the finding that the criterion of perceived non-supportiveness was found to be the most accurate predictor of those likely to do well with this type of crisis intervention. It seems that providing support for those who apparently lack it is instrumental in improving general health ratings.

Clearly, this factor may appear more important in a sample which was essentially self-selected and therefore was possibly in greater need of support than the general widowed population would be. But also relevant is the finding that although the non-intervention control group had poorer health outcomes on many indices, the major area of difference to that of the intervention group is in the category of doctors' visits. Significantly more of the controls than of the intervention group visited the doctor for general symptoms. One might speculate that, in the absence of the intervention given to the experimental group, the control subjects were seeking support (someone to talk to) rather than specifically medical help. Raphael drew attention to the possibility of a "doctoring need" which might be present among the widowed in general, and which might have been fulfilled in the experimental group through the contact with the intervenor.

Gerber, Wiener, Battin and Arkin (1975) also conducted a longitudinal investigation of a non-patient sample of aged bereaved in New York, to evaluate the effects of unsolicited crisis intervention on medical outcomes. Therapy was conducted by trained psychiatric workers. The bereaved were assigned randomly to treatment or to non-treatment control groups.

Besides providing support for affective aspects of the grief response, the therapist acted as "a 'primer or programmer' of some of the activities of the bereaved. . . organizing among suitable friends and relatives a flexible modest scheme for the same

purpose" (p. 314). Instrumental support was offered too: the bereaved were helped in dealing with legal and financial matters and household chores. They were also advised about future plans. The support offered seems, then, to have covered the various different types indentified earlier. Of the bereaved subjects, 116 were assigned to the treatment group, and 53 to non-treatment. Subjects were interviewed at home, just over half of the interviews being conducted by telephone. Comparisons in outcome were made at approximately 3,5,8 and 15 months after bereavement, though intervention was terminated after the first half year of bereavement.

Even though subjects were not selected according to a criterion of high risk, results were somewhat similar to those of Raphael (1977). On five out of the seven measures of outcome dealing with illness behavior and medication use there were some beneficial effects of the therapy on health. At 5 and 8 months (when the therapy could be expected to be most effective due to the greater amount of therapeutic contact) the intervention appeared to be particularly effective. The intervention group received fewer drug prescriptions, consulted doctors less frequently and reported feeling ill without consultation less often than those not in intervention.

Although these two studies provide fairly solid evidence for the usefulness of professional support for the bereaved on a one-to-one bases, more negative results were reported in a study by Polak, Egan, Vandebergh and Williams (1975). However, as Raphael (1977) and Parkes (1980) pointed out, these may be explained in terms of methodological problems of Polak's study and procedural differences between the studies.

Evaluation of Intervention Programs

Evidence from the voluntary aid and professional help programs for the bereaved shows that most of the support programs reduce the risk of psychological or physical debility: those who participate have better ratings on these measures than those who do not. These programs seemed to be particularly effective for "high risk" groups. It is important to note, however, that high risk was not determined by physical health examinations (other than self-ratings). Nor did the bereaved subjects meet criteria for psychiatric referral. Perceived lack of social support was one of the major criteria used to establish high risk, and in the study that looked at risk factors separately (Raphael, 1977), it was the most accurate predictor of those whose outcome would improve. Thus, at least partially, intervention is filling a gap in support which the informal social network of the bereaved person does not (or cannot) provide.

As is evident from the studies reviewed above, not only the professional services provided by trained persons such as social workers, psychologists or doctors, but also those conducted by

volunteers or by other bereaved persons who have recovered from their own grief, have positive effects on bereavement outcome. It is difficult to compare the relative effectiveness of these different types of support due to differences between studies. These include differences in samples, in sampling techniques, differences with regard to amount and type of interaction, and differences in timing and duration of support. Yet, regardless of these differences, most of these programs have been seen to be very effective. From the description of the interventions offered it can be seen that all, to greater or lesser extents, provided emotional, validation, and instrumental support. Thus, the conclusion to be drawn from these findings is that the provision of support, in whatever form, is the crucial factor in mediating the positive effects of bereavement.

It is, however, necessary to add one qualification to this general conclusion. Whereas grief counseling may be a valuable supplement to more traditional interventions for those among whom grief runs an uncomplicated course, in cases of pathological grief, or in cases where a history of psychiatric illness existed prior to bereavement, social support intervention of the type provided in these studies may not be appropriate as a therapeutic method. Ramsay (1979) outlined the pathological response:

"... the 'normal' reactions of shock, despair, and recovery are often distorted, exaggerated, prolonged, inhibited or delayed." (p. 225)

In instances where recovery from loss has gone severely wrong in one of these ways, psychiatric advice and guidance is called for. Social support such as is provided in the programs described above may be the last thing that a severely depressed widow needs, wants or would find helpful; a bereaved person who has failed to grieve for a number of years will not make a spontaneous recovery through attendance at such meetings. It is not possible here to elaborate on the distinction between normal and pathological grief (in fact, the line is hard to draw). Nor is it possible to evaluate the effectiveness of professional therapy in alleviating complicated grief reactions. Detailed discussions of pathological grief are available in the literature (see, for example, Bowlby, 1980; Freud, 1917; Lindemann, 1944; Parkes, 1972; Ramsay, 1979; Schoenberg, 1980; Worden, 1982).

DOES SOCIAL SUPPORT HELP THE BEREAVED?

Like all scientists, social scientists are reluctant to give definite answers to clear questions. However, having declared at the outset that the purpose of this chapter was to "examine the hypothesis that social support furthers the recovery from grief", we now feel obligated to reply to this question.

Two types of evidence were presented in an attempt to address this issue. First, studies were reviewed which related the

availability of, and satisfaction with, social support to bereavement outcome. Although most of these investigations reported an association of poor bereavement outcome with living alone, with having little contact with family and friends, and with perceptions of non-supportiveness from the social network, these findings are ambiguous with regard to causal interpretations. Although they are consistent with the assumption that lack of perceived or actual support leads to poor health outcome among the bereaved, they are equally consistent with several alternative interpretations which were discussed earlier.

The conclusion that social support has an ameliorative effect on bereavement is therefore mainly based on the evidence from therapeutic intervention studies. By systematically manipulating social support across conditions and by assigning subjects randomly to these conditions, the design of these studies permits one to eliminate most of the rival explanations. In particular, the plausible alternative interpretation of the association of poor bereavement outcome and low social support as due to some personality trait can be excluded. In the non-intervention studies the association observed between poor bereavement outcome and low social support could have been due to a personality trait like neuroticism increasing both the probability of the social environment being (or being perceived as) non-supportive and the probability of a breakdown occurring under the stress of bereavement. In the intervention studies the random assignment of subjects to conditions would have precluded systematic differences between conditions in subjects' personality traits.

Unfortunately, this increase in experimental control also has one disadvantage. The gain in internal validity over the network research described earlier was paid for by a decrease in external validity. Participation in intervention studies is largely based on self-selection. Thus, although one can now be fairly confident that the types of social support provided by therapeutic intervention furthered recovery from bereavement, it is unclear whether such intervention would have had the same positive impact on those individuals who did not choose to participate in these programs. As mentioned earlier, there is some evidence to suggest that individuals with a high need for social support are most likely to participate in such intervention programs. It follows that the provision of social support is effective in reducing risk of poor outcome only among those bereaved who actually lack social support, or who perceive their environments as being deficient in this respect.

Nevertheless, if one assumes that the perception of a lack of social support reflects an actual deficit caused by loss of support from the spouse, the finding that intervention programs were particularly effective for "high risk" groups would be consistent with the deficit model of bereavement proposed earlier. According to this model, the availability of an additional source of social support should further recovery from bereavement only if

some support deficit occurred due to loss, for which compensation was provided from an alternative source. Since one major criterion for considering subjects "low risk" was that they did not perceive any support deficits, the finding that intervention did not significantly affect their health outcome is also consistent with the deficit model. Thus, in addition to supporting the general conclusion that social support furthers recovery and lowers risk, the pattern of findings so far is also consistent with more specific predictions derived from the deficit model of loss.

Clearly we have given an answer only to the general question of the impact of social support on the health of bereaved persons. While in historical perspective the inclusion of interpersonal processes in the study of bereavement reactions has stimulated much research, we are well aware of the limitations of available data. Current research on such topics as specific needs of bereaved persons for the various types of social support according to the duration of their bereavement, and on reciprocity in helping relationships, will enable closer examination of the association between social support and health of the bereaved and a better test of the deficit model.

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REACTIONS TO VICTIMS OF LIFE CRISES:
SUPPORT ATTEMPTS THAT FAIL

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We became interested in the topic of social support while studying how people cope with a variety of life crises such as loss of a loved one, life-threatening illness, and physical disability. In most discussions of social support, it is generally assumed that support attempts made by the provider will be valued and appreciated by the receiver. There is growing awareness that in many cases, however, others' well-intentioned efforts to provide support may be regarded as unhelpful by the recipient, may result in negative consequences, or both (Dunkel-Schetter & Wortman, 1982a; 1982b; House, 1981; Thoits, 1982). In a study we recently completed on coping with the loss of a spouse or child (Lehman, Wortman & Williams, in press), respondents reported that others frequently tried to support them by making statements like, "I know exactly how you feel," "It was God's will," or "It's a good thing you have other children." Such statements were commonly judged by respondents to be unhelpful (see also Glick, Weiss & Parkes, 1974; Maddison & Walker, 1967). In research with Roxane Silver on coping with the loss of an infant through the Sudden Infant Death Syndrome, we found that many of the statements made to parents, such as "You're young--you can always have another one," or "It was only a baby whom you didn't know" are not regarded as helpful (see also Helmrath & Steinitz, 1978).

There are two ways in which the approach taken in this paper differs from most past work in the area of social support. First, to date, most of the empirical studies on social support have focused on the consequences of receiving social support. In analyzing the support problems that may be encountered by those who have experienced undesirable life events, this paper will focus instead on determinants of support. Specifically, we

attempt to identify situational and personal factors that influence whether support attempts are made to victims of life crises, and whether these attempts are beneficial. Second, empirical research on social support has focused primarily on the recipient. In most cases, the recipient is asked a number of questions about the types of support desired, or the adequacy of support provided. We believe that it is equally important to explore support from the perspective of the provider. In this paper, an attempt is made to develop a more complete understanding of the feelings and beliefs that providers hold toward victims of life crises, and the support attempts that are based on these beliefs.

This paper will focus on three basic reasons why people may respond to victims of life crises in ways that are unsupportive:

(1) For many reasons, people appear to hold negative feelings about others who are suffering or distressed. There is evidence in the social psychological literature that contact with victims of life crises makes others feel threatened and vulnerable, and thus reduces the likelihood that effective support will be provided. In fact, the more unfortunate the victim's plight or the more distress that he or she shows, the more threatened (and therefore unsupportive) others may become. Consequently, individuals in greatest need of social support may be least likely to get it (see Coates & Wortman, 1980; Dunkel-Schetter & Wortman, 1982a, 1982b, for a more detailed discussion). (2) Most people have relatively little experience dealing with others who are in the throes of a life crisis. For this reason, they may experience a great deal of uncertainty about what to say or do. Their anxiety may be heightened by the awareness that the victim is extremely vulnerable, and that inappropriate behaviors on the support provider's part might intensify the victim's distress. (3) Others appear to hold a number of misconceptions about how people should react to undesirable life events (see Silver & Wortman, 1980). Beliefs about how much distress should be experienced and displayed, and how long the effects of the crisis should last, are likely to influence the kinds of support attempts that are made. To the extent that these beliefs are in error, others' support attempts may be inappropriate or counter-productive.

Below, we attempt to illustrate how each of these factors operates to produce and sustain support attempts that are unhelpful.

Feelings Elicited by Victims of Life Crises

There is a strong and consistent body of research in the field of social psychology suggesting that people often hold negative feelings about others who are suffering, unhappy, or in need of help (see Coates, Wortman & Abbey, 1979, for a more detailed discussion). In this section of the paper, we discuss

several different reasons why negative feelings may be evoked. Contact with victims of life crises may generate negative feelings because this shatters our own illusions of invulnerability, or because it evokes strong feelings of helplessness. If the victim is a family member or close friend, interactions may also elicit fears and anxieties about what lies ahead, and whether the support provider will be able to handle the increased demands that may be placed on him or her.

Contact with victims of life crises can sometimes make us feel more vulnerable to a similar fate. When we encounter a person who has lost a loved one or endured a serious illness, our assumptions about the world, as well as our beliefs about our ability to control and influence future outcomes, may be threatened (Wortman, 1983). Such events force us to realize that these negative outcomes can happen to anyone at any time. The theorist who has provided the most cogent discussion of this possibility is Lerner (1975; Lerner, Miller & Holmes, 1976), who has argued that we are motivated to believe that the world is a just place where "people get what they deserve and deserve what they get." According to Lerner, if we can believe that people do not suffer unless something is wrong with them or with their behavior, we will feel protected from undeserved suffering in the future (see Wortman, 1976, for a more detailed discussion).

Lerner has suggested that in order to reduce the distress caused by exposure to others who are suffering from a life crisis, we will look for weaknesses in the victim's behavior to explain the incident. For example, if we learn that an acquaintance has been seriously injured in a car accident, we may sift through the available evidence in search of shortcomings on his or her part. Was the person driving too fast? Under bad conditions? Without a seat belt? Under the influence of alcohol? Alternatively, we may attempt to identify weaknesses in the person's character that would help account for his or her fate. Lerner and his associates have conducted a number of laboratory experiments that support these predictions (see e.g., Lerner & Matthews, 1967).

In addition to this heightened sense of vulnerability, contact with victims of life crisis may evoke strong feelings of helplessness. As Caine (1974) has expressed in writing about bereavement, "No one knows what to do with a tumultuous, angry, sobbing woman who is railing at the fates" (p. 30). Interactions with those who are in the throes of a life crisis may leave others feeling that there is little that they can say or do to alleviate the victim's pain or improve his or her situation. What can a person do that will really help someone who has a life-threatening illness or who has lost a spouse? Moreover, the magnitude and scope of the problem may overwhelm others. For example, conjugal bereavement affects the remaining spouse's psychological, social, and financial well-being (Glick, Weiss & Parkes, 1974; Parkes & Weiss, 1983; Lehman, Wortman & Williams, in press).

Particularly when the victim is a family member or close

friend, exposure to his or her distress may produce negative feelings because of its unsettling implications for the future. Support providers may worry that they will be called upon for help on a regular basis and may wonder whether they can handle repeated interactions that make them feel helpless and afraid. Anxieties about the future may be especially prevalent among the spouse or family members of the victim. They may not want to believe that the victim's problems are serious or that his or her distress is deep and long-lasting. Such an acknowledgement forces them to recognize that the victimization may significantly and permanently disrupt their relationship with the victim, as well as their own lives.

In summary, interaction with individuals who have suffered a life crisis are powerful in their ability to arouse negative feelings. Such interactions may be distressing because they force us to think about things we would rather not contemplate, and confront us with problems that we are unable to solve (cf. Dunkel-Schetter & Wortman, 1982b).

Uncertainties about Appropriate Behavior

Because of the negative feelings described earlier, most potential supporters may experience intense anxiety about the prospect of interacting with a person who has experienced a life crisis. They may feel confused and uncertain about how to behave or what to say, and may have little past experience to serve as a guide to them in this difficult situation. The support provider's anxieties may be heightened by the awareness that the victim is extremely vulnerable, and that inappropriate behaviors might even intensify the victim's distress. Some support for this reasoning has been provided by Peters-Golden (1982), who found that disease-free respondents were unsure of what comments might be helpful in conversations with cancer patients. These respondents were also afraid that they would say the wrong thing and do harm to the cancer patient (see also Krant & Johnston, 1977, 1978). Peters-Golden's (1982) results suggest that the respondents' fears were justified, since for the most part, they demonstrated a limited understanding of the true concerns of the person experiencing the crisis. The healthy respondents she studied assumed that the loss of the breast is the major concern of the patient. In fact, the breast cancer patients in her sample were much more concerned with fear of recurrence or treatment side effects.

Even when support providers are knowledgeable about the victim's concerns, or about the types of support that are desired or beneficial, the anxiety that characterizes most interactions between victims and support providers may impede empathic responses. The potential supporter may be so conscious of what is happening, and so worried about responding inappropriately, that natural expressions of concern are unlikely to occur. This

anxiety may also enhance the likelihood of support attempts that are automatic or ritualized, for example, offering platitudes like, "I know how you feel," or "Things will get better soon," rather than carefully reasoned plans about how to meet the victim's needs.

It is interesting to note that all of these three factors discussed--negative feelings about victimization, anxieties about providing effective support, and misconceptions about how victims will react--are much more likely to predominate when the consequences of the victimization are serious. This suggests that just at those times in life when we are particularly in need of effective support, we may be least likely to get it.

Others' Misconceptions About the Coping Process

It appears that many people have misconceptions about the emotional impact that is associated with an undesirable life event. Most people seem to assume that when a life crisis occurs, an individual will initially experience distress as he or she attempts to cope with the crisis. However, victims of life crises are expected to "pull themselves together" rather quickly. Others seem unaware that the sequelae of undesirable life events often last for many years (see Silver & Wortman, 1980, for a more detailed discussion of these and other misconceptions that are commonly held about the coping process). In coping with life-threatening illness, for example, Vachon (1979) has suggested that "the woman with breast cancer is expected to quickly 'get back on her feet' and resume her role within her family. The assumption is that once treatment is completed, the disease shouldn't have much effect on her life" (p. 12). However, a number of studies on psychological reactions to mastectomy have found evidence that many women show severe distress long after surgery, even when the disease does not recur. In a study by Maguire, Lee, Bevington, Kuchemann, Crabtree and Cornell (1978), 39% of the breast cancer patients they surveyed were still experiencing levels of anxiety, depression and/or sexual difficulties "serious enough...to warrant psychiatric help" (p. 963) a year after their mastectomy, and showed significantly more distress than a matched control group of women with benign breast disease (see also Maguire, 1978; Morris, Greer, & White, 1977; Ray, 1978).

In reviewing literature on the loss of a loved one, Walker, MacBride and Vachon (1977) have concluded that "widows frequently report that even intimates do not support the need to mourn their loss beyond the first few days after the death" (p.39). Yet a number of studies have suggested that bereaved individuals may experience distress for quite a few years. For example, Parkes (1975) has reported that 44% of the bereaved he studied were exhibiting moderate to severe anxiety at the final interview, which occurred 2-4 years after the loss (see also

Vachon, 1979; Marris, 1958; Schwab, Chalmers, Conroy, Farris & Markush, 1975; Parkes, 1972). In fact, among those respondents who had little or no forewarning of their spouse's death, 72% were rated as moderately to severely anxious at the final interview, 67% appeared moderately to severely upset or disturbed, and 61% indicated that they occasionally or always sensed the presence of the dead person (Parkes & Weiss, 1983). In a study of individuals who have lost a spouse or a child in a motor vehicle accident, Lehman, Wortman and Williams (in preparation) found that bereaved respondents showed significantly more psychiatric symptoms, depression, and significantly poorer social functioning, general well-being and reactivity to good events than a matched control group, even though the accident had occurred 2-4 years earlier. Others may also fail to recognize that even when an individual appears to be recovered from an undesirable life event, feelings of severe distress may be precipitated by anniversaries, geographical locale, and other external events that evoke memories of the loss (Wiener, Gerber, Battin & Arkin, 1975). Parkes (1970) has suggested that during these times, "all the feelings of acute pining and sadness return and the bereaved person goes through, in miniature, another bereavement" (p. 464).

A second misconception that seems to be quite prevalent is that it is inappropriate or unhelpful for victims of life crises to discuss their feelings about what has happened. Peters-Golden (1982) found evidence for this in a study about perceptions of support among breast cancer patients and disease-free controls. The healthy population in her study assumed that discussions of the patient's disease or treatment would be harmful to the patient, and regarded those patients who wanted to discuss their situation as less adjusted. However, the cancer patients in her sample reported being "disturbed by this ban on communication and confused by the assumption that avoiding the subject would actually be better for them."

BEHAVIOR OF OTHERS TOWARD VICTIMS OF LIFE CRISES

People's feelings of vulnerability and helplessness, their uncertainty about how to behave, their misconceptions about how victims should react, may strongly influence the nature of support attempts directed toward those who have experienced life crises. We believe that because of these feelings and beliefs, individuals engage in three types of behaviors that are intended to be supportive but that are often detrimental: (1) They discourage open expression or discussion of feelings about victimization; (2) They encourage recovery or movement to the next life stage; and (3) They fall back on automatic or scripted support attempts which may seem to dismiss or trivialize the victim's problem, such as giving advice, providing a reason for what has happened, or telling the victim that they know how he/she

feels. Below, we explore each of these behavioral reactions in more detail, and discuss the potential negative impact that they may have on the victim.

Discouraging Open Communication

A prediction that can be made from the foregoing analysis is that support providers will discourage the victim from engaging in open communication about the life crisis. In part, attempts to block discussions of feelings may make it difficult for the support provider to keep his or her own feelings under control. Such expressions may also occur because support providers believe that such discussions will only serve to reinforce and perpetuate the victim's distress. It is likely that such discussions are also discouraged because they are stressful for the support provider. Expressions of distress may heighten the provider's feelings of vulnerability, and may intensify his or her anxiety about saying or doing the wrong thing. Open discussion of feelings may also be discouraged because they place implicit demands on the support provider to resolve the victim's problems, and hence intensify feelings of helplessness. Finally, displays of distress may have threatening implications for the support provider's future. Support providers may worry that they will be repeatedly faced with new demands brought about by the victim's plight, as well as with interpersonal interactions that are difficult and draining.

There is some empirical evidence to support our prediction that others may discourage victims from discussing their feelings (see, e.g., Bard, 1952; Jamison, Wellisch & Pasnau, 1978; Krant & Johnson, 1977-1978; Mitchell & Glicksman, 1977; Peters-Golden, 1982; see Dunkel-Schetter & Wortman, 1982a, for a review). In a study of mastectomy patients conducted by Quint (1965), both patients and nurses reported that nurses did not permit open communication. Instead, they would direct the conversation into "safe channels" whenever patients tried to discuss their feelings. Moreover, patients also reported that family and friends blocked them from discussing their illness. Similarly, Kastenbaum and Aisenberg (1972) asked nurses and orderlies how they responded when elderly geriatric patients attempted to discuss their feeling about death. Approximately 80% of the time, the respondents reported that they avoided the subject, denied the implications of the patient's remark, or ended the discussion. As the authors put it, "the clear tendency was to 'turn off' the patient as quickly and as deftly as possible." In their interviews with couples who had lost a newborn infant, Helmrath and Steinitz (1978) have reported that "friends and family steadfastly avoided mentioning the infant or the death" although "each [mother] described a strong need to talk about the physical characteristics of the baby, the details of the hospital course and death, and her desires, expectations, and fantasies for

the baby..." (pp. 787-788).

These attempts to prevent or discourage open communication about negative feelings may take many specific forms, including minimization of the victim's problem and forced cheerfulness. There is evidence that others frequently make comments that minimize the extent of the crisis that the loved one faces. They may try to convince the person that the problem does not exist at all, that it is not as bad as it seems, or that things could be worse. Some support for this reasoning is provided by Ginsberg and Brown (1982), who conducted a study of depressed women who were mothers of preschool children. These investigators report that "the majority of husbands responded to their wives' complaints of symptoms such as suicidal ideas, feelings of helplessness, and lethargy with such comments as 'You're imagining it all,' or 'Stop being silly'" (p.95). Half of the women also reported that others minimized their symptoms by "explaining them away" -- for example, by telling the woman that "it's just postnatal depression" or "you're bound to get this with a baby" (p. 94). Helmrath and Steinitz (1978) have reported that in attempting to comfort individuals who had lost a newborn infant, others frequently make comments which minimized the loss, such as "you can always have another one" or "it was only a baby you didn't know -- it's worse to lose a child you know" (p. 788). Finally, the nurses and orderlies studied by Kastenbaum and Aisenberg (1972) reported that when geriatric patients brought up the topic of their own death, they frequently minimized the issue by making comments like "You are not going to die. Oh, you're going to live to be a hundred" (p. 222). Although minimization may help the support provider maintain his or her assumptions about the world, there is some evidence that it is not helpful to support recipients. In a study of social support among cancer patients with breast and colo-rectal cancer (Dunkel-Schetter, 1981), two behaviors commonly reported to be unhelpful were "being told not to worry because things will work out" and "minimizing the patients' problems and feelings about cancer."

Another way that support providers can reduce the likelihood that victims will express negative feelings is by encouraging them to focus on the positive elements of their situation, and to remain as cheerful and optimistic as possible. The bereaved spouses in the previously described study by Glick et al. (1974) were frequently enjoined to look on the bright side and to remember all they had to live for; they did not find such statements to be helpful (see also Maddison & Walker, 1967). The most frequent reason given by the nurses and attendants in Kastenbaum and Aisenberg's (1972) study for avoiding the subject of death was because they wanted to "cheer up" the patient, and they felt that the best way to do so was to focus the patient's attention on something else. The healthy individuals interviewed by Peters-Golden (1982) felt that it was important for cancer patients to remain as cheerful and optimistic as possible about

their situation; 66% said that they would "go out of their way" to cheer up a cancer patient. However, patients generally perceived others' attempts to cheer them up as unhelpful. Sixty percent of the cancer patients she studied reported that they were made to feel isolated and alone by "unrelenting optimism that seemed inauthentic" (p. 486). One reason why patients reported disliking the forced cheerfulness of others was because such cheerfulness frequently "eclipsed the opportunity to reveal and discuss their true feelings."

Encouraging Recovery or Progression to the Next Life Stage

A second prediction that is suggested from the previous analysis of support providers' beliefs and feelings is that they will attempt to encourage the victim to recover from the crisis and to move on the next stage in his or her life (cf. Silver & Wortman, 1980). They may do so because they feel the victim should be recovered after a few months, or because they want the victim to recover as soon as possible. It is clearly less threatening, upsetting, and demanding to deal with a person who is "recovered" from his or her life crisis.

Some evidence concerning how victims react to others' attempts to encourage recovery is provided in a study by Maddison and Walker (1967). Widows were asked to rate several potentially supportive behaviors, indicating how helpful or unhelpful they perceived each to be, and whether they needed more of each type of support than they had received. According to these investigators, others often tried to arouse the widow's interest in new activities, in the development of new friendships, or the resumption of old hobbies or occupations. For the most part, such exhortations were not regarded as helpful. According to Maddison and Walker (1967), "only a small handful of widows found any sort of detailed discussion about their future to be of value to them" (p. 1063). Moreover, it was quite rare for any respondent to indicate that she needed more help in orienting her to the future than she received.

One way individuals may encourage recovery among the bereaved is to bring up the topic of remarriage. There is evidence that discussions of this topic are often initiated by others quite early in the bereavement period -- often within a few days or weeks of the spouse's death (Maddison & Walker, 1967; Glick, et al., 1974). The evidence is consistent that such admonitions are not regarded as helpful. Glick, et al. (1974) report that "widows invariably found early suggestions that they consider remarriage unpleasant and even jarring" (p. 22). The widows studied by Maddison and Walker (1967) almost always reacted unfavorably to such suggestions, leading them to conclude that "whatever else one might say to a widow during the height of crisis, a discussion, or even a hint, of some future romantic liaison must always be hazardous operation" (p. 1063).

Reliance on Scripted or Automatic Support Behaviors

Above, we reviewed several reasons why many people may find interactions with victims to be very anxiety-provoking, and feel very uncertain about what to say or do in this situation. Feelings of anxiety may be intensified if attempts to discourage open communication have failed, and if the victim does not appear to be recovering from the crisis as quickly or as completely as the support provider expects. This anxiety may make it difficult to be attentive to the victim's needs, and to initiate and carry out support attempts designed to meet these needs. Instead, support providers may respond to the pressure inherent in the situation by relying on scripted (Schank & Abelson, 1977) or automatic support behaviors. These are behaviors that are part of our cultural understanding of how to help others, and which can therefore be evoked automatically or with little thought. Such behaviors may include giving advice (for example, telling widows that perhaps they should move) providing a reason for what has happened (e.g., "It was God's will"), or for the distress that the victim is experiencing (e.g., "It's natural to be upset at a time like this") or attempting to identify with the victim's feelings ("I know how you feel"). Such responses may reduce the support provider's anxiety by making him or her feel that at least he/she has said or done something to help the victim. In many instances, however, we believe that such support attempts are not appreciated by victims. As we describe in more detail below, these strategies may seem to trivialize the victim's problems, and dismiss his/her feelings as "normal" and therefore not important or worthy of respect.

Although they have not been extensively studied, there is some evidence that such support attempts are not regarded as helpful. Maddison and Walker (1967) have suggested that many specific types of advice, such as telling the widow that she must control herself and pull herself together, and making suggestions about things the widow might do to find happiness in the future, were generally judged to be unhelpful. Maddison and Walker (1967) also observed that when told that the death of their spouse was the "will of God," respondents who were themselves religious "tended to regard such interventions as gratuitous and unnecessary," while those without any profound religious conviction "found such attempts at comfort meaningless and often extremely irritating" (p.1063) (see also Glick, et al., 1974).

There is also some evidence that victims do not appreciate others' attempts to provide them with reasons for their distress. Ginsberg and Brown (1982) have described how the depressed women in their study disliked being told by others that their feelings were a natural reaction to a life crisis. According to these investigators, such statements "usually appeared to be based on the assumption that since the symptoms were an expected reaction to the stress they were no cause for concern or action" (p.94).

In the study by Maddison and Walker (1967), there is also some evidence that widows did not appreciate others' attempts to minimize their grief by pointing out to them the sufferings of other widows, or telling them just how much grief was appropriate for a widow to feel. In this study, widows were also distressed when others claimed to share their grief and insisted that they understood how she was feeling, since they strongly believed such claims to be patently untrue.

IMPACT ON THE VICTIM

Above, we reviewed several reasons why others make support attempts that may not be helpful to victims of life crises. These include discouraging discussion of the victim's problems; encouraging the victim to recover more quickly, and relying on scripted or automatic supportive behaviors such as giving advice, providing an explanation for what has happened or for the victim's feelings, or suggesting that they know how the victim feels.

There are three reasons why these particular support tactics may be ineffective or harmful. First, these strategies may make victims feel isolated and alone because they curtail opportunities to share concerns with others. This is true not only for direct attempts to discourage the victim from expressing feelings, but for the other strategies discussed above as well. If others attempt to encourage recovery by telling a bereaved mother that she should have another child, she may not feel it appropriate to mention her feelings of despair about the child she has lost. Similarly, if a widow is given advice or provided with an explanation for what has happened, she may feel that further discussions of her problems would make her appear unappreciative. Even the statement, "I know how you feel," may discourage most victims from further attempts to reveal their feelings.

Although research on this topic is not extensive, there is some evidence to suggest that opportunities to discuss feelings are perceived as desirable and helpful by the victim, and that such discussions are associated with positive health outcomes (see Dunkel-Schetter & Wortman, 1982a; and Silver & Wortman, 1980, for reviews). For example, Mitchell and Glicksman (1977) report that 86% of the cancer patients they studied wished that they could discuss their situation more fully with someone. Almost 90% of the bereaved questioned by Schoenberg, Carr, Peretz, Kutscher and Cherico (1975) felt that "expression rather than repression of feelings, and crying, should be encouraged at least sometimes" (p.365). Similarly, Vachon, Freedman, Formo, Rogers, Lyall and Freeman (1977) found that over 80% of the cancer widows they interviewed who openly discussed death with their dying spouses said that this made it easier to face the bereavement (see also Maddison & Walker, 1967). Finally, the value of many intervention studies may stem from the opportunity for open discussion of

feelings that is provided. For example, Raphael (1977) conducted a randomized study in which widows at risk for post-bereavement morbidity were assigned to a treatment involving "support for the expression of grieving affects such as sadness, anger, anxiety, hoplessness, helplessness, and despair" (p. 1451). This treatment was continued for a maximum of 3 months and included an average of only four sessions per widow. Nonetheless, when assessed 13 months after their spouse's deaths, those women assigned to the intervention group reported significantly better psychological and physical health than those assigned to a no-treatment group.

A second reason why the tactics reviewed above may be unhelpful is that in many cases, they may seem to dismiss the victim's feelings as insignificant or unimportant. When others provide advice or explanations for what has happened, they may imply that the victim's problems can easily be resolved or explained away. By telling a widow that her distress is normal, or that this is the way she is supposed to feel, others may seem to dismiss her feelings as not worthy of further consideration. Such pronouncements, as well as statements like, "I know how you feel," may also upset the victim because they imply that her feelings are not unique or special.

A third reason why the aforementioned tactics may have a detrimental impact on victims is because they imply that the victim should not feel as bad as he or she does. In fact, there is evidence that victims often worry about whether their feelings of distress are normal. Kennell, Slyter, and Klaus (1970) have reported that the parents who have lost infants "were not well prepared for their own mourning responses--their reactions worried and perplexed them" (p. 347). The intensity of their fears and feelings may sometimes lead victims to believe that they are coping poorly, or even that they are losing their grip on reality (Pattison, 1977; Kaltreider, Wallace, & Horowitz, 1979). In fact, fears of insanity or of losing one's mind have frequently been reported among the bereaved (Bergman, Pomeroy & Beckwith, 1969). By enjoining the victim to look on the bright side, by stating that things are not as bad as they seem, or by suggesting that the victim should be ready for remarriage or another child, others may subtly convey that the victim's feelings and behaviors are inappropriate, or perhaps even indicative of a serious psychological problem. Providing victims with advice, or specific suggestions about the future, may also convey the implication that they are incapable of handling their own problems (cf. Brickman, Rabinowitz, Karuza, Coates, Cohn & Kidder, 1982).

We are suggesting that for the most part, victims will not feel supported by the classes of support attempts that we have delineated above. In fact, even though they may realize that others' remarks are well intentioned, they may find these attempts extremely irritating. Despite their impact on the victim, however, support providers may rarely receive feedback on the

adequacy of their support attempts. Since one of the greatest fears of many victims is that they will be rejected and abandoned by others (c.f. Dunkel-Schetter & Wortman, 1982a), they may be reluctant to mention that a specific effort to provide support was inappropriate or unhelpful (cf. Peters-Golden, 1982). Particularly if the support providers' comments have the effect of discouraging further displays of distress, providers may be very likely to conclude that the support attempt was a success. Consequently, ineffective support strategies will be used again in the future.

There are two conclusions which follow from the analysis that has been presented here. First, this discussion emphasizes the importance of permitting people who have experienced life crises to talk about their feelings if they wish to do so. When such feelings are being shared, it is important to avoid responses that close off further attempts at discussion, which trivialize or dismiss the victim's feelings, or which imply that the victim is coping poorly. Because of the feelings of vulnerability and helplessness discussed earlier, as well as beliefs about how victims should be reacting, it may be extremely difficult to listen to such feelings. This may be true even for professional helpers, who may become anxious when they are unable to act, help, or "do something" for another person to make them well or alleviate their suffering. Thus, most people may find that they have to restrain the impulse to react to each problem mentioned with a comment, a piece of advice, or a proposed solution. Such well-intentioned remarks as "I know how you feel" or "Don't worry, it will be all right" may be less helpful than simply acknowledging that the other person's concerns are being heard (see Dunkel-Schetter & Wortman, 1982a, for a more detailed discussion). In general, providing an opportunity for the expression of feelings, and attentive listening to those feelings, may constitute a powerful and frequently overlooked means of support. It may be useful to provide training to support providers that clarifies their misconceptions about the coping process, and that teaches them how to recognize and deal with their own negative feelings and vulnerabilities.

An unsettling implication of our analysis is that those who are closest to the victim, and who therefore have the greatest stake in the victim's recovery, may have the least tolerance for the victim's distress. For this reason, those who care the most about the victim may be especially likely to engage in the inappropriate support behaviors described above, such as blocking discussions of feelings or pushing the victim toward a quicker recovery. Therefore, in mobilizing support for victims of life crises, it may be important to look beyond family members and friends, with all of their misconceptions and vulnerabilities, to other potential sources of support. Others who have experienced a similar problem may be in a unique position to provide effective support. Similar others may respond to discussions of feelings

with interest rather than with anxiety or fear, and therefore may be less likely to block such feelings. Similar others may also be less likely to push the victims toward a quicker recovery, since they may have a better idea of how long distress typically lasts. Because similar others are accustomed to the victimization and its consequences, conversations with another victim may also be less anxiety-provoking for them than for other support providers. For this reason, they may be less likely to respond to every sign of distress with the automatic or scripted support attempts discussed above. When they do give advice or suggest that they know how the victim feels, the victim may be less likely to feel that his or her feelings have been trivialized or dismissed.

Supportive interventions involving similar others--both lay counselor programs and peer support groups--have become increasingly prevalent in recent years, and constitute an important movement in the mental health field (cf. Katz, 1981; Lieberman & Borman, 1979). Although systematic evaluation studies are rare, the available evidence suggests that peer support can be very effective in facilitating coping with the life crisis (Minde, Shosenberg, Marton, Thompson, Ripley & Burns, 1980; Spiegel, Bloom & Yalom, 1981; Vachon, Lyall, Rogers, Freedman-Letofsky & Freeman, 1980).

In this paper, we have discussed several reasons why others may make support attempts that are not effective in meeting the victim's needs. We suggested that these and other support attempts which block victims' attempts to discuss their feelings when they want to, which trivialize or dismiss their feelings, or which imply that the victim is coping poorly, will be judged as unsupportive. In contrast, listening to the victim's feelings, or encouraging contact with similar others may be more helpful to the victim. We recently conducted a study on the long-term effects of sudden, unexpected bereavement that provides some preliminary data in support of these assertions. Although a full description of the study is beyond the scope of this paper (see Lehman, Wortman & Williams, in press), the results that are relevant to this discussion are summarized briefly below. In this study, interviews were conducted with approximately 100 people, about half of whom had lost a spouse, and about half of whom had lost a child in a motor vehicle accident which had occurred four to seven years ago. Interviews were also conducted with a control group of respondents who had not lost a loved one; controls were matched to the bereaved respondents on a case-by-case basis on sex, age, income, education, and number and ages of children. Both groups were asked to complete previously validated instruments designed to assess psychiatric symptomatology, quality of life, perceived health status, and functioning in family and work roles. Bereaved respondents were also asked a number of questions about how the loss affected their lives, and about helpful as well as unhelpful support attempts that others had made. As part of their interview, control respondents were asked some questions designed

to tap how they would provide support to someone who had lost a spouse or child. It is from these questions on support that we collected data that is relevant to the issues raised in this paper.

Among bereaved respondents, the question attempting to tap "helpful" social support was, "In the time since your loved one has died, is there anything that other people have said or done that has been especially helpful to you in coping with (his/her) death? If so, what was that?" After a careful investigation of a random sample of responses, the following coding categories were developed: (1) Contact with similar others; (2) Provide opportunity to discuss feelings; (3) Express concern; (4) Presence ("being there"); (5) Provide a philosophical perspective; (6) Religion; (7) Involve in social activities; (8) Compliment the deceased; (9) Provide tangible support; (10) Give advice; (11) Discuss the bereaved being reunited with the deceased; (12) Tell the bereaved that his/her feelings are understood; (13) Encourage recovery; and (14) Miscellaneous. Respondents who answered "No" to the first part of the question received a code of (15) No answer. Respondents who gave an answer that did not apply to the question received a code of (16) Not applicable. Multiple mentions were allowed, so consequently, each respondent's answer was coded for as many categories as it reflected (the greatest number of coded categories for an individual response was three).

Each response was rated independently by two coders who were supplied with a list of coding categories and examples. The inter-rater reliability estimate was the percentage of agreed upon category codes for all of the responses (excluding the "No" and "Not applicable" responses). Out of a total of 111 codable responses, the two coders agreed on 92, or 82.9% of the time. When there was a discrepancy the authors decided on the appropriate code.

The question asked of our bereaved respondents to tap "attempted, but unsuccessful" social support was the following: "Sometimes, in trying to help someone who has lost a loved one, people may say or do things that are not in fact helpful. In the time since the accident, is there anything that people have said or done to help you that has not been helpful, or that has upset you, made you angry, or offended you? If so, what was that?" All of the codes from the "helpful" question (16--including Misc., No, and Not applicable) were possible codes for the "unhelpful" question. In addition, the following new categories were added as a result of the authors' examination of a random sample of responses: 17) Rude remark or behavior; 18) Minimization/forced cheerfulness; and 19) Identification with feelings ("I know how you feel"). Each response was again coded for as many categories as it reflected.

The inter-rater reliability was estimated as it had been in the "helpful" question. Out of a total of 81 codable responses, the two coders agreed on 69, or 85.2% of the time. Again, when

there was a discrepancy the authors decided on the appropriate code.

Among control respondents, the question asked was, "If you met someone who had recently lost a spouse or child in a motor vehicle accident, or if this happened to a friend of yours, and you wanted to comfort and help them, what would you do or say?" All of the codes from the "helpful" and "unhelpful" questions (19 codes) were included in the control question coding scheme. In addition, coders were supplied with the following new categories as a result of the authors' examination of a random sample of responses: (20) Showing signs of sympathy; (21) Discuss memories of the deceased; and (22) Physical closeness (touching). Each response was again coded for as many categories as it reflected (in the control sample, the greatest number of coded categories for an individual response was five).

The inter-rater reliability was estimated as it had been for the bereaved respondents' questions. Out of a total 195 codable responses, the two coders agreed on 163, or 83.6% of the time. Discrepancies were again decided upon by the authors.

The results, which are summarized in Tables 1 and 2 are, (with category examples provided in Table 2) generally consistent with the reasoning in this paper.² Approximately 60% of our bereaved respondents answered affirmatively to the question of whether others had said or done anything that was unhelpful. The four support tactics most commonly identified as unhelpful were giving advice, encouragement of recovery, minimization/forced cheerfulness, and identification with feelings. Aside from a few respondents who regarded giving advice or encouraging recovery as helpful, these strategies were overwhelmingly regarded as upsetting or offensive. In addition, the bereaved sample indicated that others frequently made remarks or engaged in behavior that was blatantly rude or unsupportive, such as calling one of them an "unfit mother" because she let her young children drive with a seventeen year-old aunt. Probably not intended to be helpful, this type of response is consistent with our argument that victimization may engender negative feelings in others.

Approximately 70% of the bereaved sample agreed that someone in their support network had provided support that was particularly helpful. Consistent with the reasoning in this article, "contact with similar others" and "provide opportunity to discuss feelings" were among the responses most frequently identified by others as especially helpful. Two other responses--mere presence (or "being there"), and expressions of concern--were also commonly judged as helpful. In no cases were any of these responses identified as unhelpful by respondents. Unlike responses commonly identified as unhelpful, one could argue that these support tactics allow the victims to express feelings if they want to, and do not dismiss their feelings or convey that they are coping poorly with the loss.

The typical bereaved respondent in our study received both

TABLE 1

Frequency of Response Types in Bereaved and Control Groups			
	Helpful Question - <u>Bereaved</u>	Unhelpful Question - <u>Bereaved</u>	Hypothetical Question - <u>Control</u>
# of People Asked Question	94	94	100
# Who Gave at least 1 Codable Response	65 (69%)	55 (58%)	99 (99%)
# Who Said "No"	29 (31%)	39 (42%)	1 (1%)
# of Total Codable Responses	111	81	195
Percentage of Codable Responses in a Given Category			
Contact with Similar Other	13%	0	0
Provide Opportunity to Ventilate	10%	0	13%
Express Concern	10%	0	7%
Presence (Being There)	9%	0	16%
Provide a Philosophical Perspective	9%	7%	7%
Religion	8%	0	8%
Involve in Social Activities	7%	0	3%
Give Advice	4%	21%	3%
Encouragement of Recovery	3%	20%	4%
Rude Remark/Bad Behavior	0	16%	0
Minimization/Forced Cheerfulness	0	10%	0
Identification With Feelings (I know how you feel)	0	8%	0
Showing Signs of Sympathy	0	0	13%
Miscellaneous Helpful or Not Helpful (Too general to code)	7%	8%	16%
Other "Helpful" (compliment the deceased, provide tangible support, discuss loved one being reunited with deceased, tell loved one their feelings are understood, discuss memories of the deceased, physical closeness or touching, discuss resurrection)	20%	0	10%
Other "Unhelpful" - (Provide unwanted tangible support, interference into loved one's life, and mentions deceased when loved one doesn't want it).	0	10%	0

TABLE 2

Examples of Codable Responses In a Given Category

Contact with Similar Other

(Helpful) - "Ten months after the death, I moved into an apartment complex where I met many widows who helped reconcile me to the loss."

Provide Opportunity to Ventilate

(Helpful) - "Friends have listened to me when I wanted to talk about it. That helps. I can pick up the phone day or night and they will always listen."

(Hypothetical) - "I would tell her if there was anything I could do or if she just wanted to talk. I would gladly just sit and let her talk."

Express Concern

(Helpful) - "The comfort that knowing that people cared and showing their love and support has helped."

(Hypothetical) - "Tell them I loved them and if there was anything I could do for them I would be honored to be able to help them."

Presence (Being There)

(Helpful) - "My sister was very supportive. She will call me every morning and if I sound upset, she is at my door -- just being there."

(Hypothetical) - "I don't think I would do or say anything. I would just be there to help. There isn't too much you can say."

Provide a Philosophical Perspective

(Helpful) - "My friend told me he is no longer suffering now and is with God."

(Unhelpful) - "People making statements like he completed his mission in life here -- God wants him now. One lady said, He's a little flower in God's garden. That really irritated me."

(Hypothetical) - "Tell them they had no control over it. Everything has a time and place and purpose."

Religion

(Helpful) - "If it wasn't for my pastor, I wouldn't have gotten through it."

(Hypothetical) - "I would tell them to turn to the Lord because it has helped me."

(TABLE 2, continued)

Involve in Social Activities

(Helpful) - "Some of my friends wanted me to go fishing with them to get me out of the house".

Give Advice

(Unhelpful) - "The person who came in and told me to take my son's picture down."

Encouragement of Recovery

(Unhelpful) - "Just recently someone said your husband has been dead a few years, now you should get married again. It made me feel bad because I am not ready yet. They think 4 years you should be ready, but you're not always."

Rude Remark/Bad Behavior

(Unhelpful) - "My brother is a police officer and I asked him to help with the paper work, to help me through it -- he never got back to me. My husband had filed for a divorce but said he wouldn't go through with it, but he did. I couldn't get help from the family."

Minimization/Forced Cheerfulness

(Unhelpful) - "You can have another child. It can't be that bad."

Identification with feelings (I know how you feel)

(Unhelpful) - "People often say, I know how you feel or I know exactly what you're going through, and they don't -- not unless you've been there."

Showing Signs of Sympathy

(Hypothetical) - "Not much I would say. I would offer my condolences."

effective and ineffective support attempts from those in their social environment. Clearly, there are some individuals who are able to provide effective support to the bereaved despite any negative feelings or misconceptions that they may have harbored. These are the people the bereaved respondents described when asked about helpful support attempts. There were others who said or did the wrong things when attempting to support our bereaved respondents -- either because they did not know what responses would be helpful, or because they knew what to do or say in hypothetical terms, but were not able to carry out their intentions in an actual encounter with a victim.

The data from our control respondents sheds some light on the source of difficulties in providing effective support. When we asked control respondents what they would do if they were attempting to provide support for someone who had lost a spouse or

child, their responses were surprisingly enlightened. Controls rarely indicated that they would engage in such "unhelpful" support tactics as minimizing the loss, encouraging recovery, or giving advice (although quite a few control respondents did indicate that they would attempt to provide a philosophical perspective). Instead, controls commonly indicated that they would "be there" for the victim, that they would express concern, and that they would provide opportunities for discussion of feelings. The control respondents knew, hypothetically, how they should behave toward the bereaved: only 11% mentioned that they would say or do things that our bereaved sample perceived as not helpful.

Taken together, the results of this study are consistent with the reasoning conveyed in this chapter, and suggest a number of possible directions for subsequent research. We have suggested that in actual situations with victims of life crises, the factors identified in this article may constitute powerful forces motivating others to say or do the wrong thing. Further evidence for our assertions could be obtained by conducting studies that focus on actual transactions between victimized people and those in their natural support network. It would be interesting to ask support providers what they feel would be helpful and what type of support they intend to provide, and then compare this with the type of support that is actually offered to the victim. It would also be interesting to study how the heightened emotionality that may characterize interactions with victims influence support attempts. Are those involved in direct interactions with victims of life crises less capable of arriving at an accurate assessment of the victim's needs than those who are merely observing the interaction? Are they therefore more likely to generate scripted support attempts that are perceived as unhelpful?

If such studies reinforce the suggestions from this study that others often know what to do but are unable to carry out support attempts effectively, the implications for intervention are straightforward. Instead of teaching people specific support strategies that have been found to be helpful, it may be more worthwhile to develop interventions that are designed to enhance the support provider's skills in actual encounters with victims. For example, perhaps support providers could be taught to control and manage the negative feelings that are evoked when the victim reveals distress, and to listen to such feelings without feeling compelled to make comments or offer advice.

In subsequent research, it would also be useful to identify those additional characteristics of the provider, recipient or the setting that enhance the likelihood that ineffective support attempts will be made. Is it really the case that those who are closest to the victim are more likely to engage in inappropriate support behaviors? Do similar others use different classes of

some types of distress on the victim's part (e.g., anger, bitterness) impede empathic responding on the part of the support providers more than other types (e.g., sorrow) (cf. Wortman, 1983)?

In this article, we suggested that ineffective support attempts stem from the provider's feelings and misconceptions about victimization and about the recovery process. To date, however, the feelings and attitudes of support providers have received almost no scientific investigation. In longitudinal studies of victims of life crises, it would be useful to interview support providers in addition to the victims themselves. Do others really expect or urge recovery before the victim is ready? Do they make erroneous assumptions about what the victim wants or needs at various points? How do their own feelings about the victimization change with time, and how are these changes related to those experienced by the victims themselves?

Studies of actual supportive transactions, in combination with longitudinal research focusing on support providers as well as victims, would also be helpful in refining our knowledge about the impact of particular support tactics. For example, we have suggested that advice is generally not appreciated by those who are victimized; however, there are clearly some types of advice, or conditions under which advice may be useful. For example, advice may be appreciated from a similar other, but found to be extremely irritating when offered by a friend or acquaintance who has no experience with the victimization. Hopefully, future studies will help to clarify the conditions under which particular types of support are likely to be offered by particular providers, and judged as beneficial by the recipient. Such knowledge may not only shed light on the process through which support may protect people from stress, but may also provide information relevant to the design of effective interventions (cf. Gottlieb, 1981).

In this paper, the major conclusions were based on judgments about the value of particular support behaviors from the perspective of the recipient. In the future, however, it is imperative that the long-term health and mental health consequences of such exhortations be carefully studied. Such information would help clarify whether those support attempts most valued by victims are indeed beneficial for them in the longer run.

NOTES

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¹The religion category was a multi-faceted code, with people mentioning either help or guidance they received from a religious figure or support they received from being in church or temple with others.

²Although there were some interesting differences between respondents who had lost a spouse and those who had lost a child, in terms of what was seen as helpful or unhelpful, they are beyond the scope of the present chapter.

³A closer examination of the responses which were coded as "provide a philosophical perspective" is needed in the future. It does appear, however, that when this type of response was viewed as "helpful" it was usually offered by a close friend or a religious figure. When this type of response was viewed as "unhelpful" it was usually offered by an acquaintance or stranger.

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THE COSTS OF CARING: A PERSPECTIVE ON THE RELATIONSHIP BETWEEN SEX AND PSYCHOLOGICAL DISTRESS

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This paper is different from the others in this volume in that we discuss the health-damaging effects of providing social support rather than the health-promoting effects of receiving it. Our topic has heretofore been neglected. Yet, as we show below, there is good reason to think that there are serious personal costs associated with being a supporter; costs that should be taken into account by those who advocate community interventions to increase access to support.

That professional helpers sometimes find themselves victims of their own efforts to help others has been well-documented in the literature on staff burnout (Edelwich & Brodsky, 1980). The consensus view in this literature is that successful helpers walk the narrow line between overinvolvement and lack of concern. Helpers must be realistic enough to set limits on what they are willing to do for clients without losing their sense of care and concern. Not all professional helpers are able to do this and for many of them the depth of their concern is so great that they lose themselves to those they help.

It is even more difficult for a layperson to cope emotionally with providing support to his or her loved ones. We know this to be the case in special situations of extreme stress--like that facing a wife whose husband returns home after being hospitalized for a heart attack (Speedling, 1982). Although little systematic research has been done on the costs of providing support in less serious crisis situations, what little we know points to the conclusion that many people in our society--especially women--are severely burdened by the demands made on them by their social networks. One study actually has documented that some social networks are so demanding that their female members are in better mental health if they are isolated from other members of the

network than closely integrated (Cohler & Lieberman, 1980).

In our recent work we have been studying the costs of network membership by using a methodology that parallels the approach taken in the literature on the stress-buffering effects of social support. Working with normal population surveys of the relationship between life events and psychological distress, we have been examining the impact of network crises on the well-being of persons who are potential providers of support. As part of a larger body of research on sex differences in psychological distress (Kessler & McRae, 1981; 1982; 1983; 1984), this investigation has been particularly concerned with the differential impact of network crises on men and women. We have found clear evidence that women are substantially more burdened than men by the costs of caring.

BACKGROUND

Our decision to approach the study of sex differences in distress from the life events perspective was motivated by a finding in the life events literature that women are more vulnerable than men to the emotional effects of life events. From the available evidence, this sex difference in vulnerability is large enough to account for a substantial part of the overall relationship between sex and distress found in the normal population (Dohrenwend, 1973; Uhlenhuth, Lipman, Balter, & Stern, 1974; Kessler, 1979; Radloff & Rae, 1981).

This result has, to date, been documented only in surveys that use highly aggregated measures of life events. As is well known, individual life events occur with low frequency in the normal population, so it is seldom possible to analyze the effects of single events. Instead, indices of exposure to any of a large number of different events are created. While this approach is the best one for the data available, it does create interpretation problems. A researcher is likely to ask whether the result is pervasive across a wide range of undesirable events or confined to only a few. This question cannot be answered when the life event index is aggregated.

Over the past year we have been trying out a different approach. We began with a number of surveys, each of which contains a few people who report the occurrence of some uncommon event like job loss, divorce, or widowhood. Models for the effects of these separate events were estimated in each survey and pooled across surveys to arrive at more stable overall estimates of the individual event effects. It is not possible to study the rarest events in this way, but it is possible to disaggregate event indices into a number of conceptual types for analysis.²

This approach has yielded stable and theoretically rich results when applied to the study of sex differences. In an

earlier paper (Kessler & McLeod, 1983) we documented that the greater vulnerability of women to life events is particularly evident when we concentrate on events that occur to someone other than the person under study. We refer to these as "loved one" events. Sex differences in the influences of these loved one events account for a very substantial part of the overall sex-distress relationship.

We turned our attention to social support on the basis of this result. An obvious interpretation is that women are more emotionally involved than men in the lives of their loved ones, so that the distress experienced by a loved one translates itself into more distress for women than men.

The literature provides a great deal of evidence consistent with this interpretation. Women have repeatedly been shown to have a stronger orientation than men to the needs and desires of their loved ones (Eisenberg & Lennon, 1983). They recognize these needs more readily and develop a greater sense of responsibility for meeting them (Gilligan, 1982). As a result, women maintain both a wider field of concern than men and a greater propensity to involve themselves in help-giving activities.

These sex differences manifest themselves in the types of relationships men and women develop. In childhood, girls establish small, intimate friendship networks while boys construct larger and more affiliative ones (Jourard, 1971). The same patterns persist in adulthood (Booth, 1972; Wheeler & Nezlock, 1977) and are echoed in men's and women's reports about their marriages. Husbands are more likely than wives to say that their spouse is appreciative, affirming, and affectionate (Vanfossen, 1981) and that their spouse understands them well (Campbell, Converse, and Rodgers, 1976), while wives are more likely than husbands to feel that their spouse talks too little about feelings (Veroff, Douvan, Kulka, 1981).

The greater sensitivity and concern of women for their loved ones can also be seen in intergenerational relationships. Women are much more likely than men to voice some concern about their aging parents (Lieberman, 1978; Brody, 1981) and their children (Menaghan, 1978).

Taken together, these considerations form a picture of women as emotionally more open than men to the concerns of their loved ones: a picture that conforms closely with the results documented in our analysis of life events. Our results suggest that support and concern may be interpreted as intervening links between stressful life events and the high levels of distress known to exist among women.

RESEARCH ON SEX DIFFERENCES IN LIFE EVENTS

In the past, most life event surveys have found that men and women do not differ significantly in overall exposure to

undesirable events (Uhlenhuth et al., 1974; Markush & Favero, 1974; Kessler, 1979). This result, however, is probably an artifact because most inventories underrepresent events that are more likely to occur to women (Makosky, 1980). Questions about entering and leaving the armed forces, for example, are included in most life event inventories while rape and spouse abuse are seldom included.

Dohrenwend (1977) made an attempt to correct for this bias by asking a sample of men and women in an urban area to describe the events that had been sources of stress in their lives. One difference in the lists generated by men and women stood out: women were much more likely to mention events that occurred to people they care about. In designing our analysis we were sensitive to this sex difference and so sought out surveys in which respondents were asked to report events that occurred to people important to them as well as events which occurred to themselves.

Five separate surveys were found that contain data of this sort. Two of these are urban surveys from New Haven (Myers, Lindenthal, & Pepper, 1974) and Kansas City (Markush & Favero, 1976). Two are separate surveys of a single rural area in Maryland (Comstock & Helsing, 1976). And the fifth is a national survey (Veroff et al., 1981). Each of the local surveys contains a parallel pair of life event inventories--one for events that "occurred to you" and the other for events that "occurred to someone important to you" in the past year. In the New Haven survey respondents were asked if the latter occurred to a son, daughter, spouse, or other important person (in the other surveys information about the identity of the person who experienced the event was not obtained).

In the national survey respondents were asked: "Now think about the last time something really bad happened to you. What was it about?" One event was recorded for each respondent and the date of its occurrence was noted. Although not as precise as a life event inventory, this open-ended question is useful as it provides information in a large national survey that parallels the procedure used by Dohrenwend to generate information about sex differences in reporting loved one events.

Differential Exposure to Life Events

We reanalyzed the data in all of these surveys to see if men and women differ in the types of life events they report--particularly in events that occurred to people important to them. Our results are consistent with those of previous literature. Most personal events are reported with equal frequency by men and women, while women consistently report more events that occurred to their loved ones.

When we looked carefully at particular types of loved one events it became obvious that most of this sex difference in

self-reported exposure can be attributed to the fact that women define more people as someone important to them than men do. At the same time, it is important to recognize that women might actually know about more events than men because people in the midst of life crises are more likely to confide in women than men. We discuss this more fully below when we examine survey data on patterns of informal help-seeking.

Differential Response to Loved One Events

Knowing that women are more aware of life crises that have occurred to their loved ones and that more people are important to women is only half the story. We also want to know if these events have an emotional impact on women they do not have on men. To determine this we estimated regression models predicting psychological distress from exposure to personal events, loved one events, and control variables separately for men and women in each of the five surveys.³

The results show that women are more highly influenced by network life crises than men. This is particularly striking given the fact that men are unlikely to report many of the network crises that women report. This underreporting takes a rather special form: men fail to report minor events and events that occurred to people at the edges of their networks. This means that a higher proportion of the loved one events reported by men as compared to women are serious life crises that occurred to their spouses or children. It is in this sense that the greater impact of loved one events, taken as whole, on women than men is particularly striking. The bias in men's reports would be expected to slant the results toward an appearance that they are more affected than women.

We also found a rather striking specification of the previously documented fact that personal events have a more pronounced emotional impact on women than men. In our multisurvey analysis we were able to break down personal events into subtypes: income losses, separation/divorce, widowhood, death of other loved ones, and personal health problems. In this way we could see precisely which personal events accounted for the overall greater responsiveness of women. The most serious personal events--income losses and marital disruptions--do not affect women more than men. Indeed, they have much the same capacity to produce distress among men as women. The largest category of personal events that affect women more in these surveys involves the death of a loved one other than a spouse. This result suggests that the previously documented vulnerability of women to personal events is actually a special case of their vulnerability to loved one events.

A Decomposition of the Sex-Distress Relationship

So far we have said that women are more likely to report

loved one events and more responsive to them than men. The combination of these two features is crucial for understanding sex differences in psychological distress. We followed the regression analyses with an examination of how important differences in levels of exposure and vulnerability are in accounting for overall sex differences in average levels of distress. We used the results of the regressions to create a series of hypothetical estimates in which the means and regression slopes of men and women were switched and predicted values of distress calculated. These estimates suggested that if the exposure and emotional response of men to loved one events were increased to the levels found among women well over half of the observed sex difference in average levels of psychological distress would disappear. In other words, the emotional costs of caring for those in one's network account for a substantial part of the pervasive mental health disadvantage of women.

This disadvantage, however, is not caused by women being deeply distressed by one or two loved one events. The situation is more complex. Women, like men, are more emotionally affected by personal events than by events that occur to their loved ones. And some distant loved one events have effects that are small enough to be called "trivial". But women are affected by these events in a way men are not, enough so that the sex difference in responsiveness to loved one events is larger than the difference in responsiveness to personal events. Furthermore, even though each of the loved one events has only a small emotional impact a great many events of this sort occur in a typical social network—far more than the number of personal events that occur. Understanding how the sex difference in response combines with the larger sex difference in exposure illuminates the observed sex difference in emotional functioning.

DISCUSSION

Alternative Interpretations

Although our interest revolves around the burdens experienced by people who provide support to their loved ones when the latter are in the midst of a life crisis, none of the surveys considered here contains explicit information about what respondents did or felt about their loved one's problem. We suspect that the differences found between men and women might be importantly determined by women's greater capacity for involvement in the affairs of their loved ones. We reasoned that if this were so, we should be able to see evidence of it in data about the frequency with which men and women provide help to people in crisis situations.

The national survey included a series of questions about the help-seeking behaviors of respondents who reported that they had experienced a personal life crisis some time in the past.

Respondents were asked to name all people who helped them during the crisis period and to describe what these helpers did. Inspection of these data suggested: (1) that women are more likely than men to become involved as supporters in a crisis situation; and (2) that the types of support men and women give are very similar. These differences lead to considerably more women than men who help their loved ones. Depending on the type of problem, women are between 10% and 40% more likely than men to support a loved one. Summed across all problem types women are 30% more likely than men to provide some type of support.

The data on which our second conclusion is based are also worth describing. Respondents in the national survey were presented with a card listing the following types of support: listened to me, cheered or comforted me, talked things out, told me who else to see, showed me a new way to look at things, gave me advice, helped me to take action. They were then asked to report which helpers did each of these things. The data show no substantial sex difference in the type of support offered. Male supporters are somewhat less likely than their female counterparts to offer comfort and a bit more likely to take action and give advice, but our overwhelming impression from the data is that men and women are very alike in the things they do for those they help. While we can detect some sex differences in the involvement of supporters, the similarities are far more striking.

These results can be used to make some inferential assessment of the hypothesis that the vulnerability of women to loved one events stems from the fact that they are more likely than men to become involved as supporters. Although we cannot evaluate this hypothesis directly (as respondents were not asked what they did or felt when they found out that the loved one event had occurred), we consider this hypothesis unlikely. Our reasoning is as follows. Since women are involved in about 30% more support transactions than men, we would expect the slopes of personal distress on loved one events to be about 30% larger among women than men if nothing more were involved than differences in providing support. The observed female slope of distress on loved one events is considerably larger than this. Indeed, we find no evidence that men are emotionally affected at all by loved one events other than those that occur to their wives or children.

On the basis of this discrepancy between observed and predicted male slopes, our inclination is to reject the view that sex differences in vulnerability can be explained by the greater involvement of women than men as supporters. Although not as likely to become involved as women, men do provide a substantial amount of support to people in crisis. Yet men are somehow able to avoid the personal distress felt by women.

One could make sense of this invulnerability by assuming that men are defective in their ability to empathize with the distress of other people, but this assumption is inconsistent with evidence that empathy can be elicited from men in experimental situations

(Eisenberg & Lennon, 1983). Our suspicion is somewhat different: men, despite their ability to empathize, structure their lives in such a way that they care about fewer people than women do.

If women have a wider field of concern than men, sex differences in vulnerability should be smallest for events that occur to closely-related loved ones and larger when we consider more distant relationships. This is exactly what we found in further analysis. In the New Haven Survey respondents were asked to identify the loved one who experienced each undesirable event. We know whether the loved one was a son, daughter, spouse, or "other" important person. We found that men are distressed when a close loved one (spouse or child) experiences a crisis, but not when a more distant loved one experiences a crisis. This is true, we might add, even though the events reported by respondents as having occurred to distant loved ones are much more serious than those reported for close loved ones. A full 66% of the distant loved one events reported involve bereavement, while 75% of the remainder involve serious illnesses. The events that respondents report as having occurred to their spouses and children are much less serious---with school-related events (among children) and minor illnesses predominating.

This pattern of results is consistent with the hypothesis that sex differences in vulnerability are generated by a difference in the range of caring. Women cast a wide net in their concern and so are emotionally affected not only by the well-being of their immediate family and most intimate friends but also by those to whom they are less intimately related.⁴ Men, by comparison, are emotionally affected only by the events that occur to the few people to whom they feel emotionally close.

There are several alternate possibilities. One is that providing support leads to distress only when the provider is overloaded with so many other demands that his or her capacities for keeping emotional distance break down. A second possibility is that providing support has a distressing effect only when the provider is emotionally overinvolved with the recipient. According to this view, because of their socialization experiences the concern of women can become so all encompassing that personal needs are neglected. Men may be more successful at preventing involvement from becoming overinvolvement, perhaps in part by their greater ability to maintain emotional distance.

These two possibilities suggest that providing support has a distressing effect under some circumstances but not others. A third possibility is that the provision of support itself is unimportant, and that emotional involvement determines whether a loved one's event will lead to personal distress. In this view, support can play a part in creating emotional involvement but it is not important in itself. If the person in crisis is one whom the supporter would not normally worry much about (a neighbor whom the supporter knows only casually), then the act of providing support may be a vehicle for creating deeper caring and,

potentially, more distress. If the person in crisis is someone with whom the supporter is already emotionally involved, support will not have the same effect. Indeed, one can even imagine a situation in which the opportunity to provide support to a very close loved one is less stress-provoking than a situation in which one is helpless (as, for example, when a loved one lives far away).

We believe that each of these possibilities has some validity. The time and energy demands placed on a supporter can themselves lead to distress, particularly when they come on top of an already demanding set of role responsibilities. This overload can be exacerbated by a personal style that promotes overcommitment and emotional overinvolvement. Deep personal concern for the well-being of a loved one can lead to distress even when no support is being provided. The precise way in which support and emotional involvement combine to determine the effect of loved one crisis events will vary depending on characteristics of the situation and the helper.

With the right kind of data it should be possible to determine empirically whether the act of providing support is itself involved in the distress of women and, if so, under what circumstances. Our intention has been more modest here: to document that life crises which occur to loved ones play a substantial part in the mental health of women. We also have suggested that the part played by support is far from obvious.

The Rewards and Costs of Caring

Up to this point we have emphasized the negative implications of caring for others. This is an important side of the support transaction to bear in mind as little attention has been devoted to negative effects like these we have examined here. As Belle (1982) has noted, the assumption implicit in most discussions is that support takes place within networks of mutual obligation. This view gives the impression that these networks have health-promoting effects for all their members, an impression which overlooks the fact that women are overrepresented among providers and so fails to appreciate the emotional costs sustained by women as a result of their role as caregivers.⁵

At the same time, we have to realize that caring can have rewards as well as costs. The opportunity to support someone who we care about can be gratifying. It is consistent with evidence available in the national survey data, where there is a negative relationship between personal distress and the amount of time one reports "doing things to help or please other people." And it is consistent with the belief that the ability to be nurturant is a fulfilling and self-validating experience (Weiss, 1974). We know, too, that married women are in somewhat better mental health than those who have never been married even though the married are subjected to more social burdens (Kessler & McRae, 1984). We also

know that involvement in large social networks is associated with good mental health even though people in large networks complain that their loved ones make too many demands on them (Fischer, 1982).⁶

It is clear from this evidence that rewards are as much a part of involvement with loved ones as costs. Whether the rewards outweigh the costs is something that varies over time and from one situation to the next. And usually the rewards dominate so that the balance has a positive effect on mental health.

Suggestions for Future Research

The research initiatives implied by the results presented here are obvious at this point. We need to obtain data of a sort not normally collected in life event surveys to test the different interpretations for the greater vulnerability of women to loved one events, data about the responses of men and women to life crises that occur to people important to them. Estimates should be obtained of the amount of time invested in providing support as well as of the additional time spent thinking or worrying about the problem. Detailed information about the types of support given will also be required, as it might be that the support that women are asked to give is more intense and personally costly than the support given by men. Interviews obtained from pairs of respondents, one a help-seeker and the other a supporter, could be useful in eliciting other important information, both about the different demands made on male and female helpers and about differences in the willingness of men and women to provide support of various sorts when asked.

The collection of these types of data could also put to a critical test the assumption that because of their roles, women are exposed to more chronic stresses than men (Gove, 1978). The typical argument cites the isolation of life as a homemaker, the overload that comes with being a working mother, or the asymmetry in giving and receiving social support (Belle, 1982) without any explicit comparison of how great these are relative to the stresses faced by men. The further research suggested here can be thought of as a first step toward developing comparative analyses of exposure and response to events and chronic stresses. We suspect that the greater ability of men to distance themselves from the many potentially stressful experiences that occur in one's network of friends and relatives protects their subjective quality of life.

Such comparative analyses of vulnerability will require a conceptually valid and psychometrically sound measure of self-other orientation. It is easy to detect variations in this orientation through in-depth interviews like those reported by Gilligan (1982), but it is far from obvious how to capture these differences in a set of structured questions. One possibility is to use vignettes. The basic procedure would be to present

respondents with a series of hypothetical situations and ask them to describe what they would do in each. The situations would be structured in such a way that a choice has to be made between some personal need and the needs of a loved one. This sort of strategy has been used for many years in research on moral development. It has more recently been introduced into survey methodology as a way of obtaining data about a wide range of social judgments (Rossi & Nock, 1982). Results are encouraging enough to suggest that this method might be an appropriate vehicle for quantifying sex differences in commitment to the needs of loved ones.

SUMMARY

Our analysis is a first step in a comprehensive consideration of sex differences in exposure to role-related stress. We have concentrated on the out-of-the-usual rather than the ordinary. In doing this we have not accounted entirely for the greater distress of women than men. We believe that women are not only exposed to more acute stresses than men, but also to more of the day-to-day chronic stresses that are associated with normal role functioning. A consideration of these will be necessary before we can comprehensively assess the part played by role stress.

At the same time, the importance of this first step should not be minimized. What we have presented is the first systematic documentation that women are more exposed than men to distressing life events and that this role-related difference in exposure is one important source of the mental health advantage of men. We have been able to elaborate these results to the point that several plausible interpretations have presented themselves. We have discussed directions that future data collection must take if we are to adjudicate among them. We have suggested that there may be multiple reasons for the vulnerability of women. Overload and overcommitment might both be involved and it is likely that their relative contributions vary from one support situation to another. These are important speculations to evaluate because they quite possibly hold the key to understanding the major role-related burdens of men and women.

NOTES

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²This approach was suggested to us by our reading of Pearlin, Lieberman, Menaghan & Mullan (1981) and Eckenrode &

Gore (1981), both of which demonstrated the utility of a disaggregated approach to the analysis of life events.

³The measures of distress varied from one study to another, but were alike in measuring nonspecific lack of well-being in a checklist format with a series of questions about depressed mood, anxiety, and psychophysiological reactions. These scales are more fully discussed in Kessler & McLeod (1983). Although the relationship between screening scales of this sort and clinical judgments of mental illness is far from perfect (Link & Dohrenwend, 1980:116), these scales do meaningfully discriminate clinically impaired people in the normal population from those who are not impaired (Wheaton, 1982).

⁴Thanks to Eudora Welty for this insight. It is important to recognize that we have not implicated the provision of support in this relationship between sex and vulnerability. Men are reported by recipients of support as providers in a sizable minority of cases. Yet men are not distressed by loved one events other than those that occur in their immediate family. Involvement as a provider of support is apparently not sufficient in itself to produce distress.

⁵Belle also suggests that women have less access to support than men. While it is true that women are somewhat less likely than men to report that their spouse is emotionally available as a confidant (Vanfossen, 1981), this difference is not great. Furthermore, in other respects there is little evidence that women are disadvantaged in their access to support. From the help-seeking data we find that women are, in fact, somewhat more likely than men to obtain help for their problems and more likely to receive emotional support. Even though it is true that husbands are not as available as providers of support for their wives as wives are for their husbands, married women usually have other women to whom they can turn when they need emotional support (Bernard, 1981). Indeed, if anything, women are probably more able to obtain intimate support since they have more intense and confiding same-sex friendships than men do.

⁶Men profit from relatedness more than women, as we might expect from the considerations above. The health-promoting effects of marriage are greater among men than women (Gove, 1978) and the feelings of social burden that come with membership in large networks are less pronounced among men than women (Fischer, 1982).

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- Abbey, A. 464
Abelson, R. 472
Abramson, L. 58, 59, 60
Adams, D.L. 445
Adams, J.S. 159
Adams, K.S. 332
Adcock, S. 51, 419
Adelson, J. 128
Ader, R. 440
Ahrns, C.R. 375, 376
Ainlay, S.L. 245, 373, 380
Ainsworth, M.D. 22, 246
Aisenberg, R. 469, 470
Akiskal, H.S., 307, 308, 309, 310
Alagna, S.W. 104
Aldous, J. 138
Aldwin, C. 395, 398, 404, 405
Almquist, E.M. 187
Altman, E. 336
Altman, I. 256
Ambelas, A. 330
Anderson, J.B. 187
Andrews, G. 51, 391, 395
Aneshensel, C.S. 51, 393
Angrist, S.S. 187
Antonovsky, A. 12, 197, 395
Antonucci, T.C. 12, 22, 23, 24, 30, 32, 150, 247, 250, 273
Antonuccio, D.O. 272
Arconad, M. 46
Arend, K. 22
Arkin, A.M. 46, 452
Arling, G. 32, 445
Armstrong, J.W. 333
van Arsdell, P.P. 287
Atchley, R.C. 32, 138
Auslander, N. 248, 276, 280,
Austin, D.M. 234
Averill, J.R. 9
Avery, A.D. 97
Bahr, H.M. 444
Baldessare, M. 119, 257
Balter, M.B. 492
Balter, M.O. 10
Bandura, A. 59, 60, 174, 175, 373,
Bard, M. 469
Barnett, R.D. 448
Barrera, M., Jr., 8, 40, 77, 163, 245, 249, 253, 259, 371, 372, 373, 379, 380
Barrett, C.J. 420
Barrett, J.E. 333
Bart, P. 57
Basham, R.B. 8, 9, 11, 39, 40, 41, 42, 104, 165, 138, 139, 143, 150, 191, 233, 275
Batten, P.G. 229, 236
Battin, D. 452, 468
Baum, A. 3
Baum, J. 276
Bebbington, P.E. 327, 337, 352, 353
Beck, A.T. 46
Becker, A. 448
Beckman, L.G. 443
Beckwith, J.B. 474
Bell, B.D. 3
Bell, R.R. 191
Belle, D. 499, 500
Bengston, V.L. 138
Benjaminsen, S. 330
Benson, C. 291
Bentler, D. 120
Bentley, A. 188
Berg, B. 378, 384
Berg, S. 279
Berger, D. 430
Bergman, A.B. 474
Bergner, M. 96
Berkman, L.S. 29
Berkowitz, S.D. 218
Bernard, J. 502
Berndt, T. 128
Berry, C.C. 97, 100, 101
Berscheid, E. 403, 405
Bevington, D.J. 467
Bhrolchain, M. 330, 397, 405, 418
Billings, A.G. 243, 404, 419
Binder, J. 324
Bingswanger, L. 439
Birkel, R.C. 11
Birley, J.L.T. 321, 322, 326, 332
Black, E.M. 336
Blalock, H.M. Jr., 351
Blau, D. 439

- Blau, Z. 270
 Blazer, D.G. 75, 297, 336
 Bleda, P.R. 194
 Blehar, 22
 Blisk, D. 378, 384
 Block, M.R. 32
 Bloom, B. 40, 131
 Bloom, J. 476
 Bobbitt, R.A. 96
 Boissevain, J. 205
 Bolk, E.W. 443
 Bond, C.F. 282
 Boomraem, C.D. 403
 Booth, A. 493
 Borman, L. 476
 Bornstein, P. 442, 443
 Bourke, G. 338
 Bowlby, J. 42, 66, 246, 247, 455
 Bowman, M., E. 375
 Boyd, J.H. 304
 Bradburn, N.M. 142, 269, 274
 Bragg, M.E. 279
 Brassard, J.A. 372, 380
 Braver, S. 377
 Brehm, J.W. 429
 Brennan, T. 248, 276, 280, 448
 Brickman, P. 474
 Briggs, S.R. 234, 236
 Broadhead, W.E. 350
 Brody, E.M. 428, 493
 Brody, J.J. 427
 Brodsky, A. 491
 Bronfenbrenner, 129
 Brook, R.H. 97, 100
 Brown, G.W. 52, 54, 58, 60, 61, 305, 307, 309, 312, 314, 315, 321, 322, 324, 326, 330, 332, 333, 334, 336, 337, 349, 350, 351, 352, 353, 358, 359, 360, 361, 393, 397, 405, 418, 419, 470, 472
 Brown, L.B. 43
 Bruce, T. 230, 276
 Brugha, T. 338
 Buchanan, W.W. 96
 Buchsbaum, N. 307
 Bultena, G.L. 32, 394, 404
 Bulter, M.B. 494
 Bunch, J. 443
 Burgess, E.W. 138
 Burke, B.W. 187
 Burns, S. 420, 476
 Bush, J.W. 1 97, 99, 100, 101
 Byrne, D.G. 8, 40, 42, 51, 336, 419
 Caine, L. 465
 Camara, K.A. 375, 377, 378
 Campbell, A. 198, 493
 Campbell, D.T. 395
 Campbell, E. 353
 Campbell, J.D. 3
 Campbell, M. 379
 Caplan, G. 4, 54, 58 249, 391, 393, 397, 398, 405
 Caplan, R.D. 175
 Carlson, R. 138
 Carnrike, C.L. 403
 Carpenter, B.N. 227, 228, 234
 Carr, A.C. 473
 Carrington, P. 216
 Carter, W. 96
 Carver, C.S. 76
 Cassel, J.C. 29, 51, 53, 58, 59, 108, 287
 Castore, C.H. 194
 Cattell, R.B. 139
 Cavert, C.W. 230, 276
 Chalmers, J. 468
 Chambers, L.W. 96
 Chandler, S.M. 392
 Chapman, J. 131
 Chapman, L. 131
 Chartier, G.M. 162
 Chen, M. 99, 100
 Cherico, D.J. 473
 Chess, S. 11
 Chiriboga, D.A. 32
 Chodoff, P. 397
 Clayton, P.J. 442, 443
 Cleminshaw, H.K. 378, 379
 Clendenin, W.W. 333
 Clum, G.A. 333
 Coates, D. 9, 62, 464, 474
 Cobb, J. 391, 398, 400, 405
 Cobb, S. 3, 4, 6, 51, 53, 58, 59, 103, 245, 287
 Cochran, M.M. 372, 379, 380,
 Cochran, C. 330
 Coddington, R.D. 157

- Cohen, C. 40, 430
 Cohen, F. 3
 Cohen, J.B. 5
 Cohen, S. 3, 5, 8, 10, 11, 54,
 62, 73, 74, 75, 76, 79, 83,
 84, 87, 89, 155, 157, 243,
 245, 251, 252, 253, 59, 273,
 309, 371, 372, 373, 374, 426,
 441
 Cohler, B. 492
 Cohn, E. 474
 Colleta, N.D. 376, 379
 Comstock, G.W. 494,
 Conger, J. 128
 Conner, K.A. 394, 404
 Conroy, R. 338
 Conroy, S. 468
 Converse, P. 493
 Cooley, C.H. 57, 58
 Cooper, B. 334
 Cooperband, A. 448, 449
 Cope, S.I. 353
 Cornell, C.E. 467
 Corty, E. 291
 Costa, P.T., Jr. 137, 138, 142,
 150
 Costello, G.G. 353
 Cowen, E.L. 403
 Cox, D.J. 163
 Cox, M. 3, 375, 376, 377, 378,
 379
 Cox, R. 3, 375, 376, 377, 378,
 379
 Coyne, J.C. 8, 42, 139, 197,
 249, 250, 253, 254, 311, 395,
 398, 404, 405
 Crabtree, R.J. 467
 Criqui, M.H. 104
 Cross, K.W., 334, 338
 Crutcher, M.M. 234
 Csikszentimihalyi, M. 272, 277
 Cullen, K. 96
 Cutler, N.E. 138
 Cutrona, 42, 227, 230, 236,
 237, 272, 274, 275, 277, 278,
 279
 Daly, L. 338
 Darell, S. 332
 David, T. 123
 Davidio, S.F. 12
 Davidson, E. 403
 Davidson, J.L. 32
 Davidson, P.O. 293
 Davidson, S. 334, 338, 350
 Davies, M.H. 334, 338
 Davies-Avery, A.R. 100
 Davis, P. 311
 Dean, A. 51, 58, 156, 245, 247,
 315, 391, 400
 deAraujo, G. 287
 deJong-Gierveld, J. 271, 274
 Delaney, W. 338
 Denny, R. 270
 DePaulo, B.M. 9, 12
 Depue, R.A. 131, 305, 307, 308,
 310, 311, 313, 314, 315, 329,
 333, 338,
 Derogatis, L.R. 127, 289
 DeVellis, B.M. 104
 DeVellis, R.F. 104
 Dewey, J. 188
 Diamant, L. 279
 Dickerson, K.G. 187
 Dienelt, M.N. 326, 329, 330
 Dijkhuizen, N. 175
 DiMatteo, M.R. 256
 Dohrenwend, B.P. 197, 303, 305,
 321, 391, 440, 502
 Dohrenwend, B.S. 3, 197, 303,
 321, 391, 395, 440, 492, 494
 Dokecki, P.R. 188
 Donald, C.A. 51, 97, 103, 303,
 304, 305, 391, 394, 395
 Dondero, E. 338
 Donson, C. 270
 Douvan, E. 128, 187, 493, 494
 Dowland, J. 352
 Droppleman, L.F. 142, 157
 DuCETTE, J.I. 162
 Duck, S. 232
 Dudley, D.L. 287
 Duff, 28
 Duncan-Jones, P. 8, 40, 42, 41,
 254, 256, 336, 338, 419
 Dunkel-Schetter, C. 9, 11, 463,
 466, 469, 470, 473, 475
 Dunner, D.L. 330
 Durkheim, E. 58, 243, 248, 256
 Eaton, W.W. 54, 371
 Eckenrode, J. 501

- Edelmich, J. 491
 Edwards, P.I. 400
 Egan, D. 454
 Eisenberg, N. 377, 493, 498
 Eisler, R.M. 327
 Embree, M.C. 9
 Emery, G. 46
 Emms, E.M. 338
 Endicott, J. 292
 Ensel, W.M. 51, 53, 156, 245, 247, 315
 Erickson, E.H. 57, 128
 Erickson, K. 3
 Ernst, J.A. 97
 Essex, M. 54
 Evans, R. 307
 Eysenck, H.J. 142
 Farber, S.S. 398, 406
 Faris, R.E.L. 243
 Farmer, R.D.T. 333
 Farr, D. 131, 305, 307, 310, 311, 314
 Fava, G.A. 329
 Favero, R.V. 494
 Feinglos, M. 104, 106
 Feld, S. 210
 Feldt, L.S. 393
 Felner, 398, 406
 Ferguson, G.A. 381
 Ferguson, M. 227, 228, 279
 Fernandez, R. 11
 Festinger, L. 60, 193, 441
 Field, S. 197
 Fieve, R.R. 330
 Filipp, S.H. 440
 Finlay-Jones, R. 333, 353
 Fischer, C.S. 119, 208, 252, 256, 257, 500, 502
 Fischer, J.R. 333
 Fisher, J.D. 9, 12, 256, 373, 377
 Flaherty, J.A. 336
 Fletcher, J. 338
 Flowers, J.V. 403
 Folkman, S. 5, 394
 Fondacaro, M.R. 288, 290
 Foner, A. 138
 Fooskas, S. 76
 Foote, N. 138
 Formo, A. 443, 444, 449, 473
 Fowler, R.C. 329
 Fowles, D.C. 307
 Frame, C.L. 291
 Frankel, B.G. 7, 8, 11
 Frankl, V. 57
 Franks, D.D. 59
 Frazier, C.L. 292
 Freedman, K. 443, 444, 449, 473
 Freedman-Letofsky, K. 420, 449, 450, 451, 476
 Freeman, J.A. 227, 228
 Freeman, S.J. 359, 420, 440, 443, 444, 445, 449, 450, 451, 473, 476
 French, J.R.P., Jr. 175, 304, 391, 395
 French, R. de S. 227
 Frerichs, R.R. 51, 393
 Freud, S. 439, 455
 Friedman, S.B. 397
 Friend, R. 91
 Frieze, I.H. 187
 Friis, H. 270, 274
 Fromm-Reichmann, F. 247, 270, 403
 Fuller, J.L. 306, 313, 314
 Fulton, J.A. 375, 381
 Futterman, S. 400
 Gaertner, S.L. 12
 Gaid, R. 332
 Galle, O.R. 244
 Garrity, T.F. 397
 Gatchel, R.S. 3
 Gecas, V. 59
 Gehlback, S.H. 350
 Geiger, J. 430
 Georges, A. 270
 Gerber, I. 453, 468
 Gergen, K.S. 441
 Gergen, M. 441
 Gersh, F.S. 307
 Gerson, A.C. 229, 272, 279
 Gerson, F. 257
 Gerson, K. 119
 Gerst, M. 10, 334
 Gesten, E.L. 403
 Gibbs, J.C. 189
 Gilligan, C. 120, 493, 500
 Gilson, B.S. 96
 Ginsberg, S.M. 470, 472

- Giuliano, T. 425
 Glasberg, M. 333
 Glass, D.C. 441
 Glassner, B. 330
 Glazer, N. 270
 Glick, I. 441, 449, 463, 465, 470, 471
 Glick, P.C. 371
 Glicksman, A.S., 469, 473
 Goldberg, D.P. 304
 Goldbourt, U. 29
 Goldenberg, C. 278
 Goldhammer, H. 138
 Goldston, S.E. 270, 279
 Gomez, J. 322
 Goode, W.J. 400
 Goodrich, W. 3
 Goplerud, E. 131, 305, 307, 310, 311, 314
 Gordon, S. 248
 Gore, S. 51, 53, 58, 59, 108, 198, 350, 418, 502
 Gormley, J. 230
 Goswick, R.A. 228, 229, 280
 Gottlieb, B.H. 6, 7, 11, 24, 76, 243, 245, 254, 349, 391, 404, 418, 424, 426, 442, 483
 Gottman, J.M. 289
 Gove, F.L. 22
 Gove, W.R. 244, 249, 250, 253, 500, 502
 Govi, L. 289
 Grace, M.C. 397
 Graef, R. 272, 277
 Graham, D. 76, 426
 Gram, J.D. 32
 Granovetter, M. 211
 Grant, I. 10, 334
 Green, B.L. 397
 Greer, H.S. 467
 Gregory, W.L. 162
 Grimson, R. 350
 Gronnerud, P. 162
 Gross, A.E. 12
 Gruder, C.L. 194
 Guidubaldi, J. 378, 379
 Gumerman, R.E. 12
 Gurin, G. 245
 Gurin, G. 197
 Gutwirth, L. 430
 Haberman, S.J. 10
 Hacker, T.A. 39, 42, 275
 Hagnell, O. 333
 Haldipar, C. 330
 Halikas, J.A. 442, 443
 Hall, A. 7, 205, 216, 219
 Hall, K.S. 330
 Hamburg, D.A. 397
 Hammer, M. 40, 430
 Hampton, P.W. 234
 Hansen, C.J. 138
 Hanson, L.R. 257
 Hansson, R.O. 62, 228, 229, 230, 234, 236, 237,
 Harris, T.O. 54, 58, 60, 61, 321, 322, 324, 326, 330, 334, 336, 337, 349, 350, 351, 352, 353, 359, 361, 363, 393, 397, 405, 418
 Harter, R.A. 392
 Harvey, C. 400, 444
 Haven, C. 54, 394, 418, 419, 421, 444
 Havighurst, R.J. 137, 138
 Hays, R. 256
 Hecht, E. 10
 Heim, M. 226, 278, 279
 Heinemann, G.D. 276
 Heitzman, C.A. 105
 Heller, K. 3, 8, 12, 29, 40, 54, 59, 243, 254, 287, 288, 289, 290, 291, 292, 294, 314, 371, 373, 447
 Helm, B. 229
 Helmrath, T.A. 463, 469, 470
 Helsing, K.J. 494
 Hendershot, G.E. 443
 Henderson, C. 379
 Henderson, S. 8, 40, 42, 51, 138, 139, 247, 336, 338, 419
 Hendrie, H.C. 332
 Henry, J.P. 7
 Hermele, S.L. 333
 Hess, B.B. 32
 Hess, R.D. 375, 377, 378
 Hetherington, E.M. 3, 371, 374, 375, 376, 377, 378, 379, 442
 Hewson, D.M. 51, 391, 395
 Heyden, S. 350
 Hill, R. 138

- Hiltz, S.R. 448
 Hingst, A.G. 377, 384
 Hinkle, L. Jr., 303, 305, 316, 392
 Hirsch, B.J. 12, 40, 119, 123, 124, 131, 245, 371, 374, 379, 384, 391, 392, 404, 405, 418
 Hirsch, S.R. 332
 Hizkey, N. 338
 Hobbs, S.A. 29, 233, 275,
 Hoberman, H.M. 8, 11, 40, 83, 84, 155, 157, 245, 253, 273, 371, 373
 Hobfoll, S.E. 393, 396, 397, 398, 399, 400, 405
 Hochreich, D.J. 174
 Hockenbury, D. 229, 233, 275
 Hoffman, B. 159
 Hojat, M. 273, 279
 Holahan, C.J. 304, 392, 394, 395
 Holmes, T.H. 60, 287, 321, 326, 391, 441, 465
 Homans, G.G. 254, 257
 Hong, 28
 Hoover-Dempsey, K.V. 188
 Hornblow, A.R. 332
 Horowitz, L.M. 227
 Horowitz, M.J. 3, 474
 House, J.S. 4, 7, 29, 53, 54, 59, 60, 66, 74, 108, 121, 131, 137, 243, 246, 249, 250, 253, 304, 391, 395, 427, 441, 442, 463
 Howarth, E. 233
 Huang, A.S. 187
 Huba, G. 120
 Hudgens, R.W. 322
 Huesmann, L.R. 403
 Hughes, M. 244, 249, 250, 253
 Hurry, J. 327, 353
 Hurst, M.W. 3, 10
 Husaini, B.A. 51, 156, 165
 Hutchinson, M. 311
 Hyman, H.H. 193
 Ilgen, D.R. 197
 Indart, M.S. 234
 Ingersoll, B. 30, 32
 Ingham, J.G. 314, 334, 338, 350, 418, 419
 Iscoe, I. 7, 40
 Isherwood, J. 332
 Israel, 30
 Jackson, C.S. 257
 Jackson, R. 119
 Jacob, T. 254
 Jacobs, J.W. 376
 Jacobs, S. 332
 Jacobson, D.S. 378
 James, W. 57, 66
 Jamison, K.R. 469
 Jenkins, C.O. 3, 10
 Jette, A.M. 98
 Joffe, P. 289
 Johnson, J.H. 3, 391
 Johnston, L. 466, 469
 Jolly, E.A. 123, 124
 Jones, L.M. 119, 257
 Jones, W.H. 227, 228, 229, 230, 233, 234, 235, 236, 237, 272, 275, 276, 280, Jourard, S.M. 402
 Kafry, D. 197
 Kahn, R.L. 12, 22, 23, 24, 74, 150, 247, 250, 257, 273
 Kalashian, M.M. 400
 Kaltreider, N.B. 474
 Kanner, A.D. 5, 197, 311
 Kanter, R.M. 121
 Kaplan, B.H. 29, 51, 53, 58, 59, 287, 350
 Kaplan, H.B. 333
 Kaplan, R.M. 97, 100, 101, 102, 104, 105, 108, 257
 Karuza, J. 474
 Kashket, E.R. 187
 Kasl, S. 121
 Kastenbaum, R. 469, 470
 Katz, A.H. 476
 Kelley, H.H. 193, 254, 257, 293
 Kellner, R. 329
 Kelly, J.B. 422, 423, 425
 Kelly, J.G. 245, 276, 277
 Kelso, D. 396, 399
 Kennedy, S. 329, 330
 Kennell, J.H. 474
 Kessler, R.C. 54, 106, 108, 492, 493, 494, 499, 502
 Kidder, L. 474
 Kiesler, C. 261

- Killilea, M. 362
 King, D.A. 292, 293
 Kissel, S. 293
 Kitt, H.S. 193
 Klaus, M.H. 474
 Kleber, H.D. 329
 Kleiman, R. 311
 Klein, D. 131, 305, 307, 310, 311, 314
 Klerman, G.L. 46, 326, 329, 330
 Kobasa, S.C. 165, 398
 Kobrin, F.E. 443
 Kochen, M. 205
 Kohlberg, L. 174
 Korn, S. 11
 Kosten, T.R. 329
 Kraag, G. 96
 Krant, M.J. 466, 469
 Krauss, S. 311
 Kubler-Ross, E. 399
 Kuchemann, C.S. 467
 Kuhn, T.S. 400
 Kulka, R.A. 493, 494
 Kuo, W. 51, 53
 Kurdek, L.A. 371, 374, 378, 384, 422
 Kutscher, A.H. 473
 Lachar, D. 332
 Lahniers, L.E. 327
 Lakey, B. 156, 163, 295, 392, 395, 405, 426
 Lancee, W.J. 359, 440, 445, 450
 Langner, T.S. 392
 LaRocco, J.M. 304, 391, 395
 Larson, M. 96, 272, 277
 Latane, B. 193
 Launier, R. 5, 6, 61
 Laws, J.L. 188
 Lawton, 237, 248, 259
 Lazarus, R.S. 3, 5, 6, 8, 61, 75, 139, 197, 249, 250, 253, 254, 311, 394, 395, 398, 404, 405
 Lee, E.G. 467
 Lee, T.R. 404
 Lefcourt, H.M. 105, 156, 159, 162, 163, 165, 233
 Leff, J.P. 327, 332
 Leff, L.P. 429
 Lehman, D.R. 463, 476
 Leighton, A. 108
 Leighton, B. 208
 Lennon, R. 493, 498
 Lennox, K. 332
 Lerner, M.J. 465
 Levenson, R.W. 289
 Levin, D. 7, 8
 Levine, H.M. 8, 11, 40, 41, 46, 104, 138, 139, 143, 150, 165, 191, 233, 275
 Levinger, G. 256, 403
 Levy, M.T. 333
 Lewin, 8
 Lewinsohn, P.M. 42, 46, 253, 292
 Lewis, C.E. 397, 419
 Liang, M.H. 96
 Lichtenstein, E. 76, 87
 Liddle, 359
 Lieberman, M.A. 6, 54, 59, 61, 393, 398, 403, 426, 449, 476, 492, 493, 501
 Liem, J.H. 198, 254
 Liem, R. 198
 Lieve, L. 187
 Lin, N. 51, 53, 54, 58, 245, 247, 315, 391
 Lindeman, E. 3, 455
 Lindenthall, J.J. 326, 329, 330, 391, 494
 Lindy, J.D. 397
 Link, B. 305, 502
 Linn, M.W. 400
 Lipman, R.S. 10, 289, 492, 494
 Liskow, B.I. 329
 London, P. 396, 397, 399, 400, 405
 Lopata, H.Z. 237, 270, 271, 276, 440, 419, 421, 444
 Lorr, M. 142, 147
 Lowenthal, M.F. 270, 394, 402, 418, 419, 421, 444
 Luscomb, R.L. 333
 Lyall, W.A.L. 359, 420, 440, 443, 444, 445, 449, 450, 451, 473, 476
 Lykken, D.T. 306
 Lynch, J.J. 441
 MacBride, A. 467
 Maccoby, E.E. 165, 168

- Maddison, D.C. 440, 444, 445,
 452, 463, 470, 471, 472, 473
 Maddox, G.L. 137
 Maeulen, L. 137
 Magnusson, D. 11, 174
 Maguire, G.P. 467
 Makosky, V.P. 494
 Mann, G. 104
 Markson, E.W. 32
 Markush, R.E. 468, 494
 Marlowe-Crowne, D. 142
 Marolla, J. 59
 Marris, P. 247, 468
 Marrow, A.J. 189
 Martin, C.J. 353
 Martin, R.A. 233
 Marton, P. 420, 476
 Marvin, R. 11
 Masciocchi, C.F. 427
 Mason, J.W. 397
 Masuda, M. 391
 Mathews, K.E. 62
 Matthews, G. 465
 Maurice, W.L. 442, 443
 Mazel, J. 137
 McAndrew, I. 11
 McAuley, H. 338
 McCall, G. 118
 McCourt, W.F.
 McCrae, R.R. 137, 138, 142, 150
 McDonald, L.A. 96
 McDonald, P. 162
 McDonald, R. 138
 McDonald, W.J. 137
 McGowan, J. 76
 McGuinness, B. 322
 McKay, G. 3, 5, 10, 54, 62, 74,
 75, 84, 86, 89, 155, 243,
 251, 253, 259, 309, 372, 373,
 374, 441, 426
 McLeod, J.D. 493, 502
 McLoughlin, 378
 McMullen, P.A. 12
 McNair, D.M. 142, 157
 McNemar, Q. 393
 McRae, J.A., Jr., 492, 499
 Mead, G.H. 57, 66, 118
 Mechanic, D. 9, 61
 Medalie, J.H. 29
 Meier, R.F. 252
 Mellstrom, D. 279
 Menaghan, E.G. 6, 54, 59, 61,
 393, 398, 403, 426, 493, 501
 Mendels, J. 330
 Mensch, I.N. 10
 Mermelstein, R. 76, 86
 Merton, R.K. 193, 404
 Mettee, D.R. 194
 Metzner, H.C. 29, 108
 Meyerowitz, B.E. 9, 399
 Meyers, J.K. 391
 Miceli, M. 278
 Michael, S.T. 392
 Michela, J.L. 278, 279
 Miell, D.E. 419
 Miles, D.L. 100
 Milhoj, P. 270, 274
 Miller, D.T. 118, 131, 465
 Miller, P. 314, 334, 338, 350,
 418, 419
 Miller, R.S. 156, 159, 165, 193
 Miller, S.J. 32, 138
 Minde, K. 420, 476
 Minuchin, 256
 Mitchell, G.W. 469, 473
 Mitchell, R.E. 12, 40, 243,
 372, 399, 405, 419
 Mitchell, T. 336
 Monroe, S.M. 10, 303, 305,
 307, 308, 311, 312, 315, 395
 Moore, J.A. 229
 Moore, K.P. 162
 Moore, M.C. 51, 156, 165
 Moos, D.S. 77
 Moos, R.H. 3, 12, 77, 243, 304,
 392, 394, 395, 399, 404, 419
 Morasch, B. 278
 Morgan, L.A. 444
 Morgan, T.S. 234
 Morris, T. 467
 Moss, G.E. 7
 Mosteller, F. 98
 Motulsky, A.G. 1313, 314
 Moulton, R. 175
 Moustakas, C.E. 272
 Mowrer, O.H. 402
 Muellen, D.P. 53
 Mullan, J.T. 6, 54, 59, 61,
 393, 398, 403, 426, 501
 Munari, F. 329

- Munoz, R.F. 245
 Murphy, E. 327, 337, 353
 Murphy, G.E. 333
 Murray, H. 243
 Mussen, P.H. 22
 Myers, J.K. 304, 326, 329, 330, 332, 494
 Naditch, M.P. 289
 Nadler, A. 9, 12, 256, 373, 377
 Nahemon, L. 248
 Navaco, R.W. 250, 260
 Neff, J.A. 51, 156, 165
 Nelson, P. 10
 Nelson, S.A. 234
 Nelson-Legall, S. 12
 Neugarten, B.L. 22, 324
 Newbrough, J.R. 51, 156, 165
 Newcomb, M. 120
 Nezelek, J. 228, 493
 Nock, S.L. 501
 Norris, A.H. 137, 142
 Nuckolls, K.B. 29
 Nuckolls, K.G. 287
 O'Connor, P. 353, 360, 419
 O'Brien, S.E.M. 333
 O'daffer, V.E. 234
 O'Hanlon, J. 338
 Olah, A. 174
 Opler, M.K. 392
 Oppegard, K.M. 234
 O'Reilly, P. 399
 Oxford, J. 399
 Parish, E.A. 229, 236
 Parker, G. 42
 Parkes, C.M. 247, 441, 442, 444, 449, 451, 454, 455, 447, 463, 465, 467, 468, 470, 471, 472
 Parks, A. 424
 Parmelee, P. 248
 Parry, G. 353
 Parson, T. 404
 Parsons, J.E. 187
 Pasnau, R.O. 469
 Pasvan, L. 329
 Patrick, D.L. 99, 100
 Patsiokas, A.T. 333
 Pattison, E.M. 474
 Paykel, E.S. 314, 321, 322, 326, 327, 329, 330, 332, 333, 334, 335, 336, 337, 338, 353, 366
 Pearlin, D. 393
 Pearlin, L.I. 4, 6, 7, 54, 59, 60, 61, 398, 403, 405, 426, 501
 Pennebaker, J.W. 7
 Peplau, L.A. 42, 226, 227, 228, 244, 246, 248, 260, 269, 270, 271, 272, 273, 274, 275, 276, 278, 279, 398
 Pepper, M.D. 391
 Pepper, M.P. 326, 329, 330, 494
 Peretz, D. 473
 Perlman, D. 226, 229, 244, 246, 248, 269, 270, 271, 272, 275, 276, 279, 398
 Perrucci, R. 399
 Perry, J.D. 378, 379
 Persad, E. 330
 Persson, G. 279
 Peters-Golden, H. 466, 468, 469, 470
 Peterson, W.J. 396, 399, 400
 Peto, J. 334
 Phares, E.J. 162
 Pihlblad, C.T. 445
 Pines, A. 197
 Plas, J.M. 188
 Platt, J.J. 46, 106
 Platt, S. 359
 Pokorny, A.D. 333
 Polak, P.R. 1 327, 454
 Pomeroy, M.A. 474
 Pool, I.S. 205
 Porritt, 28
 Poulshock, S.W. 427
 Powers, E.A. 32, 394, 404
 Primavera, J. 398, 406
 Procidano, M.E. 8, 40, 124, 291, 292, 294, 371, 373
 Prudo, R. 352, 353, 361
 Prusoff, B.A. 322, 329, 332
 Pugliesi, K.L. 55
 Quint, J.C. 469
 Quintana, D. 227
 Quinton, D. 359
 Raadschelders, J. 274

- Rabinowitz, V.C. 474
 Rabkin, J.G. 3, 198, 383, 391
 Radloff, L.S. 492
 Rae, D.S. 492
 Rahe, R.H. 60, 321, 326, 440, 441
 Ramsay, R.W. 455
 Ramsay, T.B. 8, 40, 77, 371, 373
 Rands, M. 256
 Rapaport, J. 119
 Raphael, B. 448, 452, 454, 474
 Rapkin, B.D. 402
 Rassaby, E.S. 338
 Raush, H.L. 3
 Ray, C. 467
 Reis, H.T. 228, 254
 Reischl, T.M. 125
 Reissman, F. 260
 Renders, R. 130, 131
 Rennie, T.A. 392
 Renzaglia, G.L. 9
 Reynolds, W.J. 100
 Richardson, B.Z. 62
 Rieder, R. 307
 Riesman, D. 270
 Riley, N.W. 138
 Riordan, C. 329
 Ripley, J. 420, 476
 Ritchie, K. 338
 Robbins, C. 29, 108, 137
 Robbins, E. 442, 443
 Robbins, M.L. 187
 Robertson, J.F. 32
 Robins, E. 292
 Robins, L.N. 3
 Rodgers, W. 473
 Rogers, C.R. 270
 Rogers, J. 359, 420, 440, 443, 444, 445, 449, 450, 451, 473, 476
 Rohde, P.D. 332
 Rokeach, M. 102
 Rook, K.S. 226, 237, 248, 254, 257, 269, 273
 Rorsman, B. 333
 Rose, R.M. 3, 10
 Rose, S. 76, 334, 338
 Rosenberg, H.S. 291
 Rosencranz, H.A. 445
 Rosenthal, R. 394
 Rossi, P.H. 501
 Rotter, J.B. 29, 174
 Roundtree, R. 228, 234
 Rounsaville, B.J. 329
 Roy, A. 329, 330, 337
 Rubenstein, C.M. 272, 273, 274, 276, 280
 Rubin, D.B. 394
 Ruble, D.N. 187
 Ruch, L.O. 392
 Rush, J.A. 46
 Rushing, W.A. 100
 Russell, D. 42, 226, 227, 228, 233, 270, 272, 273, 274, 275, 278, 279
 Rutter, M. 3, 358, 359
 Saleh, W.E. 233
 Salt, P. 391
 Sandler, I.N. 8, 40, 76, 77, 156, 163, 371, 372, 373, 377, 392, 395, 405, 436
 Sansone, C. 229
 Santrock, J.W. 375, 376, 377, 378, 379
 Sarason, B.R. 8, 11, 39, 40, 41, 42, 43, 45, 104, 138, 139, 143, 150, 165, 191, 233, 275
 Sarason, I.G. 3, 8, 11, 39, 40, 41, 42, 43, 45, 47, 104, 138, 139, 143, 150, 165, 191, 233, 275, 391
 Sarbin, T.R. 57, 118
 Schachter, S. 441
 Schaefer, C. 5, 8, 197, 139, 249, 250, 253, 254, 311
 Schaeffer, M.A. 3
 Schank, R. 472
 Scharfetter, C. 324
 Scheier, M.F. 76
 Schmid, I. 324
 Schmidt, N. 274
 Schoenbach, V.H. 350
 Schoenberg, B.B. 473
 Schoenberg, B.M. 455
 Schooler, C. 4, 7, 405
 Schwab, J. 468
 Schwartz, A.N. 10
 Schwartz, C.C. 332

- Scott-Jones, D. 12
Scott, R. 8, 40, 42, 51, 419
Scovern, A.W. 104, 106
Seelbach, W.C. 138
Seligman, E.P. 58, 59, 60
Selye, H. 197, 253
Sermat, V. 229, 274
Shanas, E. 270, 274
Sharbaugh, M.E. 197
Shaver, P. 272, 273, 274, 276, 280
Shaw, B.R. 46
Shaw, M.E. 193
Sheldon, A.R. 359, 440, 445, 450
Sherif, C.W. 189, 191, 193, 199
Sherman, S.E. 62
Sholomaskas, D. 329
Shosenberg, N. 420, 476
Shott, S. 59, 62
Shure, M.B. 403
Sieber, S.D. 57
Siesky, A.E. 384
Silver, R. 3, 463, 467, 471, 473
Silverman, P.R. 448, 449
Simeone, R.S. 51, 53
Simmons, J. 118
Simon, B. 237
Skeleton, J.A. 7
Sklair, F. 321, 322, 326
Slade, K.M. 234
Slater, J.F. 305, 307, 310, 311, 314, 329, 333, 338
Slyter, H. 474
Smith, G. 194
Smith, T.G. 236
Smyer, M.A. 428
Snider, R.K. 236
Snider, R.L. 230, 276
Snow, J. 392, 395
Snowden, L.R. 245
Sobieszek, B. 58
Sokolovsky, J. 40, 430
Solano, C.H. 229, 236
Speedling, E. 491
Spencer, P. 127
Spiegel, D. 476
Spielberger, C.D. 40, 402
Spinner, B. 272, 279
Spitzer, R.L. 292
Spivack, G. 46, 106, 403
Srole, L. 392
Stancer, H.C. 330
Stanley, J.C. 395
Stason, W.B. 98
Stehouwer, J. 270, 274
Stein, M. 333
Steinberg, H.R. 332
Steinitz, E.M. 463, 469, 470
Steinmetz, J.L. 292
Stephens, P.M. 7
Stern, M. 492, 494
Steuve, C. 119
Stewart, A.J. 100, 391
Stewart, J.E. 162
Stokes, J.P. 233, 379, 394
Stotland, E. 62
Stroebe, M.S. 441
Stroebe, W. 441
Stroufe, L.A. 22
Struening, E.L. 3, 198, 303, 391
Struere, C. A. 257
Stryker, S. 57
Sturt, E. 353
Sullivan, H.S. 118, 248, 270, 271, 272
Suls, J.M. 193
Surtees, P.G. 338, 349
Surwit, R.S. 104, 106
Sussman, M.B. 429
Svanborg, 279
Swann, W.B., Jr. 425
Sweetwood, H.L. 10, 334
Swindle, R.W. 3, 29, 288, 291, 293, 373
Sylph, J. 334
Syme, S.L. 29, 73, 79, 443
Talkington, J. 42
Tangri, S. 187
Tanna, V.L. 329
Tanner, J. 327
Targ, D.B. 399
Taub, S.I. 162
Taylor, D.A. 256
Teasdale, J.D. 58, 59
Teichman, Y. 1 397, 402
Tellegen, A. 306, 310, 311
Tennant, A. 306, 310, 311

- Tennant, C. 51, 327, 337, 352,
 353, 391, 395
 Thibaut, J.W. 254, 257
 Thoits, P.A. 3, 51, 53, 54, 55,
 57, 60, 66, 67, 118, 123,
 205, 248, 256, 288, 305, 372,
 398, 400, 463
 Thompson, J. 420, 4765
 Thompson, R. 329, 330
 Thompson, W.D. 304, 329
 Thompson, W.R. 306, 313, 314
 Thurlow, H.I.J. 305, 316
 Tibblin, G. 350
 Tietjen, A.M. 375
 Tillich, P. 1 271
 Titus, L.J. 1 282
 Toldsdorf, C.C. 399, 404, 405,
 430
 Toldsdorf, G. 130
 Toro, P.A. 403
 Townsend, P. 270, 274
 Trickett, E.J. 40, 372
 Troll, L.E. 32, 138
 Tugwell, P. 96
 Tunstall, J. 1 270
 Tupling, H. 42
 Turk, S. 47
 Turner, B. 32
 Turner, R.S. 7, 8, 11, 51, 54,
 66, 243
 Uhlenhuth, E.H. 10, 322, 333,
 492, 494
 Vachon, M.L.S. 359, 420, 440,
 444, 445, 449, 450, 451, 467,
 468, 476
 Vaillant, G.E. 51, 391, 395
 Vandebergh, R. 454
 Vanfossen, B.F. 1 403, 502
 Vaughn, C.E. 327, 332, 424
 Vaux, A. 250, 260
 Veit, C.T. 102
 Veroff, J. 197, 245, 493, 494
 Videka-Sherman, L. 449
 Vogel, F. 313, 314
 VonBaeyer, C.I. 163
 Wagner, E.H. 350
 Walfisch, F. 393, 396, 398, 405
 Walker, W.L. 440, 445, 452,
 463, 470, 471, 472, 473
 Wall, S. 22
 Wallace, A. 474
 Wallerstein, J.S. 374, 376,
 377, 384, 422, 423
 Wallston, B.S. 104
 Walsh, N. 338
 Walster, E. 256, 403, 405
 Walster, G.W. 256, 403, 405
 Walstrom, B.S. 188
 Walum, L. 187
 Ware, E.E. 163
 Ware, J.E., Jr. 51, 97, 100,
 102, 103, 303, 304, 305, 391,
 394, 395
 Warshak, R.A. 375, 376, 377,
 378, 379
 Waters, 22
 Watson, D. 81, 306, 310
 Webber, E.L. 443
 Webster, M. 58
 Wedderburn, D. 270, 274
 Weeks, D.G. 278, 279
 Weinstein, M.C. 98
 Weiss, J.M. 197
 Weiss, R.S. 227, 247, 248, 251,
 260, 261, 270, 271, 272, 273,
 352, 358, 376, 400, 402, 403,
 424, 441, 444, 463, 465, 468,
 470, 471, 472, 499
 Weissberg, R.P. 402
 Weissman, M.M. 46, 304, 338
 Wellisch, D.K. 469
 Wellman, B. 53, 39, 204, 208,
 216, 218, 219, 254, 371, 372,
 418
 Wells, J.A. 7
 Wentowski, 30
 Wenz, F. 443
 Werner, C. 248
 Wetherington, E. 22
 Wheaton, B. 502
 Wheeler, L. 228, 493
 Whitcher-Alagna, S. 256, 373,
 377
 White, K. 327
 White, P. 4671
 White, R.W. 59
 Whitehead, P.C. 400
 Wiener, A. 453, 468

- Wilcox, B.L. 5, 6, 7, 8, 10,
11, 391, 393, 394, 395, 400,
404, 405
- Williams, A.F. 476
- Williams, A.W. 51, 303, 304,
305, 391, 394, 395, 463
- Williams, J.G. 236
- Williams, K.B. 236
- Williams, W.V. 454
- Wills, T.A. 3, 86
- Windholz, G. 279
- Windom, C.S. 187
- Wolchik, S. 377
- Wolfstetter-Kausch, H. 131,
305, 307, 310, 311, 314
- Wolk, S. 162
- Wolpe, J. 253
- Wood, V. 32
- Worden, J.W. 448, 455
- Wortman, C.B. 3, 9, 11, 62,
463, 464, 465, 466, 467, 469,
471, 473, 475, 483
- Wortman, C.G. 254, 256
- Wright, M.H. 162
- Wylie, R. 128
- Yager, J. 10, 334
- Yalom, I. 476
- Young, D.M. 376, 377
- Young, J.E. 271, 272, 274, 279
- Young, R.D. 291
- Zarling, C. 122
- Zeller, G. 330
- Zilboorg, G. 270
- Zuckerman, M. 395